Quadratic maps satisfying $f \circ f = f + \mathrm{id}$: nilpotence, Lie reconstruction, and growth

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Abstract

The equation $f \circ f = f + \mathrm{id}$ couples three-cycles to vector addition: each nonzero orbit is a binary plane with its zero removed. We prove that every finite binary quadratic solution has a nilpotent polar Lie algebra and a quadratically stable distinguished biderivation. Its canonical restricted derived quotient preserves all linear intertwiners; finite free covers have exact restricted $\mathbb{F}_4$-structures on their derived ideals, with square $P^2 = 0$. Ordinary Lie cohomology governs central reconstruction. A rank-preserving realization of ordinary three-Engel $\mathbb{F}_4$-Lie algebras gives a two-sided comparison of class growth. It yields a quartic class bound in point rank and logarithmic maximal derived length; nonmetabelian models reach class c in dimension $O(2^{c/2})$. We determine the highest-weight central lifts of universal metabelian three-Engel algebras. For ordinary three-Engel algebras with $d \geq 6$ generators and second derived ideal in the fourth upper center, the sharp class bound remains $d+2$. Explicit low-rank towers exhibit delayed growth of class. In the metabelian range, an intrinsic change of addition gives an equivalence with metabelian three-Engel $\mathbb{F}_4$-Lie algebras. We determine sharp minimum dimensions and equality cases, together with doubly exponential near-minimum populations and a normalized-surplus threshold $\sqrt{2}$ in the strict subcritical and supercritical regimes. Finally, quadratic linearization has inverse degree bounded by polar class, while the unoriented orbit spread recovers the rule up to global reversal and determines the polar upper central series.

Keywords. Quadratic map; nilpotent Lie algebra; restricted Lie algebra; central extension; Lie cohomology; isomorphism growth; binary line spread.

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1 Introduction

Let V be a finite-dimensional vector space over $\mathbb{F}_2$. We study the equation

$$f(f(x)) = f(x) + x \qquad (x \in V) \tag{1}$$

for maps of binary algebraic degree at most two. Every nonzero orbit is $\{x, f(x), x + f(x)\}$, the three nonzero points of a binary plane. The equation ties a local cycle to the addition on the whole space. Our aim is to identify the structure forced by this interaction, reconstruct the rule from that structure, and measure its dimension and diversity.

The linear solutions already reveal a scalar field: their minimal polynomial is $X^2 + X + 1$, so they are multiplication by $\omega \in \mathbb{F}_4$, where $\omega^2 + \omega = 1$. Degree two introduces interactions between the orbit planes. They are recorded by the polar form

$$B(x, y) = [x, y] = f(x + y) + f(x) + f(y).$$

The functional equation forces this alternating bilinear form to satisfy Jacobi. It therefore equips V with an intrinsic Lie algebra $L = L_f$. The general theory starts with global nilpotence and a quadratic stability theorem for the distinguished polar biderivation. Together they give a restricted algebraic core, retain the central data needed for reconstruction, and organize the dimension and growth problems. The metabelian subcategory admits exact classification and enumeration.

1.1 Global nilpotence and general structure

Every finite quadratic solution has a nilpotent polar Lie algebra (Theorem 4.1). The proof derives biderivations and centroid constraints from (1), excludes every possible nonabelian simple polar factor, and uses Premet's Cartan-subalgebra theorem at the binary-centroid endpoint [34, Theorem 3(ii)]. This gives every solution a lower-central filtration with intrinsic $\mathbb{F}_4$-scalars on its graded factors.

To measure size, define the *point rank* to be the least number of points generating V under binary addition and f. Lemma R.7 identifies it with

$$d = d(L) = \dim_{\mathbb{F}_4}(L/L').$$

A choice of point generators gives at most $2d$ ordinary binary Lie generators, by adjoining their images under f.

The second structural result controls the nonlinearity of the distinguished biderivation

$$D(x, y) = [f(x), y] + [x, [x, y]].$$

For every quadratic refinement q_D, the composite $q_D f$ is again quadratic (Theorem 5.6). Its proof joins the restricted structure of $E = L'/(L' \cap Z(L))$ to exact mixed identities, then removes the remaining abelianization term by a finite-geometric character argument and the cyclic norm. The stability statement holds in every polar class.

On a finite free solution, this theorem gives a unique exact scalar factor $D = JB$ on the derived ideal, with restricted square $P = f + J$ and $P^2 = 0$ (Theorem 5.8). For an arbitrary solution the corresponding intrinsic map is a central quotient

$$j_f : L' \twoheadrightarrow Q_f, \qquad P_Q = f_Q + \omega\,\mathrm{id}, \qquad P_Q^2 = 0.$$

It is functorial for every binary-linear solution intertwiner (Theorem 5.10). Its central kernel, the abelianization action, and the lifting refinements carry the data beyond this restricted core. The class-four central-factorization example specifies why the quotient is needed. The

same tensor relations extend Q_f to a restricted envelope of the whole solution, with precisely this central kernel and with every linear intertwiner preserved (Theorem 5.16). When the kernel vanishes, this gives a faithful realization of the solution inside a restricted Lie algebra.

The reduced adjoint action gives a complementary matrix realization: the solution maps onto a binary ideal in a standard restricted model, with kernel of class at most three and abelian cokernel (Theorem 5.15). The point is to retain both descriptions: the functorial derived core handles scalar compatibility, and the matrix model controls the ambient action and growth.

These structures also connect the general quantitative problem to ordinary three-Engel Lie theory. Let $\kappa(d)$ be the maximum polar class at point rank at most d, and let $e(d)$ be the maximum class of a finite-dimensional $\mathbb{F}_4$-Lie algebra generated by at most d ordinary Lie generators and satisfying the three-Engel identity. Theorem 6.5 gives the same-rank comparison

$$e(d) - 1 \leq \kappa(d) \leq e(d) + d + 1.$$

For the reverse direction, a restricted adjoint envelope carries a quadratic orbit map; its binary subsolution generated by the original marked generators retains the envelope's class. Thus the realization preserves point rank without asserting an equivalence with bare Lie algebras. The forward direction uses the full derived action of a finite free cover and a short commuting action on its derived center. It follows that a linear bound for one class-growth function is equivalent to a linear bound for the other. Traustason's classical larger-field estimate [40] then yields

$$\kappa(d) \leq \left\lfloor \frac{(d+1)^4}{2} \right\rfloor + d + 1 \qquad (d \geq 2)$$

in Theorem 6.2.

Within ordinary three-Engel theory over $\mathbb{F}_4$, the first nonmetabelian layer has a sharp class law. Let $e_{\mathrm{cbm}}(d)$ be the maximum class at ordinary generator rank at most d, subject to $H'' \subseteq Z(H)$. Then

$$e_{\mathrm{cbm}}(d) = \max(5, d+2) \qquad (d \geq 2).$$

Theorem 6.7 supplies attaining objects at every exact rank. For $d \geq 4$, the construction determines all highest-weight central lifts of the universal metabelian object: a determinant-twisted symmetric pairing parametrizes the full space, and every such lift is three-Engel (Theorem 6.6). Thus this central-derived layer has an exact linear ceiling; growth beyond it requires a noncentral second derived ideal.

The ceiling persists beyond a central second derived ideal:

$$H'' \subseteq Z_s(H) \implies \operatorname{cl} H \leq d+2 \quad \text{for } (s,d) = (2,\geq 4), (3,\geq 5), (4,\geq 6).$$

Theorem 6.8 proves these bounds sharply. At rank five, three central lifts give the class pattern $7 \to 7 \to 7 \to 8$; for every $d \geq 6$, three lifts over the complete centre-by-metabelian object still cannot pass $d+2$. The proof combines explicit collecting identities, a next-to-top Koszul kernel, and nonlinear rank propagation. It therefore isolates a concrete structural barrier inside the unrestricted Engel class problem.

The trace-restricted current family has

$$d = m + 1, \qquad \operatorname{cl}(L_m) = 2m + 1, \qquad \dim_{\mathbb{F}_2} L_m = 6 \cdot 2^m - 4$$

(Theorem 6.11). Its central truncations yield $m(c) = O(2^{c/2})$, where $m(c)$ is the minimum binary dimension at polar class c. In every class $c \geq 5$, these constructions have dimension below the metabelian minimum. The maximum derived length at point rank at most d is $\Theta(\log d)$. The category of finite solutions also admits a finite free object at each finite point rank (Theorems 6.13 and 6.14).

1.2 Reconstruction and central lifting

The original rule has a canonical linear carrier over $\mathbb{F}_4$:

$$M_f = \operatorname{span}_{\mathbb{F}_2}\{(x, f(x)) : x \in V\}, \qquad \mathcal{J}(x, y) = (y, x + y).$$

Its marked graph and projection recover f, and the graph's additivity defects recover its polar bracket. The carrier retains every binary-linear intertwiner (Proposition 13.8).

The nilpotent filtration then reconstructs every solution by central lifts. With a marked coefficient line $(\mathbb{F}_4, \omega)$, a lift over g is

$$F_q(x, z) = (g(x), \omega z + q(x)), \qquad qg = \omega^2 q.$$

Its admissible class space H_g embeds in ordinary binary Lie cohomology. On the quadratic-stable subspace K_g, composition with g induces a canonical cyclic operator τ, and $H_g = (K_g)_{\omega^2}$. Correcting the antiscalar central defect of a marked Lie-extension arrow recovers all solution homomorphisms (Theorems 13.2–13.6).

Every invariant quadratic factors through abelianization. Thus the fixed cyclic sector has dimension d^2, where $d = d(L_g)$, in every polar class, and

$$b_2(g) = \operatorname{rank} \Theta_g + d^2 + 2 \dim_{\mathbb{F}_4} H_g,$$

where Θ_g detects degree leakage under composition. For free class-three quotients we determine the complete admissible spectrum, the ordinary second Betti number, and the entire leakage rank. Their nonmetabelian directions have a uniform class-four realization; almost every conjugacy type in this family is nonmetabelian as the rank grows (Section 14).

The constructions retain morphisms at the following precise levels.

Construction	Morphisms retained
Metabelian equivalence and marked graph carrier	All binary-linear solution intertwiners.
Canonical restricted quotient Q_f	All binary-linear solution intertwiners.
Reduced derived algebra E and matrix completion	Solution isomorphisms and surjective homomorphisms.
Marked central reconstruction	All marked homomorphisms, after the antiscalar central correction.

1.3 Exact metabelian dimensions and populations

When $W = L'$ is abelian, commuting square-zero operators and a cocycle encode every interaction. In suitable binary-linear coordinates, the complete normal form is

$$f(x, z) = \big(\omega x, \omega z + P(x)z + Q(x) + P(x)Q(x)\big), \tag{2}$$

where $P(x) = \sum x_i P_i$, $Q(x) = \sum_{i<j} x_i x_j a_{ij}$, the P_i commute and square to zero, and $P_i a_{jk} + P_j a_{ik} + P_k a_{ij} = 0$. The coefficients generate W under these operators. The correspondence is intrinsic: a corrected addition on the same set gives an equivalence with metabelian 3-Engel $\mathbb{F}_4$-Lie algebras, preserving all homomorphisms (Theorem 3.6).

The universal commutator module gives the sharp metabelian minimum

$$\mu(2) = 6, \qquad \mu(3) = 8, \qquad \mu(4) = 12, \qquad \mu(c) = 2^c + 2c - 6 \quad (c \geq 5). \tag{3}$$

Compatibility asks whether the prescribed scalars extend to make the *original* polar bracket $\mathbb{F}_4$-bilinear. In this subclass,

$$\nu(c) = 2^c + 2c - 2 \qquad (c \geq 2). \tag{4}$$

Consequently compatibility has an exact extra cost of four binary dimensions for $c \geq 5$. Both minima and their equality cases are determined completely. The metabelian minimum has linearly many types in c, while the compatible minimum has doubly exponentially many (Theorems 7.6 and 11.7).

The same abundance appears at the first surplus. If $A_{c,r}$ counts metabelian types in binary dimension $\mu(c) + 2r$, then for every fixed $r \geq 1$,

$$\log_4 A_{c,r} = r2^{c-1} + O_r(c^2).$$

The generic object has minimum derived dimension, faithful quotient action, and a one-dimensional $\mathbb{F}_4$-center; it is indecomposable and nonrestrictable. Theorems 8.3 and 8.8 give logarithmic and multiplicative counts.

As the surplus grows, its allocation undergoes a sharp transition. Put $s = c - 1$, and let $\pi_{s,r}$ be the proportion of binary conjugacy types having minimum derived dimension. Then, as $s \to \infty$,

$$\limsup \frac{r}{2^{s/2}} < \sqrt{2} \implies \pi_{s,r} \to 1, \qquad \liminf \frac{r}{2^{s/2}} > \sqrt{2} \implies \pi_{s,r} \to 0.$$

Theorem 9.2 proves this threshold. Beyond it, Theorem 9.5 identifies the limiting allocation, and Theorem 9.7 resolves the discrete critical staircase. The corresponding results hold over every fixed finite characteristic-two field of order greater than two, with Lie isomorphism over that field. In those counting sections the local scale $d = c-1$ is written explicitly; it is distinct from the point rank $d(L)$ of an individual object.

1.4 Classification, linearization, and orbit geometry

Global nilpotence reduces the eight-dimensional classification to the metabelian correspondence, giving exactly three binary-linear conjugacy classes. The entire stratum with binary abelianization dimension four is also classified: one class occurs in each of dimensions $4, 6, 8, 10, 12, 14, 16$, and none thereafter. Its final two members are nonmetabelian. Central descent and exact extension-space certificates establish completeness (Section 12).

For every finite solution, the lower-central graded scalars lift to a filtration-adapted operator J_0. The quadratic permutation $H_0 = f + J_0^2$ linearizes f, with inverse degree at most the polar class. In the metabelian normal form, this is precisely the change of addition defining the Lie correspondence. The linearization section also records the resulting sharp distinction between linear measurements and linear decoding (Section 15).

The unoriented orbit spread retains the polar bracket and determines its quadratic orientation up to global reversal $f \leftrightarrow f^2$. Its normal planes are exactly the central orbit planes, so recursive normal-element reduction recovers the upper central series. The invariant $x \mapsto x \wedge f(x)$ separates all orbits; in the nonlinear case the separating degree is exactly three (Section 16).

1.5 Context and related work

On complete mappings, $\mathcal{F}(g) = (\mathrm{id} + g)^{-1}$ is the classical parastrophic action of order three [30, Theorem 2.3]; its strict fixed objects satisfy (1). The arithmetic study in [22] also describes these objects by oriented binary line spreads. Triple-cycle permutations and regular complete permutation polynomials are studied in [24, 41, 26], and quadratic complete mappings in [37]. Here the orbit-sum identity, binary degree, and original binary addition are considered together. Example 2.8, from the switching construction of [24, Theorem 22], shows that simultaneous three-cycle behavior of f and $f + \mathrm{id}$ alone does not give the orbit-sum identity.

The binary operation $x * y = x + f(x + y)$ is a Mendelsohn quasigroup satisfying $(x * y) * y = y * x$. This is the *quaternary* identity in Zhang's terminology [42, Definition 7.8]. Zhang

develops central representations and recursions in that setting, including the central six-dimensional example in Lemma 10.3(1). We retain the external binary addition and its degree filtration, and measure nilpotency in the original polar Lie algebra. The global theorem supplies that nilpotency for every finite quadratic solution.

The Lie-theoretic tools have established sources. Traustason's partial linearization identifies the metabelian 3-Engel category [39, Section 1.2], and his collection estimates provide the rank bounds used in general growth [40]. The trace-current construction uses the classical simple Lie algebra and derivations in [38], and restricted squares and quadratic maps are related in [31, 32]. Centroid descriptions of biderivations in the perfect, centerless setting are developed in [6]. The present stability argument instead uses the orbit equation and binary degree to reach exact scalars on free derived ideals and a functorial central quotient. Its finite-geometric step uses restrictions to low-dimensional subspaces, a method also present in lifting of function codes [16]. Cohn's construction [8, Section 2] contains the regular compatible model and the zero-λ slice of the metabelian extremizers; Section 7.4 gives the finite identifications and distinguishes the nonzero-λ models.

The universal commutator module has the classical Koszul presentation [33, Theorem 6.2]; see also the characteristic-two methods of [20]. Ordinary Lie extensions [5], restricted-square constructions [17], and the nilpotent implicit-function principle [3] enter the reconstruction arguments. Lie-ring operations for automorphic loops provide a related change-of-operation framework [18, Section 5]. The degree-four polar identity is the classical standard Lie polynomial [12, Theorem 9]; its forced occurrence is one part of the passage from the orbit equation to Lie structure.

For enumeration, Bahturin–Olshanskii [4, Section 8] study generic algebras in a fixed central primitive class as rank grows. Our populations instead occupy dimension layers above a class-dependent minimum. The commutator module also has central elements below its last lower-central term, so its bounded-defect analysis uses a different central structure. The resulting counts track full isomorphism layers and their dimension allocation.

A companion paper [23] studies fixed-rank ideal growth of metabelian restricted Lie algebras without bounding restricted-power height. It determines the quadratic main term, its rank-dependent correction, and the three-term rank-two expansion. It also describes typical quotient layers and proves that, for two generators, the fraction of ideals with a p-nilpotent quotient is $q^{-n/2+O(\sqrt{n})}$. The connection is the commutator-module and restricted-hull interface; the companion develops the full ideal-growth and probability proofs. The present paper retains the orbit equation, its structural realization, and the near-minimum three-Engel populations. These are distinct source classes and counting regimes.

1.6 Conventions and reading routes

Composition powers are denoted by f^j. Quadratic means binary degree at most two, including the linear cases. Unless a field is specified, dimensions are binary. Equivalence means binary-linear conjugacy; a solution homomorphism is a binary-linear intertwiner. Lie brackets are alternating and hence symmetric in characteristic two. We use $\gamma_1(L) = L$, $\gamma_{i+1}(L) = [L, \gamma_i(L)]$, and $L^{(0)} = L$, $L^{(i+1)} = [L^{(i)}, L^{(i)}]$. A nonzero abelian algebra has class one. Symbols attached to a normal form or matrix action are local to that construction. The distinguished biderivation is always D; the functional defect operator is Δ_g, and the counting parameter dimension is D_{par}.

Symbol	Meaning
$L, W = L', U = L/W$	Original polar Lie algebra, derived ideal, and abelianization.
$d(L), w, w_B$	Point rank $\dim_{\mathbb{F}_4} U$; metabelian derived dimension over $\mathbb{F}_4$; binary derived dimension.
$m(c), \mu(c), \nu(c)$	Minimum binary dimensions at class c: general, metabelian, and compatible metabelian.
E, J, P	Reduced derived algebra $W/(W \cap Z(L))$, its scalar operator, and restricted square.
D, Q_f, P_Q	Distinguished biderivation, functorial restricted derived quotient, and its square.
$\kappa(d), e(d)$	Maximum polar class at point rank at most d, and maximum ordinary three-Engel $\mathbb{F}_4$-Lie class at at most d generators.
$e_{\mathrm{cbm}}(d)$	Maximum ordinary three-Engel $\mathbb{F}_4$-Lie class with $H'' \subseteq Z(H)$, at at most d ordinary generators.
$D_{d,s}, M_j$	Free ordinary three-Engel central-depth quotient and the successive actions on its second derived ideal.
$\ell_{\max}(s)$	Maximum polar derived length at point rank at most s.
$N, \mathcal{T}, I$	Scalar adjoint envelope, its restricted quotient, and the kernel of $L \to \mathcal{T}$.
$R, \mathfrak{m}$	Square-zero polynomial ring and its augmentation ideal.
$E_c, A_{c,r}, N_c$	Binary conjugacy counts at $\mu(c)$, $\mu(c) + 2r$, and $\nu(c)$.
$B_{c,r}(q), \pi_{s,r}(q)$	Metabelian 3-Engel isomorphism count and minimum-derived proportion over $\mathbb{F}_q$; $s = c - 1$.
H_g, K_g, τ, Θ_g	Admissible central lifts, quadratic-stable ordinary cohomology, its cyclic action, and degree leakage.
Δ_g, δ	Functional defect $T + \omega^2 \,\mathrm{id}$ and linear section gauge $T + \omega \,\mathrm{id}$.
$D_{\mathrm{par}}, \varphi(\alpha)$	Counting parameter dimension $2^d - d - 3$ and the macroscopic population profile.
b_2, i_g, h_g	Ordinary binary second Betti number, fixed zero-constant quadratic dimension, and admissible lift dimension; the last two use $\mathbb{F}_4$.

The *structure route* passes through Sections 2–6, then Sections 13–16. The *exact classification and counting route* starts with the metabelian correspondence in Section 3 and continues through Sections 7–12, including fixed-surplus populations in Section 8 and the critical transition in Section 9. The following table locates the longer proofs.

Main results	Detailed proofs
Global nilpotence; reduced restricted structure	Appendices N and O.
Global quadratic stability; canonical restricted core	Appendix P.
Same-rank Engel comparison, the central-derived boundary and general growth	Appendices R, S, and W.
Metabelian and compatible equality cases	Appendices B and A.
Near-minimum counts and critical transition	Appendices C, E, and I.
Central reconstruction and fixed-sector rigidity	Appendices J and T.
Quadratic linearization and orbit geometry	Appendices L and U.

Exact certificates and computational controls are collected in Appendix X; the general proofs specify the interfaces they use.

2 The polar Lie algebra

Proposition 2.1. *Every solution of* (1) *satisfies*

$$f^3 = \mathrm{id}, \qquad f(0) = 0, \qquad f(x + f(x)) = x.$$

Its only fixed point is zero. Every nonzero orbit spans a binary plane, and these planes partition $V \setminus \{0\}$*. In particular* $\dim V$ *is even.*

Proof. Evaluate (1) at $f(x)$. It gives $f^3(x) = f^2(x) + f(x) = x$, so f is bijective. If $a = f(0)$, then $f(a) = a$; applying f^3 at zero gives $a = 0$. A fixed point in (1) must be zero. The remaining identity follows from $x + f(x) = f^2(x)$. The nonzero orbits have length three and sum zero, and $3 \mid 2^{\dim V} - 1$ implies the parity assertion. □

Theorem 2.2. *Let* f *be a quadratic solution. Its polar form* $[x, y] = f(x + y) + f(x) + f(y)$ *is a Lie bracket. It satisfies*

$$[x, f(x)] = 0, \tag{5}$$

$$f([x, y]) = [f(x), f(y)] + [x, y] + [f(x) + f(y), [x, y]], \tag{6}$$

$$[[a, b], [c, d]] + [[a, c], [b, d]] + [[a, d], [b, c]] = 0. \tag{7}$$

Proof. The polar is alternating and bilinear because f is quadratic and $f(0) = 0$. Expand $f(x + f(x)) = x$ by polarization. The terms $f(x) + f^2(x)$ sum to x, leaving (5). Take the third additive difference at zero of $x \mapsto [x, f(x)]$. The terms involving individual values of f cancel, leaving

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

This is Jacobi.

Next polarize $f^2 = f + \mathrm{id}$. Since $f(x + y) = f(x) + f(y) + [x, y]$, the expansion of the left side gives (6).

For (7), introduce Boolean variables $t_1, \ldots, t_4$ and vectors $x_1, \ldots, x_4$. The exact expansion is

$$f\Big(\sum_i t_i x_i\Big) = \sum_i t_i f(x_i) + \sum_{i<j} t_i t_j [x_i, x_j].$$

In applying f again, the coefficient of $t_1 t_2 t_3 t_4$ can only come from a polar cross term between two disjoint pair summands. There are three such pairings, giving the left side of (7). The composition has degree at most two, so this coefficient vanishes. All coefficient comparisons take place in the Boolean function algebra, with $t_i^2 = t_i$. □

The two polarization steps have distinct roles. Binary degree two gives an alternating bilinear form; the commuting identity (5) forces its Jacobi identity. The fourth polarization then supplies (7), an additional restriction on the resulting Lie algebra. The polynomial in (7) is the associative standard polynomial of degree four expressed as a Lie element in characteristic two [12, Theorem 9]. Its forced occurrence here gives the explicit exclusions in Section 12.

Proposition 2.3. *All terms of the derived, lower central, and upper central series of the polar Lie algebra* $L = (V, [\ ,\ ])$ *are* f*-invariant. The center is* f*-invariant and* f *is linear on it. Every nonzero factor of any of these series has even binary dimension.*

Proof. Let I be an f-invariant ideal. In (6), take $x \in I$. Each term on the right belongs to $[I, L]$, which is an ideal. Quadratic expansion extends invariance from bracket generators to their span: the extra polar terms also lie in that ideal. The same argument with $x, y \in I$ proves invariance of $[I, I]$, using that it is an ideal of L. Induction starts with $I = L$.

Polarization of (5) gives

$$[x, f(y)] + [y, f(x)] + [x, [x, y]] + [y, [x, y]] = 0. \tag{8}$$

If $x \in Z(L)$, this says $[y, f(x)] = 0$ for every y. Thus $Z(L)$ is invariant; its polar is zero, so f is linear there. Pass successively to quotients by invariant central ideals to obtain the upper central assertion. Every series factor inherits a solution of (1), so Proposition 2.1 proves its parity. □

Lemma 2.4 (Ideal descent)**.** *For every Lie ideal I of L,*

$$\{f(x) + I : x \in I\} \subseteq Z(L/I).$$

Every perfect ideal is f-invariant. Every ideal I for which $Z(L/I) = 0$ is also f-invariant.

Proof. For $x \in I$, all terms in (8) except $[y, f(x)]$ belong to I. This proves the inclusion. The map $\phi : I \to L/I$, $x \mapsto f(x) + I$, is binary-linear because its polar takes values in I. Equation (6) gives $\phi([x, y]) = [\phi(x), \phi(y)]$ for $x, y \in I$. Its image is central, so $\phi([I, I]) = 0$. This proves the perfect-ideal assertion. The centerless-quotient assertion follows immediately from the inclusion. Bijectivity and finiteness turn each invariant inclusion into equality. □

Corollary 2.5. *If a quadratic solution has a nonsolvable polar algebra, then there is a quadratic solution whose polar algebra is nonabelian simple, of even binary dimension no larger than the original dimension.*

Proof. The stabilized derived ideal P is nonzero and perfect, hence invariant. Choose a maximal proper Lie ideal M of P. The quotient P/M is nonabelian simple and centerless. Apply Lemma 2.4 inside P to see that M is invariant. Restriction and quotient therefore give the required quadratic solution. □

Lemma 2.6. *A nonabelian nilpotent polar Lie algebra has binary abelianization dimension at least four.*

Proof. The abelianization is a nonzero even-dimensional space with the linear action induced by f. If its dimension were two, any nonzero coset $x + L'$ and its image $f(x) + L'$ would form a binary basis. The subalgebra H generated by $x, f(x)$ is abelian by (5). Moreover $L = H + L'$. Inductively, $L = H + \gamma_j(L)$ implies $L = H + \gamma_{j+1}(L)$, because $[H + \gamma_j, H + \gamma_j] \subseteq H + \gamma_{j+1}$. Nilpotency then gives $L = H$, a contradiction. □

Proposition 2.7 (Canonical graded scalars)**.** *For any quadratic solution, the operators induced by f make $\mathrm{gr}_\gamma L$ canonically an $\mathbb{F}_4$-Lie algebra. This assertion concerns the lower-central graded algebra and requires neither metabelianity nor nilpotency.*

Proof. Write F_i for the induced operator on γ_i/γ_{i+1}. Proposition 2.3 and $[\gamma_i, \gamma_i] \subseteq \gamma_{i+1}$ show that F_i is linear and $F_i^2 + F_i = \mathrm{id}$. For $x \in \gamma_i$, $y \in \gamma_j$, reduce (8) modulo γ_{i+j+1} to get $[F_i\bar{x}, \bar{y}] = [\bar{x}, F_j\bar{y}]$. The double brackets have depths at least $i + j + 1$. Equation (6) on the same quotient gives

$$F_{i+j}[\bar{x}, \bar{y}] = [F_i\bar{x}, F_j\bar{y}] + [\bar{x}, \bar{y}].$$

Apply the first identity to $(\bar{x}, F_j\bar{y})$, then use $F_j^2 = F_j + \mathrm{id}$. The right side is $[\bar{x}, F_j\bar{y}]$, proving $[F_i\bar{x}, \bar{y}] = [\bar{x}, F_j\bar{y}] = F_{i+j}[\bar{x}, \bar{y}]$. Thus the graded bracket is $\mathbb{F}_4$-bilinear. □

The induced scalars are intrinsic. A lift to the original space need not make its polar bracket $\mathbb{F}_4$-bilinear; that stronger condition is the subject of Section 10. Theorem 4.1 shows that this filtration exhausts the whole algebra for every finite solution.

Example 2.8. Let $\mathbb{F}_{16}$ contain $\mathbb{F}_4$, let $\eta \in \mathbb{F}_4 \setminus \mathbb{F}_2$, and write $\chi(x) = 1$ off $\mathbb{F}_4$ and $\chi(x) = 0$ on $\mathbb{F}_4$. The switching construction of [24, Theorem 22], with seed $x \mapsto \eta x$, direction $\gamma = 1$, and Boolean selector χ, gives the following map; it is also included in [41, Corollary 6]:

$$G(x) = \eta x + \chi(x) = \eta x + \mathrm{Tr}_{\mathbb{F}_{16}/\mathbb{F}_4}(x)^3.$$

The trace is binary-linear, and its cube is binary-quadratic. Both $\mathbb{F}_4$ and its complement are invariant under multiplication by η and under translation by one. Hence G is a permutation with $G^3 = \mathrm{id}$ and only zero fixed. The same holds for $G + \mathrm{id}$. Nevertheless

$$G^2(x) + G(x) + x = \eta\chi(x),$$

so G does not satisfy (1). This separates the equation studied here from simultaneous triple-cycle complete permutation behavior even within binary degree two.

3 A complete metabelian correspondence

Assume in this section that the polar Lie algebra is metabelian. Then $W = L'$ is abelian, $f|_W$ is linear, and the induced map on $U = V/W$ is linear. Both operators satisfy $T^2 + T = \mathrm{id}$. We give U and W the $\mathbb{F}_4$-structures in which these operators are multiplication by ω.

Lemma 3.1 (The operator pencil)**.** *In any binary-linear splitting of $V \to U$, write*

$$f(x, z) = (\omega x, \omega z + L_x z + q(x)). \tag{9}$$

The binary-linear pencil $x \mapsto L_x$ satisfies

$$L_x\omega = \omega L_x, \quad L_x^3 = 0, \quad L_{\omega x} = \omega L_x + L_x^2, \quad L_x L_y = L_y L_x, \quad L_x^2 L_y = 0.$$

Proof. Put $J = \omega\,\mathrm{id}_W$ and $M_x = J + L_x$. Both M_x and $\mathrm{id} + M_x$ are invertible: they are the restrictions on coset differences of the permutations f and $f + \mathrm{id} = f^2$. Substitution in (1) gives

$$M_{\omega x} M_x = \mathrm{id} + M_x.$$

For $A = M_x$, the three matrices along the ω-orbit are $A, \mathrm{id} + A^{-1}, (\mathrm{id} + A)^{-1}$. Their sum is J, because the three L-terms sum to zero. Thus J is a rational function of A, and in particular commutes with it. Clearing the denominators gives

$$JA(\mathrm{id} + A) = A^3 + A + \mathrm{id}\,.$$

Since $J^2 + J = \mathrm{id}$ and $J^3 = \mathrm{id}$, this is $(A + J)^3 = 0$. It follows that L_x commutes with J and is cube zero. Expanding $(J + L_x)^{-1}$ now gives $L_{\omega x} = JL_x + L_x^2$. The additivity of the left side forces the L_x's to commute.

For $P = L_x, Q = L_y$, the identity $(P + Q)^3 = 0$ yields $P^2Q = PQ^2 =: T$. Apply the same identity to $P' = JP + P^2 = L_{\omega x}$ and Q. It gives $(J^2 + J)T = P^2Q^2$, hence $T = P^2Q^2 = PT$. As P is nilpotent, this implies $T = 0$. □

Theorem 3.2 (Normal form and equivalence)**.** *Every metabelian quadratic solution is binary-linearly conjugate to*

$$f(x, z) = \big(\omega x, \omega z + P(x)z + Q(x) + P(x)Q(x)\big) \quad \text{on } \mathbb{F}_4^d \oplus W, \tag{10}$$

where

$$P(x) = \sum_i x_i P_i, \qquad Q(x) = \sum_{i<j} x_i x_j a_{ij}, \tag{11}$$

$$P_i P_j = P_j P_i, \qquad P_i^2 = 0, \tag{12}$$

$$P_i a_{jk} + P_j a_{ik} + P_k a_{ij} = 0 \quad (i<j<k), \tag{13}$$

$$W = \sum_i P_i W + \operatorname{span}_{\mathbb{F}_4}\{a_{ij} : i<j\}. \tag{14}$$

Conversely these conditions define a quadratic solution with intrinsic polar image W.

The category of these maps with all binary-linear intertwiners is equivalent to the category of finite-dimensional metabelian $\mathbb{F}_4$-Lie algebras $\mathfrak{l}$ satisfying

$$(\operatorname{ad} u)^2|_{\mathfrak{l}'} = 0 \qquad (u \in \mathfrak{l}). \tag{15}$$

with all $\mathbb{F}_4$-Lie homomorphisms as morphisms. For corresponding objects, $C_{\mathrm{GL}(V,2)}(f) \cong \mathrm{Aut}_{\mathbb{F}_4}(\mathfrak{l})$. Their lower central dimensions agree after multiplication by two.

Proof. Lemma 3.1 provides the preliminary actions. We first normalize a section and separate the scalar weights of the cocycle (Steps 1–2). The decisive step is to combine binary degree two with intrinsic generation to force square zero (Step 3). We then verify sufficiency, construct the auxiliary Lie algebra and compare lower central series (Steps 4–5). Finally, tracking changes of coordinates upgrades the normal form to an equivalence on morphisms (Step 6).

Step 1: a normalized section. Choose an $\mathbb{F}_4$-basis e_i of U, with arbitrary lifts v_i in V. Define a binary-linear section by $e_i \mapsto v_i$, $\omega e_i \mapsto f(v_i)$. The images form a basis modulo W. Since $f(v_i + f(v_i)) = v_i$, the cocycle q in (9) vanishes on each coordinate $\mathbb{F}_4$-line.

Step 2: scalar weights. Initially put $P_i = L_{e_i}$, without yet asserting $P_i^2 = 0$. Lemma 3.1 gives

$$L_{\alpha e_i} = \alpha P_i + (\alpha + \alpha^2) P_i^2 \quad (\alpha \in \mathbb{F}_4).$$

Set $N(x) = \sum_i x_i (P_i + P_i^2)$. Then $L_x = N + N^2$, $N^3 = 0$, and $L_{\omega x} = \omega N + \omega^2 N^2$. Decompose $q = q_0 + q_1 + q_2$ into the three scalar weights $q_j(\omega x) = \omega^j q_j(x)$. Projection to each weight is a linear combination of $q(x), q(\omega x), q(\omega^2 x)$, so preserves binary degree and the vanishing on coordinate lines.

The cocycle equation is

$$q(\omega x) = \omega^2 q(x) + (\omega N + \omega^2 N^2) q(x).$$

Its weight components give

$$q_0 = N q_2 + \omega N^2 q_1, \qquad q_1 = \omega N q_0 + \omega^2 N^2 q_2.$$

Using $N^3 = 0$ yields

$$q_0 = N q_2, \qquad q_1 = N^2 q_2. \tag{16}$$

Step 3: binary degree forces square zero. The reduced $\mathbb{F}_4$-polynomial of a weight-two binary-quadratic map has the form $\sum_i b_i x_i^2 + \sum_{i<j} a_{ij} x_i x_j$. Line normalization removes the first sum, so $q_2 = Q$. Consequently

$$q_1 = N^2 Q = \sum_{k, i<j} x_k^2 x_i x_j P_k^2 a_{ij}.$$

This is where the degree bound becomes decisive. For $k = i$ or j, the exponent pattern is $(3,1)$; otherwise it is $(2,1,1)$. Both patterns have binary degree three, whereas the weight projection

q_1 has degree at most two. The cubed or squared coordinate identifies k, and the remaining coordinates identify $\{i, j\}$. Thus each displayed term has its own reduced monomial, forcing

$$P_k^2 a_{ij} = 0 \qquad \text{for all } k, i, j. \tag{17}$$

For completeness, the one-coordinate function basis $1, x, x^2, x^3$ has binary degrees $0, 1, 1, 2$. Its leading classes form a basis in the associated graded degree filtration. Tensoring over the coordinates proves that distinct reduced monomials of the displayed degree three cannot cancel to degree two. This justifies the comparison in the binary degree filtration.

We now extend this vanishing from the cocycle coefficients to the whole intrinsic derived ideal. Let $\mathfrak{m}W = \sum_i P_i W$. This is an $\mathbb{F}_4$-subspace because the P_i are $\mathbb{F}_4$-linear. By the preliminary identities, every P_k^2 annihilates $\mathfrak{m}W$. Modulo $\mathfrak{m}W$, the polar values in (9) are exactly the values of the polar of Q. For $i < j$ and $\alpha \in \mathbb{F}_4$,

$$\mathrm{pol}(Q)(e_i, \alpha e_j) = \alpha a_{ij}.$$

All polar values of Q lie in the $\mathbb{F}_4$-span of the a_{ij}, and the displayed values contain all their scalar multiples. Their binary span is therefore exactly that $\mathbb{F}_4$-span. Since W is the binary span of all polar values, this proves (14). Together with (17), it gives $P_k^2 = 0$ on all of W. Now $L_x = P(x)$ and $q = Q + PQ$. The degree-three terms of PQ with three distinct indices vanish precisely when (13) holds. The other terms have exponent patterns $(2, 1)$, of binary degree two. This proves necessity.

Step 4: sufficiency. Conditions (12) and (13) make $P(x)^2 = 0$ and make PQ binary-quadratic. Under $x \mapsto \omega x$, the functions Q and PQ have weights two and zero. Thus

$$(Q + PQ)(\omega x) = \omega^2 Q + PQ = \omega^2 (Q + PQ) + \omega P(Q + PQ).$$

Together with $(\omega\,\mathrm{id} + \omega P)(\omega\,\mathrm{id} + P) = \omega^2\,\mathrm{id} + P$, this verifies $f^2 = f + \mathrm{id}$ by substitution. Mixed polar values generate $\mathfrak{m}W$; modulo that space, the values of the polar of Q generate the a_{ij}'s over $\mathbb{F}_4$. Equation (14) therefore makes the polar image exactly W.

Step 5: the associated Lie algebra. Put $\beta(x, y) = Q(x + y) + Q(x) + Q(y)$ and define

$$[(x, z), (y, t)]_0 = (0, P(x)t + P(y)z + \beta(x, y)). \tag{18}$$

Jacobi follows from commuting operators and (13). This is a metabelian $\mathbb{F}_4$-Lie algebra with derived ideal W, and it satisfies (15). Conversely, splitting the abelianization of any algebra with these properties gives exactly such operators and an alternating cocycle. Choose its cross-term refinement Q to recover (10).

For both the original polar bracket and (18), the action on W is $P(x)$. Hence, with $\mathfrak{m}$ the ideal generated by these actions,

$$\gamma_2 = W, \qquad \gamma_k = \mathfrak{m}^{k-2} W \quad (k \geq 3). \tag{19}$$

This proves the lower central comparison.

Step 6: morphisms. A binary-linear intertwiner maps the intrinsic space W into W'. Its diagonal blocks $A : U \to U'$ and $C : W \to W'$ commute with the intrinsic scalar actions, hence are $\mathbb{F}_4$-linear. Write its off-diagonal block as $t_s + t_a$, where $t_s(\omega x) = \omega t_s(x)$ and $t_a(\omega x) = \omega^2 t_a(x)$. Direct substitution in $Sf = f'S$ gives the two conditions

$$CP(x) = P'(Ax)C, \tag{20}$$

$$Q'(Ax) = CQ(x) + t_a(x) + P'(Ax)t_s(x). \tag{21}$$

Indeed the unseparated cocycle equation reduces to the second one by multiplication with $\mathrm{id} + P'(Ax)$, whose inverse is itself. Polarization over $\mathbb{F}_4$ eliminates t_a and gives

$$\beta'(Ax, Ay) = C\beta(x, y) + P'(Ax)t_s(y) + P'(Ay)t_s(x). \tag{22}$$

Equations (20) and (22) say exactly that $(x, z) \mapsto (Ax, Cz+t_s(x))$ is an $\mathbb{F}_4$-Lie homomorphism.

Conversely, for any such Lie homomorphism the function $Q'(Ax) + CQ(x) + P'(Ax)t_s(x)$ has zero polar and is an ordinary homogeneous quadratic. It is therefore a unique sum of squares $\sum b_i x_i^2$. Taking this as t_a gives a unique intertwiner. Scalar parts respect composition because A and C are $\mathbb{F}_4$-linear. Thus the correspondence is an equivalence on both objects and all morphisms, proving also the centralizer assertion. No inverse of A, C, or the intertwiner was used. Thus the argument includes maps with nontrivial kernel or cokernel. □

Corollary 3.3. *Every finite-dimensional metabelian polar algebra is nilpotent and satisfies* $(\operatorname{ad} v)^3 = 0$ *for every* v. *If* $w = \dim_{\mathbb{F}_4} W$, *its class is at most* $w + 1$.

Proof. The commuting square-zero operators generate a nilpotent associative ideal. They have a common nonzero kernel whenever $W \neq 0$; repeating on the quotient gives a strictly triangular flag, so all products of w operators vanish. Apply (19). Finally $\operatorname{ad} v$ maps V into W, and its square on W is zero. □

Theorem 3.4. *A finite-dimensional polar Lie algebra is solvable if and only if it is nilpotent. Its solvable radical is invariant under* f *and is nilpotent.*

Proof. Induct on derived length. The abelian case is immediate. Put $W = L'$, $W' = L''$. The restricted solution on W has nilpotent polar algebra by induction. The quotient L/W' is metabelian, hence nilpotent by Corollary 3.3. For fixed $x \in L$, the derivation $D = \operatorname{ad} x|_W$ is therefore nilpotent on W/W'.

We use the following elementary derivation fact. If N is nilpotent and D is nilpotent on N/N', then it is nilpotent on N. Each factor $\gamma_i(N)/\gamma_{i+1}(N)$ is a quotient of $(N/N')^{\otimes i}$. The induced derivation is the sum of the i commuting operators acting in individual tensor positions. If $D^s = 0$ on N/N', the sum has power $i(s-1)+1$ equal to zero, by the pigeonhole principle on its expanded products. The finite lower central filtration proves the assertion on N, in every characteristic.

Apply this fact to $N = W$. Since $\operatorname{ad} x$ maps L into W, it is nilpotent on L. Engel's theorem now gives nilpotency of L. The converse is immediate. If R is the solvable radical, L/R has zero center. Lemma 2.4 makes R invariant, and the result just proved applies to the restricted solution. □

3.1 An intrinsic change of addition

Let $\mathcal{Q}$ have finite binary-quadratic solutions $f^2 = f + I$ with metabelian polar algebra as objects, and all binary-linear intertwiners $Tf = gT$ as morphisms. Let $\mathcal{E}$ have finite-dimensional metabelian $\mathbb{F}_4$-Lie algebras satisfying $(\operatorname{ad} u)^2\mathfrak{l}' = 0$, with all $\mathbb{F}_4$-Lie homomorphisms.

Proposition 3.5 (An involutive twist of binary Lie structures)**.** *Let* $(E, +, B)$ *be a metabelian binary Lie algebra with derived ideal* W, *satisfying* $(\operatorname{ad} u)^2 W = 0$ *for every* u. *Set* $P_u = \operatorname{ad} u|_W$ *and*

$$C(u, v) = (I + P_u + P_v)B(u, v), \qquad u \star v = u + v + C(u, v).$$

Then $(E, \star, C)$ *is another binary Lie algebra in the same category. Applying this operation twice recovers* $(+, B)$. *It is a canonical involutive functor on this precise Lie category, with the same underlying morphism functions.*

Proof. The operators P_u commute because W is abelian, and square to zero by hypothesis. For $a = P_x, b = P_y, c = P_z$, put $w = C(x, y) = (I + a + b)B(x, y)$. Direct expansion gives

$$C(x \star y, z) + C(x, z) + C(y, z) = bB(x, z) + aB(y, z) + (I + a + b + c)c(I + a + b)B(x, y).$$

The last operator is c, so this is zero by the original Jacobi identity. Thus C is additive for $\star$ in both arguments, before associativity is assumed. Both associations of three inputs now equal $x+y+z+C(x,y)+C(x,z)+C(y,z)$. The operation is therefore an abelian group operation, with identity zero and exponent two. On W, it is the old addition, and $C(x,w) = B(x,w)$ for $w \in W$. The Jacobi sum for C is the old Jacobi sum plus $\sum_{\text{cyc}} P_x(P_y + P_z)B(y,z)$; the latter vanishes by multiplying the old Jacobi identity by P_x, P_y, P_z and summing.

The new derived ideal is W. One inclusion follows from $C(E,E) \subseteq W$; for the other use

$$B(u,v) = C(u,v) + C(u \star v, C(u,v)).$$

The actions on W are unchanged, so the new square condition holds. If $S = u \star v$, $w = C(u,v)$, then $S \star w \star C(S,w) = u+v$: the last extra term is $P^2_{u+v}w = 0$. The second twisted bracket is $(I + P_u + P_v)^2 B(u,v) = B(u,v)$. This proves involutivity. Every original Lie homomorphism preserves the defining formulas; involutivity gives the converse as well. □

There is a choice-free realization of that equivalence. In the next formulas $+$ and B denote the original binary addition and polar bracket of a source object. Define on its underlying set

$$C(u,v) = B(u,v) + B(u+v, B(u,v)), \qquad u \oplus v = u + v + C(u,v), \qquad \omega \cdot u = f(u). \quad (23)$$

The other two $\mathbb{F}_4$ scalars are zero and the identity, and $\omega^2 u = f^2(u)$.

Theorem 3.6 (Canonical all-morphism equivalence)**.** *The operations* (23) *make the underlying set a vector space over* $\mathbb{F}_4$*, and* C *makes it an object of* $\mathcal{E}$*. On morphisms the correspondence uses the same set map. It is fully faithful and has a canonical inverse: for an object* $(L, \oplus, C)$ *of* $\mathcal{E}$*, define*

$$u + v = S \oplus C(u,v) \oplus C(S, C(u,v)), \qquad S = u \oplus v, \qquad f(u) = \omega u. \quad (24)$$

These are the binary vector addition and map of an object of $\mathcal{Q}$*. The two constructions are inverse on objects and on their underlying morphism functions.*

Proof. It suffices to establish the formulas in any normal-form coordinates; their right sides are already intrinsic. Put

$$H_f(x,z) = (x, (I + P(x))(z + Q(x))), \qquad H_f^{-1}(x,z) = (x, (I + P(x))z + Q(x)).$$

For the morphisms in Step 6 of Theorem 3.2, direct use of its cocycle equation gives the naturality identity

$$H_g T_f = T_L H_f. \quad (25)$$

Indeed the second coordinate inside $I + P'(Ax)$ becomes $C(z + Q(x)) + (I + P'(Ax))t_s(x)$, whose image is $C(I + P(x))(z + Q(x)) + t_s(x)$.

For completeness, write $u = (x,z)$, $v = (y,t)$, $a = P(x)$, $b = P(y)$, and $\beta = Q(x+y) + Q(x) + Q(y)$. A calculation using $a^2 = b^2 = 0$, $ab = ba$, gives both

$$H_f^{-1}(H_f u + H_f v) + u + v = [H_f u, H_f v]_0 = B(u,v) + (a+b)B(u,v).$$

Here H_f^{-1} is the identity on W, and $(a+b)B(u,v) = B(u+v, B(u,v))$. Thus H_f transports $\oplus, C, f$ to ordinary vector addition, the auxiliary Lie bracket, and multiplication by ω. This proves that (23) defines the claimed structure in every class, not merely in a low-class truncation.

To recover the old addition put $w = C(u,v) \in W$ and $S = u \oplus v$. On $V \times W$, the two bracket formulas agree and their actions square to zero. Hence $S \oplus w = S + w + C(S,w)$, and subsequently adding $C(S,w)$ by $\oplus$ cancels that last term. The result is $u + v$, proving (24).

For an arbitrary object of $\mathcal{E}$, choose a vector-space splitting only to verify the construction: its usual action and cocycle data give the existing normal-form solution, and the preceding calculation transports that solution to exactly (24). Thus this formula is associative, has exponent two, and gives the required quadratic solution; no splitting enters its definition. Applying the two formulas successively recovers the original operations.

Every source intertwiner preserves $+, f, B$, hence $\oplus, C$ and the $\mathbb{F}_4$ scalars. Conversely, a target Lie homomorphism preserves the right side of (24) and $f = \omega$, so it is an original binary-linear intertwiner. This proves full faithfulness and the asserted inverse functor. □

Exactness and central extensions. The canonical functor changes operations, not the underlying sets or morphism functions. Kernels and images therefore have the same point sets. Short exact sequences are preserved and reflected, with $\dim_{\mathbb{F}_2} = 2\dim_{\mathbb{F}_4}$ on corresponding objects. Even ranks are also immediate from the original orbit-parity argument and are not a separate new invariant.

The original derived ideal is the same set W as the new derived ideal. On W the additions coincide, and on $V \times W$ the brackets coincide; hence all lower central terms agree as sets. The original polar bracket can be recovered as

$$B(u, v) = C(u, v) \oplus C(u \oplus v, C(u, v)).$$

Thus the two centers coincide as sets. An original f-stable binary subspace is the same set as a new $\mathbb{F}_4$-Lie subalgebra; original f-stable polar Lie ideals correspond to new $\mathbb{F}_4$-Lie ideals. No claim is made about original ideals lacking f-invariance. These assertions follow in both directions directly from the two operation formulas. In particular, central short exact sequences whose middle object remains metabelian correspond to central Lie extensions. This does not cover central extensions that leave the metabelian category.

Scope. The formula uses the metabelian normal-form identities, especially commuting square-zero actions on the derived ideal. Section 14 gives a nonmetabelian model for which the same short formula is not associative. The canonical change of addition preserves the original classification through its full morphism correspondence.

4 Global nilpotence of the polar algebra

The polar identities do more than equip a quadratic orbit system with a Lie bracket: they exclude a nonsolvable polar algebra altogether. Thus nilpotence is a consequence of the orbit equation, rather than an additional hypothesis on the solutions considered below.

Theorem 4.1 (Global polar nilpotence)**.** *Let V be a finite-dimensional binary vector space and let $f : V \to V$ have binary algebraic degree at most two and satisfy $f \circ f = f + \mathrm{id}$. Its polar Lie algebra L_f is nilpotent.*

The proof combines identities specific to the functional equation with classical finite-dimensional Lie theory. Here is the structural step that makes the combination possible. Put $B(x, y) = [x, y]$ and, in the ordinary universal enveloping algebra of L_f, define

$$\begin{aligned} R_x = x^2 + f(x), \quad D(x, y) = [R_x, y] = [f(x), y] + [x, [x, y]], \\ G(x, y) = [R_x, R_y], \qquad S(x, y) = G(x, y) + B(x, y) + D(x, y). \end{aligned} \tag{26}$$

Although f is generally nonlinear, R is binary-linear. The three maps D, G, S take values in L'_f and are alternating biderivations. Moreover,

$$G([x, y], z) = [G(x, y), z]. \tag{27}$$

Consequently, on a perfect centerless polar algebra, G factors through the bracket by an element of the centroid.

Proof of Theorem 4.1: roadmap. By Corollary 2.5, a nonsolvable polar algebra would have a nonabelian simple polar subquotient. On such a simple algebra the finite centroid is a field K, and (27) gives $G = \kappa B$ for some $\kappa \in K$. The biderivation identities impose polynomial constraints on $\operatorname{ad} x$. If $|K| \geq 8$, they give $(\operatorname{ad} x)^4 = 0$; if $K = \mathbb{F}_4$, they give $(\operatorname{ad} x)^5 = 0$. Engel's theorem excludes both cases.

The remaining case $K = \mathbb{F}_2$ requires a different argument. A square-zero associative product built from D first shows that there are no nonzero ad-nilpotent elements, and then eliminates the nilpotent part of every adjoint operator. The resulting identity

$$(\operatorname{ad} x)^4 + (\operatorname{ad} x)^2 + \operatorname{ad} x = 0$$

has square-free annihilating polynomial. Every adjoint is therefore semisimple, and Premet's theorem over perfect C_1-fields forces the algebra to be commutative [34, Theorem 3(ii)]. This excludes the last simple case. The polar algebra is consequently solvable, and Theorem 3.4 makes it nilpotent. Appendix N supplies all enveloping-algebra, centroid, and square-zero-product details. □

The use of Premet's result concerns ordinary finite-dimensional Lie algebras; no restricted structure on the original polar algebra is assumed. The proof of Theorem 4.1 also precedes, and does not use, the finite two-generator model or the universal Engel identities established in the next section. This order separates the global nilpotence argument from the subsequent structure theory.

5 Canonical structure beyond the metabelian range

Global nilpotence permits a uniform passage from the orbit equation to restricted Lie theory. The free two-generator solution supplies universal identities, and these identities give canonical scalars on a reduced derived ideal. We then prove quadratic stability of the special biderivation in every finite solution. On finite free covers this produces an exact scalar on the whole derived ideal; on arbitrary quotients it produces a universal restricted derived quotient, natural for every linear intertwiner. A reduced adjoint representation gives a complementary matrix realization with a kernel of class at most three and an abelian cokernel. The quotient structures below retain the original polar bracket.

5.1 The free two-generator solution and universal identities

For a solution (G, g), let G' denote its polar derived ideal. The quotient G/G' has its intrinsic $\mathbb{F}_4$-scalar, given by the map induced by g. For $k = \mathbb{F}_4$, write

$$H_g(k) = \frac{\{q : G \to k : \deg_2 q \leq 2,\ q(0) = 0,\ qg = \omega^2 q\}}{\{tg + \omega t : t \in \operatorname{Hom}_{\mathbb{F}_2}(G, k)\}}. \tag{28}$$

This is the elementary central-extension space, whose general construction is given in Proposition 13.1. Theorem 4.1 has already established nilpotence for every finite target. Thus the following criterion applies to the entire finite class.

Lemma 5.1 (A freeness criterion from central extensions)**.** *Suppose a finite solution (G, g) is generated by marked points $\xi_1, \dots, \xi_s$, their classes form an $\mathbb{F}_4$-basis of G/G', and $H_g(k) = 0$. Every assignment of the marked points to points of a finite solution extends uniquely to a binary-linear solution homomorphism.*

The proof is a central splitting argument, given in Appendix O. The sixteen-dimensional model in (115) satisfies the hypotheses with marked points a, b: its full central-extension space vanishes, as certified by the final row of Table 3. It is therefore the free two-generator solution among finite solutions. The vanishing certificate can be used directly, before classifying the intermediate stages of its central-cover construction. The dependency is global nilpotence, followed by this central-splitting criterion and its finite certificate, followed by the identities below.

Retain D, G from (26). The free solution supplies identities valid in every finite quadratic solution.

Theorem 5.2 (Universal two-orbit identities)**.** *Every finite quadratic solution satisfies*

$$G(x, y) = [x, y] + D(x, y), \tag{29}$$

$$[x, [x, [x, y]]] = [[x, y], [x, [x, y]]], \tag{30}$$

$$(\operatorname{ad} x)^4 = 0. \tag{31}$$

The last bound is sharp: the fourteen-dimensional model has $(\operatorname{ad} a)^3 b \neq 0$*. One also has*

$$[x, [[x, y], [x, z]]] = 0. \tag{32}$$

The exact identity (30) explains why the fourth adjoint vanishes without making the third vanish. Its right side lies deeper in the lower central series, and retains information that would disappear in a homogeneous Engel approximation. Appendix O gives a sparse coordinate verification on the free solution and proves the propagation to arbitrary pairs. The three-variable identity (32) is derived directly from the unconditional biderivations; it does not require a three-generator classification.

5.2 A Lie pencil and a canonical restricted derived quotient

Theorem 5.3 (The polar Lie pencil)**.** *The special biderivation D is a Lie bracket and satisfies*

$$D([x, y], z) = [D(x, y), z]. \tag{33}$$

In particular it is an adjoint Lie two-cocycle, and $B + tD$ is a Lie bracket after any extension of the binary coefficient field. Its derivations satisfy

$$[D_x, D_y] = \operatorname{ad}_{[x,y]+D(x,y)}. \tag{34}$$

Put

$$W = L', \qquad C = W \cap Z(L), \qquad E = W/C. \tag{35}$$

Here the center is that of the whole polar algebra. Equation (33) makes the following prescription independent of the bracket expression:

$$J\left(\sum_i [x_i, y_i] + C\right) = \sum_i D(x_i, y_i) + C. \tag{36}$$

Theorem 5.4 (Canonical restricted structure)**.** *The operator J on E satisfies $J^2 + J = \mathrm{id}$, belongs to the centroid of its original Lie bracket, and commutes with every ambient adjoint action induced by L. Thus J gives E a canonical $\mathbb{F}_4$-Lie structure. The map f descends to $\bar{f}$ on E, and*

$$P = \bar{f} + J \tag{37}$$

is a canonical restricted square on this $\mathbb{F}_4$-Lie algebra, with $P^2 = 0$. Explicitly,

$$P(u+v) = P(u) + P(v) + [u,v], \qquad P(Ju) = J^2 P(u),$$
$$\mathrm{ad}_{P(u)} = (\mathrm{ad}_u)^2, \qquad P(P(u)) = 0.$$

These constructions are preserved by solution isomorphisms and by surjective solution homomorphisms.

This theorem produces scalars for the original quotient bracket, with no metabelian hypothesis. The central quotient is essential:

Proposition 5.5 (A whole-derived central obstruction)**.** *There is a quadratic solution of binary dimension 22, with metabelian polar algebra of class four, for which no binary-linear map $J_W : W \to W$ satisfies*

$$J_W[x,y] = D(x,y) \qquad (x,y \in L).$$

Its obstruction is supported on $W \cap Z(L)$. The canonical structure of Theorem 5.4 on E remains valid.

The explicit model and a two-cycle detecting the obstruction appear in Appendix O. This separates the intrinsic reduced-derived scalar from the auxiliary scalar structure furnished by the metabelian normal form.

5.3 Global quadratic stability

An alternating binary-bilinear cochain A is *quadratically stable* if $q_A \circ f$ has binary degree at most two for a zero-constant quadratic refinement q_A of A. The condition is independent of the refinement: a linear adjustment composes with f to a quadratic function.

Theorem 5.6 (Global stability of the special biderivation)**.** *For every finite quadratic solution, the special biderivation D is quadratically stable. Thus, for every quadratic refinement q_D,*

$$\deg_2(q_D \circ f) \le 2. \tag{38}$$

Theorem 5.7 (The ambient cube annihilates the derived ideal)**.** *Every finite quadratic solution satisfies*

$$(\mathrm{ad}_z)^3(W) = 0 \qquad (z \in L). \tag{39}$$

The fourth-adjoint bound on the whole space remains sharp, as in Theorem 5.2.

The proof first transports the restricted structure of E through its faithful ambient adjoint action. This controls compositions with derived inputs. The remaining high-degree part lies on the intrinsic $\mathbb{F}_4$-abelianization; scalar-character projections and the cyclic norm eliminate it. The complete argument is given in Appendix P. Quadratic stability controls composition degree; the central CE class detected in Proposition 5.5 may still be nonzero. The same construction retains a canonical central square-height class, described in Proposition P.3; it records central data before passage to the universal derived quotient.

5.4 Exact restricted structure on finite free covers

Theorem 6.14 constructs a finite free solution $\mathfrak{F}_s^{\mathrm{fin}}$ at every finite point rank. Its admissible central quadratic extension space vanishes. Global stability now converts that projectivity into an exact scalar structure on its derived ideal.

Theorem 5.8 (Exact free-derived scalar and restricted square)**.** *Let (F, g) be a finite free solution and put $W_F = F'$. There is a unique binary-linear map $J_F : W_F \to W_F$ satisfying*

$$J_F B_F(x, y) = D_F(x, y) \qquad (x, y \in F). \tag{40}$$

It obeys $J_F^2 + J_F = \mathrm{id}$, is a centroid of the original bracket on W_F, and commutes with every ambient adjoint action of F. The map

$$P_F = g|_{W_F} + J_F \tag{41}$$

is a restricted square on this $\mathbb{F}_4$-Lie algebra, with $P_F^2 = 0$. In particular $g|_{W_F} = J_F + P_F$.

Every finite solution is a quotient of such an F. Under a quotient, central relations can fail to be stable under J_F. The next construction records exactly the additional relations needed to retain the restricted structure. The proof of Theorem 5.8, including the use of quadratic-stable cohomology, is in Appendix P.

5.5 A universal restricted derived quotient

Write $B, D : \Lambda^2_{\mathbb{F}_2} L \to W$ for the induced linear maps and fix $k = \mathbb{F}_4$, with $\omega^2 + \omega = 1$.

Definition 5.9 (The scalar-compatible derived quotient)**.** Define

$$\begin{gathered} Q_f = \frac{k \otimes_{\mathbb{F}_2} W}{\langle 1 \otimes D(c) + \omega \otimes B(c) : c \in \Lambda^2_{\mathbb{F}_2} L \rangle_k}, \\ j_f(w) = 1 \otimes w. \end{gathered} \tag{42}$$

Let $N_f = \ker j_f$.

Theorem 5.10 (Canonical restricted quotient and all linear arrows)**.** *The map $j_f : W \to Q_f$ is binary-linearly surjective, and $N_f \subseteq W \cap Z(L)$ is f-stable. The original bracket and $f|_W$ descend to Q_f, with*

$$\begin{gathered} [j_f(u), j_f(v)] = j_f([u, v]), \qquad f_Q(j_f(w)) = j_f(f(w)), \\ J_Q(j_f(B(c))) = j_f(D(c)), \end{gathered} \tag{43}$$

where J_Q is multiplication by ω. The map $P_Q = f_Q + J_Q$ is a restricted square on the original quotient bracket, and $P_Q^2 = 0$.

For every k-space T, a binary-linear map $h : W \to T$ with $hD = \omega hB$ factors uniquely through a k-linear map $Q_f \to T$. Thus N_f is the least kernel required by the scalar-factor relation. In particular there is a canonical surjective restricted k-Lie map $Q_f \to E$.

Every binary-linear solution homomorphism $T : (L, f) \to (L_1, g)$ induces a restricted k-Lie homomorphism

$$Q(T) : Q_f \longrightarrow Q_g, \qquad Q(T) j_f(w) = j_g(Tw). \tag{44}$$

These maps preserve identities and composition. Hence Q is a functor on all finite solutions and all their linear intertwiners.

Unlike the quotient by the full ambient center, this construction does not require an arrow to preserve that center. The tensor relation defines the map directly, while the free-cover description proves that its kernel is stable under the quadratic rule.

Theorem 5.11 (Exact free-presentation formula)**.** *Let $\pi : (F, g) \twoheadrightarrow (L, f)$ be any finite free presentation, and put $H = \ker \pi \cap F'$. Then*

$$Q_f \simeq F'/(H + J_F H), \qquad N_f \simeq (H + J_F H)/H. \tag{45}$$

The subspace $H + J_F H$ is the least J_F-stable subspace containing H; it is automatically stable under g and P_F, and is an ideal. Its image modulo H is central.

The formula shows that Q_f retains the central directions that are compatible with the scalar relation. On the example of Proposition 5.5, it removes the detected central obstruction. It is a derived quotient with an exact universal property; the ambient solution is still described by its quotient, action, and extension data. Proofs of both quotient theorems are in Appendix P.

Corollary 5.12 (Reversal of the orbit orientation)**.** *If a binary-linear map satisfies $Tf = g^2T$, then $j_f(w) \mapsto j_g(Tw)$ induces a Frobenius-semilinear Lie map $Q_f \to Q_g$ preserving the restricted square. Replacing f by f^2 leaves N_f unchanged, conjugates J_Q, and leaves P_Q unchanged. The zero map is the same underlying map for either orientation.*

5.6 A standard restricted presentation with a low-class kernel

Use $k = \mathbb{F}_4$ and write $\omega = J$ on E. The canonical adjoint representation is

$$\rho : L \longrightarrow \mathrm{Der}_k(E), \qquad \rho(x)(w + C) = [x, w] + C. \tag{46}$$

Theorem 5.13 (Reduced adjoint representation)**.** *The map ρ is a binary-linear Lie homomorphism satisfying*

$$\rho(fx) = \rho(x)^2 + \omega\rho(x), \qquad \rho(x)^3 = 0.$$

Its kernel K is an f-stable ideal with

$$[K, W] \subseteq C, \qquad \gamma_3(K) \subseteq C, \qquad \gamma_4(K) = 0.$$

Every $\rho(x)$ is a restricted derivation of (E, P):

$$\rho(x)(P(u)) = [u, \rho(x)u].$$

The cube in this statement is the associative matrix cube. It follows from Theorem 5.7; the Lie-adjoint powers in the scalar-envelope formula below concern a different operation.

Let $X = \rho(L)$. It is a binary subspace stable under $A \mapsto \omega A + A^2$. Passing to its scalar span gives the restricted structure that will carry the realization.

Theorem 5.14 (The scalar envelope)**.** *The space $N = kX = X + \omega X$ is the restricted k-Lie algebra generated by X in $\mathrm{Der}_{\mathrm{res}}(E, P)$. It is an ordinary three-Engel k-Lie algebra, and*

$$N' = X' = \mathrm{Inn}(E).$$

Define

$$\mathcal{M} = N'' \cap Z(N') = \mathrm{ad}_E\big(E' \cap Z_2(E)\big), \tag{47}$$

where $Z_2(E) = \{e \in E : [e, E] \subseteq Z(E)\}$. Then $\mathcal{M}$ is a restricted ideal, any two of its matrices have zero product, and

$$T^4 = 0 \qquad (T \in N). \tag{48}$$

More precisely, for $A, B \in X$,

$$(A + \omega B)^4 = (\mathrm{ad}_A)^3 B + \omega(\mathrm{ad}_B)^3 A. \tag{49}$$

The right side is zero in every finite solution.

Already $T \mapsto \omega T + T^2$ is a quadratic solution on all of N. The quotient $\mathcal{T} = N/\mathcal{M}$ is a further restricted Lie reduction, with square induced from matrices, and

$$F(\bar{T}) = \omega\bar{T} + \bar{T}^{[2]} \tag{50}$$

is a quadratic solution on its entire k-space.

Theorem 5.15 (Bounded-class-kernel realization)**.** *The binary-linear map*

$$q : L \longrightarrow \mathcal{T}, \qquad q(x) = \rho(x) + \mathcal{M}$$

intertwines f and F. Its kernel $I = \rho^{-1}(\mathcal{M})$ is an f-stable ideal with

$$\gamma_3(I) \subseteq C \subseteq Z(L), \qquad \gamma_4(I) = 0.$$

Its image $Y = q(L)$ is an F-stable binary Lie ideal containing $\mathcal{T}'$, and $kY = \mathcal{T}$. There are exact sequences of quadratic solutions

$$0 \longrightarrow I \longrightarrow L \longrightarrow Y \longrightarrow 0, \qquad 0 \longrightarrow Y \longrightarrow \mathcal{T} \longrightarrow \mathcal{T}/Y \longrightarrow 0,$$

where $\mathcal{T}/Y$ is abelian. In particular the kernel is metabelian of class at most three, uniformly in the dimension and class of L.

The realization places the interactions outside a class-three kernel inside a standard restricted presentation; the extra scalar directions contribute an abelian cokernel. Its map is onto Y, which need not be all of $\mathcal{T}$. It is canonical under isomorphisms and surjective solution maps, the maps that preserve the center quotients used in its construction.

Equation (49) and the full-derived cube identity make the scalar envelope itself a height-two restricted presentation. The further quotient retains the bounded-kernel realization above. Section 6 uses the envelope directly to control class growth and its comparison with ordinary three-Engel Lie algebras.

The same pencil also connects the orbit equation to Bockstein-closed quadratic group extensions: D and $G = B + D$ are the classical binding maps for f^2 and f, respectively. The resulting applications, with their classical sources, are developed in Appendix Q.

5.7 A canonical restricted envelope of the whole solution

The canonical quotient Q_f also determines a restricted Lie algebra containing the entire solution modulo exactly the same central obstruction. This construction applies without a metabelian hypothesis and respects every binary-linear intertwiner. Put $k = \mathbb{F}_4$ and let B, D be as in Definition 5.9. Define

$$\mathcal{E}_f^+ = \frac{k \otimes_{\mathbb{F}_2} L}{\langle \omega \otimes B(c) + 1 \otimes D(c) : c \in \Lambda^2_{\mathbb{F}_2} L \rangle_k}, \qquad i_f(x) = 1 \otimes x.$$

Theorem 5.16 (The whole-solution tensor envelope)**.** *The space $\mathcal{E}_f^+$ carries a canonical restricted k-Lie structure with square P such that*

$$P(i_f x) = i_f(fx) + \omega i_f x, \qquad P^2(i_f x) = 0 \quad (x \in L).$$

Its derived ideal is $i_f(W) \simeq Q_f$, and $\ker i_f = N_f \subseteq W \cap Z(L)$. In particular, if $n = \dim_{\mathbb{F}_2} L$, $w = \dim_{\mathbb{F}_2} W$ and $n_0 = \dim_{\mathbb{F}_2} N_f$, then

$$\dim_k \mathcal{E}_f^+ = n - \frac{w + n_0}{2}.$$

Every binary-linear solution homomorphism induces a restricted k-Lie homomorphism between the corresponding envelopes. Hence $N_f = 0$ gives a faithful binary-linear realization of (L, f) inside a restricted k-Lie algebra, with $f = \omega I + P$ on its image.

Proof. Let R be the displayed relation space in $k \otimes L$. The identity $D([x,y],z) = [D(x,y),z]$ from Theorem 5.3 shows that R is a Lie ideal, so the scalar-extension bracket descends. Choose a binary complement V of W in L. As $R \subseteq k \otimes W$, the quotient is $(k \otimes V) \oplus Q_f$ as a k-space. Theorem 5.10 therefore gives $\ker i_f = N_f$, $i_f(W) = Q_f$, and the stated dimension. The derived ideal equals $i_f(W)$ because it is the scalar span of the bracket values and $i_f(W) = Q_f$ is already a k-space.

On Q_f use its restricted square P_Q. Choose a binary basis $v_1, \dots, v_s$ of V, prescribe $P(i_f v_j) = i_f(f v_j) + \omega i_f v_j$, and extend by $P(au) = a^2 P(u)$ and $P(u+v) = P(u) + P(v) + [u,v]$. These rules agree on the derived summand by the definition of P_Q. Both sides of the displayed formula for $P(i_f x)$ have the same polar form, so agreement on the chosen basis and on W proves it for every $x \in L$.

For source vectors, the defining relation and the definition of D give

$$[P(i_f x), i_f y] = i_f([fx, y]) + \omega i_f([x,y]) = i_f([x,[x,y]]).$$

Together with the restricted identity on Q_f, scalar extension and the Jacobi identity give $\mathrm{ad}_{P(u)} = (\mathrm{ad}_u)^2$ for all u in the envelope. Thus P is a restricted square. Finally, $f^2 x = x + fx$ and $[x, fx] = 0$ imply $P^2(i_f x) = 0$ by direct expansion. The tensor relation is preserved by every solution homomorphism, and the defining formula for P proves compatibility with the induced map. Since the source image spans the envelope over k, this also proves uniqueness and naturality of the square. □

The assertion $P^2(i_f x) = 0$ concerns the marked binary image. It does not assert that P^2 vanishes on the entire scalar envelope. The construction extends the quotient Q_f from the derived ideal to the whole solution while retaining the central obstruction N_f exactly.

6 General growth and nonmetabelian dimension compression

The general structure gives quantitative control of every finite quadratic solution. The exact metabelian theory in Section 7 will provide an exponential comparison scale. There are two complementary conclusions. The functional equation bounds polar class quartically in the number of point generators and connects its optimal growth to the ordinary three-Engel problem, whereas an explicit nonmetabelian family reaches every class in dimension $O(2^{c/2})$. Together these give logarithmic maximal derived length and finite free objects for all finite targets.

Write $L = L_f$ and

$$d = d(L) = \dim_{\mathbb{F}_4}(L/L'),$$

using the intrinsic scalar induced by f on the abelianization. As defined in Section 1, the point rank is the minimum number of generators under binary addition and f. It equals $d(L)$ by Lemma R.7; ordinary binary Lie generation uses the corresponding orbit pairs. For $c \geq 2$, let $m(c)$ be the least binary dimension of a solution of polar class c, and let $\mu(c)$ be the corresponding metabelian minimum. Section 7 determines $\mu(c)$ exactly.

6.1 A quartic bound on the general polar class

The class estimate separates a scalar Lie action from its remaining central module. The scalar action is controlled by the classical larger-field 3-Engel bound; the central module has a much shorter commuting action. For $d \geq 2$, put

$$q(d) = \left\lfloor \frac{(d+1)^4}{2} \right\rfloor, \qquad C_0(d) = q(d) + d + 2, \qquad C(d) = q(d) + d + 1. \tag{51}$$

Set $C_0(0) = C(0) = 0$ and $C_0(1) = C(1) = 1$.

Proposition 6.1 (A bound independent of higher-rank free objects)**.** *Every finite quadratic solution of point rank d has polar class at most $C_0(d)$.*

This first estimate uses the canonical action on $E = L'/(L' \cap Z(L))$, the full-derived cube identity, and paired point generation. It supplies finite free objects without assuming their exact whole-derived scalar in advance.

Theorem 6.2 (A uniform quartic class bound)**.** *Every finite quadratic solution of point rank d satisfies*

$$\operatorname{cl} L \le C(d). \tag{52}$$

In particular the bound is $\lfloor (d+1)^4/2 \rfloor + d + 1$ for $d \ge 2$. Rank zero is the zero solution; a nonzero rank-one solution is the linear scalar plane and has class one.

Proof outline. Proposition 6.1 first gives the finite free solution F at point rank d. On $W = F'$, Theorem 5.8 supplies the exact scalar $J = \omega$, commuting with all ambient adjoints. The actual binary action image consists of cube-zero operators and is closed under $A \mapsto \omega A + A^2$. Its scalar envelope is a 3-Engel $\mathbb{F}_4$-Lie algebra with the same class as an ordinary d-generator subalgebra. The larger-field estimate of [40] bounds this class by $q(d)$.

After that many lower-central steps, the remaining term lies in $Z(W)$. Ambient actions on this space commute; the two companion cube identities give $a_i^3 = a_i^2 a_j = 0$. Consequently $d+1$ further actions kill it for $d \ge 2$. Every finite target of point rank at most d is a quotient of F. Appendix R proves both estimates in this order, including the independent construction of F. □

The quantitative 3-Engel input is the larger-field improvement after Theorem 1 of [40]. The application here uses the precise orbit-to-action interface and the separate central-module bound; no bound on arbitrary representation words is inferred from Lie class alone.

Proposition 6.3 (The action on an invariant ideal)**.** *Let I be an f-stable Lie ideal and let $M = I/I'$. The map $J = f_M$ gives M its intrinsic $\mathbb{F}_4$-scalar structure. For every $x \in L$, the induced operator $A_x = \operatorname{ad}_x|_M$ satisfies*

$$A_x J = J A_x, \qquad A_x^3 = 0, \qquad A_{f(x)} = \omega A_x + A_x^2. \tag{53}$$

The actual binary image is an ordinary 3-Engel Lie algebra.

6.2 The same-rank comparison with ordinary three-Engel algebras

For $d \ge 1$, let $\kappa(d)$ be the largest polar class of a finite quadratic solution of point rank at most d. Let $e(d)$ be the largest class of a finite-dimensional ordinary 3-Engel Lie algebra over $\mathbb{F}_4$, generated by at most d ordinary Lie generators. These maxima exist: the preceding bound controls κ, and the classical larger-field estimate controls e.

Proposition 6.4 (An Engel algebra realized without increasing point rank)**.** *Let $H \ne 0$ be a finite-dimensional ordinary 3-Engel $\mathbb{F}_4$-Lie algebra, generated by at most r elements and of class c. There is a finite binary quadratic solution of point rank at most r and polar class $c-1$. For an abelian input, the resulting solution may be zero.*

Theorem 6.5 (Same-rank class-growth comparison)**.** *For every $d \ge 1$,*

$$e(d) - 1 \le \kappa(d) \le e(d) + d + 1. \tag{54}$$

In particular $\kappa(d) = \Theta(e(d))$. For every $\alpha \ge 1$, the bound $\kappa(d) = O(d^\alpha)$ holds if and only if $e(d) = O(d^\alpha)$. If $e(d)/d \to \infty$, then $\kappa(d)/e(d) \to 1$.

For the upper comparison, retain the actual class of the scalar action in the preceding proof instead of replacing it by $q(d)$. For the reverse construction, take the restricted matrix envelope of $\operatorname{ad} H$. Classical 3-Engel operator identities make every fourth power zero. The standard map $A \mapsto \omega A + A^2$ then gives an orbit solution; the binary subsolution generated by the original adjoints retains the full scalar span and hence the class. Appendix R gives this construction and the rank-preserving argument. The restricted-square viewpoint is classical [31]; the comparison keeps track of the original point rank and polar class.

Thus the linear-growth question for the orbit category has the same difficulty as the ordinary $\mathbb{F}_4$ three-Engel class problem. Unbounded principal-ideal class already occurs in compatible orbit solutions, by the classical current construction discussed in Remark R.4. The comparison concerns growth and explicit realizations, rather than an equivalence of the two underlying categories.

6.3 Top central covers and a sharp centre-by-metabelian boundary

The comparison leads to a precise boundary inside the ordinary three-Engel problem. Put $k = \mathbb{F}_4$, $U = k^d$, and use the classical commutator module W_d of Section 7. The ordinary Lie algebra

$$H_d = U \oplus W_d, \qquad [x_i, x_j] = v_{ij}, \quad [x_i, w] = t_i w, \quad [W_d, W_d] = 0 \tag{55}$$

is the universal metabelian 3-Engel algebra on d ordinary generators. Give x_i Lie weight one and v_{ij} weight two. For $d \geq 4$, its top weight is $d + 1$.

Theorem 6.6 (The full top central cover)**.** *Let $d \geq 4$, and give a central coefficient space Z weight $d + 2$. There is a natural isomorphism*

$$H^2_{\mathrm{CE}}(H_d, Z)_{d+2} \simeq \operatorname{Hom}_k\big(\det(U) \otimes \operatorname{Sym}^2 U, Z\big), \tag{56}$$

where the subscript denotes total input weight. Every class in this homogeneous space defines an ordinary 3-Engel central extension. If $\dim_k Z = 1$, every nonzero class gives an extension of exact ordinary generator rank d, class $d + 2$, and second derived ideal equal to Z.

The top commutator space is naturally $\det(U) \otimes U$. The CE equations determine a cocycle from its pairing of that space with U, and the pairing is exactly a symmetric bilinear form with the displayed determinant twist. The symmetry is bilinear symmetry in characteristic two, so its diagonal coefficients remain independent.

Let $e_{\mathrm{cbm}}(d)$ be the largest class of a finite-dimensional ordinary 3-Engel k-Lie algebra of generator rank at most d satisfying $L'' \subseteq Z(L)$. Thus $L/Z(L)$ is metabelian.

Theorem 6.7 (A sharp centre-by-metabelian class law)**.** *For every $d \geq 2$,*

$$e_{\mathrm{cbm}}(d) = \max(5, d + 2). \tag{57}$$

The maximum is also attained at exact ordinary generator rank d.

The upper bound is the classical metabelian ceiling $\max(4, d + 1)$, obtained from W_d, followed by one central step. The top covers attain it for $d \geq 4$. A direct two-generator calculation supplies the low-rank endpoints and proves $e(2) = 5$ (Lemma R.6). Consequently any superlinear mechanism for $e(d)$ must have noncentral second derived ideal.

The commutator module and the CE/Hopf central-cover framework are classical; the free centre-by-metabelian setting is developed in [20]. Here the square-zero module relations determine the full top quotient and make its cocycles automatically 3-Engel. Appendix R gives the quotient calculation, the ordinary metabelian ceiling, and the attaining models.

6.4 Central depth and delayed growth of class

How many class-preserving central lifts can occur before a new class appears? The answer depends on generator rank. The sharp ceiling above persists through several larger central-action layers, with distinct low-rank endpoints.

Throughout this subsection the field is $k = \mathbb{F}_4$, and generation and the three-Engel identity refer to ordinary Lie algebras. For the relatively free three-Engel algebra $\mathcal{F}_d$ on d generators, put

$$M_0 = \mathcal{F}_d'', \qquad M_{j+1} = [M_j, \mathcal{F}_d], \qquad D_{d,s} = \mathcal{F}_d/M_s, \qquad E_d = D_{d,1}.$$

Thus $D_{d,s}$ is universal under the condition $H'' \subseteq Z_s(H)$. Here $Z_s(H)$ denotes the sth upper central term, with $Z_0(H) = 0$.

Theorem 6.8 (Sharp class ceilings at central depths two, three, and four)**.** *Let H be a finite-dimensional ordinary three-Engel k-Lie algebra generated by at most d elements. Then*

$$H'' \subseteq Z_s(H) \quad \Longrightarrow \quad \operatorname{cl} H \leq d+2 \quad \begin{cases} s = 2, & d \geq 4, \\ s = 3, & d \geq 5, \\ s = 4, & d \geq 6. \end{cases} \tag{58}$$

Every bound is attained at exact ordinary generator rank d. Equivalently, $\operatorname{cl} D_{d,s} = d+2$ in these ranges.

The second-layer bound follows from a balanced symmetric-square module. At the third layer, two polarized Engel instances and thirteen explicit depth relations give a five-variable collecting identity; nonlinear substitution propagates it to every larger rank. The fourth layer uses the next-to-top part of the same module and a separate suspension argument. Its rank-six starting step is computer-assisted: three complete multihomogeneous systems are replayed independently. The symbolic arguments and the finite starting systems are given in Appendix W.

Proposition 6.9 (Boundary ranks)**.** *The complete relatively free objects satisfy*

	$D_{d,1}$	$D_{d,2}$	$D_{d,3}$	$D_{d,4}$
$d = 4$	6	6	7	*not determined here*
$d = 5$	7	7	7	8

where entries are nilpotency classes. At rank five there are concrete central epimorphisms

$$A_{178} \twoheadrightarrow A_{167} \twoheadrightarrow A_{139} \twoheadrightarrow A_{118}, \tag{59}$$

with classes $8, 7, 7, 7$, respectively. Subscripts give dimensions over k; the three kernels have dimensions $11, 28, 21$. Every object has exact ordinary generator rank five. At rank four there is a corresponding $42 \to 41 \to 40$ dimensional chain of classes $7, 6, 6$.

These concrete quotients certify the lower bounds; they are not identified with the entire relatively free objects. The upper bounds follow from Theorem 6.8 and the fact that a central extension raises class by at most one.

Corollary 6.10 (A lower bound on the number of unlocking lifts)**.** *For $d \geq 6$, three successive central epimorphisms of at-most-d-generated ordinary three-Engel algebras over the complete object E_d cannot produce class greater than $d+2$. At least four such lifts are required to exceed this ceiling. For $d = 5$, three suffice, as in* (59)*.*

More generally, for $d \geq 6$ and $s \geq 4$,

$$H'' \subseteq Z_s(H) \quad \Longrightarrow \quad \operatorname{cl} H \leq d+s-2. \tag{60}$$

Proof. The three lifts place the second derived ideal in Z_4; apply the theorem. For the last assertion apply its fourth-layer case to $H/Z_{s-4}(H)$ and add the removed central depth. □

The fourth-layer result gives a structural obstruction inside the ordinary function $e(d)$: for $d \geq 6$, class greater than $d+2$ forces $[H'', H, H, H, H] \neq 0$. Four unlocking lifts are not asserted sufficient in every rank. The passage from these ordinary k-Lie towers to admissible quadratic orbit-map lifts is a further interface, rather than part of this statement.

The classical centre-by-metabelian framework includes the balanced pairing and index-transfer identities of Kuz'min [21]; the corrected module treatments in [20, 27] fix the characteristic-two context. Here the balanced module controls successive central-action layers after the three-Engel relations are imposed.

6.5 An explicit family below the metabelian dimension scale

Let $k = \mathbb{F}_4$, and let A_m be the augmentation ideal of

$$k[t_1, \dots, t_m]/(t_1^2, \dots, t_m^2).$$

Thus $\dim_k A_m = 2^m - 1$, and every element of A_m has square zero. We take $A_0 = 0$.

Theorem 6.11 (The trace-restricted current family)**.** *On the binary vector space $k \oplus A_m^3$, define*

$$\begin{aligned} f_m(a,u,v,w) = \big(&\omega a,\ \omega u + av + (a+a^2)w + uv,\\ &\omega v + aw + uw,\ \omega w + vw\big). \end{aligned} \tag{61}$$

This is a quadratic solution of $f_m^2 = f_m + \mathrm{id}$, with

$$\dim_{\mathbb{F}_2} L_m = 6 \cdot 2^m - 4, \qquad d(L_m) = m+1, \qquad \mathrm{cl}\, L_m = 2m+1. \tag{62}$$

It is nonmetabelian for $m \geq 2$. For $m \geq 1$, its original binary polar algebra admits no restricted square.

Construction and proof outline. Use the classical three-dimensional algebra $S = \langle U, V, W \rangle_k$, with $[U,V] = U, [U,W] = V, [V,W] = W$, and the commuting derivations $\delta = \mathrm{ad}_U, \delta^2$. The restricted current presentation

$$\widehat{L}_m = (kx \oplus ky) \ltimes (S \otimes A_m)$$

has restricted square

$$P(a,b,u,v,w) = (0, a^2, av + bw + uv, aw + uw, vw), \qquad P^2 = 0. \tag{63}$$

Consequently $\widehat{f}_m = \omega\,\mathrm{id} + P$ is a quadratic solution. The binary graph $b = a + a^2$ is $\widehat{f}_m$-stable; its restriction is (61). A positive grading gives the exact class, and a direct derived-ideal calculation gives the intrinsic rank. The graph is binary-linear but is not closed under the ambient k-scalar or P. Appendix S proves all these assertions, including the nonrestrictability statement. □

The simple algebra and its derivations are classical [38, Proposition 3.3(1)]; the current multiplication is the one used in [39, Proposition 2.1]. The passage from restricted squares to quadratic maps is also classical [31, Lemma 9.5 and Example 9.6]. Here the binary invariant restriction and its exact polar class and point rank give the following dimension comparison.

Corollary 6.12 (Dimension compression in every class)**.** *Let $m(c)$ denote the general minimum from Corollary 12.3. Then*

$$\begin{aligned} m(2j+1) &\le 6\cdot 2^j - 4 \quad (j \ge 1),\\ m(2j) &\le 6\cdot 2^j - 6 \quad (j \ge 1). \end{aligned} \tag{64}$$

In particular $m(c) = O(2^{c/2})$, with $m(7) \le 44$ and $m(8) \le 90$. For every $c \ge 5$, the displayed constructions are nonmetabelian and have dimension strictly below $\mu(c) = 2^c + 2c - 6$.

Proof. Use L_j for odd class. For even class, quotient by its last lower-central term $kU_{[j]}$, an invariant central binary plane. The quotient has exact class $2j$ and loses two dimensions. The bracket $[U_{\{1\}}, V_{\{2\}}] = U_{\{1,2\}}$ survives for every resulting class at least five. The comparison with $\mu(c)$ follows immediately from $\lceil (c-1)/2 \rceil \le c - 3$ for $c \ge 5$. □

The construction gives the general bound $m(c) = O(2^{c/2})$. Its odd-class model has center equal to its last lower-central plane, so every nontrivial solution quotient lowers its class (Proposition S.3). Determining the optimal general dimension scale is a separate problem in Section 17.

The same family has a quadratic linearizing permutation with a quadratic inverse (Proposition S.5). Thus unbounded polar class and derived length do not force large inverse degree. More generally, central lifts of an involutively linearized base have a quadratic linearizer with inverse degree at most three (Proposition S.6), with the involutivity hypothesis imposed at the base stage.

6.6 Logarithmic derived length and finite universal objects

Let $\ell_{\max}(s)$ be the maximum polar derived length among finite solutions generated by at most s points, with $L^{(0)} = L$, $L^{(j+1)} = [L^{(j)}, L^{(j)}]$.

Theorem 6.13 (Maximal derived length)**.** *For $s \ge 1$,*

$$\lceil \log_2(3s) \rceil - 1 \le \ell_{\max}(s) \le 4\log_2(s+1) + 1. \tag{65}$$

In particular $\ell_{\max}(s) = \Theta(\log s)$. The same maximum is obtained at exact point rank s.

Proof outline. The upper bound follows from (52) and $L^{(j)} \subseteq \gamma_{2^j}(L)$. The family in Theorem 6.11 has

$$\mathrm{dl}(L_m) = \lceil \log_2(3m+3) \rceil - 1$$

by its exact support recurrence (Proposition S.4). Set $m = s - 1$. Adding linear scalar factors raises a smaller point rank to s without changing nonzero derived length. Full details are in Appendix R. □

The logarithmic order controls the whole finite class, including its nonmetabelian members. It leaves both the optimal logarithmic coefficient and exact values of $\ell_{\max}(s)$ open. A further consequence of the class bound is a growing dimension cost: if $a(c) = \min\{d \ge 2 : c \le C(d)\}$, then

$$m(c) \ge 2(c-1) + 2a(c) \ge 2c + 2(2c)^{1/4} - 6 \qquad (c \ge 2). \tag{66}$$

This bound follows from parity of the lower-central factors and provides a lower comparison to (64).

Theorem 6.14 (Finite free solutions and complete stem towers)**.** *For every $s \geq 0$, there is a finite solution $\mathfrak{F}_s^{\mathrm{fin}}$, generated by marked points $\xi_1, \dots, \xi_s$, such that every assignment of these points to a finite solution extends uniquely to a binary-linear intertwiner. Every such target has nilpotent polar algebra by Theorem 4.1. Its abelianization is $\mathbb{F}_4^s$, and $H_{\mathfrak{F}_s^{\mathrm{fin}}}(\mathbb{F}_4) = 0$.*

Starting with the linear solution $\mathbb{F}_4^s$, iteratively take the full quadratic central cover with kernel $H_g(\mathbb{F}_4)^*$ whenever $H_g(\mathbb{F}_4) \neq 0$. Every step is stem, meaning that its kernel lies in the derived ideal. The process terminates after finitely many nontrivial steps, and its terminal marked solution is $\mathfrak{F}_s^{\mathrm{fin}}$.

The reason for finiteness is quantitative: paired Lie generation and Proposition 6.1 bound the dimension of every s-point-generated finite solution. The free object is obtained by a finite product of marked representatives. The central-cover description gives an independent recognition criterion: a finite s-generated object with marked abelianization basis is free exactly when its full quadratic extension space vanishes. The proofs, including all noninvertible arrows, are in Appendix R.

The universal property covers finite targets and, by passage to the subsolution generated by the marked images, locally finite targets. The stem-cover construction terminates precisely when its full extension space vanishes. Its quantitative dimension bound ensures termination; explicit free objects in higher point rank remain a computational problem.

7 Sharp metabelian bounds and equality cases

The metabelian correspondence turns the dimension problem into a quotient problem for one multigraded commutator module. We first use that module to determine the least dimension at each polar class, then classify every equality case. Section 8 examines the much larger populations immediately above this minimum.

7.1 A multigraded commutator module

For $d \geq 2$, put

$$R_d = \mathbb{F}_4[t_1, \dots, t_d]/(t_1^2, \dots, t_d^2), \qquad \mathfrak{m} = (t_1, \dots, t_d).$$

Let W_d be the R_d-module generated by v_{ij}, $i < j$, subject to

$$t_i v_{jk} + t_j v_{ik} + t_k v_{ij} = 0 \qquad (i < j < k). \tag{67}$$

We set $v_{ji} = v_{ij}$ and $v_{ii} = 0$. This is the classical free metabelian commutator-module presentation, base-changed to R_d; compare [33, Theorem 6.2] and [20, Sections 2–3].

Every derived module in Theorem 3.2 is a quotient of W_d: send t_i to P_i and v_{ij} to a_{ij}. The intrinsic-generation condition makes this homomorphism surjective. Conversely, every quotient of W_d, with these images, gives the normal form (10).

Grade R_d by coefficient degree, and put the generators v_{ij} in coefficient degree zero. We also use the finer grading $\deg t_i = e_i$, $\deg v_{ij} = e_i + e_j$.

Proposition 7.1. *The coefficient-degree Hilbert polynomial of W_d is*

$$H_d(T) = \sum_{k=2}^{d} \binom{d}{k} \big((k-1)T^{k-2} + kT^{k-1}\big) + \binom{d}{2} T^2. \tag{68}$$

Consequently

$$\dim_{\mathbb{F}_4} W_d = (d-1)2^d + 1 - d + \binom{d}{2}.$$

The universal solution on $\mathbb{F}_4^d \oplus W_d$ has class $\max(4, d+1)$.

Proof. A raw multihomogeneous monomial is $t_S v_{ij}$, where S is square-free. Its multidegree has entries zero, one, or two, and at most two entries are doubled. Fix its support of size $k \geq 2$.

If no index is doubled, the raw coordinates are the edges of the complete graph on the support. Relations (67) are its triangle sums. The quotient edge space has dimension $k-1$, in coefficient degree $k-2$.

If exactly one index j is doubled, only edges incident with j occur. Triangle relations through j identify every pair of these edges, leaving dimension one. There are k choices of j, in coefficient degree $k-1$.

If exactly two indices i, j are doubled, there is one raw coordinate. A third index in the support kills it by a triangle relation, since the other two terms contain a squared variable. The coordinate survives exactly when the support is $\{i, j\}$, giving the final term of (68). These cases exhaust the multigrading.

The maximum coefficient degree is two for $d = 2, 3$, and $d-1$ for $d \geq 4$. Equation (19) gives the class. □

In particular, binary degree two permits unbounded polar nilpotency class. For the exact minimum dimension we must retain a terminal commutator in as small a quotient of W_d as possible.

7.2 Terminal tensors and contraction modules

Lemma 7.2. *Let $\mathfrak{l}$ be a metabelian $\mathbb{F}_4$-Lie algebra satisfying* (15)*, of class $c \geq 3$. Some nonzero terminal commutator has a repeated input between its first bracket pair and an operator position. If $c \geq 5$, there is a subalgebra of the same class generated by exactly $c-1$ elements modulo its derived ideal.*

Proof. The last lower central ideal is spanned by

$$T(x_1, x_2; x_3, \dots, x_c) = P(x_3) \cdots P(x_c)\beta(x_1, x_2).$$

Indeed $W = \operatorname{span}\beta + \mathfrak{m}W$, and multiplication by $\mathfrak{m}^{c-2}$ kills the second summand. The operator positions are symmetric, and a repeated input within them gives zero.

Suppose every value $T(x, y; x, z_4, \dots, z_c)$ were zero. Polarizing the repeated input gives

$$T(x, y; z, \dots) = T(z, y; x, \dots).$$

Together with symmetry of the first pair and the operator positions, this makes T fully symmetric. Jacobi on the first three inputs then gives $3T = T = 0$, a contradiction.

Choose a nonzero repeated-input value. Its inputs span a subspace U_0 of the abelianization of dimension at most $c-1$. Choose a linear section on a basis of U_0 and use its linear lifts for all the inputs. Changing a lift inside W does not alter the terminal commutator: the additional term belongs to $\mathfrak{m}^{c-1}W = 0$. These basis lifts generate a subalgebra H of class c, with abelianization exactly U_0. Indeed, any linear combination of the chosen generators whose image in U_0 vanishes is zero, and all remaining Lie words lie in H'. Its module is a quotient of W_{d_0}, where $d_0 = \dim U_0$. By Proposition 7.1, class $c \geq 5$ requires $d_0 \geq c-1$. Thus $d_0 = c-1$. □

For $d \geq 4$, the top coefficient degree of W_d is $d-1$. It has basis $T_1, \dots, T_d$, where

$$T_j = \Big(\prod_{h \neq i} t_h\Big) v_{ij} \quad (i \neq j). \tag{69}$$

Triangle relations identify the choices of i.

Lemma 7.3. *The top component is the* $\mathrm{GL}(d,4)$*-module* $\det(U)\otimes U$*. For every nonzero functional* λ *on that component, extended by zero to the other degrees,*

$$\dim_{\mathbb{F}_4} R_d\lambda = 2^d - 2.$$

If a quotient M *of* W_d *retains a nonzero top component, then* $\dim_{\mathbb{F}_4} M \geq 2^d - 2$*, whether or not the quotient is graded.*

Proof. Under a diagonal basis change A, T_j scales by $\det(A)A_{jj}$. A transvection $e_j \mapsto e_j + \alpha e_k$ sends T_j to $T_j + \alpha T_k$, and fixes the other top basis vectors. To check the formula choose $i \notin \{j,k\}$ in (69); any extra product has a squared variable. For the remaining vectors an index outside the affected entries can be chosen since $d \geq 4$. Together with permutations these transformations prove the module assertion. They act transitively on nonzero dual vectors, including scalar multiples.

Normalize $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \neq j$. We claim

$$\mathrm{Ann}_{R_d}(\lambda) = \Big(\prod_{i\neq j} t_i\Big). \tag{70}$$

The annihilator is multigraded: contractions by distinct monomials have distinct resulting dual multidegrees. For a square-free monomial t_S not containing j, the remaining multidegree is of the one-doubled-index type. It survives unless S contains every other index. If $j \in S$, the remaining multidegree has no doubled index. It survives whenever an index is omitted, as witnessed by an edge joining that index to j; it vanishes when all indices are used. These are exactly the two monomials in the ideal on the right of (70). This proves the claim and the dimension formula.

For an arbitrary quotient M, pull back a functional nonzero on the top image to $\phi \in W_d^*$. Its highest homogeneous component ϕ_{d-1} is nonzero. Independent homogeneous contractions of ϕ_{d-1} lift to independent contractions of ϕ, by comparison of their leading dual degrees. Hence

$$\dim M \geq \dim R_d\phi \geq \dim R_d\phi_{d-1} = 2^d - 2.$$

The filtered contraction argument applies to every quotient $W_d \to M$, including those with nongraded kernel. □

7.3 Exact minima

Theorem 7.4. *Among metabelian quadratic solutions of polar class* $c \geq 2$*, the minimum binary dimension is*

$$\mu(2) = 6, \quad \mu(3) = 8, \quad \mu(4) = 12, \qquad \mu(c) = 2^c + 2c - 6 \quad (c \geq 5).$$

Proof. Write $d = \dim_{\mathbb{F}_4}(V/W)$, $w = \dim_{\mathbb{F}_4} W$. Nonabelian objects have $d \geq 2$ by Lemma 2.6. A nonzero product $P_1 \cdots P_k v$, for commuting square-zero operators, yields 2^k independent vectors

$$v_S = \Big(\prod_{i\in S} P_i\Big)v \qquad (S \subseteq \{1,\dots,k\}).$$

In a relation, choose a term with nonzero coefficient of smallest support and multiply by its complementary operators. Every other surviving term repeats an operator and vanishes, while the chosen term does not. Taking $k = c-2$ gives $w \geq 2^{c-2}$. For $c = 2,3,4$ this gives the respective lower bounds $6, 8, 12$.

They are attained by cyclic quotients of $W_2 = R_2 v_{12}$: use $R_2/\mathfrak{m}$, $R_2/(t_2)$, and R_2, respectively, as the derived module in (10). The first two representatives are

$$(a, b, z) \longmapsto (\omega a, \omega b, \omega z + ab),$$
$$(a, b, y, z) \longmapsto (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab).$$

The class-four representative appears in (115) below.

Let $c \geq 5$. Proposition 7.1 gives $d \geq c - 1$. Lemma 7.2 supplies a same-class subalgebra with $d_0 = c-1$. Its derived module retains the top of W_{c-1}, so Lemma 7.3 gives $w \geq 2^{c-1} - 2$. Therefore

$$\dim_{\mathbb{F}_2} V = 2(d + w) \geq 2^c + 2c - 6.$$

To attain the bound, set $d = c - 1 \geq 4$, choose a top functional λ as in Lemma 7.3, and put

$$M_d = W_d / \{w : \lambda(rw) = 0 \text{ for every } r \in R_d\}. \tag{71}$$

The denominator is an R_d-submodule. The quotient has dimension $2^d - 2$ and retains the terminal value, so its solution has class $d + 1 = c$ and binary dimension $2(d + 2^d - 2)$. □

The change of formula at class five comes from the top degree of the universal module. For two and three quotient generators,

$$H_2(T) = 1 + 2T + T^2, \qquad H_3(T) = 3 + 8T + 6T^2.$$

In these cases the degree-two double-endpoint terms contribute at the top, allowing class four even with two quotient generators. For $d \geq 4$, they lie below the top degree $d - 1$; the terminal component $\det(U) \otimes U$ then controls class $d + 1$ uniformly. This is why the contraction argument yields one formula from class five onward, while the cyclic quotients of W_2 give the three smaller values.

The extremal quotient M_d in (71) has coefficient Hilbert polynomial

$$(d - 1) + \sum_{k=1}^{d-1} \binom{d}{d - 1 - k} T^k.$$

For example, the class-five extremizer has derived Hilbert vector $(3, 6, 4, 1)$ and binary lower central dimensions $(36, 28, 22, 10, 2, 0)$. The contraction construction supplies an explicit finite module description at every class, without an orbit search.

7.4 All minimum-dimensional metabelian solutions

The bound $\mu(c)$ also admits an equality classification. Let E_c denote the number of binary-linear conjugacy classes of metabelian quadratic solutions of class c and dimension $\mu(c)$. The three small classes close directly in the cyclic module W_2.

Lemma 7.5 (The three small minima)**.** *There is exactly one metabelian extremizer in each of classes two, three, and four. Thus* $E_2 = E_3 = E_4 = 1$.

Proof. Equality in the separate bounds $d \geq 2$ and $w \geq 2^{c-2}$ used in Theorem 7.4 forces $d = 2$ and $w = 1, 2, 4$, respectively. Intrinsic generation makes W cyclic over $R_2 = k[t_1, t_2]/(t_1^2, t_2^2)$, $k = \mathbb{F}_4$, generated by a_{12}: iterate $W = ka_{12} + \mathfrak{m}W$ and use $\mathfrak{m}^3 = 0$. Hence $W = R_2/I$, with the cocycle represented by its cyclic generator. For $w = 1$ the ideal is $I = \mathfrak{m}$, and for $w = 4$ it is zero.

If $w = 2$, then $I \subseteq \mathfrak{m}$ has dimension two. Every nonzero ideal of R_2 contains its socle kt_1t_2: multiply a nonzero linear part by a suitable generator, or start with a socle element.

Thus I also contains a nonzero linear form u, after subtracting a multiple of t_1t_2, and $I = (u)$. A change of basis in $\mathrm{GL}_2(k)$ takes (u) to (t_2). The accompanying nonzero determinant factor on a_{12} is absorbed by a rescaling of the cyclic module generator. These are precisely the three quotients displayed above, up to the isomorphisms of Theorem 3.2. □

For the uniform range, equality first fixes the derived module. We then classify its surviving cocycles under all changes of quotient basis and splitting; this last step distinguishes equality objects with identical lower central dimensions. Fix $c \geq 5$, and put $k = \mathbb{F}_4$, $d = c - 1$, $m = d - 1$, and

$$U = ke_0 \oplus H, \qquad H = \mathrm{span}_k\{e_1, \dots, e_m\}, \qquad R = k[t_0, \dots, t_m]/(t_0^2, \dots, t_m^2).$$

Write $K = (t_1, \dots, t_m)$, $\tau = t_0 \cdots t_m$, and $\tau_{ij} = \prod_{\ell \notin \{i,j\}} t_\ell$. For $\kappa = (\kappa_i) \in H^*$ and $\lambda = (\lambda_{ij})_{0 \leq i < j \leq m} \in k^{\binom{d}{2}}$, define an alternating cocycle $\beta_{\kappa,\lambda} : \Lambda^2 U \to K$ by

$$\begin{aligned} \beta_{\kappa,\lambda}(e_0, e_i) &= t_i + \lambda_{0i}\tau_{0i} && (1 \leq i \leq m), \\ \beta_{\kappa,\lambda}(e_i, e_j) &= t_0(\kappa_i t_j + \kappa_j t_i) + \lambda_{ij}\tau_{ij} && (1 \leq i < j \leq m). \end{aligned} \tag{72}$$

Let $F_{\kappa,\lambda}$ be the map in (10) with

$$W = K, \qquad P(x)w = \Big(\sum_{i=0}^{m} x_i t_i\Big)w, \qquad Q(x) = \sum_{i<j} x_i x_j \beta_{\kappa,\lambda}(e_i, e_j).$$

Thus these representatives require only multiplication in a square-zero polynomial ring.

Theorem 7.6 (Unrestricted metabelian extremizers)**.** *Every metabelian quadratic solution of class $c \geq 5$ and binary dimension $\mu(c)$ is binary-linearly conjugate to some $F_{\kappa,\lambda}$. Its derived module is isomorphic to K.*

The exact equivalence relation on the displayed parameters is the following ordinary Lie-cohomology orbit relation:

$$[\beta_{\kappa',\lambda'}] = \big[a\,\sigma_B\beta_{\kappa,\lambda}(B^{-1}\cdot, B^{-1}\cdot)\big], \qquad B = \begin{pmatrix} a & 0 \\ v & M \end{pmatrix}, \tag{73}$$

where $a \in k^\times$, $M \in \mathrm{GL}(H)$, $v \in H$, $\sigma_B(t_j) = \sum_i B_{ij}t_i$, and brackets denote classes in $H^2(U, K)$ for the action $e_i w = t_i w$. In particular, no nonlinear substitutions or Galois identifications are required.

Their number E_c is

$$E_c = \begin{cases} (7c - 13)/2, & c \geq 5 \text{ odd}, \\ 7(c-2)/2 + 2\mathbf{1}_{6|c}, & c \geq 6 \text{ even}. \end{cases} \tag{74}$$

Proof. Equality in the contraction bound forces the derived module to be the dual of $R/(t_1 \cdots t_m)$, hence to be K. An ordinary cohomology calculation gives a terminal star together with the κ- and λ-classes in (72); intrinsic generation is equivalent to the star being nonzero, and it has been normalized to one. Module intertwining maps give exactly (73). Theorem 3.2 then transfers the auxiliary-Lie classification to binary conjugacy. Appendix B proves the common-module assertion, the cohomology and morphism statements, and the complete orbit count. □

For $d \geq 5$, the orbit relation is especially explicit. If $A^{[2]}$ means entrywise squaring, its action is

$$\kappa' = a^2 M^{-T}\kappa, \qquad \lambda' = a\det(B)(\Lambda^2 B^{-T})^{[2]}\lambda. \tag{75}$$

After setting $\Omega = \lambda^{[2]}$ and writing $\Omega = e_0^* \wedge \ell + b$, the classification reduces to an alternating form b on H, two covectors on its radical, and one zero/nonzero flag. In the corank-two stratum an additional three-valued scalar invariant survives exactly when $6 \mid c$; this accounts for the correction in (74). At $c = 5$, two cohomological degrees coincide and the parameter representation is nonsplit. A symmetric-matrix description nevertheless gives all eleven classes, including this endpoint in the same count.

Models inside Cohn's construction. The regular Lie model and the zero-λ slice have a concrete classical source. Take characteristic two and the finite ground field $k = \mathbb{F}_4$ in Cohn's construction [8, Section 2, pp. 402–403]. Reindex its transcendental elements as $\xi_0, \xi_1, \ldots$, put $u = u(0)$, and let $D_i = \operatorname{ad} \xi_i|_M$. For $S \subseteq \{0, \ldots, d-1\}$, set

$$v_S = \Big(\prod_{i \in S} D_i\Big) u = \Big(\prod_{i \in S} \xi_i\Big) \sum_{T \subseteq S} u(\mathbf{1}_T).$$

Here $\mathbf{1}_T$ is the indicator sequence indexing Cohn's module basis. The displayed expansion is triangular in those basis vectors, so the v_S's are independent even over Cohn's larger coefficient field. The operators commute and square to zero, and $D_i v_S = v_{S \cup \{i\}}$ if $i \notin S$, and zero otherwise. Consequently the k-span of the ξ_i's and v_S's is the regular-module semidirect product $k^d \ltimes R_d$, of dimension $d + 2^d$ and class $d + 1$. Its polar realization is (105), or the compatible model F_0 after writing a ring element as its constant term plus an element of J.

The zero-λ models in (72) also occur as finite subalgebras. Choose

$$e_0 = \xi_0 + v_\varnothing, \qquad e_i = \xi_i + \kappa_i v_{\{0\}} \quad (1 \le i < d).$$

Their derived module is spanned by the v_S's with $S \cap \{1, \ldots, d-1\} \neq \varnothing$, and is therefore K. Their action and brackets are multiplication by t_i, $\beta(e_0, e_i) = t_i$, and $\beta(e_i, e_j) = t_0(\kappa_i t_j + \kappa_j t_i)$. Thus this finite subalgebra has dimension $d + 2^d - 2$, class $d + 1$, and gives $F_{\kappa,0}$ through Theorem 3.2. These finite identifications are derived here from Cohn's construction; his paper establishes the infinite-dimensional example rather than an extremal dimension formula.

Cohn's full characteristic-two algebra satisfies

$$[[[x, y], x], y] = 0.$$

Indeed, for $x = a + u'$, $y = b + v'$, its bracket is $D_b u' + D_a v'$; the commuting operators D_a, D_b square to zero, so applying $D_a D_b$ kills it. In the present family,

$$[[[e_i, e_j], e_i], e_j] = t_i t_j \beta(e_i, e_j) = \lambda_{ij} \tau.$$

Hence the nonzero-λ models lie outside every Lie subquotient of Cohn's characteristic-two example. The sharp bounds and full orbit classifications determine how the classical slice and these additional models populate the extremal layers. The same finite-subalgebra identifications hold over any finite field of characteristic two; the binary-map interpretation uses $k = \mathbb{F}_4$.

8 Fixed-surplus populations

At the sharp minimum the metabelian types have linear growth in the polar class. The first two added binary dimensions already produce a doubly exponential population. This section identifies the mechanism and gives the fixed-surplus law and exact principal orbits.

For orientation, put $n = 2^{c-1}$, and let r be the surplus in k-dimension, corresponding to $2r$ binary dimensions at $q = 4$. The three population scales are

$r \geq 1$ fixed :	fixed-surplus law (Theorem 8.3);
$r/\sqrt{n} \to \alpha \in (0, \infty)$:	growth and allocation (Theorem 9.5);
$r/\sqrt{n} \to \infty$:	minimum-derived density tends to zero (Theorem 9.1).

The critical staircase then resolves the transition on the integer defect scale in Section 9.5.

Counting convention. Throughout this section, $d = c - 1$ is the class-dependent core parameter. It is distinct from the point rank $d(L)$ of an individual solution: in a surplus branch the quotient can have dimension $d + s$, where s is the quotient surplus. The corresponding proof appendices use the same local convention. At $q = 4$, ordinary $\mathbb{F}_4$-Lie isomorphism counts agree with binary-linear conjugacy counts through the metabelian correspondence. We specify the field whenever the count is stated over a general $\mathbb{F}_q$.

8.1 Two more binary dimensions

The linear class count at $\mu(c)$ suggests a stability question: does a small dimension surplus preserve the rigidity of the minimum? The next theorem gives a sharp answer. Already the first admissible increase in dimension supports a much larger family, while keeping the same derived module as a metabelian extremizer.

Theorem 8.1 (A sharp jump above the minimum)**.** *For each $c \geq 5$, there is a family of indecomposable, noncompatible metabelian quadratic solutions of polar class c and binary dimension*

$$\mu(c) + 2 = 2^c + 2c - 4.$$

Put $d = c-1$ and $D_{\mathrm{par}} = 2^d - d - 3$. If T_c counts the binary-linear conjugacy classes represented by this family, and A_c counts all metabelian classes of polar class c and binary dimension $\mu(c) + 2$, then

$$\frac{4^{D_{\mathrm{par}}} - 1}{3\, 4^d |\operatorname{GL}_d(4)|} \leq T_c \leq 4^{D_{\mathrm{par}}} - 1, \qquad T_c \leq A_c \leq E_c\, 4^{2^d - 2} + 4^{(3d^2-d)/2}. \tag{76}$$

In particular

$$\log_4 A_c = 2^{c-1} + O(c^2) \qquad (c \longrightarrow \infty).$$

Thus doubly exponential class growth occurs at the first admissible dimension above the metabelian minimum.

Proof. We construct the family, identify an intrinsic line in its quotient, and use that line to bound how many parameters can become conjugate. The key point is parameter recovery under arbitrary binary conjugacy.

Construction. Use

$$R = k[t_0, \ldots, t_{d-1}]/(t_i^2), \qquad K = (t_1, \ldots, t_{d-1}), \qquad k = \mathbb{F}_4.$$

Write $z = t_1 \cdots t_{d-1}$, $\tau = t_0 z$, and $I = (z) = kz \oplus k\tau = \operatorname{Ann}_R K$. Let $\mathcal{P}_d$ be the space of sums of square-free monomials of degree at least two, excluding z and τ. It is a fixed representative space for $\operatorname{rad}(R/I)^2$, has dimension $D_{\mathrm{par}} = 2^d - d - 3$, and is contained in K.

For $p \in \mathcal{P}_d$, take $U = \operatorname{span}_k\{e_0, \ldots, e_{d-1}, v\}$, with actions on K and cocycle

$$P(e_i) = t_i \cdot, \qquad P(v) = p\cdot,$$
$$\beta(e_0, e_i) = t_i \quad (i \geq 1), \qquad \beta(e_0, v) = p,$$

all other quotient-basis pair values being zero. The actions commute and square to zero. The only additional triangle equation is $t_i p + p t_i = 0$, and the t_i, $i \geq 1$, generate K. These are therefore data for Theorem 3.2. Explicitly, put $b_p(X, V) = \sum_{i=1}^{d-1} X_i t_i + Vp$. The map is

$$G_p(a, X, V, w) = \big(\omega a, \omega X, \omega V, \omega w + (at_0 + b_p)w + ab_p + a^2 t_0 b_p\big), \quad w \in K. \tag{77}$$

Indeed $Q = ab_p$ and $PQ = a^2 t_0 b_p$, since $b_p^2 = 0$. All displayed terms have binary degree at most two. The quotient has k-dimension $d+1$, and $\dim_k K = 2^d - 2$, giving the stated binary dimension. As $p \in \mathfrak{m}^2$, the radical action ideal is still $\mathfrak{m} = (t_0, \ldots, t_{d-1})$, and

$$\mathfrak{m}^{d-1} K = k\tau \neq 0, \qquad \mathfrak{m}^d K = 0.$$

Thus every model has polar class $d + 1 = c$.

Parameter recovery. The unital algebra generated by the actions on the derived ideal is $\mathcal{A} = R/I$, acting faithfully on K. Its radical $\mathfrak{r}$ is intrinsic. The map

$$U \longrightarrow \mathfrak{r}/\mathfrak{r}^2, \qquad u \longmapsto P(u) \pmod{\mathfrak{r}^2},$$

has kernel exactly kv. A binary conjugacy between G_p and G_q induces k-linear maps on U and K, since their intrinsic scalar actions are multiplication by ω. It must therefore induce a quotient map of the form

$$Ae_i = \sum_j B_{ji} e_j + \alpha_i v, \qquad Av = \delta v,$$

where $B \in \mathrm{GL}_d(k)$, $\alpha \in k^d$, and $\delta \in k^\times$. There are $3\,4^d |\,\mathrm{GL}_d(4)|$ possible such maps.

Let C be the map on the derived ideal. Mixed polar brackets give $CP_p(u)C^{-1} = P_q(Au)$. Hence conjugation by C induces an algebra isomorphism $\Phi : \mathcal{A} \to \mathcal{A}$ satisfying

$$\Phi(t_i) = \ell_i + \alpha_i q, \qquad \ell_i = \sum_j B_{ji} t_j, \qquad \Phi(p) = \delta q.$$

In this algebra every radical element squares to zero. Substituting the fixed polynomial representative of p therefore gives the exact identity

$$\delta q = p(\ell + \alpha q) = p(\ell) + qh, \qquad h = \sum_i \alpha_i (\partial_i p)(\ell) \in \mathfrak{r}.$$

The derivatives are taken on that representative in R, before evaluation in R/I; no derivation on the quotient is assumed. Since p has no linear part, h has zero constant term. Consequently $\delta + h$ is a unit, and

$$q = (\delta + h)^{-1} p(\ell) = (\delta^{-1} + \delta^{-2} h) p(\ell) \quad \text{in } R/I. \tag{78}$$

For a fixed quotient map A, this determines the canonical representative q uniquely. Thus a conjugacy class meets $\mathcal{P}_d$ in at most $3\,4^d |\,\mathrm{GL}_d(4)|$ parameters. This necessary condition suffices for the counting bound; it does not require deciding which quotient maps lift to full conjugacies.

Indecomposability and counting. Restrict to $p \neq 0$. The action map $U \to \mathcal{A}$ is then injective: the t_i have independent linear parts and p is a nonzero element of $\mathfrak{r}^2$. A central vector of the original polar algebra must consequently lie in K. There its centralizer condition is $t_i w = 0$ for all i, so its center is exactly $k\tau$, of binary dimension two. In a nontrivial direct sum of solutions, each nilpotent polar summand has a nonzero center of even binary dimension (Proposition 2.3). Such a direct sum would have center dimension at least four. Hence every G_p, $p \neq 0$, is indecomposable. None is compatible, because $\mu(c) + 2 < \nu(c)$.

There are $4^{D_{\mathrm{par}}} - 1$ nonzero parameters, and the fiber bound proves the bounds on T_c in (76).

An upper bound for the entire dimension layer. The separate dimension bounds from Theorem 7.4 leave only two possibilities for the k-dimensions of quotient and derived ideal:

$$(d+1, 2^d - 2) \quad \text{or} \quad (d, 2^d - 1).$$

In the first case, Lemma 7.2 supplies a same-class subalgebra with d quotient generators. Its derived ideal already has dimension at least $2^d - 2$, so equals that of the full algebra. It is therefore one of the E_c metabelian extremizers. After choosing its representative, the extra quotient action belongs to $\mathrm{End}_R K = R/I$. Its constant term is zero, and its linear part can be removed by adding core quotient directions. At most $4^{D_{\mathrm{par}}}$ actions remain. For a fixed action p, the mixed cocycle values $a_i = \beta(e_i, v)$ satisfy

$$t_i a_j + t_j a_i = p\beta(e_i, e_j).$$

Their differences are ordinary one-cocycles. Changing the lift of v by $w \in K$ adds the one-coboundary $(t_i w)_i$. Lemma B.3 gives $\dim_k H^1(U_0, K) = d + 1$, where U_0 is the core quotient. This case therefore contributes at most $E_c 4^{D_{\mathrm{par}}+d+1} = E_c 4^{2^d-2}$ classes.

In the second case the derived module is a quotient of W_d retaining the top. Lemma B.2 gives at most $4^{\binom{d}{2}}$ possible module types and $\dim_k H^2(U, W) \le d^2$. For each fixed module, changes of section identify cocycles differing by a two-coboundary. Thus at most $4^{\binom{d}{2}+d^2} = 4^{(3d^2-d)/2}$ classes occur in this case. These are upper bounds: no uniqueness of the chosen core or the codimension-one module presentation is required.

Combining the two cases proves the bound on A_c. Since $E_c = O(c)$ and $\log_4 |\mathrm{GL}_d(4)| = d^2 + O(1)$, the upper and lower bounds give the asserted logarithmic asymptotic. Finally all solution dimensions are even by Proposition 2.1, so two is the first possible positive surplus above $\mu(c)$. □

The two upper bounds locate the source of the jump. Adding one $\mathbb{F}_4$-dimension to the derived ideal allows at most $4^{(3d^2-d)/2}$ classes. Keeping that ideal at its minimum and adding one quotient direction already produces the doubly exponential contribution. The location of the extra dimension therefore matters as well as its size.

The contrast is already quantitative at class seven. The minimum dimension $\mu(7) = 136$ has exactly $E_7 = 18$ classes, whereas (76) gives at least 32 485 958 indecomposable noncompatible classes in dimension 138. This lower bound comes from the uniform recovery argument, rather than an enumeration of the models.

Lie-algebra interpretation. The associated auxiliary $\mathbb{F}_4$-Lie algebras have the same one-dimensional center $k\tau$, so they too are indecomposable. They admit no restricted square. Indeed, $(\mathrm{ad}\, e_0)^2$ is zero on K. If it were $\mathrm{ad}\, y$, injectivity of the quotient action would force $y \in K$. But

$$(\mathrm{ad}\, e_0)^2 e_1 = t_0 t_1, \qquad [y, e_1] = t_1 y,$$

and the latter has zero $t_0 t_1$-coefficient because $y \in K$ has no t_0-term. Thus $(\mathrm{ad}\, e_0)^2$ is not inner, contradicting the restricted-square adjoint axiom. The correspondence therefore also gives the stated number of distinct indecomposable, nonrestrictable Lie algebras of class c and $\mathbb{F}_4$-dimension $2^{c-1} + c - 2$.

The construction uses the classical abelian-extension framework [5]. Its additional feature is that an intrinsic quotient line and (78) control all binary conjugacies despite the large group acting on the derived module. The upper bound also controls the full first-surplus layer. Explicit orbit representatives and an exact finite count for every c remain separate classification questions.

8.2 A law for every fixed surplus

The first-surplus transition belongs to a uniform counting law. It is convenient to formulate it for ordinary Lie algebras over a finite field $k = \mathbb{F}_q$ of characteristic two, with $q \geq 4$. Dimensions in this subsection are over k, unless explicitly called binary. A Lie algebra is *3-Engel* if $(\operatorname{ad} x)^3 = 0$ for every x. The following classical linearization identifies the relevant metabelian category; it is the metabelian consequence of [39, Section 1.2, identity (6)]. We include the short argument to specify the field-size hypothesis.

Lemma 8.2. *Let L be metabelian over a characteristic-two field with at least four elements, and put $W = L'$. Then L is 3-Engel if and only if $(\operatorname{ad} x)^2|_W = 0$ for every $x \in L$.*

Proof. The restricted operators $P_x = \operatorname{ad}(x)|_W$ commute, since $[L', L'] = 0$. If their squares vanish, then $(\operatorname{ad} x)^3 z = P_x^2[x, z] = 0$. Conversely, apply the 3-Engel identity to $x + ty$ and z. Commutativity and characteristic two give

$$0 = P_x^2[x, z] + tP_x^2[y, z] + t^2 P_y^2[x, z] + t^3 P_y^2[y, z].$$

This polynomial has degree at most three, below the field size. Its coefficient of t vanishes, so $P_x^2[y, z] = 0$. Brackets span W, proving the assertion. □

Write $\ell(c)$ for the minimum dimension of a metabelian 3-Engel algebra of class $c \geq 5$. Lemma C.1 gives $\ell(c) = 2^{c-1} + c - 3$. Let $B_{c,r}(q)$ count ordinary k-Lie isomorphism classes of class c and dimension $\ell(c) + r$ in this category.

Theorem 8.3 (The fixed-surplus law)**.** *Put $d = c - 1 \geq 4$ and $D_{\mathrm{par}} = 2^d - d - 3$. For $1 \leq r \leq D_{\mathrm{par}}$,*

$$B_{c,r}(q) \;\geq\; \frac{\displaystyle\prod_{j=0}^{r-1} (q^{D_{\mathrm{par}}} - q^j)}{q^{dr} |\operatorname{GL}_d(q)| \, |\operatorname{GL}_r(q)|}. \tag{79}$$

The lower bound is realized by indecomposable algebras with one-dimensional center and no restricted square. For each fixed q and $r \geq 1$,

$$\log_q B_{c,r}(q) = r\, 2^{c-1} + O_{q,r}(c^2) \qquad (c \longrightarrow \infty). \tag{80}$$

Proof. Put

$$R = k[t_0, \ldots, t_{d-1}]/(t_i^2), \quad K = (t_1, \ldots, t_{d-1}), \quad I = (t_1 \cdots t_{d-1}) = \operatorname{Ann}_R K.$$

Put $\mathfrak{m} = (t_0, \ldots, t_{d-1})$, so $\dim K = 2^d - 2$. Take canonical representatives $\mathcal{P}_d$ of $\operatorname{rad}(R/I)^2$, excluding the two monomials in I; their dimension is D_{par}. Choose an ordered independent frame $\mathbf{p} = (p_1, \ldots, p_r)$ in $\mathcal{P}_d$. On $U \oplus K$, where $U = \langle e_0, \ldots, e_{d-1}, v_1, \ldots, v_r \rangle_k$, define commuting actions and an alternating cocycle by

$$P(e_i) = t_i \cdot, \quad P(v_a) = p_a \cdot, \qquad \beta(e_0, e_i) = t_i \ (i \geq 1), \quad \beta(e_0, v_a) = p_a,$$

with all other basis pair values zero. In particular, the Jacobi equation on e_0, v_a, v_b is $p_a p_b + p_b p_a = 0$. The cocycle values generate K, so the derived ideal is exactly K. All radical elements of R square to zero; hence the algebra is 3-Engel by Lemma 8.2. Its lower central ideals are $\gamma_j = \mathfrak{m}^{j-2} K$ for $j \geq 2$, with terminal term $k(t_0 \cdots t_{d-1})$. Its class is $d + 1$, and its dimension is $d + r + 2^d - 2 = \ell(c) + r$.

The faithful generated action algebra is $\mathcal{A} = R/I$. The kernel of the intrinsic map $U \to \operatorname{rad}(\mathcal{A})/\operatorname{rad}(\mathcal{A})^2$ is $V_r = \langle v_1, \ldots, v_r \rangle_k$. Any isomorphism between two models therefore induces

$$Ae_i = \sum_j B_{ji} e_j + \sum_b \alpha_{bi} v_b, \qquad Av_a = \sum_b \Delta_{ba} v_b,$$

where $B \in \mathrm{GL}_d(k)$, $\Delta \in \mathrm{GL}_r(k)$. There are $q^{dr}|\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ possible quotient maps. For target frame $\mathbf{p}'$, conjugation on the derived ideal induces an algebra isomorphism satisfying

$$\Delta^T \mathbf{p}' = \mathbf{p}(\ell + \alpha^T \mathbf{p}'), \qquad \ell_i = \sum_j B_{ji} t_j. \tag{81}$$

All polynomials here are evaluated on their fixed representatives.

For a fixed quotient map this equation has at most one target frame. Indeed, compare two solutions and let e be their difference. Every source polynomial has degree at least two, and all its arguments lie in $\mathrm{rad}(\mathcal{A})$. Exact telescoping of each monomial therefore gives $\Delta^T e = Me$, with every entry of M in that radical. Thus $e = Ne$, where $N = (\Delta^T)^{-1}M$. Since $\mathrm{rad}(\mathcal{A})^d = 0$, also $N^d = 0$, and iteration gives $e = 0$. The exact telescoping identity retains all products involving distinct target parameters. This uniqueness is an instance of the nilpotent implicit-function principle of Bahturin–Olshanskii [3, Sections 1.2–1.3]; the intrinsic quotient subspace and the equation forced by isomorphism specify its application here. Each isomorphism class consequently contains at most $q^{dr}|\mathrm{GL}_d(q)||\mathrm{GL}_r(q)|$ ordered frames from this family, proving (79).

Frame independence makes $U \to \mathcal{A}$ injective. The center is therefore contained in K, where it is the one-dimensional socle $k(t_0 \cdots t_{d-1})$. This implies indecomposability. If $(\operatorname{ad} e_0)^2 = \operatorname{ad} y$, its zero action on K forces $y \in K$. But $(\operatorname{ad} e_0)^2 e_i = t_0 t_i$, whereas $t_i y$ has zero $t_0 t_i$-coefficient. Hence no restricted square exists.

For fixed q, r, the numerator of (79) has logarithm $rD_{\mathrm{par}} + O_{q,r}(1)$, while the denominator has logarithm $d^2 + dr + O_{q,r}(1)$. This proves the required lower estimate. Proposition C.3 gives the matching upper estimate, including all possible derived-module defects, and proves (80). □

Corollary 8.4 (Generic structure)**.** *With isomorphism classes counted uniformly, the proportion in $B_{c,r}(q)$ failing any of the following properties is at most $q^{-2^d+O_{q,r}(d^2)}$: the derived ideal has minimum dimension 2^d-2 and is the standard core module K; the action $L/L' \to \mathrm{End}_k(L')$ is injective; the center is one-dimensional; the algebra is indecomposable and has no restricted square.*

Proof. In the upper-bound proof, write the quotient and derived dimensions as $d + s$ and $2^d - 2 + h$, where $s + h = r$. The branches $h \geq 1$ together contribute at most $q^{(r-1)2^d+O_{q,r}(d^2)}$ classes. When $h = 0$, the derived module is K. Removing the linear parts of the extra actions gives an r-tuple in $\mathcal{P}_d$. Rank-deficient tuples number at most $q^{(r-1)D_{\mathrm{par}}+r(r-1)}$: choose $r-1$ spanning vectors and their coefficient matrix. The remaining cocycle choices cost only $q^{O_{q,r}(d^2)}$. Thus these exceptions have the same bound. Injectivity of the full quotient action is intrinsic, so different choices of core do not affect this division into cases.

For a full-rank tuple, the center and indecomposability follow as above. In a normalized extremal core, (166) gives $\beta(e_0, e_i) \equiv t_i \pmod{\mathfrak{m}K}$. Hence $(\operatorname{ad} e_0)^2 e_i$ still has coefficient one at $t_0 t_i$. The same inner-derivation obstruction excludes a restricted square, independently of the mixed cocycle choices. Dividing the exceptional bound by (79) proves the asserted proportion. □

Corollary 8.5 (Binary fixed-surplus law)**.** *Let $A_{c,r}$ count all metabelian quadratic solutions of polar class c and binary dimension $\mu(c) + 2r$, up to binary-linear conjugacy. Then $A_{c,r} = B_{c,r}(4)$, and, for each fixed $r \geq 1$,*

$$\log_4 A_{c,r} = r\, 2^{c-1} + O_r(c^2).$$

Almost all have binary polar center of dimension two and are indecomposable; their auxiliary $\mathbb{F}_4$-Lie algebras admit no restricted square.

Proof. Use Theorem 3.2, Lemma 8.2, and $\mu(c) = 2\ell(c)$. For the generic objects the faithful quotient action forces the original polar center into the derived ideal, where it is the same one-dimensional $\mathbb{F}_4$-socle. Its binary dimension is two, excluding a nontrivial direct sum by Proposition 2.3. The lower-bound family is noncompatible for every r: the compatibility criterion would require $t_i b_0 = t_0 t_i$ with $b_0 \in K$, whose $t_0 t_i$-coefficients are respectively zero and one. □

Corollary 8.6 (Compatibility at fixed surplus)**.** *For every fixed $r \geq 2$, the proportion of compatible objects among the $A_{c,r}$ classes is at most $4^{-2^d+O_r(d^2)}$. At $r = 2$, their dimension is precisely $\nu(c)$, and the compatible classes are precisely the extremizers counted by N_c. With $D_{\mathrm{par}} = 2^d - d - 3$,*

$$\frac{N_c}{A_{c,2}} \leq \frac{60(1+\eta_d)\,4^{-2^d+3d+4}}{(1-4^{-D_{\mathrm{par}}})(1-4^{1-D_{\mathrm{par}}})} = 4^{-2^d+O(d)}. \tag{82}$$

Here η_d is the explicit bound from Corollary 11.13.

Proof. Lemma 11.2 gives $\dim_{\mathbb{F}_4} L' \geq 2^d - 1$, so compatible objects lie in the $h \geq 1$ upper-bound branches. This proves the first assertion. For the sharper $r = 2$ estimate, combine

$$N_c \leq \frac{4^{D_{\mathrm{par}}+1}}{3|\,\mathrm{GL}_d(4)|}(1+\eta_d)$$

with (79). The $\mathrm{GL}_d(4)$ factors cancel, and $|\,\mathrm{GL}_2(4)| = 180$ gives (82). □

8.3 Exact principal orbits at every fixed surplus

The minimum-derived stratum has a complete orbit model for any number of additional quotient directions. Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and put

$$d = c - 1 \geq 4, \quad C_d = \binom{d}{2}, \quad D_{\mathrm{par}} = 2^d - d - 3, \quad w_0 = 2^d - 2.$$

Use R, K, I from (165), and let $\mathcal{P}_d$ be the canonical monomial representatives of $\mathrm{rad}(R/I)^2$. On $U = \langle e_0, \dots, e_{d-1}, v_1, \dots, v_r \rangle_k$, let $P_{\mathbf{p}}(e_i) = t_i \cdot$ and $P_{\mathbf{p}}(v_a) = p_a \cdot$, for $\mathbf{p} \in \mathcal{P}_d^r$. Let η denote the degree-one star coefficient, and set

$$\mathcal{Y}_{d,r} = \coprod_{\mathbf{p} \in \mathcal{P}_d^r} \{[\beta] \in H^2(U, K; P_{\mathbf{p}}) : \eta = 1\}.$$

Each fiber has dimension

$$f_{d,r} = (r+1)d + C_d + r + \binom{r}{2} - 1.$$

Write $\mathcal{Y}_{d,r}^{\mathrm{fr}}$ for its restriction to linearly independent tuples $p_1, \dots, p_r$.

Let P_H be the parabolic preserving $H = \langle e_1, \dots, e_{d-1} \rangle_k$. The effective quotient group is

$$G_{d,r} = \left\{ A_g = \begin{pmatrix} B & 0 \\ \alpha & \Delta \end{pmatrix} : B = \begin{pmatrix} a & 0 \\ b & M \end{pmatrix} \in P_H,\ \alpha \in k^{r \times d},\ \Delta \in \mathrm{GL}_r(k) \right\}, \tag{83}$$

with

$$|G_{d,r}| = (q-1)q^{rd+d-1}|\,\mathrm{GL}_{d-1}(q)||\,\mathrm{GL}_r(q)|.$$

For $\ell_i = \sum_j B_{ji} t_j$, its action is

$$\Delta^T \mathbf{p}' = \mathbf{p}(\ell + \alpha^T \mathbf{p}') \qquad \text{in } R/I, \tag{84}$$

$$[\beta'] = [a\sigma_g \beta(A_g^{-1} \cdot, A_g^{-1} \cdot)], \qquad \sigma_g(t_i) = \ell_i + \sum_a \alpha_{ai} p'_a. \tag{85}$$

The implicit system has a unique solution in $\mathcal{P}_d^r$.

Theorem 8.7 (Complete minimum-derived orbits)**.** *The isomorphism classes of metabelian 3-Engel k-Lie algebras of class c, dimension $\ell(c)+r$, and minimum derived dimension w_0, are exactly $\mathcal{Y}_{d,r}/G_{d,r}$. The faithful-action subfamily is exactly $\mathcal{Y}^{\mathrm{fr}}_{d,r}/G_{d,r}$. Equations* (84)–(85) *are sufficient as well as necessary for equivalence.*

Theorem 8.8 (Multiplicative enumeration at fixed surplus)**.** *For every fixed $q \geq 4$ of characteristic two and fixed $r \geq 1$,*

$$B_{c,r}(q) \sim \frac{\prod_{j=0}^{r-1}(q^{D_{\mathrm{par}}} - q^j)\, q^{C_d+r+\binom{r}{2}}}{(q-1)|\operatorname{GL}_{d-1}(q)||\operatorname{GL}_r(q)|} \qquad (c \longrightarrow \infty). \tag{86}$$

Almost all isomorphism classes have full automorphism-group order

$$|\operatorname{Aut}_k(L)| = q^{2w_0+d+r-1} = q^{2^c+c+r-6}. \tag{87}$$

For $q = 4$, these are respectively the binary conjugacy count $A_{c,r}$ and its generic binary-linear centralizer order.

The proof retains all cocycle coefficients, including brackets between different added directions. Nontrivial linear parts are controlled by an exterior-rank estimate. For a pure matrix shear, specializing its active variables to zero gives a necessary affine multiplication constraint. This retains the higher cross-products in the implicit substitution and controls the summed Burnside error for every fixed r. Appendix D proves both statements. At $r = 1$, (86) reduces exactly to

$$B_{c,1}(q) \sim \frac{(q^{D_{\mathrm{par}}} - 1)q^{C_d+1}}{(q-1)^2|\operatorname{GL}_{d-1}(q)|} \sim \frac{q^{2^d-C_d-5}}{(1-q^{-1})^2\prod_{j=1}^{\infty}(1-q^{-j})}.$$

9 The critical transition

The fixed-surplus law persists while the surplus is small relative to the square root of the core dimension. At that scale the whole population changes its typical derived dimension. We first locate the transition, then determine its constant, allocation profile, and discrete staircase. Let $B^{\min}_{c,r}(q)$ count the classes in $B_{c,r}(q)$ whose derived ideal has the minimum possible dimension $2^{c-1}-2$, and put

$$\pi_{d,r}(q) = \frac{B^{\min}_{d+1,r}(q)}{B_{d+1,r}(q)}, \qquad d = c-1.$$

Both the numerator and denominator count unlabelled, k-linear Lie isomorphism classes. In particular this is not a parameter-weighted proportion.

Theorem 9.1 (A transition scale for minimum-derived objects)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and let $d \to \infty$. Uniformly over positive integers r,*

$$\log_q B_{d+1,r}(q) = \Theta\big(r2^d + r^3\big). \tag{88}$$

For integer sequences $r = r(d) \geq 1$,

$$r = o(2^{d/2}) \quad \Longrightarrow \quad \log_q B_{d+1,r}(q) \sim r2^d, \qquad \pi_{d,r}(q) \longrightarrow 1, \tag{89}$$

$$r/2^{d/2} \longrightarrow \infty \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \tag{90}$$

In the second regime, $\log_q \pi_{d,r}(q) \leq -(2/27 - o(1))r^3$.

The small-surplus assertion follows by retaining the derived defect h in Proposition C.3; its loss of $h2^d$ dominates the remaining terms. For the opposite regime, two-step alternating-tensor families supply a cubic population. Their contribution can be counted through the characteristic quotient $L/\gamma_3(L)$, whose dimension discards the large minimum core. Appendix C.1 gives the precise bounds and proof.

Thus $2^{(c-1)/2}$ separates the two off-window behaviors of the minimum-derived population. Theorem 9.2 below sharpens this scale to an explicit critical constant. The finer allocation and staircase theorems then localize the remaining undetermined densities to shrinking boundary bands. The large-surplus lower family uses direct sums, but the conclusion concerns only the derived dimension; it makes no assertion that a typical object in that regime is decomposable. At $q = 4$, the same conclusions apply to binary conjugacy classes in dimension $\mu(c)+2r$, with minimum binary derived dimension $2^c - 4$, by Corollary 8.5.

9.1 The critical constant for minimum derived dimension

The transition scale admits a sharp constant away from its boundary. Two effects determine it. Normalizing the subspace spanned by the extra actions removes a quadratic coordinate redundancy from the minimum-derived count. A single additional central derived direction, coupled to those same quotient directions, supplies enough alternating forms to outweigh that count on the other side. The latter family consists of indecomposable algebras.

Theorem 9.2 (The critical constant)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. For every integer sequence $r = r(d) \geq 1$, as $d \to \infty$,*

$$\limsup_{d\to\infty} \frac{r}{2^{d/2}} < \sqrt{2} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \tag{91}$$

$$\liminf_{d\to\infty} \frac{r}{2^{d/2}} > \sqrt{2} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \tag{92}$$

In particular, if $r/2^{d/2} \to \alpha \neq \sqrt{2}$ with $0 \leq \alpha \leq \infty$, the limit is one for $\alpha < \sqrt{2}$ and zero for $\alpha > \sqrt{2}$. These are proportions of unlabelled k-linear isomorphism types.

Proof. Write $n = 2^d$, $D_{\mathrm{par}} = n - d - 3$, and use B_h for the branch with derived dimension $n - 2 + h$, so $\pi_{d,r}(q) = B_0/\sum_h B_h$. For the first implication there is an $\epsilon > 0$ such that eventually $r^2/2 \leq (1-\epsilon)n$. In particular $r \leq D_{\mathrm{par}}/2$, and

$$Z(d, r) = 2d^2 + 3dr + 2r + 6 = o(n).$$

Proposition E.3 gives, with $a = n - r^2/2 - Z(d, r) \geq (\epsilon - o(1))n$,

$$1 - \pi_{d,r}(q) \leq \frac{q^{-a}}{1 - q^{-a}} \longrightarrow 0.$$

For the second implication, first consider the indices with $r \leq D_{\mathrm{par}}/2$. Here $r \geq 3$ eventually and Proposition E.4 yields

$$\pi_{d,r}(q) \leq \frac{B_0}{B_1} \leq q^{\,n - r^2/2 + dr + d^2 + r/2 + \log_q(\frac{3}{2}(r+1))}.$$

The strict lower limit assumption gives an $\epsilon > 0$ with $n - r^2/2 \leq -\epsilon r^2$ eventually. Since $r \gg d$, every remaining exponent term is $o(r^2)$; the last display tends to zero along these indices. On the remaining indices, $r > D_{\mathrm{par}}/2$, one has

$$\frac{r}{\sqrt{n}} > \frac{n - d - 3}{2\sqrt{n}} \longrightarrow \infty.$$

The all-surplus upper-side assertion (90) of Theorem 9.1 therefore applies along this subsequence and also gives $\pi_{d,r}(q) \to 0$. Both subsequences have the same limit. This argument uses no monotonicity of $\pi_{d,r}(q)$ as the dimension changes. □

For $q = 4$, Theorem 3.2 transfers the same critical constant to binary-linear conjugacy classes of metabelian quadratic solutions in binary dimension $\mu(c) + 2r$, where $d = c - 1$. The counted minimum derived dimension is $2^c - 4$. The coupled lower family gives indecomposable quadratic solutions by Remark E.5. Theorems 9.5 and 9.7 refine this constant-scale statement to the macroscopic allocation and the discrete critical profile. The relative boundary weights inside the remaining shrinking bands are separate questions.

The following initial localization is sharpened in Corollary 9.8.

Corollary 9.3 (An initial outer localization of the transition)**.** *Fix a finite characteristic-two field of order $q \geq 4$, put $n = 2^d$, and let $r = r(d) \geq 1$ be integers. Then, as $d \to \infty$,*

$$\begin{aligned} r \leq \sqrt{2n} - 3d - 3 \quad \textit{eventually} \quad &\Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \\ r \geq \sqrt{2n} - d + 1 \quad \textit{eventually} \quad &\Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \end{aligned}$$

The two thresholds are separated by $2d + 4$, giving an outer localization of order $d = c - 1$ around $\sqrt{2n}$. This does not assert that the actual transition has that width or determine its boundary density.

Proof. Set $t = \sqrt{2n}$. The exponent

$$a(d, r) = n - \tfrac{1}{2}r^2 - 2d^2 - 3dr - 2r - 6$$

in Proposition E.3 is decreasing in real $r \geq 0$. Its value at the lower endpoint is

$$a(d, t - 3d - 3) = t + \tfrac{5}{2}d^2 + 6d - \tfrac{9}{2} =: A_d.$$

For all sufficiently large d, the assumed lower-side range is contained in $1 \leq r \leq D_{\mathrm{par}}/2$. Hence

$$1 - \pi_{d,r}(q) \leq \frac{q^{-A_d}}{1 - q^{-A_d}} \longrightarrow 0.$$

For the upper side, Corollary E.7 proves the stronger endpoint by combining a uniform count of the minimum-derived stratum with the full-central-cocycle lower family. That comparison is independent of the present lower-side argument. No monotonicity of $\pi_{d,r}(q)$ is assumed. □

9.2 Uniform enumeration through the transition scale

Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$. The minimum-derived stratum admits a uniform multiplicative count even when the surplus grows. Put $d = c - 1$, $n = 2^d$, $D_{\mathrm{par}} = n - d - 3$, $C_d = \binom{d}{2}$, and $w_0 = n - 2$. Write $B_0 = B^{\min}_{d+1,r}(q)$. The range in the following theorem is

$$2d \leq r \leq \frac{D_{\mathrm{par}}}{4d}.$$

It includes every sequence $r/\sqrt{n} \to \alpha$ with $0 < \alpha < \infty$.

Theorem 9.4 (Multiplicative principal enumeration at growing surplus)**.** *Uniformly for the integer range* (193)*, as $d \to \infty$,*

$$B_0 \sim \frac{\prod_{i=0}^{r-1}(q^{D_{\mathrm{par}}} - q^i)\, q^{C_d + r + \binom{r}{2}}}{(q-1)\,|\,\mathrm{GL}_{d-1}(q)|\,|\,\mathrm{GL}_r(q)|}. \tag{93}$$

Almost all classes in this stratum have full automorphism order $q^{2w_0+d+r-1}$. These statements concern the whole minimum-derived stratum, not the complete dimension layer above the critical constant.

The proof reparametrizes a fixed pure shear by its induced ring automorphism σ and the action subspace Q:

$$\operatorname{span}\{\sigma(t_i) - t_i\} \subseteq Q \subseteq \operatorname{Fix}(\sigma).$$

A displacement-rank inequality and Gaussian incidence counts then control the complete stabilizer sum. This avoids a loss depending on the number of added directions in the fixed-surplus estimate. Appendix E.3 gives the proof and the sharper upper-side localization used in Corollary 9.3. The full dimension layer has the same multiplicative asymptotic on the subrange $2d \le r \le \sqrt{2n} - 3d - 3$, by the independent minimum-derived dominance bound. Above the critical constant, (93) counts only the minimum-derived stratum.

9.3 Exact growth and the typical surplus allocation

At a fixed ratio of surplus to the square-root core scale, an explicit profile determines both the growth of the entire population and its typical allocation. Continue to put $n = 2^d$, $d = c - 1$, and count ordinary k-linear Lie isomorphism types uniformly. For $\alpha > 0$, define

$$\gamma_\alpha = \begin{cases} \alpha, & 0 < \alpha \le \sqrt{2}, \\ (\alpha + \sqrt{\alpha^2 + 6})/3, & \alpha > \sqrt{2}, \end{cases} \qquad \beta_\alpha = \alpha - \gamma_\alpha, \qquad \varphi(\alpha) = \gamma_\alpha + \frac{\beta_\alpha \gamma_\alpha^2}{2}. \tag{94}$$

Theorem 9.5 (Exact logarithmic growth and allocation)**.** *Fix a finite field $k = \mathbb{F}_q$ of characteristic two, $q \ge 4$, and suppose $r/\sqrt{n} \to \alpha \in (0, \infty)$. Then*

$$\log_q B_{d+1,r}(q) = \big(\varphi(\alpha) + o(1)\big) n^{3/2}. \tag{95}$$

For every $\varepsilon > 0$, there is an $\eta > 0$ such that, for all sufficiently large d, all but a proportion $q^{-\eta n^{3/2}}$ of these types satisfy

$$\left| \frac{h}{\sqrt{n}} - \beta_\alpha \right| < \varepsilon, \qquad h = \dim_k L' - (n - 2). \tag{96}$$

Thus for $\alpha > \sqrt{2}$, the typical derived share is

$$\frac{h}{r} \longrightarrow \frac{2 - \sqrt{1 + 6/\alpha^2}}{3}, \qquad \frac{s}{r} = 1 - \frac{h}{r}.$$

The core contributes the term γ, while alternating central brackets contribute $\beta\gamma^2/2$. Their competition gives the profile φ and its unique maximizing allocation. The main upper-bound issue is that arbitrary nonsplit derived modules may have additional actions. Restricting those actions to the core socle separates their visible part from a controlled kernel. A common rational flag bounds the visible tuples, and ordinary cohomology charges their rank against the available cocycles. This coupled bound covers every derived module, so the result counts the whole layer. Appendix G proves the theorem using the classical triangular square-zero count and ordinary Chevalley–Eilenberg tools.

At $q = 4$, the same statement counts metabelian quadratic solutions in binary dimension $\mu(c) + 2r$. At the critical ratio it implies $h = o(\sqrt{n})$ in type-probability; the finer question whether $h = 0$, and the density inside the remaining critical band, requires additional information.

There is also a finer comparison at the first positive derived defect. Writing B_h for the branch with derived dimension $n - 2 + h$, Corollary I.5 gives, uniformly for $a \le r/\sqrt{n} \le b$ with fixed $0 < a < b < \infty$,

$$\log_q B_1 = (r-1)n - 3(r-1) + O_{q,a,b}(d),$$
$$\log_q (B_1/B_0) = -n + \binom{r+d}{2} + O_{q,a,b}(d).$$

The proof classifies the three possible one-defect core modules and identifies which extra actions survive Jacobi. This locates the B_1-versus-B_0 comparison near $\sqrt{2n}-d+\frac{1}{2}$; controlling all defects on this finer scale is the subject of Section 9.5.

9.4 Explicit comparison bounds above the threshold

The exact allocation law identifies the limiting derived share. The next estimate gives explicit interior margins and an exponential error bound for each fixed finite supercritical ratio. Continue to put $d = c - 1$, $n = 2^d$, and $w_0 = n - 2$. In the layer of dimension $\ell(c) + r$, write

$$\dim_k L' = w_0 + h, \qquad \dim_k(L/L') = d + s, \qquad h + s = r.$$

Thus h and s measure the parts of the surplus spent on the derived ideal and on the quotient, respectively. Proportions below give equal weight to each k-linear isomorphism type.

Proposition 9.6 (Explicit interior margins)**.** *Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \geq 4$, and a finite constant $\alpha > \sqrt{2}$. Suppose $r/2^{d/2} \to \alpha$ as $d \to \infty$. There are constants $\epsilon_h, \epsilon_s, \eta > 0$ such that the proportion of all types in $B_{d+1,r}(q)$ satisfying*

$$\epsilon_h 2^{d/2} < h < r - \epsilon_s 2^{d/2} \tag{97}$$

is at least $1 - q^{-\eta 2^{3d/2}}$ for all sufficiently large d. One may take

$$\delta = \alpha^2/2 - 1, \qquad \epsilon_h = \frac{\delta}{16\alpha}, \qquad \epsilon_s = \frac{\alpha}{4(1+6\alpha^2)}, \qquad \eta = \min\left\{\frac{\delta^2}{16\alpha}, \frac{\alpha}{4}\right\}.$$

In particular, both h and s have order r for almost all isomorphism types in this regime.

The comparison family couples any central derived defect h to the same faithful high-class core. Every member of that family is indecomposable. Comparing its count with the complete branch upper bounds gives the explicit constants above; the limiting allocation is determined by Theorem 9.5. At $q = 4$, the estimate concerns binary conjugacy classes in dimension $\mu(c) + 2r$, with binary derived dimension $2^c - 4 + 2h$. Appendix F gives the comparison construction and proof.

The constants depend on the fixed finite ratio $\alpha > \sqrt{2}$. The off-window regimes, including $r/2^{d/2} \to \infty$, are covered separately by Theorem 9.1.

9.5 The discrete critical transition

The macroscopic allocation law of Theorem 9.5 determines $h/\sqrt{n}$. On a finer scale, the entire population is governed by an integer defect profile. Put

$$T = \sqrt{2n}, \qquad R_d^* = T - d + \tfrac{1}{2}, \qquad x_{d,r} = r - R_d^*, \qquad M(x) = \max_{h \in \mathbb{Z}_{\geq 0}} \{hx - h(h-1)\}.$$

Write $B_h = B_{d+1,r}^{(h)}(q)$ for the number of types with derived dimension $n - 2 + h$; thus $B_0 = B_{d+1,r}^{\min}(q)$ depends on d, r, q. The centering by d matters here: bounded changes in $x_{d,r}$ can change the typical integer derived defect even though they disappear in $r/\sqrt{n}$.

Theorem 9.7 (The critical profile and the defect staircase)**.** *Fix $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$. For each fixed $A > 0$, uniformly for integer r with $|x_{d,r}| \leq A$,*

$$\log_q \frac{B_{d+1,r}(q)}{B_0} = TM(x_{d,r}) + O_{q,A}(d). \tag{98}$$

If $x_{d,r} \to x$, then all but a proportion $q^{-\kappa T}$ of the types have the following derived defects, for some $\kappa > 0$:

(i) $h = 0$ *if* $x < 0$*;*

(ii) $h = j$ *if* $2j - 2 < x < 2j$*, for an integer* $j \geq 1$*;*

(iii) $h \in \{j, j+1\}$ *if* $x = 2j$*, for an integer* $j \geq 0$*.*

Near an adjacent-branch boundary,

$$\log_q(B_{j+1}/B_j) = T(x_{d,r} - 2j) + O_{q,j}(d). \tag{99}$$

The relative weights inside the remaining shrinking boundary band are not determined by this estimate.

Thus the typical defects are successively $0, 1, 2, \ldots$, with changes at offsets $0, 2, 4, \ldots$. The score difference $f_x(h+1) - f_x(h) = x - 2h$ is elementary; the counting content is a uniform bound on every module and defect branch, including the growing-defect tail.

Corollary 9.8 (A shrinking outer band for minimum derived dimension)**.** *For every integer sequence* $r = r(d) \geq 1$*,*

$$\begin{aligned} r \leq R_d^* - 4d/T \quad &\textit{eventually} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 1, \\ r \geq R_d^* + 4d/T \quad &\textit{eventually} \quad \Longrightarrow \quad \pi_{d,r}(q) \longrightarrow 0. \end{aligned}$$

The two outer endpoints have separation $8d/T = o(1)$*.*

Appendix I proves both statements. The key structural step is an exact ordinary-cohomology obstruction: when the derived center has its maximum dimension $h + 1$, the core module is a diagonal central extension E_Q of K, and Jacobi removes precisely $h(d-1)$ nonpolynomial action parameters per extra direction. The linear stabilizer of $\ker Q$ then makes the first-defect logarithmic error $O_q(d)$. The socle-sensitive bounds discard the complementary modules and the growing-defect tail. Diagonal Frobenius terms are standard in characteristic-two extension algebra; compare [20, Section 2]. Their use here is in the complete nearcritical census and its resulting discrete distribution.

For odd d, $T = 2^{(d+1)/2}$ is an integer. At each fixed integer offset ℓ, $r = T - d + \ell$, almost every type has

$$h = \max\{0, \lceil \ell/2 \rceil\}.$$

In particular, the first change occurs between the two consecutive integer surpluses $T - d$ and $T - d + 1$. The boundary weights in the theorem concern limits of the real offsets $x_{d,r}$ along integer surplus sequences.

10 When the original polar bracket is $\mathbb{F}_4$-bilinear

Return to the normal form (10) and keep the intrinsic scalar actions on $U = V/W$ and W. Its auxiliary bracket (18) is always $\mathbb{F}_4$-bilinear. The following definition concerns the original polar bracket of f.

Definition 10.1. A metabelian solution is *compatible* if V admits an $\mathbb{F}_4$-vector space structure that restricts to the intrinsic structure on W, induces the intrinsic structure on U, and makes the original polar bracket $\mathbb{F}_4$-bilinear.

In the criterion below we use the symmetric convention $a_{ji} = a_{ij}$ and $a_{ii} = 0$.

Theorem 10.2. *For a normal form with data* P_i, a_{ij}*, the following are equivalent:*

(i) *the solution is compatible;*

(ii) *there are* $b_1, \dots, b_d \in W$ *satisfying*

$$P_i b_i = 0, \qquad P_i b_j = P_j a_{ij} \quad (i \neq j); \tag{100}$$

(iii) *the auxiliary* $\mathbb{F}_4$*-Lie algebra has a restricted square* r *with* $r(W) = 0$ *and* $r(\mathfrak{l}) \subseteq W$.

If a compatible scalar lift exists, the number of such scalar actions on the fixed binary space, with the prescribed actions on U, W*, is* 4^{dk}*, where* $k = \dim_{\mathbb{F}_4} \bigcap_i \ker P_i$.

Proof. Every binary-linear lift of multiplication by ω has the form

$$T(x, z) = (\omega x, \omega z + t(x)).$$

The relation $T^2 + T = \mathrm{id}$ is exactly $t(\omega x) = \omega^2 t(x)$. Thus $t(x) = b(x) = \sum_i x_i^2 b_i$. The involutory binary shear $S(x, z) = (x, z + b(x))$ conjugates T to standard scalar multiplication. It changes the cocycle of f to

$$Q + b + P(Q + b).$$

The first two terms have $\mathbb{F}_4$-bilinear polars. The last term has scalar weight zero. Its polar could be $\mathbb{F}_4$-bilinear only if that polar were zero, since an $\mathbb{F}_4$-bilinear polar has weight two. An additive zero-constant function of weight zero is zero: additivity and invariance under ω give its value at $\omega^2 x = x + \omega x$ equal to zero, and then invariance gives its value at x equal to zero. Compatibility is therefore equivalent to

$$P(x)(Q(x) + b(x)) = 0. \tag{101}$$

After the triangle cancellations, comparison of x_i^3 and $x_i x_j^2$ gives precisely (100).

We recall explicitly the restricted-square axioms in characteristic two: $r(\alpha u) = \alpha^2 r(u)$, $r(u+v) = r(u) + r(v) + [u, v]_0$, and $\operatorname{ad} r(u) = (\operatorname{ad} u)^2$. Any square with the range and vanishing conditions in (iii) must have the form

$$r(x, z) = (0, P(x)z + q(x)), \qquad q(x) = Q(x) + b(x). \tag{102}$$

Its first two axioms are automatic. For $u = (x, z), v = (y, t)$, the two adjoint expressions have final coordinates

$$P(x)P(y)z + P(x)\beta(x, y), \qquad P(x)P(y)z + P(y)q(x),$$

respectively. Condition (101) implies $P(x)\beta(x, y) = P(y)q(x)$: take the coefficient of s in $P(x + sy)q(x + sy) = 0$. Its degree in s is at most three, below $|\mathbb{F}_4|$, so the comparison is valid for functions over $\mathbb{F}_4$. This proves the third restricted axiom. Conversely that axiom, applied to u, gives $[u, r(u)]_0 = 0$, hence (101).

Finally two solutions of (100) differ by independent choices of d vectors in $\bigcap_i \ker P_i$. The displayed count follows. It counts scalar actions, not their conjugacy classes. □

Example 10.3. Write the eight-dimensional class-three model over $\mathbb{F}_4$ as

$$N(a, b, y, z) = (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab).$$

On $W = \operatorname{span}_{\mathbb{F}_4}\{y, z\}$, its data are $P_1 z = y$, $P_1 y = 0$, $P_2 = 0$, and $a_{12} = z$. The equation with $(i, j) = (2, 1)$ in (100) would require

$$0 = P_2 b_1 = P_1 a_{21} = y,$$

which is impossible. Its auxiliary bracket is $\mathbb{F}_4$-bilinear, but the original polar bracket admits no compatible scalar action. Thus the class-three metabelian minimum $\mu(3) = 8$ lies outside the compatible subclass, whose minimum is $\nu(3) = 12$. The general dimension cost is proved in the next section.

Remark 10.4. The restricted square in Theorem 10.2 is a square on the auxiliary algebra. The theorem imposes both its range and its restriction to the derived ideal. General correspondences between quadratic maps and restricted Lie structures, such as [31, Section 9], carry their own additional identities. Here the exact additional condition is the solvable linear system (100).

11 Sharp compatible bounds and extremizers

11.1 The abelianization bound

Lemma 11.1. *A nonabelian compatible solution of class c, with $d = \dim_{\mathbb{F}_4}(V/W)$, satisfies $c \le d$.*

Proof. Use a restricted square from Theorem 10.2. The identity $P(x)\beta(x,y) = P(y)r(x)$, with a lift of x understood, implies

$$P(x)P(y)\beta(x,y) = 0. \tag{103}$$

In the universal module this kills the submodule K spanned by $t_i t_j v_{ij}$, $i < j$. It has dimension $\binom{d}{2}$ and lies in coefficient degree two. Every multiplication by an endpoint kills it by a square, and multiplication by a third index kills it by a triangle relation.

The space K is $\mathrm{GL}(U)$-invariant: the expression $P(x)P(y)\beta(x,y)$ is additive in each variable and homogeneous of degree two in each. For additivity in x, the cross term is

$$P(y)\big(P(x)\beta(z,y) + P(z)\beta(x,y)\big) = P(y)^2\beta(x,z) = 0$$

by Jacobi. Its basis values therefore span precisely K.

Put $\overline{W}_d = W_d/K$. The multigrading in Proposition 7.1 shows that its highest coefficient degree is $d-1$, for every $d \ge 2$, and that its top has the basis T_j of (69). This top is $\det(U)\otimes U$ also for $d = 2, 3$: the same diagonal and transvection calculations apply, with any additional double-endpoint term now in K. For $d = 2$ this is immediate from $T_1 = t_1 v_{12}, T_2 = t_2 v_{12}$.

Every compatible derived module M is a quotient of $\overline{W}_d$, so $c \le d+1$. Suppose $c = d+1$. Choose a functional on M nonzero on its terminal component. A change of quotient basis normalizes its top pullback to $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \ne j$. Using $P_j a_{ij} = P_i b_j$, we have, for $i \ne j$,

$$\mathrm{im}(T_j) = \Big(\prod_{h\ne j} P_h\Big) b_j. \tag{104}$$

Write $b_j = \sum c_{kl} a_{kl} \pmod{\mathfrak{m}M}$, using intrinsic generation. The product on the right kills $\mathfrak{m}M$. If j is an endpoint of v_{kl}, then $(\prod_{h\ne j} t_h)v_{kl}$ is a top vector whose doubled endpoint is different from j. If j is not an endpoint, the vector has both endpoints doubled and is in K. Thus the normalized top functional is zero on the right side of (104), but one on the left. This contradiction proves $c \le d$. □

11.2 Counting preimages of an operator cube

Lemma 11.2. *A compatible solution of class $c \ge 2$ satisfies*

$$\dim_{\mathbb{F}_4} W \ge 2^{c-1} - 1.$$

Proof. The case $c = 2$ says $W \ne 0$. Let $c \ge 3$. By Lemma 7.2, choose a nonzero terminal value

$$P(x)P(z_1)\cdots P(z_{c-3})\beta(x,y).$$

Restrictedness gives $P(x)\beta(x,y) = P(y)b$, where $b = r(x) \in W$. Put $m = c-2$, $u_1 = y, u_2 = z_1, \ldots, u_m = z_{c-3}$. Then $P(u_1)\cdots P(u_m)b \neq 0$. The 2^m vectors

$$b_S = \Big(\prod_{i\in S} P(u_i)\Big)b \qquad (S \subseteq \{1,\ldots,m\})$$

are independent by the complementary-operator argument used in Theorem 7.4. Let C be their span. Since $[x, r(x)]_0 = 0$, $P(x)$ kills b, hence all of C.

For each nonempty S, choose $i \in S$ and set

$$v_S = \Big(\prod_{j\in S\setminus\{i\}} P(u_j)\Big)\beta(x,u_i).$$

These vectors lie in the intrinsic derived ideal, and restrictedness gives $P(x)v_S = b_S$. Thus

$$\dim\ker(P(x)|_W) \geq 2^m, \qquad \operatorname{rank}(P(x)|_W) \geq 2^m - 1.$$

Rank-nullity yields $\dim W \geq 2^{m+1} - 1 = 2^{c-1} - 1$. □

Theorem 11.3. *For every $c \geq 2$, the minimum binary dimension among compatible solutions of polar class c is*

$$\nu(c) = 2^c + 2c - 2.$$

Proof. Lemmas 11.1 and 11.2 give the lower bound by addition. For attainment put $s = c-1$, $R_s = \mathbb{F}_4[t_1,\ldots,t_s]/(t_i^2)$, and use

$$f(x,z) = \left(\omega x, \omega z + \Big(\sum_i x_i t_i\Big)z\right) \quad \text{on } \mathbb{F}_4^s \oplus R_s. \tag{105}$$

The square of $\sum x_i t_i$ is zero, so direct substitution gives (1). The polar bracket is $\mathbb{F}_4$-bilinear and has intrinsic derived ideal $\mathfrak{m}R_s$. The scalar actions induced by f on that ideal and on the abelianization are the displayed ones, so the map is compatible. Its lower central ideals are $\gamma_k = \mathfrak{m}^{k-1}R_s$ for $k \geq 2$, giving class $s+1 = c$. The derived and quotient dimensions are $2^s - 1$ and $s+1$. The binary dimension is therefore $2(s + 2^s) = 2^c + 2c - 2$. □

The regular Lie model in (105) is a finite subalgebra of Cohn's characteristic-two construction [8]; Section 7.4 gives the identification. The preceding bounds establish its precise extremal dimension.

For $c \geq 5$, the extra four binary dimensions have an exact structural meaning. The sharp bounds in the two subclasses are

	d	w
metabelian	$c-1$	$2^{c-1}-2$
compatible metabelian	c	$2^{c-1}-1$

where each entry is the corresponding lower bound, attained simultaneously. Compatibility therefore requires one additional $\mathbb{F}_4$-dimension in the quotient and one in the derived ideal, giving four additional binary dimensions. The restricted square explains the strengthened bounds: it constrains terminal commutators and supplies independent preimages for the nonempty faces of the operator cube. The elementary semidirect construction attains this cost.

11.3 A common module and a full parameter family

The equality cases of the two bounds have a rigid module structure, although the resulting maps admit many inequivalent parameters. Throughout this part put $k = \mathbb{F}_4$, $s = c - 1 \geq 2$, and

$$R = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \quad J = (t_1, \dots, t_s), \quad \tau = t_1 \cdots t_s, \quad Z = k\tau. \tag{106}$$

The socle $Z = \operatorname{Ann}_R(J)$ is one-dimensional. We use the parameter space

$$\mathcal{V}_s = J^2/Z = \bigoplus_{j=2}^{s-1} \Lambda^j(k^s), \qquad D_s = \dim_k \mathcal{V}_s = 2^s - s - 2. \tag{107}$$

In characteristic two the exterior algebra is the displayed commutative square-zero algebra. Each parameter is represented by its unique sum of terms of degrees two through $s-1$. These are degrees of fixed ring coefficients; the maps below still have binary degree at most two.

Proposition 11.4 (Canonical extremal models)**.** *For $p \in \mathcal{V}_s$, put*

$$a_p(X, V) = \sum_{i=1}^{s} X_i t_i + Vp \quad (X \in k^s,\ V \in k).$$

Then

$$F_p(X, V, z) = \big(\omega X, \omega V, \omega z + a_p(X, V)(z + V)\big), \qquad z \in J, \tag{108}$$

is a compatible solution of class $c = s + 1$ and binary dimension $\nu(c)$. Its intrinsic derived ideal is J, and

$$\gamma_j = J^{j-1} \quad (j \geq 2), \qquad \dim_k \gamma_j = \sum_{h=j-1}^{s} \binom{s}{h}.$$

Proof. Every element of J squares to zero. Write $r_p = F_p - \omega\,\mathrm{id}$. At the image under F_p, the expressions a_p and $z + V$ become ωa_p and $(\omega + a_p)(z + V)$, respectively. Hence $r_p(F_p(y)) = \omega^2 r_p(y)$, proving $F_p^2 = F_p + \mathrm{id}$. The displayed expression is quadratic over k, and thus has binary degree at most two. Its original polar bracket is k-bilinear:

$$[(X, V, z), (Y, T, w)] = (0, 0, a_p(X, V)(w + T) + a_p(Y, T)(z + V)).$$

With quotient basis $e_1, \dots, e_s, v$, the action and cocycle are

$$P(e_i) = t_i \cdot, \quad P(v) = p \cdot, \quad \beta(e_i, e_j) = 0, \quad \beta(e_i, v) = t_i. \tag{109}$$

The square values of r_p are $b_i = 0$, $b_v = p$. The cocycle values generate J, and the core actions generate its augmentation filtration, proving all assertions about the derived ideal and lower central series. The dimension is $2((s + 1) + (2^s - 1)) = 2^c + 2c - 2$. □

Lemma 11.5 (The common extremal module)**.** *Every compatible solution of class $c \geq 3$ and dimension $\nu(c)$ has derived module isomorphic to J under $s = c - 1$ independent quotient directions acting by $t_1, \dots, t_s$. There is one remaining quotient direction, whose action, after a change of basis, is multiplication by a parameter in $\mathcal{V}_s$. In particular all these extremizers have the lower central dimensions in Proposition 11.4.*

Proof. Equality in the two separate bounds forces $d = c$ and $\dim_k W = 2^{c-1} - 1$. In the proof of Lemma 11.2, put $m = c - 2$, $T = P(x)$, $A = k[t_1, \dots, t_m]/(t_i^2)$, and $C = Ab$. The complementary-operator argument identifies C with the regular module A. Every nonempty face lies in $\operatorname{im} T$, whereas $C \subseteq \ker T$. Equality of dimensions now gives

$$\ker T = C \cong A, \qquad \operatorname{im} T = I_A b, \quad I_A = (t_1, \dots, t_m).$$

Here and below kernels and images are taken on W.

For completeness, the required splitting follows directly from the Frobenius pairing. Let ℓ_A take the top coefficient in A, and extend $ab \mapsto \ell_A(a)$ from C to a linear functional ℓ on W. Nondegeneracy of $(a, a') \mapsto \ell_A(aa')$ defines an A-linear retraction $\rho : W \to A$ by $\ell_A(a\rho(w)) = \ell(aw)$ for every $a \in A$. Thus $W = C \oplus N$, and T identifies N with $I_A b$. In these coordinates $W = A \oplus I_A$, $T(a, q) = (q, 0)$. The map $(a, q) \mapsto t_0 a + q$ identifies this with the augmentation ideal of $A[t_0]/t_0^2$, giving the claimed J.

The directions $x, u_1, \dots, u_m$ are independent: applying an action relation to b first kills all coefficients of the u_i, and $T \neq 0$ kills the last one. There is therefore exactly one remaining direction v. The top-coefficient pairing makes R self-injective: $R \cong \mathrm{Hom}_k(R, k)$, and $\mathrm{Hom}_R(-, \mathrm{Hom}_k(R, k)) \cong \mathrm{Hom}_k(-, k)$ is exact. Every R-endomorphism of J extends to multiplication on R, so $\mathrm{End}_R(J) = R/Z$. Accordingly $P(v)$ is multiplication by some $p \in R/Z$. Its square is zero, forcing its constant coefficient to vanish. Subtracting its linear part from v leaves $p \in J^2/Z$, as required. □

Theorem 11.6 (All compatible extremizers)**.** *Every compatible solution of class $c \geq 3$ and dimension $\nu(c)$ is binary-linearly conjugate to a model F_p in* (108)*.*

Proof. Theorem 10.2 allows us first to write the solution as $\omega\,\mathrm{id} + r$, with a compatible restricted square r. Use the common module from Lemma 11.5. The remaining data are core cocycle values $a_{ij} = \beta(e_i, e_j)$, core square values $b_i = r(e_i)$, mixed values $A_i = \beta(e_i, v)$, and $b_v = r(v)$. We normalize them in three steps. All changes of section take values in the intrinsic ideal J.

Core cohomology. Lemma A.1 computes the relevant low-degree complex explicitly. A section change puts the core data in the form

$$b_i = \kappa_i t_i, \qquad a_{ij} = \kappa_i t_j + \kappa_j t_i \quad (\kappa_i \in k). \tag{110}$$

The restricted identities give $t_i A_i = p b_i$, so $A_i = \kappa_i p + t_i z_i$, with $z_i \in R$. Mixed Jacobi gives $t_i t_j (z_i + z_j) = 0$; hence all z_i have the same constant η. If $\eta = 0$, the mixed values have no linear terms, and the linear core values span at most the proper hyperplane $\ker(\sum \kappa_i t_i^*)$ of J/J^2 (or zero if $\kappa = 0$). This contradicts intrinsic generation of J. Replace v by $\eta^{-1} v$, scaling p, A_i by η^{-1} and b_v by η^{-2}, to make $\eta = 1$. Also $t_i b_v = p A_i \in J^3$ for all i; since $s \geq 2$, b_v has no linear part.

Removing the core class. Replace e_i by $e_i + \kappa_i v$. The new square values $b_i + \kappa_i^2 b_v + \kappa_i A_i$ have no linear terms. The new actions are multiplication by $t_i + \kappa_i p$. The substitution $\phi(t_i) = t_i + \kappa_i p$ is an automorphism of R, because its linear part is the identity. Reidentify J by ϕ^{-1}; in particular the parameter must be changed simultaneously to

$$p_{\mathrm{new}} = \phi^{-1}(p) \pmod{Z}. \tag{111}$$

The core square values still have no linear terms. By the same cohomology calculation their class is zero, so a J-valued section change kills every b_i and a_{ij}.

Removing the mixed freedom. Now (A_i) is a one-cocycle. Its linear part is still (t_i), so Lemma A.1 gives $A_i = t_i + t_i w$ for a common $w \in J$. Replacing the section of v by $v + w$ makes $A_i = t_i$. Restrictedness then gives $t_i(b_v + p_{\mathrm{new}}) = 0$ for every i, whence $b_v = p_{\mathrm{new}} + \zeta\tau$. The last term is removed by the binary-linear central shear $z \mapsto z + \zeta\tau V^2$. We have reached precisely (109), with $b_i = 0$, $b_v = p_{\mathrm{new}}$, proving the theorem.

All ring substitutions in this argument reidentify the finite-dimensional k-space J; they are linear maps on that space. No nonlinear change of the input vector is used. □

11.4 The exact parameter orbits

Theorem 11.7 (Binary conjugacy of extremizers)**.** *Let $s \geq 2$. The maps F_p, F_q are binary-linearly conjugate if and only if*

$$\sigma_B(p) = \delta q \pmod{Z} \quad \text{for some } B \in \mathrm{GL}_s(k),\ \delta \in k^\times, \qquad \sigma_B(t_i) = \sum_j B_{ji} t_j. \tag{112}$$

Thus the compatible extremizers of class $c = s+1$ are in bijection with the orbits of $\mathcal{G}_s = \mathrm{GL}_s(k) \times k^\times$ on $\mathcal{V}_s$, acting by $p \mapsto \delta^{-1}\sigma_B(p)$.

Proof. Let $SF_p = F_q S$. The intrinsic ideal J is preserved, so $S(u,z) = (Au, Cz + h(u))$. The diagonal blocks commute with the intrinsic action of ω, and are therefore k-linear; h is initially only binary-linear. The mixed polar relations show that C conjugates the unital action algebras, both equal to $H = R/Z$. The kernel of $U \to \mathrm{rad}(H)/\mathrm{rad}(H)^2$ is kv. Hence

$$Av = \delta v', \qquad Ae_i = \sum_j B_{ji} e'_j + \alpha_i v', \quad B \in \mathrm{GL}_s(k),\ \delta \neq 0.$$

Compare the zero core brackets modulo J^2. All terms from h lie in J^2, and the remaining equation is $\alpha_j B_i + \alpha_i B_j = 0$ for $i \neq j$. Independence of the columns of B and $s \geq 2$ force $\alpha_i = 0$ for every i. The induced algebra automorphism therefore sends each t_i to its linear substitution $B(t_i)$. The action of v gives (112). This proves necessity without assuming the off-diagonal block is k-linear.

For the chosen representatives, the condition is equality in R, because linear substitutions preserve degree. The map

$$S(X, V, z) = (BX, \delta V, \delta\sigma_B(z))$$

then intertwines the two models directly. For arbitrary representatives with $\theta = \sigma_B(p) + \delta q \in Z$, the same conclusion holds after adding $\delta\theta V^2$ to the last coordinate. Thus passing modulo the socle loses no invariant. □

The final equivalence group consists of linear variable substitutions and one common scalar. Arbitrary nonlinear automorphisms of R do not preserve this normal form with its mixed cocycle; nor is an extra Galois action allowed, since the diagonal blocks commute with the fixed intrinsic ω.

Corollary 11.8 (Extremal diversity)**.** *Let N_c be the number of compatible extremal conjugacy classes of polar class $c \geq 3$, and put $s = c-1$. Then*

$$N_c = \frac{1}{3|\,\mathrm{GL}_s(4)|} \sum_{B \in \mathrm{GL}_s(4)} \sum_{\delta \in k^\times} 4^{\sum_{j=2}^{s-1} \dim_k \ker(\Lambda^j B - \delta I)}, \tag{113}$$

$$N_c \geq \left\lceil \frac{4^{D_s}}{3|\,\mathrm{GL}_s(4)|} \right\rceil > \frac{1}{3} 4^{D_s - s^2}, \qquad N_c \leq 4^{D_s}. \tag{114}$$

In particular $\log_4 N_c = 2^{c-1} + O(c^2)$.

Proof. Apply Burnside's lemma to Theorem 11.7. The parameter set has 4^{D_s} points and every orbit has size at most $3|\,\mathrm{GL}_s(4)|$. Finally $|\,\mathrm{GL}_s(4)| = \prod_{j=0}^{s-1}(4^s - 4^j) < 4^{s^2}$. □

Corollary 11.9. *For every $c \geq 4$, at least $c-2$ compatible extremizers are given by $p = 0$ and the monomials $p = t_1 \cdots t_j$, $2 \leq j \leq s-1 = c-2$.*

Proof. The parameter action preserves degree and distinguishes zero from nonzero parameters. The listed models are therefore inequivalent. □

The growth in Corollary 11.8 is double exponential in the polar class c. In the extremal binary dimension $n = \nu(c) = 2^{s+1}+2s$, the bound is instead exponential, $N_c > 2^{n-2s^2-4s-4}/3$. For example, class seven and dimension 140 give at least 532249921743 classes by the displayed formula; no enumeration of that many models is involved.

Classical tools and the extremal interface. The algebra R is the group algebra of an elementary abelian two-group, and J is its augmentation ideal, or first syzygy of the trivial module. These are classical objects; see Dade's treatment of the syzygy modules and their endo-trivial properties [9, Part II, Section 7 and Theorem 10.1]. The low-degree complex used above is the ordinary periodic tensor resolution for this algebra. Its interpretation as restricted cohomology agrees with the Ext convention in [14, Section 2]. The extremal conclusions here require the equality-forced common module, elimination of every remaining cocycle and square parameter, and the necessity of the precise binary conjugacy criterion. They retain the mixed interactions that are absent from a classification of the module J alone.

11.5 Automorphisms of every extremal model

The general lifting problem for automorphisms of an abelian Lie extension is governed by its module action and extension class; see [5, Theorems 1.1–1.2]. For the extremal models these conditions can be solved explicitly, including the unique binary correction needed to commute with the map.

Theorem 11.10 (Automorphisms of compatible extremizers)**.** *Put $k = \mathbb{F}_4$, and let $c \geq 3$, $s = c - 1$,*

$$R_s = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \qquad J_s = (t_1, \dots, t_s), \qquad \tau_s = t_1 \cdots t_s.$$

Choose $p \in J_s^2/k\tau_s$, represented by its unique sum of terms of degrees $2, \dots, s-1$. For the canonical extremizer

$$F_p(X, V, w) = \big(\omega X, \omega V, \omega w + P_p(X, V)w + VP_p(X, V)\big), \qquad P_p(X, V) = \sum_i X_i t_i + Vp,$$

where $X \in k^s$, $V \in k$, $w \in J_s$, and $\omega^2 + \omega = 1$, define

$$\mathcal{S}_p = \{(B, \delta) \in \mathrm{GL}_s(k) \times k^\times : \sigma_B(p) = \delta p \text{ in } J_s^2/k\tau_s\}, \qquad \sigma_B(t_i) = \sum_j B_{ji} t_j.$$

The binary-linear centralizer fits into a split exact sequence

$$1 \longrightarrow K_s \longrightarrow C_{\mathrm{GL}(2(s+2^s),2)}(F_p) \longrightarrow \mathcal{S}_p \longrightarrow 1.$$

The kernel K_s is independent of p up to group isomorphism. It has $2^{s+1} + s - 2$ free k-parameters, with the group law described in Appendix A.1; in particular,

$$\big|C_{\mathrm{GL}(2(s+2^s),2)}(F_p)\big| = 4^{\,2^{s+1}+s-2}|\mathcal{S}_p| = 4^{\,2^c+c-3}|\mathcal{S}_p|.$$

The splitting sends (B, δ) to $(X, V, w) \mapsto (BX, \delta V, \delta\sigma_B(w))$.

Proof. The intrinsic derived ideal identifies the action algebra with $R_s/k\tau_s$. Its radical modulo its square identifies the core quotient directions and the remaining line. Comparing core brackets modulo J_s^2 forces the induced ring automorphism to be the linear substitution σ_B, giving the displayed map to $\mathcal{S}_p$. The stated splitting is verified by substitution. Appendix A.1 solves its kernel explicitly and proves that each auxiliary k-Lie automorphism has exactly one binary-linear lift commuting with F_p. No additional choices of the antiscalar block enter the order. □

For $p = 0$, one has $\mathcal{S}_0 = \mathrm{GL}_s(4) \times \mathbb{F}_4^\times$. For $p \neq 0$, projection onto B identifies $\mathcal{S}_p$ with the projective stabilizer of kp in $\mathrm{GL}_s(4)$; there is no further factor of three in this latter description.

Remark 11.11 (The class-two boundary). For the six-dimensional central model $F(X, V, z) = (\omega X, \omega V, \omega z + XV)$, the action algebra has no radical directions. The preceding sequence is replaced by

$$C_{\mathrm{GL}(6,2)}(F) \cong (k^2, +) \rtimes \mathrm{GL}_2(k), \qquad |C_{\mathrm{GL}(6,2)}(F)| = 2880.$$

This follows from the automorphisms of the three-dimensional Heisenberg algebra and the same unique binary lifting argument.

11.6 Asymptotic enumeration and generic symmetry

Lemma 11.12 (A uniform fixed-space bound). *Let $k = \mathbb{F}_4$, $s \geq 2$, and put*

$$\mathcal{E}_s = \Lambda(k^s), \qquad \mathcal{V}_s = \bigoplus_{j=2}^{s-1} \Lambda^j(k^s), \qquad \mathcal{G}_s = \mathrm{GL}_s(k) \times k^\times.$$

Let (B, δ) act on $\mathcal{V}_s$ by $p \mapsto \delta^{-1}(\Lambda B)p$. For every $(B, \delta) \neq (I, 1)$,

$$\operatorname{codim}_{\mathcal{V}_s} \operatorname{Fix}(B, \delta) \geq r_s := 2^{s-2} - s - 2.$$

Proof. Put $T = \Lambda B$, $A = T - \delta I$, and let L_v denote left exterior multiplication by $v \in k^s$. The identity

$$AL_v - L_{Bv}A = \delta L_{Bv - v}$$

follows from $TL_v = L_{Bv}T$. If $B \neq I$, choose $Bv \neq v$. Multiplication by a nonzero exterior generator has rank 2^{s-1}, so subadditivity of rank gives

$$2^{s-1} \leq \operatorname{rank}(AL_v) + \operatorname{rank}(L_{Bv}A) \leq 2 \operatorname{rank} A.$$

If $B = I$ and $\delta \neq 1$, then A is invertible. Finally, deleting degrees $0, 1, s$ loses at most $s + 2$ dimensions of rank. Since fixed points are the kernel of $A|_{\mathcal{V}_s}$, the claimed bound follows. □

Corollary 11.13 (Asymptotic enumeration and generic automorphisms). *Write*

$$D_s = 2^s - s - 2, \qquad g_s = 3|\,\mathrm{GL}_s(4)|, \qquad \eta_s = 3\,4^{\,s^2+s+2-2^{s-2}},$$

and let $N_{s+1} = |\mathcal{V}_s/\mathcal{G}_s|$. Then

$$N_{s+1} = \frac{4^{D_s}}{g_s}(1 + \epsilon_s), \qquad 0 \leq \epsilon_s \leq (g_s - 1)4^{-r_s} \leq \eta_s.$$

For a uniformly chosen parameter $p \in \mathcal{V}_s$,

$$\Pr\big(\mathrm{Stab}_{\mathcal{G}_s}(p) \neq 1\big) \leq \epsilon_s.$$

The proportion of parameter orbits having nontrivial stabilizer is at most $2\epsilon_s/(1 + \epsilon_s)$. Consequently

$$N_{s+1} \sim C_4\, 4^{\,2^s - s^2 - s - 2}, \qquad C_4 = \frac{1}{3\prod_{j=1}^{\infty}(1 - 4^{-j})}.$$

Under the canonical classification, these are the compatible extremizers of polar class $c = s+1$. Thus asymptotically almost all, whether counted by parameters or by conjugacy classes, have full binary-linear centralizer K_s from Theorem 11.10. For every $s \geq 9$, such extremizers exist, since $\eta_s \leq 3\,4^{-36} < 1$.

Proof. Burnside's lemma gives the stated formula with

$$\epsilon_s = \sum_{g \in \mathcal{G}_s \setminus \{1\}} 4^{-\operatorname{codim} \operatorname{Fix}(g)}.$$

The fixed-space bound and $g_s < 3\,4^{s^2}$ give the error estimate. The union bound gives the assertion for parameters. Let β_s be the proportion of parameters with nontrivial stabilizer. Counting stabilizer orders shows that the proportion of nonfree orbits is $(\beta_s + \epsilon_s)/(1 + \epsilon_s)$, which is at most the claimed bound because $\beta_s \le \epsilon_s$. Finally,

$$|\operatorname{GL}_s(4)| = 4^{s^2} \prod_{j=1}^{s} (1 - 4^{-j}),$$

the infinite product has positive limit, and $\eta_s \to 0$. At $s = 9$ its exponent is -36, and it decreases thereafter. The centralizer assertion now follows from Theorem 11.10. □

It follows in particular that, for every $c \ge 10$, the minimum binary-linear centralizer order among compatible extremizers is exactly 4^{2^c+c-3}. It is attained by asymptotically almost all such classes. This is a minimum of the symmetry group size within the already minimal-dimensional family.

11.7 The first three extremal classifications

The canonical parameters give complete classifications at three low-class compatible minima.

Corollary 11.14. *The numbers of binary-linear conjugacy classes of compatible extremizers in polar classes* $3, 4, 5$ *are* $1, 2, 7$*, respectively. Their binary dimensions are* $12, 22, 40$*. For class five, write* $p = \alpha + \beta$*, with* $\alpha \in \Lambda^2 k^4$*,* $\beta \in \Lambda^3 k^4$*, and use the notation* $t_{ij} = t_i t_j$*,* $t_{ijk} = t_i t_j t_k$*. The seven parameter representatives and their distinguishing conditions are listed in Table 1.*

Table 1: The seven compatible class-five extremizers. The conditions refer to the quadratic and cubic parts of the parameter, not to the binary degree of the map, which remains at most two.

Parameter representative	Distinguishing condition
0	$\alpha = \beta = 0$
t_{123}	$\alpha = 0,\ \beta \neq 0$
t_{12}	$\operatorname{rank} \alpha = 2,\ \beta = 0$
$t_{12} + t_{123}$	$\operatorname{rank} \alpha = 2,\ 0 \neq \beta \in \alpha \wedge k^4$
$t_{12} + t_{134}$	$\operatorname{rank} \alpha = 2,\ \beta \notin \alpha \wedge k^4$
$t_{12} + t_{34}$	$\operatorname{rank} \alpha = 4,\ \beta = 0$
$t_{12} + t_{34} + t_{134}$	$\operatorname{rank} \alpha = 4,\ \beta \neq 0$

Proof. Put $s = c - 1$. The canonical orbit correspondence reduces the assertion to the action of $\operatorname{GL}_s(k) \times k^*$ on J^2/Z, where the second factor scales the whole parameter. For $s = 2$ this parameter space is zero. For $s = 3$ it is $\Lambda^2 k^3$; the usual alternating-bivector normal form gives the two ranks zero and two.

For $s = 4$, the homogeneous decomposition is $J^2/Z = \Lambda^2 k^4 \oplus \Lambda^3 k^4$. If $\alpha = 0$, the cubic part is either zero or a nonzero decomposable three-vector, giving the first two rows. Suppose $\operatorname{rank} \alpha = 2$, and normalize $\alpha = t_{12}$. Write $k^4 = A \oplus B$, with $A = \langle t_1, t_2 \rangle$, $B = \langle t_3, t_4 \rangle$. The

stabilizer of the line $k\alpha$ consists of block matrices $\left(\begin{smallmatrix} M & N \\ 0 & P \end{smallmatrix}\right)$; the common parameter scale is then $\det M$. There is a decomposition

$$\Lambda^3 k^4 = (\Lambda^2 A \wedge B) \oplus (A \wedge \Lambda^2 B).$$

If the second component is zero, the first is either zero or nonzero, and $\mathrm{GL}(B)$ is transitive in the latter case. If the second component is nonzero, the block N removes the first component, and $\mathrm{SL}(A)$ is transitive on the remaining nonzero vector in A. These are exactly the three rank-two rows.

Finally normalize the rank-four part to $\alpha = t_{12}+t_{34}$. The map $u \mapsto \alpha \wedge u$ is an isomorphism $k^4 \to \Lambda^3 k^4$, as is seen directly on the displayed basis. It intertwines the symplectic stabilizer of α with its action on k^4. A nonzero vector extends to a symplectic basis, so that group is transitive on nonzero vectors. Thus the cubic part gives precisely the last two rows. All conditions in the table are invariant, so the listed orbits are pairwise distinct and exhaustive. □

12 Two closed classification ranges

The polar framework supports two complementary classification questions: fix the ambient dimension, or fix the binary abelianization dimension while allowing the ambient dimension to grow. Global polar nilpotence reduces dimension eight to the metabelian correspondence. Central descent and complete extension spaces then close the entire stratum with binary abelianization dimension four. Its last two members are nonmetabelian, showing the reach of the extension method beyond the correspondence of Section 3. The unrestricted minimum at class five is $m(5) = 14$, compared with the metabelian minimum $\mu(5) = 36$ (Corollary 12.3). Independent matrix and Witt illustrations of the polar identities are collected in Appendix V.

12.1 All eight-dimensional solutions

Theorem 12.1. *In binary dimension eight there are exactly three linear-conjugacy classes of quadratic solutions. Representatives on $\mathbb{F}_4^4$ are*

$$\begin{aligned} L(a,b,c,d) &= (\omega a, \omega b, \omega c, \omega d), \\ C(a,b,c,d) &= (\omega a, \omega b, \omega c, \omega d + ab), \\ N(a,b,y,z) &= (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab). \end{aligned}$$

Their polar-image dimensions are $0, 2, 4$, their center dimensions are $8, 4, 2$, and their classes are $1, 2, 3$, respectively.

Proof. Theorem 4.1 makes every polar algebra nilpotent. A nonabelian nilpotent polar algebra has dimension at least six, by Lemma 2.6 and parity of its nonzero derived ideal. In dimension eight its derived ideal has dimension at most four and is therefore abelian. The metabelian correspondence applies.

If $W = 0$, there is the single linear class. If $\dim_{\mathbb{F}_4} W = 1$, all square-zero actions vanish; the cocycle is an alternating scalar form on $\mathbb{F}_4^3$. Its nonzero rank is two, giving C. If $\dim_{\mathbb{F}_4} W = 2$, the quotient has dimension two. A zero action could not generate W from its single alternating coefficient. For a nonzero square-zero action P_1, its commuting square-zero partners are scalar multiples of it. Change quotient basis so $P_2 = 0$, and normalize $a_{12} = z, P_1 z = y, P_1 y = 0$. This gives N. The displayed intrinsic dimensions distinguish the three classes. □

12.2 Central descent with the smallest nontrivial abelianization

We now fix $\dim_{\mathbb{F}_2}(V/W) = 4$. This fixes the smallest abelianization dimension of a nonabelian nilpotent polar algebra. The following theorem closes the stratum in all ambient dimensions.

Theorem 12.2 (A closed seven-class chain)**.** *Among all quadratic solutions with binary abelianization dimension four, there is exactly one conjugacy class in each of dimensions* $4, 6, 8, 10, 12, 14, 16$*, and no class in any other dimension. Their polar classes are* $1, 2, 3, 3, 4, 5, 5$*. The extension-space ranks in the proof are exact finite computations, certified as described in Appendix X.*

The representatives are the successive prefixes of the following map, with coordinates ordered as (a, b, c, d, e, h, r, s):

$$\begin{aligned} a' &= \omega a, & b' &= \omega b, \\ c' &= \omega c + ab, & & \\ d' &= \omega d + ac + a^2 b, & & \\ e' &= \omega e + bc + ab^2, & & \\ h' &= \omega h + ae + bd, & & \\ r' &= \omega r + A, & A &= ac^2 + a^2 e + ah + cd, \\ s' &= \omega s + B, & B &= bc^2 + b^2 d + bh + ce. \end{aligned} \tag{115}$$

Each prefix is binary-quadratic. Every new cocycle q satisfies $q(g(x)) = \omega^2 q(x)$ for the preceding prefix g, proving (1) recursively. For the last two expressions this follows by substitution; the extra terms cancel in pairs or have coefficient $1 + \omega + \omega^2 = 0$.

Table 2: Intrinsic invariants of the seven prefixes. All dimensions are binary.

n	$\dim W$	$\dim Z$	Class	Lower central dimensions
4	0	4	1	$(4, 0)$
6	2	2	2	$(6, 2, 0)$
8	4	2	3	$(8, 4, 2, 0)$
10	6	4	3	$(10, 6, 4, 0)$
12	8	2	4	$(12, 8, 6, 2, 0)$
14	10	2	5	$(14, 10, 8, 4, 2, 0)$
16	12	4	5	$(16, 12, 10, 6, 4, 0)$

Proof of Theorem 12.2. The proof proceeds from descent to extensions and then to closure. Every member descends to the abelian four-dimensional solution. At each prefix we compute the complete quadratic extension quotient; kernel scalings and quotient automorphisms put all its nonzero vectors into a single orbit. This gives a unique next class until the final extension quotient vanishes. Descent then turns that last calculation into a bound on every ambient dimension.

Descent. Global polar nilpotence supplies a nonzero center. In a nonabelian member of this stratum, $Z(L) \subseteq W$. Otherwise the linear scalar action on $Z(L)$ would give a central invariant $\mathbb{F}_4$-line disjoint from W. Its quotient would be a nonabelian nilpotent polar algebra with binary abelianization dimension two, contradicting Lemma 2.6. Quotient by any central $\mathbb{F}_4$-line in W. The dimension drops by two, the abelianization dimension stays four, and a quadratic solution is inherited. Repetition ends at the unique abelian four-dimensional solution. Every member is therefore reached by successive central $\mathbb{F}_4$-extensions.

The complete extension space. For a fixed quotient solution g, every such extension has the form

$$F(x,z) = (g(x), \omega z + q(x)), \qquad q(g(x)) = \omega^2 q(x), \quad q(0) = 0,$$

where q is binary-quadratic. Let E_g be this $\mathbb{F}_4$-vector space and define

$$H_g = E_g / \delta \operatorname{Hom}_{\mathbb{F}_2}(V, \mathbb{F}_4), \qquad \delta(t) = t \circ g + \omega t. \tag{116}$$

This quotient accounts for every binary-linear change of splitting. Kernel scalings and quotient automorphisms act by scaling and pullback. The zero class splits off a central scalar line and increases the binary abelianization dimension to six. Conversely, if an extension has its new line disjoint from its derived ideal, choose an $\mathbb{F}_4$-complement to that line in the abelianization and take its inverse image. It is an invariant complement containing the derived ideal, so the extension splits and its class in H_g is zero. Thus exactly the nonzero classes remain in our stratum.

Ranks and representatives. The complete results are in Table 3. The equation for q has binary degree at most four, so values at inputs of binary weight at most four recover every coefficient by Boolean Möbius inversion. The certificates verify the ranks and the spanning of each displayed quotient. The splitting-image dimension also follows directly: $\delta(t) = 0$ precisely for the linear maps t that annihilate W and intertwine the intrinsic ω-actions. This kernel has binary dimension four, so $\dim \operatorname{im} \delta = 2n - 4$.

Table 3: Exact central-extension spaces. Here n is the dimension of g.

n	Unknowns	Rank	$\dim_{\mathbb{F}_2} E_g$	$\dim_{\mathbb{F}_2} \operatorname{im} \delta$	$\dim_{\mathbb{F}_4} H_g$	Basis of H_g
4	20	14	6	4	1	ab
6	42	30	12	8	2	$ac + a^2b,\ bc + ab^2$
8	72	58	14	12	1	$bc + ab^2$
10	110	92	18	16	1	$ae + bd$
12	156	132	24	20	2	A, B
14	210	184	26	24	1	B
16	272	244	28	28	0	–

Projective orbits. When H_g is one-dimensional, kernel scalings identify all its nonzero classes. There are two two-dimensional cases. For $n = 6$, put $u = ac + a^2b, v = bc + ab^2$. The following automorphisms of g act on their classes by a swap, a nontrivial diagonal scaling, and a shear:

$$\begin{aligned} &(a,b,c) \mapsto (b,a,c), \\ &(a,b,c) \mapsto (\lambda a, b, \lambda c), \qquad \lambda \in \mathbb{F}_4^*, \\ &(a,b,c) \mapsto (a+b, b, c+b^2). \end{aligned}$$

Their actions are $(u,v) \mapsto (v,u)$, $(\lambda^2 u, \lambda v)$, and $(u+v, v)$. They are transitive on the five projective points of $\mathbb{F}_4^2$.

For $n = 12$, the corresponding automorphisms are

$$\begin{aligned} &(a,b,c,d,e,h) \mapsto (b,a,c,e,d,h), \\ &(a,b,c,d,e,h) \mapsto (\lambda a, b, \lambda c, \lambda^2 d, \lambda e, \lambda^2 h), \\ &(a,b,c,d,e,h) \mapsto (a+b, b, c+b^2, d+e, e, h). \end{aligned}$$

They send (A,B) to (B,A), $(A, \lambda^2 B)$, and $(A + B + ab + b^2, B + b^2)$. The extra terms are splitting changes: $ab = \delta(c)$, $b^2 = \delta(b^2)$. The projective action is again transitive.

Closure. Starting with the abelian four-dimensional map, every nonzero extension class therefore yields the unique next prefix, through dimension sixteen. At that point $H_g = 0$, so no next member exists. Descent rules out every higher dimension as well. The coordinate brackets give Table 2. Their second derived dimensions in dimensions fourteen and sixteen are two and four; these are the two nonmetabelian members. □

The six-dimensional prefix is the central example already present in [42, Lemma 10.3(1)]. General central recursion also appears there. The quadratic eigenspaces, linear-conjugacy orbits, and the vanishing of the final extension quotient are the ingredients that close the present seven-class range.

Corollary 12.3 (The first general minima without a metabelian hypothesis)**.** *Let $m(c)$ be the least binary dimension of a solution whose polar algebra has class $c \geq 2$. Then*

$$m(2) = 6, \qquad m(3) = 8, \qquad m(4) = 12, \qquad m(5) = 14, \qquad m(c) \geq 2c + 4 \quad (c \geq 6).$$

Proof. Each nonzero lower central factor has even dimension, and a nonabelian nilpotent polar algebra has abelianization of dimension at least four (Proposition 2.3 and Lemma 2.6). Consequently $\dim V \geq 2c + 2$. This gives the lower bounds in classes two and three, attained by the six- and eight-dimensional models. A class-four model in dimension ten would have abelianization four, but the unique such member of Theorem 12.2 has class three. Thus class four needs at least dimension twelve, which is attained. Likewise a class-five model in dimension twelve would have abelianization four, whereas that chain member has class four. The fourteen-dimensional member attains class five. Finally an abelianization-four solution has class at most five by the complete chain. At class $c \geq 6$ the abelianization therefore has dimension at least six, yielding $2c + 4$. □

12.3 The full automorphism group of the fourteen-dimensional member

The extension quotient also determines automorphisms of an individual member, rather than only its conjugacy class. Let g be the twelve-dimensional prefix in (115), and write

$$F_{14}(x, r) = (g(x), \omega r + A(x)), \qquad x = (a, b, c, d, e, h).$$

Here and below $\mathrm{Aut}(F_{14})$ means the group of binary-linear maps commuting with F_{14}; it is not the full automorphism group of its polar Lie algebra.

Proposition 12.4. *Every element of $\mathrm{Aut}(F_{14})$ is uniquely of the form*

$$S_{\phi,\lambda,t}(x, r) = (\phi(x), \lambda r + t(x)), \qquad t \circ g + \omega t = A \circ \phi + \lambda A,$$

where $\phi \in \mathrm{Aut}(g)$, $\lambda \in \mathbb{F}_4^*$, and t is binary-linear. The allowed ϕ are precisely those whose induced matrix on $(a, b)^T$ belongs to

$$P = \left\{ \begin{pmatrix} \alpha & 0 \\ \gamma & \eta \end{pmatrix} : \alpha, \eta \in \mathbb{F}_4^*,\ \gamma \in \mathbb{F}_4 \right\}.$$

For such ϕ, necessarily $\lambda = \eta^2$, and there are exactly 16 possible t. Consequently

$$|\,\mathrm{Aut}(F_{14})| = 36 \cdot 4^{10} = 37\,748\,736, \qquad \mathrm{Aut}(F_{14}) \cong N_{14} \rtimes P, \quad |N_{14}| = 4^{10}, \quad P \cong C_3 \times A_4.$$

There is also an exact sequence

$$1 \longrightarrow \mathrm{Hom}_{\mathbb{F}_4}(\mathbb{F}_4^2, \mathbb{F}_4) \longrightarrow \mathrm{Aut}(F_{14}) \longrightarrow \mathrm{Stab}_{\mathrm{Aut}(g)}\langle [A] \rangle \longrightarrow 1.$$

The stabilizer description separates the quotient automorphism from its central lifts. The proof in Appendix M obtains the entire group from the extension quotient and its projective action. In particular, the factor 16 counts all binary-linear lifts; restricting the splitting map t to be $\mathbb{F}_4$-linear would omit automorphisms.

13 Central reconstruction and cyclic Lie cohomology

Central extensions provide a route from the metabelian correspondence to solutions with nonabelian derived ideal. The Lie bracket records the polar part of an extension, while the functional equation also constrains its quadratic refinement. We determine both constraints and retain the linear maps between the resulting extensions.

Throughout this section, $g : V \to V$ is a quadratic solution, $n = \dim_{\mathbb{F}_2} V$, $B = \operatorname{pol} g$, and $W = L'_g$. Initially the coefficient line is $A = \mathbb{F}_4$, with fixed scalar $\omega^2 + \omega + 1 = 0$. The notation $H^2_{\mathrm{CE}}(L_g, A)$ always means ordinary *binary* Lie cohomology with trivial coefficients; its scalar structure comes from A. No nilpotency or metabelian hypothesis is imposed on g here. The constructions tensor with finite $\mathbb{F}_4$-coefficient spaces when these are needed below.

13.1 Quadratic lifts and their two obstructions

Let $\mathcal{Q}_{\le j}(V, A)$ be the zero-constant functions of binary degree at most j, and put $\mathcal{L}(V, A) = \operatorname{Hom}_{\mathbb{F}_2}(V, A)$. Composition acts on the full function space by $Tq = q \circ g$. Define

$$\Delta_g = T + \omega^2 \operatorname{id}, \qquad \delta = T + \omega \operatorname{id}, \qquad N = \operatorname{id} + T + T^2. \tag{117}$$

The operator Δ_g measures the functional defect, whereas δ describes linear changes of section.

Proposition 13.1 (One-step central lifts)**.** *In a binary-linear section, a quadratic solution with marked central kernel $(A, \omega \operatorname{id})$ and quotient g is precisely*

$$F_q(x, a) = (g(x), \omega a + q(x)), \qquad q \in \mathcal{Q}_{\le 2}(V, A), \quad \Delta_g q = 0. \tag{118}$$

Its marked extension classes form the vector space

$$H_g(A) = \frac{\ker(\Delta_g|_{\mathcal{Q}_{\le 2}(V,A)})}{\delta \mathcal{L}(V, A)}. \tag{119}$$

Polarization induces a natural injection $H_g(A) \hookrightarrow H^2_{\mathrm{CE}}(L_g, A)$.

Proof. Centrality makes the mixed polar term with a vanish, giving the displayed form. Substitution into $F_q^2 = F_q + \operatorname{id}$ gives $q \circ g = \omega^2 q$. A section change by linear t replaces q by $q + \delta t$, whose polar changes by the CE coboundary tB. If $\operatorname{pol} q = tB$, then $r = q + \delta t$ is linear and satisfies $Tr = \omega^2 r$. Thus $\delta r = r$, and $q = \delta(t + r)$. This proves the injection. □

For a CE cocycle α, choose a quadratic refinement q with $\operatorname{pol} q = \alpha$. Such a refinement exists on every binary basis. The first obstruction is

$$\Theta[\alpha] = \Delta_g q \bmod \mathcal{Q}_{\le 2}(V, A) \quad \text{in } \mathcal{Q}_{\le 4}(V, A)/\mathcal{Q}_{\le 2}(V, A), \qquad K_g = \ker \Theta. \tag{120}$$

This is well defined: a refinement change is linear, and a CE coboundary is refined by δt, with $\Delta_g \delta t = 0$. Thus K_g consists of the ordinary cohomology classes whose quadratic refinements remain quadratic after composition with g. For $[\alpha] \in K_g$, define the remaining obstruction

$$\Psi[\alpha] = [\operatorname{pol}(\Delta_g q)]. \tag{121}$$

Theorem 13.2 (Complete central lifting criterion)**.** *The map $\Psi : K_g \to K_g$ is well defined. An ordinary central Lie extension class $[\alpha]$ admits a quadratic lift with the prescribed coefficient scalar if and only if*

$$\Theta[\alpha] = 0, \qquad \Psi[\alpha] = 0.$$

The two tests distinguish high-degree leakage from an incorrect quadratic weight. Their sufficiency includes the freedom to change the linear part of a refinement. For example, when $g = \omega \operatorname{id}$ on $\mathbb{F}_4^2$, the cocycle $\alpha((a, b), (c, d)) = (ad + bc)^2$ has $\alpha(x, gx) = 0$ and satisfies the four-variable obstruction. Its refinement $q = a^2 b^2$ has $\Delta_g q = q$, whose nonzero polar cannot be removed by a linear adjustment. This class fails the second test. The full correction argument is given in Appendix J.

13.2 The cyclic action and the fixed quadratic sector

Theorem 13.3 (A canonical cyclic action on quadratic-stable cohomology)**.** *On K_g, the formula*

$$\tau[\alpha] = [\mathrm{pol}(q \circ g)] \tag{122}$$

defines a natural $\mathbb{F}_4$-linear automorphism with $\tau^3 = \mathrm{id}$. Moreover

$$\Psi = \tau + \omega^2 \,\mathrm{id}, \qquad H_g(A) = (K_g)_{\omega^2}, \qquad \Pi = \mathrm{id} + \omega\tau + \omega^2\tau^2 \tag{123}$$

is the projector onto the admissible extension space.

Here g need not be additive. The action is constructed through quadratic refinements and their CE classes, on the specified domain K_g. The projector selects a cohomology class; a choice of actual lift still amounts to a choice of refinement and section. The construction commutes with pullback by every binary-linear solution homomorphism and with coefficient maps.

There is also a canonical part visible on all ordinary cohomology. Write

$$I_g(A) = \{r \in \mathcal{Q}_{\le 2}(V, A) : r \circ g = r\}.$$

Proposition 13.4 (The fixed-sector splitting)**.** *The map*

$$O_0 : H^2_{\mathrm{CE}}(L_g, A) \longrightarrow I_g(A), \qquad O_0[\alpha](x) = \alpha(x, g(x)) \tag{124}$$

has the canonical section $r \mapsto [\mathrm{pol}\, r]$. Its section image is exactly $(K_g)_1$, and

$$H^2_{\mathrm{CE}}(L_g, A) = \mathrm{pol}(I_g(A)) \oplus \ker O_0. \tag{125}$$

Proof. For any quadratic refinement q, $Nq(x) = \alpha(x, gx)$. Its third additive difference is $d_{\mathrm{CE}}\alpha$, so it is quadratic when α is a cocycle. It is invariant, and a coboundary contributes $tB(x, gx) = 0$. For invariant quadratic r, $Nr = r$; the same difference identity shows that its polar is CE-closed and $O_0[\mathrm{pol}\, r] = r$. Applying the norm to a refinement of any τ-fixed class proves the assertion about $(K_g)_1$. □

For the coefficient line put

$$b_2 = \dim_{\mathbb{F}_2} H^2_{\mathrm{CE}}(L_g, \mathbb{F}_2) = \dim_{\mathbb{F}_4} H^2_{\mathrm{CE}}(L_g, A), \quad i_g = \dim_{\mathbb{F}_4} I_g(A), \quad h_g = \dim_{\mathbb{F}_4} H_g(A).$$

Corollary 13.5 (A sharp universal lift-space bound)**.** *If $d = \dim_{\mathbb{F}_4}(L_g/L_g')$ with its intrinsic scalar structure, then*

$$\dim K_g = i_g + 2h_g, \qquad b_2 = \mathrm{rank}\,\Theta + i_g + 2h_g, \qquad i_g \ge d^2. \tag{126}$$

Consequently

$$h_g \le \left\lfloor \frac{b_2 - d^2}{2} \right\rfloor. \tag{127}$$

The last bound is attained for every d by $g = \omega\,\mathrm{id}$ on $\mathbb{F}_4^d$.

Proof. The cyclic action and its degree condition descend to $\mathbb{F}_2$. Coefficient Frobenius therefore exchanges its two nontrivial eigenspaces, giving equal dimensions h_g. The fixed-space dimension is i_g by Proposition 13.4. On the abelianization, the d^2 independent functions $u_i u_j^2$, for ordered pairs (i, j), are invariant binary quadratics; their pullbacks prove the lower bound for i_g. In the linear example, $\Theta = 0$, $i_g = d^2$, $b_2 = \binom{2d}{2}$, and $h_g = \binom{d}{2}$, proving equality. For an m-dimensional coefficient space, all the cohomological and invariant dimensions scale by m. □

The dimension law separates three contributions to failed lifting: high-degree leakage H^2_{CE}/K_g, the fixed quadratic weight, and the opposite nontrivial weight. The fixed part splits by (125). The other two parts have the natural exact sequence

$$0 \longrightarrow (K_g)_\omega \longrightarrow (\ker O_0)/H_g(A) \longrightarrow H^2_{\mathrm{CE}}(L_g, A)/K_g \longrightarrow 0. \tag{128}$$

13.3 All marked homomorphisms and recursive reconstruction

For finite $\mathbb{F}_4$-coefficient spaces A, A', consider two lifts $F_q, F_{q'}$. A binary-linear map covering a solution homomorphism $Sg = g'S$ and a coefficient map $C : A \to A'$ has the form

$$R(x, a) = (Sx, Ca + t(x)).$$

Its polar Lie condition and its remaining functional defect are

$$C\alpha + tB + \alpha'(S\cdot, S\cdot) = 0, \tag{129}$$

$$d_R = Cq + tg + \omega t + q'S. \tag{130}$$

Here t is binary-linear, $\alpha = \operatorname{pol} q$, and $\alpha' = \operatorname{pol} q'$. On the abelianization put

$$K^+_{g,A'} = \operatorname{Hom}_{\mathbb{F}_4}(V/W, A'), \qquad K^-_{g,A'} = \{k : V/W \to A' : k \text{ additive},\ k(\omega u) = \omega^2 k(u)\}. \tag{131}$$

Theorem 13.6 (Full marked reconstruction)**.** *If R satisfies* (129)*, then $d_R \in K^-_{g,A'}$. Replacing t by $t + d_R$ gives the unique solution homomorphism in its affine $K^-_{g,A'}$-coset. This correction respects composition, including noninvertible base and coefficient maps.*

Let the marked Lie-extension category retain the base solution, the coefficient scalar, and an admissible CE cocycle, and quotient its Hom sets by these K^--cosets. Polarization defines an equivalence from actual quadratic central extensions to this quotient category. A quasi-inverse may be obtained by choosing admissible refinements; different such choices give naturally isomorphic equivalences.

Thus the section gauge retains its effective and ineffective parts. With quotient and coefficient fixed, the full extension groupoid is the action groupoid of $\mathcal{L}(V, A)$ on admissible q by δ; its stabilizer is $K^+_{g,A}$. In particular, $H_g(A)$ is its set of isomorphism classes, while K^+ records the residual automorphisms.

Corollary 13.7 (The canonical central tower)**.** *Every finite quadratic solution is obtained recursively from its linear abelianization by the central extensions*

$$X_i = L/\gamma_{i+1}(L) \longrightarrow X_{i-1}, \qquad A_i = \gamma_i(L)/\gamma_{i+1}(L).$$

Global nilpotency is Theorem 4.1. All solution homomorphisms preserve this tower. At each step, Theorem 13.6 recovers the full Hom sets from admissible Lie-extension data, with the next Hom set formed over the already corrected base arrow.

13.4 The graph carrier and the section gauge

The same gauge has a direct realization on the canonical linear span of the graph. This construction also connects the refinement description to finite-dimensional linear representations of periodic maps [2].

Proposition 13.8 (The marked graph carrier)**.** *Put*

$$\Gamma_g = \{(x, gx) : x \in V\}, \quad M_g = \operatorname{span}_{\mathbb{F}_2} \Gamma_g, \quad \mathcal{J}(x, y) = (y, x + y), \quad p(x, y) = x.$$

Then $\mathcal{J}^2 + \mathcal{J} = \mathrm{id}$, $\mathcal{J}\Gamma_g = \Gamma_g$, and

$$M_g = \{(x, gx + w) : x \in V,\ w \in W\}, \qquad \dim_{\mathbb{F}_2} M_g = n + \dim_{\mathbb{F}_2} W. \tag{132}$$

Moreover

$$\ker p = 0 \oplus W, \quad \ker(p\mathcal{J}) = W \oplus 0, \qquad M_g/(W \oplus W) \simeq V/W$$

as $\mathbb{F}_4$-spaces under $\mathcal{J}$ and the induced g, respectively. The marked data $(M_g, \mathcal{J}, \Gamma_g, p)$ recover g: $p|_{\Gamma_g}$ is bijective and $g(p(s)) = p\mathcal{J}(s)$.

Write $\mathrm{Hom}_{\mathrm{anti}}(M_g, A)$ for additive maps ℓ satisfying $\ell\mathcal{J} = \omega^2\ell$.

Proposition 13.9 (Exact graph–gauge bridge)**.** *There is a natural exact sequence*

$$0 \longrightarrow K^+_{g,A} \longrightarrow \mathcal{L}(V, A) \xrightarrow{t\mapsto\ell_t} \mathrm{Hom}_{\mathrm{anti}}(M_g, A) \longrightarrow 0, \qquad \ell_t(x, y) = t(y) + \omega t(x). \tag{133}$$

On Γ_g, the functional ℓ_t restricts to the section coboundary δt.

The graph and projection retain the original binary degree condition. The carrier makes the scalar action linear, and (133) identifies its antilinear functionals with the effective section changes. Proofs of these interfaces and of the full marked reconstruction are given in Appendix J.

14 Uniform central lifts beyond the metabelian range

The central-lifting framework of Section 13 can be made uniform when the quotient is free through class three. In this family the admissible extension space, its nonmetabelian quotient and its complete quadratic-stable spectrum have polynomial dimension formulas. The same family also separates parameter counts from actual binary-linear conjugacy: after factoring the base action, almost every extension type is nonmetabelian, while its polar class remains four.

14.1 The quotient family and its first integrals

Put $k = \mathbb{F}_4$, let $U = k^d$, and define

$$Z = \Lambda^2_k U, \qquad T = (U \otimes Z)/\operatorname{im}\partial_3,$$

where

$$\partial_3(x \wedge y \wedge z) = x \otimes (y \wedge z) + y \otimes (z \wedge x) + z \otimes (x \wedge y).$$

Write $u\cdot z$ for the class of $u\otimes z$, and choose the cross-term refinement $Q(u) = \sum_{i<j} u_i u_j (e_i \wedge e_j)$. The free class-three auxiliary algebra has the functional normal form

$$g_d(u, z, t) = \big(\omega u, \omega z + Q(u), \omega t + u \cdot z + u \cdot Q(u)\big). \tag{134}$$

Jacobi cancels the terms with three distinct indices in $u \cdot Q(u)$; its remaining monomials are binary-quadratic. Thus (134) is an instance of the complete metabelian correspondence. Its derived ideal is $W = Z \oplus T$, its third lower-central term is T, and

$$\dim_k Z = \binom{d}{2}, \qquad \dim_k T = \frac{d(d^2-1)}{3}, \qquad \dim_{\mathbb{F}_2} g_d = \frac{d(d+1)(2d+1)}{3}. \tag{135}$$

The fixed cyclic sector from Section 13 has a rigid form in every nilpotency class. Global nilpotency makes the following statement apply to every finite quadratic solution.

Theorem 14.1 (Quadratic first-integral rigidity)**.** *Let g be any finite quadratic solution, with intrinsic k-abelianization dimension d. Every zero-constant binary-quadratic function $n : L_g \to k$ satisfying $n \circ g = n$ factors through L_g/L'_g. Consequently*

$$\dim_k I_g = d^2, \qquad I_g = \{n : \deg_{\mathbb{F}_2} n \le 2,\ n(0) = 0,\ ng = n\}.$$

For every such g, writing $h_g = \dim_k H_g$, one has

$$\dim_k K_g = d^2 + 2h_g, \qquad b_2(g) = \operatorname{rank}_k \Theta_g + d^2 + 2h_g.$$

Proof. By Theorem 4.1, the polar algebra is nilpotent. The maximal lower-central-depth component of the polar of an invariant quadratic is a fixed CE two-cocycle on the graded scalar Lie algebra. Splitting its two scalar eigenspaces shows that this component annihilates the derived algebra; maximality then forces every pairing with the derived ideal to vanish. The function descends to abelianization, where its basis consists of u_i^3 and $u_i^2u_j, u_iu_j^2$, $i<j$. Their number is d^2. The complete filtration argument is Theorem T.2 in Appendix T. The cyclic decomposition of Theorem 13.3 gives the two dimension identities. □

14.2 All admissible directions and their cyclic weights

Use $H_{g_d} = H_{g_d}(k)$ for the admissible central-extension space of g_d, and K_{g_d} for its quadratic-stable ordinary binary CE cohomology. The coefficient line has its prescribed scalar action ω. An admissible class is represented by a quadratic q with $qg_d = \omega^2 q$.

For $\alpha = \operatorname{pol} q$, restriction to $W \times W$ descends to an alternating k-bilinear form on $Z = W/T$. This gives the map

$$\rho : H_{g_d} \longrightarrow \Lambda^2(\Lambda^2 U^*). \tag{136}$$

In characteristic two, wedge of two two-forms is alternating in those forms, so there is a natural map

$$\mu : \Lambda^2(\Lambda^2 U^*) \longrightarrow \Lambda^4 U^*, \qquad \xi \wedge \eta \longmapsto \xi \wedge \eta. \tag{137}$$

Theorem 14.2 (Uniform admissible space and spectrum)**.** *For every d, put*

$$r_d = 3\binom{d}{3} + 2\binom{d}{4}, \qquad h_d = \binom{d}{2} + 6\binom{d}{3} + 5\binom{d}{4}.$$

Then

$$\operatorname{im}\rho = \ker\mu, \qquad \dim_k \operatorname{im}\rho = r_d, \tag{138}$$

$$\dim_k H_{g_d} = h_d, \qquad \dim_k \ker\rho = \binom{d}{2} + 3\binom{d}{3} + 3\binom{d}{4}. \tag{139}$$

The kernel of ρ is exactly the subspace of metabelian extensions. On K_{g_d}, the canonical order-three action has eigenspace dimensions

$$\dim_k (K_{g_d})_1 = d^2, \qquad \dim_k (K_{g_d})_\omega = \dim_k (K_{g_d})_{\omega^2} = h_d. \tag{140}$$

In particular $\dim_k K_{g_d} = d^2 + 2h_d$*, and the canonical admissibility projector has rank h_d.*

Proof. The four-variable polar identity gives $\mu\rho = 0$. A three-generator central lift realizes one shared-index wedge pair. The $\mathrm{GL}(U)$-action then spans all shared pairs and, by shears, the two differences among the three disjoint pairings on each four-element support. These span precisely $\ker\mu$, giving $3\binom{d}{3} + 2\binom{d}{4}$ dimensions.

For the metabelian kernel, central extensions of the free class-three base are determined by bracket degree four in the universal module of Section 7. Its Hilbert polynomial gives $\binom{d}{2} + 3\binom{d}{3} + 3\binom{d}{4}$ dimensions. Adding the derived-commutator quotient proves (139). The first-integral theorem identifies the fixed cyclic sector. Coefficient Frobenius exchanges the two nontrivial sectors, one of which is H_{g_d}, proving (140). Appendix K supplies the extension, symmetry and realization details. □

The first values h_2, h_3, h_4, h_5 are $1, 9, 35, 95$. The corresponding derived-commutator dimensions are $0, 3, 14, 40$. Thus the two-generator quotient has only metabelian lifts, while three generators already allow nonmetabelian class-four lifts. These formulas concern the specified central-extension family; the ordinary binary CE space can contain additional classes with nonzero degree leakage.

Example 14.3 (A class-four nonmetabelian lift)**.** Use coordinates

$$(a, b, c, z, w, v, h, k, r) \in k^9.$$

Take

$$\begin{aligned}
a' &= \omega a, & b' &= \omega b, & c' &= \omega c,\\
z' &= \omega z + ab, & w' &= \omega w + ac, & v' &= \omega v + bc,\\
h' &= \omega h + av + cz, & k' &= \omega k + aw + a^2 c,\\
r' &= \omega r + zw + ah + bk + a^2 v.
\end{aligned}$$

Each new cocycle has the required eigenvalue ω^2. In the last substitution, the extra a^2bc-coefficient is $1 + \omega + \omega^2 = 0$. Its polar algebra has lower-central dimensions $(18, 12, 6, 2, 0)$, and $[z, w] = r \neq 0$ with $z, w \in L'$. It is therefore nonmetabelian of class four.

This polar algebra is also intrinsically nonrestrictable. Indeed $(\operatorname{ad} a)^2 c = k$, while for every vector u, the w- and k-coordinates of $[u, c]$ are u_a and u_a^2. No inner adjoint can equal $(\operatorname{ad} a)^2$. This distinguishes a restricted square on the example itself from its possible realization in a larger tensor-algebra construction.

Remark 14.4 (The tensor-algebra realization)*.* The same derived-commutator quotient has a complementary realization using the classical restricted-square map $x \mapsto \omega x + x^2$ in a truncated associative algebra with $x^4 = 0$. Passing to an appropriate f-stable binary subspace of its class-three quotient, then pulling back and pushing out the central extension, realizes $\ker \mu$. The tensor-word proof is included in Appendix K. This description situates the realization within the classical associative and universal-module constructions; the formulas above identify their image and cyclic weights in the quadratic lifting problem. An f-stable binary subspace need not be closed under the ambient scalar multiplication or restricted square, consistently with Example 14.3.

14.3 From extension parameters to actual conjugacy types

Let N_d count binary-linear conjugacy classes among the nonzero one-dimensional k-central extensions of g_d. Their common binary dimension is

$$\frac{d(d+1)(2d+1)}{3} + 2.$$

The intrinsic fourth lower-central ideal supplies the marking needed to turn this family into an actual conjugacy problem.

Theorem 14.5 (Actual orbit count and typical nonmetabelianity)**.** *For $d \geq 2$, every nonzero one-line central extension of g_d has class exactly four and kernel γ_4. Its actual binary-linear conjugacy class is determined by its orbit in*

$$\big(H_{g_d} \setminus \{0\}\big)/(\mathrm{GL}(d, 4) \times k^*).$$

Consequently

$$\frac{4^{h_d} - 1}{3|\,\mathrm{GL}(d, 4)|} \leq N_d \leq \frac{4^{h_d} - 1}{3}, \qquad \log_4 N_d = h_d + O(d^2). \tag{141}$$

The proportion of these conjugacy types that are metabelian is at most

$$3|\,\mathrm{GL}(d, 4)|\, 4^{-r_d}, \tag{142}$$

which tends to zero as $d \to \infty$.

Proof. A nonzero extension cannot still have class at most three: it would be metabelian, and lifting the free generators would split the corresponding class-three auxiliary extension. Hence its class is four and its one-line kernel is the characteristic ideal γ_4.

The base action on extension cohomology factors through $\mathrm{GL}(U)$. To see this, use the universal central extension with kernel $H^*_{g_d}$. Its fourth lower-central term is the entire kernel; otherwise a nonzero scalar pushout would have class at most three. Every base automorphism lifts with the dual coefficient action. An IA automorphism lifts to an automorphism acting identically on abelianization, hence on all fourfold commutators. It therefore fixes the universal kernel and acts trivially on H_{g_d}.

Since the scalar kernel is intrinsic, all conjugacies are captured by the remaining base and kernel actions. Counting vectors and bounding orbit sizes gives (141); applying the same comparison to the invariant subspace $\ker \rho$ gives (142). The full argument in Appendix K uses no freeness assumption for the GL-action. □

The parameter d grows in this theorem while the nilpotency class stays four. Thus it gives a uniform family of genuine nonmetabelian types, complementary to the growing-class metabelian counts earlier in the paper.

14.4 The complete ordinary cohomology and degree leakage

The quadratic-stable spectrum can now be compared with the entire ordinary binary CE space. The comparison requires keeping track of the central Frobenius deformation in the original polar bracket.

Theorem 14.6 (Complete free-quotient obstruction)**.** *For the free class-three base g_d, for every $d \geq 1$,*

$$b_2(g_d) = \dim_k H^2_{\mathrm{CE}}(L_{g_d}, k) = \frac{d^4 + d^2}{2},$$
$$\operatorname{rank}_k \Theta_d = 4\binom{d}{2} + 6\binom{d}{3} + 2\binom{d}{4}.$$

Proof. The binary polar algebra is a central deformation of the restriction of scalars of the free class-three k-Lie algebra. Its CE equations have a surjective degree-three wedge block, so this deformation preserves the second Betti number. Scalar extension, CE Künneth, and the ordinary Lie Hopf formula give $b_2 = 2(d^4 - d^2)/4 + d^2$. Subtracting $\dim K_{g_d} = d^2 + 2h_d$ gives the leakage rank. Appendix T proves the deformation step and the homological calculation in full. □

The four-dimensional two-generator leakage sectors remain independent under pair retractions (Appendix K). The complete formula additionally accounts for the three- and four-generator contributions. It determines a dimension, while the individual $\mathrm{GL}(d, 4)$-orbits of the admissible extension family remain a separate classification problem.

15 Quadratic linearization

Every finite quadratic solution admits quadratic coordinates in which its dynamics is a linear three-cycle. Global nilpotency (Theorem 4.1) supplies a lower-central filtration that constructs the conjugator and controls its inverse degree. The same structure also gives sharp two-frame measurement bounds: recovering actual states and decoding the whole graph carrier have different dimension costs.

15.1 A quadratic conjugator along the lower-central filtration

For a nonzero solution let $c \geq 1$ be its polar class, and write $\Gamma_i = \gamma_i(L)$. Proposition 2.7 assigns an intrinsic scalar operator F_i to every factor Γ_i/Γ_{i+1}. Choose a binary-linear operator J_0 preserving this filtration, inducing F_i, and satisfying $J_0^2 + J_0 = \mathrm{id}$. Such an operator exists: lift an $\mathbb{F}_4$-basis of each factor by pairs $v, f(v)$, and put $J_0 v = f(v)$, $J_0 f(v) = v + f(v)$. This is a filtration-adapted lift of the graded scalars; it is not required to be compatible with the original Lie bracket in the sense of Section 10.

Theorem 15.1 (Filtered quadratic linearization)**.** *For any such J_0, set*

$$N(x) = f(x) + J_0 x, \qquad H_0(x) = f(x) + J_0^2 x = x + N(x).$$

Then H_0 is a permutation of binary degree at most two, and

$$H_0 f = J_0 H_0, \qquad H_0 f H_0^{-1} = J_0, \qquad \deg_{\mathbb{F}_2} H_0^{-1} \leq c.$$

For a target y, its inverse is computed by

$$x_0 = y, \qquad x_{j+1} = y + N(x_j) \quad (0 \leq j < c-1), \qquad H_0^{-1}(y) = x_{c-1}.$$

Proof. The functional equation and $J_0^2 + J_0 = \mathrm{id}$ give $H_0 f = J_0 H_0$ directly. For $u \in \Gamma_i$,

$$N(x+u) + N(x) = N(u) + [x, u] \in \Gamma_{i+1}.$$

Consequently $H_0(x+u) + H_0(x) \equiv u \pmod{\Gamma_{i+1}}$. A nonzero difference cannot disappear, proving injectivity. The same identity moves the residual of the displayed inverse iteration down one layer at each stage, and the last residual is zero. In a weighted basis of the filtration, each inverse coordinate of weight i has weighted Boolean degree at most i. This gives the degree bound; Appendix L supplies the weighted calculation. □

The intertwining expression is the order-three averaging formula $\sum_{j=0}^{2} J_0^{-j} f^j = f + J_0^2$. Its bijectivity here comes from the filtration. An arbitrary order-three operator need not work: $f = \omega^2\,\mathrm{id}$, $J_0 = \omega\,\mathrm{id}$ gives $H_0 = 0$. For a metabelian normal form, $J_0 = \omega\,\mathrm{id}$ gives

$$H_0(x, z) = (x, (\mathrm{id} + P(x))(z + Q(x))), \qquad H_0^{-1}(x, z) = (x, (\mathrm{id} + P(x))z + Q(x)).$$

Both have binary degree at most two. This is exactly the map H_f in (25): it transports the changed addition and auxiliary bracket there, and linearizes the original permutation here. In the fourteen-dimensional prefix of (115), this choice instead has a cubic inverse, whose last coordinate contains $a^3 b^2$. Thus a quadratic conjugator can require nonlinear decoding of higher degree. Linear conjugacy remains the equivalence relation for all classifications in this paper.

15.2 Two-frame measurements and linear decoding

Let $H : V \to Y = \mathbb{F}_2^m$ be binary-linear and define

$$\mathcal{O}_H(x) = (H(x), H(f(x))), \qquad \mathcal{J}(u, v) = (v, u + v).$$

The equation itself gives, for every such readout,

$$\mathcal{O}_H f = \mathcal{J} \mathcal{O}_H, \qquad H(f^2(x)) = H(x) + H(f(x)). \tag{143}$$

The third frame is determined by the first two. The polar filtration makes the state bound sharp under the original-linear restriction on H; compare finite-field function-space observability in [1].

Theorem 15.2 (Optimal state readout)**.** *For every finite quadratic solution of binary dimension n, the least m for which a binary-linear $H : V \to \mathbb{F}_2^m$ makes $\mathcal{O}_H$ injective is*

$$m_{\text{state}} = n/2.$$

An optimal readout has a triangular decoder.

Proof. Injectivity requires $2^n \le 2^{2m}$. To attain the bound, choose successively a central f-invariant $\mathbb{F}_4$-line and pass to the quotient. Binary-linear sections give coordinates

$$f_i(x) = \omega x_i + q_i(x_1, \dots, x_{i-1}).$$

Write $x_i = u_i + \omega v_i$, and read all u_i's. The next frame reads $v_i + \operatorname{coeff}_1(q_i(x_{<i}))$. Starting at the first coordinate, the two frames recover each v_i and then x_i. The readout has $n/2$ bits and is injective. Theorem 4.1 supplies nilpotency in every case. □

The cost of linear decoding. For the graph carrier M_f of Proposition 13.8, put $L_H(x, y) = (H(x), H(y))$. In this subsection use $w_B = \dim_{\mathbb{F}_2} W$, to distinguish binary dimension from the metabelian notation $w = \dim_{\mathbb{F}_4} W$. Then $d = \dim_{\mathbb{F}_4}(V/W) = (n - w_B)/2$.

Theorem 15.3 (Optimal carrier readout and linear decoding)**.** *For every quadratic solution, the least output dimension for which $L_H|_{M_f}$ is injective is*

$$m_{\text{linear}} = m_{\text{carrier}} = n - d = \frac{n + w_B}{2}.$$

For a fixed H, carrier injectivity is equivalent to the existence of a binary-linear decoder $R_{\text{dec}} : Y \oplus Y \to V$ satisfying $R_{\text{dec}}\mathcal{O}_H(x) = x$ for every state. If $K = \ker H$ and $\pi : V \to V/W$, these conditions are equivalent to

$$K \cap W = 0, \qquad \pi K \cap \omega \pi K = 0. \tag{144}$$

Proof. The carrier is $M_f = \{(x, y) : y + W = \omega(x + W)\}$. Thus its intersection with $K \times K$ is zero exactly under (144). The disjoint binary spaces $\pi K, \omega\pi K$ then give $\dim K \le d$, hence $m \ge n - d$. Equality is attained by lifting the binary span of an $\mathbb{F}_4$-basis of V/W to a kernel K. An injective carrier map gives a linear decoder on its image, extended to $Y \oplus Y$. Conversely, a linear decoder on all states satisfies $R_{\text{dec}} L_H = p$ on their span M_f. The kernel of L_H is $\mathcal{J}$-stable. Both p and $p\mathcal{J}$ vanish on that kernel, so both its coordinates are zero. □

Combining the two optima gives a dimension interpretation of the nonlinearity:

$$m_{\text{linear}} - m_{\text{state}} = w_B/2 \qquad \text{for every finite quadratic solution.} \tag{145}$$

At any optimal state readout, the induced map from M_f onto $Y \oplus Y$ has $\mathbb{F}_4$-kernel dimension $w_B/2$, yet no two actual states collide. For the fourteen-dimensional model, $w_B = 10$: seven bits per frame recover the state, whereas twelve are necessary for a linear decoder. The five remaining carrier directions are formal linear combinations of states, not state ambiguities.

The triangular decoder uses the known rule and the chosen flag. The transition $\mathcal{J}$ describes the measured dynamics; identifying the original rule also requires the observation embedding.

16 Orientation rigidity and geometric recovery of the central series

The original equation partitions the nonzero vectors into binary planes. Forgetting the cyclic direction gives a line spread $\mathcal{S}_f$. The quadratic degree bound makes this loss of information particularly small: the directions cannot be reversed independently from plane to plane.

Theorem 16.1 (Orientation rigidity and spread morphisms)**.** *Let f, g be finite quadratic solutions on the same binary vector space. If their unoriented orbit spreads coincide, then $g = f$ or $g = f^2$. More generally, for a binary-linear map $T : V \to V'$, put $P_f(x) = x \wedge f(x)$. Then*

$$(\Lambda^2 T)P_f = P_g T \quad \Longleftrightarrow \quad Tf = gT \text{ or } Tf = g^2 T.$$

For $T \neq 0$ the orientation is unique. In either case T is a homomorphism of the original polar Lie algebras.

Proof. The quadratic difference between two orientations is parallel to the input, and hence has the form $(a + \ell(x))x$. Orbit consistency forces the linear functional ℓ to be invariant, so it is zero. The relative argument for a map T, including its kernel, is given in Appendix U. □

Consequently the unoriented spread determines the original polar bracket. The spread stabilizer contains $\mathrm{Aut}(f)$ with index one or two; the second coset exists exactly when a binary-linear map conjugates f to f^2.

A spread plane ℓ is called *normal* if $\ell + m$ is partitioned by spread planes for every other spread plane m. This is a classical incidence condition [36].

Theorem 16.2 (Geometric recovery of the central series)**.** *For a quadratically oriented spread, its normal planes are exactly the orbit planes contained in the polar center. Their union with zero is therefore $Z(L_f)$, and their number is*

$$\frac{2^{\dim_{\mathbb{F}_2} Z(L_f)} - 1}{3}.$$

Passing to the quotient spread by this subspace and repeating the construction recovers the entire upper central series. It exhausts the space in exactly the polar nilpotency class many steps.

Proof. A central plane has zero bracket with every other plane, making their sum invariant. Conversely a normal plane restricts with any other plane to a four-dimensional quadratic solution, whose polar bracket is zero. This identifies the center. Iterate in the quotient and apply Theorem 4.1; full details are in Appendix U. □

This translates the polar structure into an orientation-free geometric obstruction. In particular a spread with no normal plane has no quadratic orientation. In $\mathrm{PG}(5, 2)$, exactly two spread classes have quadratic orientations: the Desarguesian spread and that of $(a, b, c) \mapsto (\omega a, \omega b, \omega c + ab)$. The comparison uses the classical enumerations of all 131044 spread classes and nine book-spread classes [28, 29], together with the solution classification proved here.

Proposition 16.3 (The exact invariant separating degree)**.** *The invariant map $P_f : x \mapsto x \wedge f(x)$ has binary degree at most three and separates exactly the three-cycle orbits (and the zero orbit). For a nonzero solution, the least degree of a family of invariant functions separating all orbits is two when f is linear and three otherwise.*

Proof. A nonzero decomposable binary two-vector determines its plane, giving separation. In the nonlinear case the derived ideal is nonzero, whereas every invariant quadratic factors through abelianization by Theorem 14.1. It cannot separate zero from a nonzero derived vector. The explicit cubic map attains the resulting lower bound. Appendix U also identifies its genuine cubic directions with the image of the ordinary CE differential on alternating two-forms. □

17 Further structural questions

17.1 Reconstruction and higher central lifting

How does the restricted core determine the whole object? Can the marked reconstruction of Theorem 13.6 be made coordinate-free, combining the core's central kernel, the abelianization action, and admissible lifts while preserving every linear intertwiner?

How does degree leakage develop under central lifting? Every finite solution satisfies $b_2 = \operatorname{rank}\Theta + d(L)^2 + 2h$. Theorem 14.6 determines these terms for the free class-three family. Determine the leakage quotient, its degree filtration, and its representation action under successive admissible lifts.

17.2 General dimension laws and linearization

What is the optimal general dimension scale? The gap $m(5) = 14 < \mu(5) = 36$ distinguishes the general problem. Between the bounds $m(c) \geq 2c + 2(2c)^{1/4} - 6$ and $m(c) = O(2^{c/2})$, are polynomial-size models possible? Theorem 6.5 equates the linear polar-class question with that for the ordinary $\mathbb{F}_4$ function $e(d)$; sharper bounds would also improve the derived-length leading constant. Theorem 6.8 extends the sharp linear ceiling through fourth central depth. Thus, for $d \geq 6$, a mechanism giving class greater than $d+2$ must have $[H'', H, H, H, H] \neq 0$.

What happens to the Jacobian sector at the fifth central layer? For $H'' \subseteq Z_5(H)$, the next peeling source involves $(B_d)_d \otimes U^{\otimes 3}$. Its fully multilinear part has a classical one-dimensional Jacobian kernel under Koszul multiplication [21, 20]. The first-order suspensions used in the fourth-layer proof vanish on this sector by the index-transfer identity. Does its actual Lie image vanish under the remaining Engel relations, or yield a uniform family of higher-class quotients? A nonzero tensor source alone does not decide this question.

What controls the inverse degree of a quadratic linearizer? Every solution has a quadratic linearizer of inverse degree at most its polar class. The metabelian and trace-current models have quadratic inverses; a specified linearizer of the fourteen-dimensional prefix has cubic inverse. Is a cubic inverse always attainable, or is there an intrinsic obstruction? The choice of linearizer is part of the problem.

17.3 Orbit geometry and fine population

Which spreads admit a quadratic orientation? Normal-element reduction gives necessary geometric conditions and recovers the polar central series of every orientable spread. A characterization of orientability would give an incidence-theoretic description of the whole solution class.

What is the complete fine-scale population law? Theorem 9.4 covers $2d \le r \le D_{\mathrm{par}}/(4d)$. Beyond this range, high-order displacements and nontrivial linear parts remain to be controlled. At each staircase boundary, two adjacent defects have logarithmic ratio known up to $O_{q,j}(d)$. Multiplicative E_Q-counts would determine finer centering, limiting weights, and integer-lattice effects.

A Extremal rigidity and automorphism calculations

This appendix gives the low-degree elimination and full kernel calculations used in the compatible equality cases. Put $k = \mathbb{F}_4$ and retain R, J, Z, τ from (106).

Lemma A.1. *Let $k = \mathbb{F}_4$,*

$$R = k[t_1, \dots, t_s]/(t_1^2, \dots, t_s^2), \qquad J = (t_1, \dots, t_s), \qquad \tau = t_1 \cdots t_s, \quad s \geq 2.$$

Give $k = R/J$ its augmentation-module structure, and write $H^n(M) = \operatorname{Ext}_R^n(k, M)$. Then

$$H^1(J) \cong k, \qquad H^2(J) \cong k^s.$$

These isomorphisms have the following explicit descriptions.

(i) *If $a_i \in J$ satisfy*

$$t_i a_i = 0, \qquad t_i a_j + t_j a_i = 0 \quad (i \neq j), \tag{146}$$

then there are a unique $\eta \in k$ and some $w \in J$ such that

$$a_i = \eta t_i + t_i w \qquad (1 \leq i \leq s).$$

Thus the class of $(t_i)_i$ generates $H^1(J)$.

(ii) *Let $b_i, a_{ij} \in J$, with $a_{ij} = a_{ji}$ and $a_{ii} = 0$, satisfy*

$$t_i b_i = 0, \tag{147}$$

$$t_i a_{ij} = t_j b_i, \qquad t_j a_{ij} = t_i b_j \quad (i \neq j), \tag{148}$$

$$t_i a_{jk} + t_j a_{ik} + t_k a_{ij} = 0 \quad (i < j < k). \tag{149}$$

Then there are unique $\kappa_1, \dots, \kappa_s \in k$ and some $w_1, \dots, w_s \in J$ such that

$$b_i = \kappa_i t_i + t_i w_i, \tag{150}$$

$$a_{ij} = \kappa_i t_j + \kappa_j t_i + t_i w_j + t_j w_i. \tag{151}$$

In particular κ_i is read directly from $b_i \equiv \kappa_i t_i \pmod{J^2}$. The class corresponding to $\kappa = (\kappa_i) \in k^s$ has representative

$$b_i = \kappa_i t_i, \qquad a_{ij} = \kappa_i t_j + \kappa_j t_i.$$

Proof. For $A_i = k[t_i]/(t_i^2)$, the complex

$$\cdots \xrightarrow{t_i} A_i \xrightarrow{t_i} A_i \xrightarrow{t_i} A_i \longrightarrow k \longrightarrow 0$$

is an augmented free resolution: at each positive degree both kernel and image are (t_i). Tensor the s resolutions over k and take the total complex. Exact complexes of vector spaces split, so this tensor complex has homology k in degree zero and zero in positive degrees. It is a free R-resolution $P_\bullet \to k$, explicitly

$$P_n = \bigoplus_{|\alpha| = n} R E_\alpha, \qquad dE_\alpha = \sum_{\alpha_i > 0} t_i E_{\alpha - \varepsilon_i}. \tag{152}$$

These are tensor-periodic differentials, with no coefficient α_i. Their square is zero because $t_i^2 = 0$ and the mixed terms occur twice in characteristic two.

Put $C^n(M) = \mathrm{Hom}_R(P_n, M)$. A zero-cochain is $w \in M$, a one-cochain is a tuple (w_i), and a two-cochain has components b_i at $E_{2\varepsilon_i}$ and a_{ij} at $E_{\varepsilon_i+\varepsilon_j}$. The first differentials are

$$(\delta^0 w)_i = t_i w, \qquad (\delta^1(w_i))_{ii} = t_i w_i, \qquad (\delta^1(w_i))_{ij} = t_i w_j + t_j w_i. \tag{153}$$

Consequently the one-cocycle equations are (146). Evaluating the next differential at $E_{3\varepsilon_i}$, $E_{2\varepsilon_i+\varepsilon_j}$, $E_{\varepsilon_i+2\varepsilon_j}$, and $E_{\varepsilon_i+\varepsilon_j+\varepsilon_k}$ gives precisely (147)–(149).

Let $\lambda : R \to k$ take the coefficient of τ. The pairing $(r, a) \mapsto \lambda(ra)$ is nondegenerate: the square-free monomial t_S pairs with t_{S^c}. It identifies R with $\mathrm{Hom}_k(R, k)$ as an R-module, and hence identifies the complexes

$$\mathrm{Hom}_R(P_\bullet, R) \cong \mathrm{Hom}_k(P_\bullet, k).$$

The latter is exact in positive cohomological degrees, since the underlying augmented vector-space resolution is exact. Thus $H^n(R) = 0$ for $n > 0$.

Apply $C^\bullet$ to $0 \to J \to R \to k \to 0$. Each P_n is free, so the result is a short exact sequence of cochain complexes. All differentials of $C^\bullet(k)$ vanish, giving $H^0(k) = k$ and $H^1(k) = k^s$. Moreover $H^0(R) = \mathrm{Ann}_R(J) = k\tau$, and its augmentation map to $H^0(k)$ is zero. The long exact sequence therefore gives the connecting isomorphisms

$$H^0(k) \xrightarrow{\sim} H^1(J), \qquad H^1(k) \xrightarrow{\sim} H^2(J).$$

Lifting $1 \in H^0(k)$ to $1 \in R$ gives the one-cocycle $(t_i)_i$. Lifting $\kappa \in H^1(k)$ to the constant tuple $(\kappa_i) \in C^1(R)$ and applying (153) gives the displayed two-cocycle representative. Subtracting these representatives leaves coboundaries with coefficients in J, proving both descriptions. Such a two-coboundary has $b_i = t_i w_i \in J^2$, so its homogeneous degree-one part vanishes; this also proves the stated identification and uniqueness of κ_i. □

For the extension data used in the extremal normal form, replacing the lift of e_i by $e_i + w_i$, with $w_i \in J$, changes the square value and bracket cocycle by

$$b_i \longmapsto b_i + t_i w_i, \qquad a_{ij} \longmapsto a_{ij} + t_i w_j + t_j w_i.$$

These are exactly the legal section coboundaries in the lemma. The constants $\kappa_i \in R$ in the connecting-map calculation are cochain lifts through $R \to k$; they are not section shifts in $W = J$. Thus the lemma distinguishes the removable section terms from the degree-one class that must be eliminated by the separate change of quotient basis in the normal-form proof.

A.1 The automorphism kernel of a compatible extremizer

Retain the notation of Theorem 11.10, and abbreviate $R = R_s$, $J = J_s$, $\tau = \tau_s$, and $\tau_i = \prod_{j \neq i} t_j$. Thus $\mathrm{Soc}(R) = \mathrm{Ann}_R(J) = k\tau$. An element of $R/k\tau$ is always represented with zero τ-coefficient. The auxiliary algebra $E(p)$, on $U \oplus J$ with $U = \langle e_1, \dots, e_s, v \rangle_k$, has

$$[e_i, e_j] = 0, \quad [e_i, v] = t_i, \quad [e_i, w] = t_i w, \quad [v, w] = pw, \quad [J, J] = 0.$$

This is also the original k-bilinear polar bracket of F_p.

The quotient and its splitting. The derived ideal is J, and its unital action algebra is $H = R/k\tau$. An automorphism induces maps $A : U \to U$, $C : J \to J$, and a section change $h : U \to J$. The kernel of

$$U \longrightarrow \mathrm{rad}(H)/\mathrm{rad}(H)^2$$

is precisely kv. Consequently $Av = \delta v$ and $Ae_i = \sum_j B_{ji} e_j + \alpha_i v$, with B invertible. The source core brackets vanish. The section-change terms lie in J^2, so their images modulo J^2

give $\alpha_j B_i + \alpha_i B_j = 0$ for $i \neq j$. Since $s \geq 2$, independence of the columns of B forces all $\alpha_i = 0$. It follows that the induced automorphism of H is σ_B, and that $\sigma_B(p) = \delta p$ modulo $k\tau$. The specified representatives make this an equality in R.

The standard Frobenius pairing of R, given by the highest coefficient of a product, yields $\mathrm{End}_R(J) = R/k\tau$. Thus $C = u\sigma_B$ for a unit $u \in H$. The mixed brackets modulo J^2 give $u(0) = \delta$. The maps $(X, V, w) \mapsto (BX, \delta V, \delta\sigma_B(w))$ therefore provide a homomorphic section of the quotient map.

All kernel elements. In the kernel, A is the identity and C is multiplication by $1 + a$, for a unique $a \in J/k\tau$. Write $h_i = h(e_i)$ and $h_v = h(v)$. Bracket preservation is equivalent to

$$t_i h_j + t_j h_i = 0 \qquad (i \neq j), \tag{154}$$

$$t_i(a + h_v) + p h_i = 0 \qquad (1 \leq i \leq s). \tag{155}$$

There is no additional equation $t_i h_i = 0$: the chosen restricted square is not a marked part of the automorphism problem.

Multiplying (154) by t_i shows that $t_i h_i = \lambda_i \tau$ for a unique $\lambda_i \in k$. Then $g_i = h_i + \lambda_i \tau_i$ satisfies $t_i g_i = 0$ and $t_i g_j + t_j g_i = 0$. These relations imply $g_i = t_i z$ for a common $z \in R/k\tau$. Here is a direct coefficient proof. The first relation allows only monomials containing t_i in g_i. For every proper subset $A \subsetneq \{1, \ldots, s\}$, define the coefficient of t_A in z to be the coefficient of $t_{A \cup \{i\}}$ in g_i, choosing any $i \notin A$. The second relation makes the answer independent of the choice. Only the highest coefficient remains free, and it is removed modulo $k\tau$.

Since $p\tau_i = 0$, equation (155) now gives

$$\begin{gathered} h_i = t_i z + \lambda_i \tau_i, \qquad h_v = a + pz + \zeta\tau, \\ a \in J/k\tau,\ z \in R/k\tau,\ \lambda \in k^s,\ \zeta \in k. \end{gathered} \tag{156}$$

These parameters are unique, and every choice satisfies the two kernel equations. Since $a^2 = 0$, multiplication by $1 + a$ is invertible. There are $(2^s - 2) + (2^s - 1) + s + 1 = 2^{s+1} + s - 2$ free parameters.

The group law. Let $\mathrm{pr} : R \to R$ remove the highest term, and let $[t_A]r$ denote a monomial coefficient. For left parameters (a, z, λ, ζ) and right parameters $(a', z', \lambda', \zeta')$, the composite has coordinates

$$\begin{aligned} a'' &= \mathrm{pr}(a + a' + aa'), \\ z'' &= \mathrm{pr}\left(z + z' + az' + \sum_i \lambda_i'[t_i]a\,\tau_i\right), \\ \lambda'' &= \lambda + \lambda', \\ \zeta'' &= \zeta + \zeta' + [\tau](aa'). \end{aligned}$$

Indeed block composition is $(u, h)(u', h') = (uu', h + uh')$, and $a\tau_i = [t_i]a\,\tau$. The formula contains no p, proving that the abstract kernel is independent of the parameter. It has exponent four: a square has $a = \lambda = \zeta = 0$, so its square is the identity; taking $a = t_1$, $z = 1$, and $\lambda = \zeta = 0$ gives an element whose square is nonidentity.

For completeness, conjugation by the section corresponding to (B, δ) is explicitly described in block coordinates by

$$u \longmapsto \sigma_B(u), \qquad h \longmapsto \delta\sigma_B \circ h \circ A^{-1}, \quad A = \mathrm{diag}(B, \delta).$$

Thus the kernel and the stabilizer action specify the full semidirect product, not only its order.

The unique binary lift. Any binary-linear map commuting with F_p preserves the intrinsic derived ideal. Its diagonal blocks commute with multiplication by ω, so they are k-linear. Write its off-diagonal block as h_s+h_a, with h_s linear over k and $h_a(\alpha\xi) = \alpha^2 h_a(\xi)$. Comparing the polar identity at ξ and $\omega\xi$ shows that h_a takes values in the common action kernel $k\tau$. It therefore contributes nothing to the polar bracket, and the k-linear part is an automorphism of $E(p)$.

Conversely, let S_0 be such an auxiliary automorphism and put $r_p = F_p - \omega\,\mathrm{id}$. The difference

$$D(\xi) = r_p(S_0(\xi,0)) - S_0(r_p(\xi,0))$$

is Frobenius semilinear: its polar vanishes, and the two restricted squares have the same adjoint square. Its values belong both to J and to the Lie center, hence to $k\tau$. It vanishes on the derived ideal. Since $\omega^2+\omega = 1$, adding the central off-diagonal map D to S_0 gives a map commuting with F_p. The lift is unique; the difference of two lifts would satisfy $(\mathrm{id} + P_p(A\xi))h_a(\xi) = 0$, and the first factor is invertible. Taking the scalar part respects composition, so this is a group isomorphism.

For the kernel parameters above, the actual binary transformation is

$$\begin{aligned}(X,V,w) \longmapsto \Big(X, V, (1+a)w + \sum_i X_i(t_i z + \lambda_i \tau_i) \\ + V(a + pz + \zeta\tau) + \tau \sum_i \lambda_i X_i^2\Big).\end{aligned}$$

The last term is forced by λ; it is not an additional family of choices. Indeed $D(e_i) = \lambda_i\tau$ and $D(v) = 0$.

The class-two boundary. For $s = 1$, the auxiliary algebra is the three-dimensional Heisenberg algebra. Its automorphisms are $(u,z) \mapsto (Bu, \det(B)z + h(u))$, with $B \in \mathrm{GL}_2(k)$ and $h \in \mathrm{Hom}_k(k^2,k)$. For $B = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$, the corresponding binary lift with $h = 0$ is

$$(X,V,z) \longmapsto (aX+bV, cX+dV, \det(B)z + acX^2 + bdV^2).$$

The kernel is $(k^2,+)$, giving order $4^2|\,\mathrm{GL}_2(4)| = 2880$, as asserted in Remark 11.11.

A.2 Pure quadratic parameters and an intrinsic action rank

Pure quadratic parameters form a readable all-class subfamily of the canonical extremizers. Their ordinary polynomial degree refers to the fixed ring coefficients, and does not increase the binary degree of the associated map.

Proposition A.2. *Let $s = c-1 \geq 3$. Among the canonical models with $p \in \Lambda^2 k^s$, there are exactly $\lfloor s/2\rfloor + 1$ binary-linear conjugacy classes, represented by*

$$p_0 = 0, \qquad p_r = t_1t_2 + t_3t_4 + \cdots + t_{2r-1}t_{2r} \quad (1 \leq r \leq \lfloor s/2\rfloor).$$

For $r \geq 1$, multiplication by p_r on the intrinsic derived module J has k-rank

$$\rho_{s,r} = 2^{s-1} - 2^{s-r-1} - 1. \tag{157}$$

The corresponding binary rank is $2\rho_{s,r}$.

Proof. Linear substitutions preserve the homogeneous degree. On $\Lambda^2 k^s$, the standard alternating-bivector normal form is determined by its even rank $2r$, giving the displayed representatives. Common nonzero scalar multiplication creates no further identifications

between different ranks. The canonical orbit correspondence therefore gives the stated complete classification of this subfamily.

For the rank calculation, regard multiplication by p_r as a square-zero differential on R. Each paired factor $k[a,b]/(a^2,b^2)$, with differential given by multiplication by ab, is the direct sum of the acyclic block $k1 \to k(ab)$ and the zero differential on $ka \oplus kb$. Tensoring the r paired factors gives homology dimension 2^r. The remaining $s-2r$ square-zero variables have zero differential and multiply that dimension by 2^{s-2r}. Thus

$$\dim_k H(R, p_r\cdot) = 2^{s-r}.$$

Since a square-zero operator has homology dimension equal to its ambient dimension minus twice its rank, multiplication by p_r on R has rank $2^{s-1} - 2^{s-r-1}$. Moreover $p_r R = kp_r \oplus p_r J$: the first summand is homogeneous of degree two, whereas the second has degree at least three. Subtracting this one dimension proves (157). □

In the canonical model, kv is the intrinsic kernel of the quotient action modulo the square of the radical of its generated operator algebra. Consequently the rank of $P(v)$, up to rescaling v, is an intrinsic invariant. Formula (157) makes the alternating rank of a pure quadratic parameter visible directly in the extremal map. For example, at $s = 4$, parameters of ranks two and four give binary action ranks six and ten.

B The unrestricted metabelian equality classification

We prove Theorem 7.6. Throughout, $k = \mathbb{F}_4$, $c \geq 5$, $d = c-1$, $m = d-1$,

$$U = ke_0 \oplus H, \quad H = \operatorname{span}_k\{e_1, \dots, e_m\}, \quad R = k[t_0, \dots, t_m]/(t_0^2, \dots, t_m^2), \quad K = (t_1, \dots, t_m).$$

Put $z = t_1 \cdots t_m$, $I = (z) = \operatorname{span}_k\{z, t_0 z\}$, and $\tau = t_0 z$. In particular I is two-dimensional, whereas $\dim_k K = 2^d - 2$. All cohomology below is ordinary Chevalley–Eilenberg cohomology of the abelian Lie algebra U, acting by $e_i w = t_i w$. The Koszul complex, Frobenius pairing, contraction homotopies and alternating-form reductions used here are standard linear-algebraic tools; see also [7, Theorem 1 and p. 279] for the classical Koszul background. We give the required calculations explicitly.

B.1 Equality forces a common derived module

Lemma B.1. *Every minimum-dimensional metabelian solution of class $c \geq 5$ has $d = c-1$ quotient generators and derived R-module*

$$W \simeq D(R/I) \simeq K, \qquad D = \operatorname{Hom}_k(-, k).$$

Moreover $\operatorname{Ann}_R K = I$ *and* $\operatorname{End}_R K \simeq R/I$.

Proof. Equality in Theorem 7.4 gives $\dim_k W = 2^d - 2$ and $d = c-1$. The quotient $W_d \to W$ retains a nonzero top component. Pull back a functional on W nonzero there to $\phi \in W_d^*$. By Lemma 7.3, a linear change of generators normalizes its highest homogeneous component to a top functional λ_0 with $\operatorname{Ann}_R \lambda_0 = I$. Write $\phi = \lambda_0 + \psi$, where the dual coefficient degrees of ψ are at most $d-2$. Since z has degree $d-1$, it annihilates ψ, and it annihilates λ_0 as well. Hence $I\phi = 0$, and $\dim_k R\phi \leq 2^d - 2$. The filtered contraction argument of the same lemma gives the reverse inequality. Thus $\operatorname{Ann}_R \phi = I$. As $R\phi \subseteq W^*$ already has dimension $\dim_k W$, $W^* = R\phi \simeq R/I$.

The pairing $(r,s) \mapsto [\tau](rs)$ is nondegenerate on R. It identifies $D(R/I)$ with $\operatorname{Ann}_R I = K$: a polynomial annihilates z precisely when its constant and t_0 coefficients vanish. The same observation gives $\operatorname{Ann}_R K = I$. Finally, duality identifies $\operatorname{End}_R K$ with the opposite of $\operatorname{End}_R(R/I) = R/I$, which is commutative. □

Lemma B.2 (One-dimensional derived-module defect)**.** *Let $d \geq 4$. Suppose a quotient W of the universal module W_d retains a nonzero top component and has $\dim_k W = 2^d - 1$. After a linear change of quotient generators, there is an exact sequence of R-modules*

$$0 \longrightarrow k \longrightarrow W \longrightarrow K \longrightarrow 0.$$

After this normalization, the number of possible R-module isomorphism types is at most $4^{\binom{d}{2}}$, and their ordinary cohomology satisfies $\dim_k H^2(U, W) \leq d^2$.

Proof. Pull back a functional nonzero on the top to $\phi \in W_d^*$, and normalize its highest component as in Lemma B.1. The same degree argument gives $I\phi = 0$, while the filtered contraction bound gives $\dim_k R\phi = 2^d - 2$. Thus $R\phi \simeq R/I$ is a codimension-one submodule of W^*. Its quotient is the trivial module k, since each t_i acts nilpotently. Dualizing gives the displayed sequence.

Such dual extensions are classified by $\operatorname{Ext}^1_R(k, R/I)$. Finite duality identifies this with $D \operatorname{Tor}^R_1(k, K)$. To compute the latter, put $S = k[t_1, \ldots, t_{d-1}]/(t_i^2)$ and $J_S = (t_1, \ldots, t_{d-1})$. Then $K = k[t_0]/t_0^2 \otimes_k J_S$. The tensor-periodic minimal resolution of k over S, as in (152), has free rank $\binom{d+j-2}{j}$ in degree j. Deleting its final copy of S gives a resolution of J_S; extending scalars gives a minimal R-resolution of K. Its degree-one free rank is $\binom{d}{2}$. Hence

$$\dim_k \operatorname{Ext}^1_R(k, R/I) = \binom{d}{2},$$

giving the asserted bound on module types. Finally the ordinary CE long exact sequence of $0 \to k \to W \to K \to 0$ and Lemma B.3 give

$$\dim_k H^2(U, W) \leq \dim_k H^2(U, k) + \dim_k H^2(U, K) = \binom{d}{2} + \left(d + \binom{d}{2}\right) = d^2.$$

Here module Ext_R and ordinary CE cohomology have different complexes; each is used with its stated meaning. □

B.2 Ordinary cocycles and intrinsic generation

Let $\tau_{ij} = \prod_{\ell \notin \{i,j\}} t_\ell$. Write $\beta_\eta(e_0, e_i) = t_i$, with all its other basis values zero. For $\kappa \in H^*$, let

$$\beta_\kappa(e_i, e_j) = t_0(\kappa_i t_j + \kappa_j t_i) \quad (i, j \geq 1), \qquad \beta_\kappa(e_0, e_i) = 0.$$

For $\lambda = (\lambda_{ij})$, set $\beta_\lambda(e_i, e_j) = \lambda_{ij}\tau_{ij}$. Thus $\beta_{\kappa,\lambda} = \beta_\eta + \beta_\kappa + \beta_\lambda$ is exactly (72).

Lemma B.3. *Every class in $H^2(U, K)$ has a unique expression*

$$\eta[\beta_\eta] + [\beta_\kappa] + [\beta_\lambda], \qquad \eta \in k, \quad \kappa \in H^*, \quad \lambda \in k^{\binom{d}{2}}.$$

Consequently $\dim_k H^2(U, K) = d + \binom{d}{2}$. Also $\dim_k H^1(U, K) = d + 1$. For a closed cocycle β, the condition

$$K = \mathfrak{m}K + \operatorname{span}_k\{\beta(x, y) : x, y \in U\}, \qquad \mathfrak{m} = (t_0, \ldots, t_m), \tag{$\dagger$}$$

holds if and only if $\eta \neq 0$. In that case the associated solution has class $d + 1$.

Proof. The ordinary CE complex for the R-module R is the tensor product of the one-variable complexes $k[t_i]/t_i^2 \xrightarrow{t_i} k[t_i]/t_i^2$. Their zeroth cohomology is kt_i, and their first cohomology has representative 1. Hence $H^n(U, R)$ has basis

$$\Big(\prod_{i \notin S} t_i\Big) e_S^*, \qquad |S| = n.$$

In particular its coefficient degree is $d - n$.

Put $A = R/K = k[t_0]/t_0^2$. The H-action on A is zero, so $H^1(U, A)$ has representatives

$$h(e_0) = 1,\ h(H) = 0; \qquad h(e_i) = \kappa_i t_0,\ h(e_0) = 0.$$

Its dimension is $1 + m = d$. Its elements have coefficient degree zero or one; the same is true of $H^2(U, A)$. Since $d \geq 4$, the maps $H^1(U, R) \to H^1(U, A)$ and $H^2(U, R) \to H^2(U, A)$ vanish by coefficient degree. The long exact sequence therefore gives

$$0 \longrightarrow H^1(U, A) \longrightarrow H^2(U, K) \longrightarrow H^2(U, R) \longrightarrow 0. \tag{158}$$

In degree zero, $H^0(U, R) = k\tau \to H^0(U, A) = kt_0$ is zero. Together with the vanishing map on H^1 above, this gives $0 \to k \to H^1(U, K) \to H^1(U, R) \to 0$, proving its dimension $d + 1$. The connecting map gives β_η and β_κ. The displayed $H^2(U, R)$-basis lifts to the closed K-valued cocycles β_λ, proving the basis claim. Equivalently, the coefficient of τ_{ij} in component (i, j) extracts λ_{ij}, because a coboundary in that component is divisible by t_i or t_j.

The quotient $K/\mathfrak{m}K$ has basis $t_1, \ldots, t_m$. The only degree-one closed cocycles are multiples of β_η; there are no degree-one coboundaries, since K has no constant component. Thus the images of the cocycle values in this quotient are precisely those of $\eta\beta_\eta$. They span it exactly when $\eta \neq 0$. This proves (†); one may also lift the spanning statement by Nakayama's lemma. Finally, $\mathfrak{m}^{d-1}K = k\tau \neq 0$ and $\mathfrak{m}^d K = 0$. Equation (19) gives class $d + 1$. □

Multiplication by a nonzero scalar normalizes η to one. Lemmas B.1 and B.3, followed by Theorem 3.2, prove the asserted coverage by $F_{\kappa,\lambda}$.

B.3 All morphisms and the normalized parabolic

Lemma B.4. *On the slice $\eta = 1$, auxiliary-Lie isomorphism is exactly* (73).

Proof. A Lie isomorphism preserves the intrinsic derived module K. Let $B \in \mathrm{GL}(U)$ be its induced quotient map and let C be its restriction to K. Intertwining the actions gives $C(rw) = \sigma_B(r)C(w)$, and hence $\sigma_B(I) = I$. The degree-m line of I is kz, and $\{u \in U : uz = 0\} = H$. Thus $B(H) = H$, so

$$B = \begin{pmatrix} a & 0 \\ v & M \end{pmatrix}.$$

Conversely every such B preserves I and K. By Lemma B.1, all intertwining C have the form $u\sigma_B$, with $u \in (R/I)^\times$. Changes of section add exactly ordinary CE coboundaries.

Positive-degree multiplication is null-homotopic on this CE complex. Indeed, if ι_i is contraction with e_i, then $\delta\iota_i + \iota_i\delta = t_i$. Writing a positive-degree element as $\sum_i t_i r_i$ gives the homotopy $\sum_i r_i\iota_i$. Consequently the unit u acts on cohomology by its constant term u_0 alone. The star transforms as $\eta' = u_0\eta/a$, as is seen either in degree one or on its lift $h(e_0) = 1$ in A. The normalization $\eta = \eta' = 1$ therefore forces $u_0 = a$. This proves necessity of (73). For sufficiency, use $C = a\sigma_B$; equality of cohomology classes supplies the required change of section.

There are no further quotient substitutions: B is the actual linear map induced on U. For the original quadratic solutions, Theorem 3.2 identifies these Lie isomorphisms with binary conjugacies, including their uniquely determined off-diagonal binary-linear parts. In particular their diagonal blocks are k-linear, not Frobenius-semilinear. □

B.4 The graded action when $d \geq 5$

The coefficient degrees of the three parts of Lemma B.3 are $1, 2, d-2$. For $d \geq 5$ these are distinct, so the displayed splitting is equivariant. On A, the substitution is $t_0 \mapsto at_0$. Pulling back the lift $h(e_i) = \kappa_i t_0$ gives $\kappa' = a^2 M^{-T}\kappa$; its extra e_0-component is a multiple of $t_0 e_0^*$, which is a boundary on A. For the other part, the complementary-minor identity gives

$$[\tau_{ij}]\sigma_B(\tau_{kl}) = \det(B)(\Lambda^2 B^{-T})_{ij,kl}. \tag{‡}$$

The pullback on the two cochain inputs contributes the same minor. Extracting the τ_{ij}-coefficient in component (i,j) therefore proves (75). For example, take $B = \operatorname{diag}(b_0, \ldots, b_m)$, so $a = b_0$. The complementary monomial contributes

$$\sigma_B(\tau_{ij}) = \frac{\det B}{b_i b_j}\tau_{ij},$$

and the two pulled-back inputs contribute a further $(b_i b_j)^{-1}$. Consequently

$$\lambda'_{ij} = a\det(B)(b_i b_j)^{-2}\lambda_{ij}.$$

The same inverse minor occurs once from each source, explaining the entrywise square in (75). Identity (‡) also proves the action on the quotient $H^2(U, R)$ when $d = 4$; only its splitting fails at that endpoint.

Put $\Omega = \lambda^{[2]}$. This is a bijective reparametrization of coefficients, not an additional Galois identification of models. Since squaring twice is the identity on k, and $a^4 = a$,

$$\Omega' = \chi\Lambda^2 B^{-T}\Omega, \qquad \chi = a(\det M)^2.$$

Writing $\Omega = e_0^* \wedge \ell + b$, with $\ell \in H^*$, $b \in \Lambda^2 H^*$, and $w = M^{-1}v$, gives the complete action

$$\begin{aligned} \kappa' &= a^2 M^{-T}\kappa, \\ b' &= \chi M^{-T} b M^{-1}, \\ \ell' &= (\det M)^2 M^{-T}(\ell + \iota_w b). \end{aligned} \tag{159}$$

B.5 Complete rank and radical-pair reduction

Fix $\operatorname{rank} b = 2r$, put $Z = \operatorname{rad} b$, $n = \dim Z = m - 2r$, and choose $H = S \oplus Z$, with b standard symplectic on S. The parameter v removes exactly $\operatorname{im} b$ from ℓ. Its residual is thus $\ell_Z = \ell|_Z$. Write $\kappa = (\kappa_S, \kappa_Z)$. For $r > 0$, the stabilizer of the chosen b has

$$M = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}, \qquad A^T b A = \rho b, \quad D \in \mathrm{GL}(Z), \quad C \text{ arbitrary}.$$

The Pfaffian identity gives $\det A = \rho^r$. The condition $\chi = \rho$ is equivalent to

$$a = \rho^{1-2r}(\det D)^{-2}. \tag{160}$$

Every $\rho \in k^\times$ and every $D \in \mathrm{GL}(Z)$ occur. When $r = 0$, the same formulas hold with S absent and the freely selectable parameter $\rho = a(\det D)^2$. Because $x^3 = 1$ for $x \in k^\times$, the radical-pair action is

$$\kappa'_Z = \rho^{2-r}(\det D)^{-1} D^{-T}\kappa_Z, \qquad \ell'_Z = \rho^{-r}(\det D)^{-1} D^{-T}\ell_Z. \tag{161}$$

If $\kappa_Z \neq 0$, the free block C removes κ_S. If $\kappa_Z = 0$, it cannot change κ_S, and the symplectic group is transitive on its nonzero values. Thus there is exactly one zero/nonzero flag when $r > 0$ and $\kappa_Z = 0$, and no such flag otherwise.

The radical pair has five possible incidence types:

$$(0,0), \quad (0,\neq 0), \quad (\neq 0,0), \quad \text{nonzero proportional}, \quad \text{independent}. \tag{$*$}$$

The last requires $n \geq 2$. In the proportional case the nonzero ratio is multiplied by ρ, so gives no further invariant. For $n \geq 2$, $\mathrm{SL}_n(k)$ is transitive on nonzero covectors. For $n \geq 3$, it is also transitive on ordered independent two-frames: adjust the determinant of a frame extension on an unused coordinate. These observations prove transitivity for all types in $(*)$, except possibly the last at $n = 2$. For $n = 1$, putting $D = (d_0)$ in (161) gives a freely selectable factor d_0 on a nonzero covector, so the same conclusion holds.

For $n = 2$ and an independent pair, let $\Delta = \det[\kappa_Z, \ell_Z]$. Equation (161) gives

$$\Delta' = \rho^{2-2r}\Delta, \tag{162}$$

since $(\det D)^3 = 1$. If $r \not\equiv 1 \pmod 3$, this character covers $k^\times$. If $r \equiv 1 \pmod 3$, the three nonzero values of Δ are separate invariants. These are complete: choose ρ using (162) when permitted, and then use $\mathrm{SL}_2(k)$, which is transitive on ordered bases of a fixed determinant.

This proves the following exhaustive count at each rank:

Rank condition	Radical condition	Number of orbits
$r = 0$	$n = m \geq 4$	5
$r > 0$	$n = 0$	2
$r > 0$	$n = 1$	6
$r > 0$	$n \geq 2$, except the next row	7
$r > 0,\ r \equiv 1 \pmod 3$	$n = 2$	9

Indeed, when $n \geq 1$ the first two types in $(*)$ each acquire the extra κ_S-flag for $r > 0$; the other types do not. For $n = 0$ only the first type remains, with its two flags. For $r = 0$ all five types occur and there is no flag. Summation gives

$$E_c = \begin{cases} 7h + 2\mathbf{1}_{h\equiv 2\ (3)}, & m = 2h \geq 4, \\ 7h + 4, & m = 2h + 1 \geq 5. \end{cases}$$

Since $m = c - 2$, these are exactly (74) for $c \geq 6$.

B.6 The scalar invariant at $6 \mid c$

The three-valued case has a coordinate-free relative-invariant description. Suppose $m = 2r+2$ and $\operatorname{rank} b = 2r$. Let $\gamma_r(b)$ be the divided exterior power: its coefficient on a $2r$-subset is the Pfaffian of the corresponding submatrix. This polynomial definition uses no factorial division. It is natural under linear maps and homogeneous of degree r. Since $\gamma_r(b)$ spans the top exterior power of $\operatorname{im} b$, $(\iota_w b) \wedge \gamma_r(b) = 0$. Consequently the coefficient J of

$$\kappa \wedge \ell \wedge \gamma_r(b) \in \Lambda^m H^*$$

in a fixed volume form satisfies

$$J' = a^{r+2}(\det M)^{2r+1} J. \tag{163}$$

For $r \equiv 1 \pmod 3$ this character is trivial on $k^\times$; equivalently $c = 2r + 4$ is divisible by six. For example,

$$b = \sum_{j=1}^{r} e^*_{2j-1} \wedge e^*_{2j}, \quad \kappa = e^*_{2r+1}, \quad \ell = t e^*_{2r+2}, \qquad t \in \{1, \omega, \omega^2\},$$

gives three inequivalent parameters with identical rank and incidence data. The divided power is essential in characteristic two: ordinary b^r is zero for $r \geq 2$.

B.7 The nonsplit endpoint $d = 4$

Now $m = 3$. Define a symmetric matrix S and vector q by

$$S = \begin{pmatrix} \lambda_{23} & \kappa_3 & \kappa_2 \\ \kappa_3 & \lambda_{13} & \kappa_1 \\ \kappa_2 & \kappa_1 & \lambda_{12} \end{pmatrix}, \qquad q = \begin{pmatrix} \lambda_{01} \\ \lambda_{02} \\ \lambda_{03} \end{pmatrix}, \qquad D(S) = \begin{pmatrix} 0 & S_{33} & S_{22} \\ S_{33} & 0 & S_{11} \\ S_{22} & S_{11} & 0 \end{pmatrix}.$$

For $B = \left(\begin{smallmatrix} a & 0 \\ v & M \end{smallmatrix}\right)$ and $w = M^{-1}v$, the full action is

$$S' = a^2(\det M)^{-1} M S M^T, \qquad q' = \det M (M^{-T})^{[2]} \big(q + D(S) w^{[2]}\big). \tag{164}$$

Here is a direct derivation. The coefficients of t_0 in the H-pair cocycles, with columns ordered as $(23), (13), (12)$, form a linear map $H^* \to H$; Jacobi is exactly symmetry of its matrix S. These coefficients cannot be altered by a K-valued degree-one coboundary. The change of volume on H, the factor $t_0 \mapsto a t_0$, and the normalizing scalar a give the first formula. Factoring $B = \operatorname{diag}(a, M) \left(\begin{smallmatrix} 1 & 0 \\ w & I \end{smallmatrix}\right)$, the unipotent factor leaves these t_0-coefficients unchanged. On q, the complementary-minor calculation gives exactly $D(S)w^{[2]}$; the diagonal factor gives $\det M (M^{-T})^{[2]}$. Thus no splitting of the colliding degrees has been assumed.

A single shear shows the collision explicitly. Take $a = 1$, $v = 0$, and $M = I + E_{21}$, so $e_1 \mapsto e_1 + e_2$. Start with $\lambda_{23} = 1$ and every other λ- and κ-coefficient zero. Then $S = E_{11}$, and

$$M S M^T = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $\lambda'_{23} = \lambda'_{13} = \kappa'_3 = 1$, with all other coefficients zero. A pure λ-class has acquired a κ-component, so the split action for $d \geq 5$ cannot be used at this endpoint.

In fact the degree-two part of (158) is nonsplit already on the Levi $\mathrm{GL}_3(k)$. For $a = 1, v = 0$, the S-space is $\det(H)^{-1} \otimes \Gamma^2 H$, the symmetric tensors. Its alternating, or zero-diagonal, subspace is the κ-space; its diagonal quotient is the corresponding Frobenius-natural twist. A diagonal-torus-equivariant splitting would have to lift each quotient basis vector to its diagonal symmetric matrix, since the weights x_i^2 and $x_i x_j$ are distinct characters of $(k^\times)^3$, also after the common determinant twist. But a transvection $e_i \mapsto e_i + e_j$ sends that diagonal matrix to the two diagonal matrices plus a nonzero off-diagonal symmetric matrix. This contradicts equivariance. Projecting to S shows that the separate q-summand cannot split the full sequence either.

It remains to classify (164). Over the perfect field k, symmetric 3×3 matrices have five congruence types: zero, alternating rank two, and nonalternating ranks one, two and three. One may obtain this list by Gram–Schmidt; if an alternating plane remains after a nonsingular line, the identity $\langle 1 \rangle \perp \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) \simeq I_3$ diagonalizes the final nonalternating case. These are also the S-orbits in (164): any ordinary congruence M is realized by choosing $a^2 = \det M$. For each type the remaining complete invariants are

S	Remaining invariant	Orbits
0	$q = 0$ or $q \neq 0$	2
Alternating, rank 2	$q = 0$; $q \neq 0$, $Sq^{[2]} = 0$; $Sq^{[2]} \neq 0$	3
Nonalternating, rank $1, 2$, or 3	$\operatorname{diag}(S)^T q = 0$ or $\neq 0$	2 each

For $S = 0$, $\mathrm{SL}_3(k)$ is transitive on nonzero q. For alternating $S = J_2 \oplus 0$, its stabilizer has

$$M = \begin{pmatrix} A & b \\ 0 & a^2 \end{pmatrix}, \qquad A \in \mathrm{GL}_2(k), \quad b \in k^2.$$

If the first two coordinates of q are nonzero as a vector, the free block b removes the third coordinate and $\mathrm{SL}_2(k)$ is transitive on the remaining vector. If they vanish, $\det A$ scales the third coordinate freely. This proves the three cases in the table. Finally, if $u = \operatorname{diag} S \neq 0$, then $\operatorname{im} D(S) = \ker u^T$, so translations identify exactly the q with equal $u^T q$. Formula (164) gives $u'^T q' = a^2 u^T q$. The simultaneous choice $M = a^{-1} I_3$ fixes every S and multiplies q by a^2. All nonzero residues are therefore equivalent, proving the two cases for each nonalternating rank. There are exactly $2 + 3 + 2 + 2 + 2 = 11$ classes. This is E_5, and completes the proof of Theorem 7.6.

C Fixed surplus over finite characteristic-two fields

We prove the upper bound in Theorem 8.3. Throughout this appendix, k is a finite field of characteristic two, $|k| = q \geq 4$, and all Lie isomorphisms are k-linear. By Lemma 8.2, the metabelian 3-Engel identity is equivalent to

$$(\operatorname{ad} x)^2 L' = 0 \qquad (x \in L).$$

Put

$$d = c - 1 \geq 4, \qquad C_d = \binom{d}{2}, \qquad w_0 = 2^d - 2,$$

and use the ring and modules

$$R = k[t_0, \dots, t_{d-1}]/(t_i^2), \quad \mathfrak{m} = (t_0, \dots, t_{d-1}), \quad K = (t_1, \dots, t_{d-1}), \quad I = (t_1 \cdots t_{d-1}). \tag{165}$$

The ideal I has dimension two. We write $D = \operatorname{Hom}_k(-, k)$, and ordinary cohomology is the Chevalley–Eilenberg cohomology of the indicated abelian quotient. Module Ext_R is used separately.

Lemma C.1 (The minimum and its common module over k)**.** *Among metabelian 3-Engel k-Lie algebras of class $c \geq 5$, the minimum k-dimension is*

$$\ell(c) = 2^{c-1} + c - 3 = d + w_0.$$

Every equality case has quotient dimension d. After a linear change of quotient generators, its derived R-module is K. For the core quotient $U_0 = k^d$, acting by $e_i w = t_i w$,

$$\operatorname{Ann}_R K = I, \qquad \operatorname{End}_R K \simeq R/I,$$
$$\dim_k H^1(U_0, K) = d + 1, \qquad \dim_k H^2(U_0, K) = d + C_d.$$

The degree-one free rank in a minimal R-resolution of K is C_d, and

$$\dim_k \operatorname{Ext}^1_R(k, K) = 1.$$

In every minimum core the degree-one cocycle can be normalized to

$$\beta(e_0, e_i) \equiv t_i \pmod{\mathfrak{m}K} \quad (i \geq 1), \qquad \beta(e_i, e_j) \in \mathfrak{m}K \quad (i, j \geq 1). \tag{166}$$

In particular, the number of minimum-dimensional k-Lie isomorphism classes is at most q^{d+C_d}.

Proof. The universal commutator module has the same square-free presentation and triangle relations as in Section 7, with k in place of $\mathbb{F}_4$. The multigrading calculation in Proposition 7.1, the repeated-terminal argument in Lemma 7.2, and the contraction calculation in Lemma 7.3 use only characteristic two. The top module is $\det(U_0) \otimes U_0$, and $\mathrm{SL}_d(k)$ is transitive on

its nonzero dual vectors. A normalized top functional has annihilator I and contraction dimension $2^d - 2$. Thus every class-c object has quotient dimension at least d and derived dimension at least w_0. The normalized contraction quotient attains both bounds.

For equality, the proof of Lemma B.1 applies over k: a top functional generates the whole dual derived module as R/I. The top-coefficient Frobenius pairing on R gives $D(R/I) \simeq K$, $\operatorname{Ann}_R K = I$, and $\operatorname{End}_R K \simeq R/I$. The ordinary Koszul calculation in Lemma B.3 likewise has no field-size dependence. It gives the displayed H^1 and H^2 dimensions. For clarity, its degree-one normalization can also be read directly from Jacobi. The degree-one coefficients lie in $\operatorname{span}_k\{t_1, \dots, t_{d-1}\}$. Jacobi at $(0, i, j)$ first makes the degree-one part of $\beta(e_i, e_j)$ zero: its t_0-multiple cannot cancel against the other two terms. Comparing the remaining monomials $t_i t_l$, using $d - 1 \geq 3$, then gives $\beta(e_0, e_i)_{[1]} = \eta t_i$ with one common scalar η. Generation of $K/\mathfrak{m}K$ forces $\eta \neq 0$, and rescaling K normalizes it to one. Two-coboundaries have coefficients in $\mathfrak{m}K$, so no degree-one term is lost in this comparison. This proves (166). For a fixed derived module, cocycles modulo changes of section form $H^2(U_0, K)$; these classes cover all minimum-dimensional objects. This proves the last count.

For the module-resolution assertions, set $S = k[t_1, \dots, t_{d-1}]/(t_i^2)$ and $J_S = (t_1, \dots, t_{d-1})$. The tensor-periodic minimal resolution of k over S, as in (152), has degree-j free rank $\binom{d+j-2}{j}$. Its shifted tail resolves J_S, and extension of scalars resolves $K = R \otimes_S J_S$. Its degree-one free rank is therefore C_d. Finally finite duality gives

$$\operatorname{Ext}^1_R(k, K) \simeq D \operatorname{Tor}^R_1(k, R/I).$$

The cyclic presentation of R/I has the principal minimal relation ideal I, so $\operatorname{Tor}^R_1(k, R/I) \simeq I/\mathfrak{m}I$ is one-dimensional. □

Lemma C.2 (Bounded derived defect)**.** *Let L be a metabelian 3-Engel k-Lie algebra of class c, with derived dimension w_0+h and quotient dimension at least d. There is a class-c core with d quotient generators. After choosing normalized core quotient coordinates, the full derived ideal $W = L'$, as an R-module, has a filtration with one factor K and h factors k. The possible normalized R-module types are covered by at most*

$$M_h(d) = (h+1)q^{\,hC_d + dh^2 + h} \tag{167}$$

representatives. Every such W satisfies

$$\dim_k \operatorname{End}_R(W) \leq w_0 + hd + h^2, \qquad \dim_k \operatorname{Soc}_R(W) \leq h + 1, \tag{168}$$

$$\dim_k H^1(U_0, W) \leq (h+1)d + 1, \qquad \dim_k H^2(U_0, W) \leq d + (h+1)C_d. \tag{169}$$

Here the core quotient U_0 acts through R.

Proof. Lemma 7.2 supplies the core. Its generators may be chosen as linear lifts of a basis of a d-dimensional quotient subspace. Changing a lift inside W leaves a terminal commutator unchanged, since $\mathfrak{m}^{c-1}W = 0$. Thus the chosen core H still has class c, and its abelianization is that quotient subspace. Write $W_0 = H'$. Its dimension is $w_0 + t$ for some $0 \leq t \leq h$, and it is a quotient of the universal module W_d retaining its top component.

Choose a functional on W_0 nonzero on that top component, and pull it back to $\phi \in W_d^*$. Normalize its highest component to have annihilator I. Write this component as λ, so the remaining dual degrees of $\phi - \lambda$ are at most $d - 2$. Multiplication by $t_1 \cdots t_{d-1}$ kills those lower components and kills λ. Therefore $I\phi = 0$. The filtered contraction bound and this annihilator bound give $\dim_k R\phi = w_0$, hence $R\phi \simeq R/I$. Its quotient in W_0^* has dimension t. Dualizing gives

$$0 \longrightarrow F_t \longrightarrow W_0 \longrightarrow K \longrightarrow 0, \qquad 0 \longrightarrow W_0 \longrightarrow W \longrightarrow N_a \longrightarrow 0, \qquad a = h - t, \tag{170}$$

where $\dim_k F_t = t$ and $\dim_k N_a = a$. The local algebra R has residue field k; refining the two small modules by composition series gives the claimed filtration. In particular the full W is handled through these two extensions; only W_0 is required to be a quotient of W_d.

A j-dimensional R-module can be specified by d matrices of size j, so there are at most q^{dj^2} possible module types. For fixed F_t, the degree-one free rank in Lemma C.1 gives

$$\dim_k \operatorname{Ext}^1_R(K, F_t) \le tC_d.$$

The first extension in (170) is thus covered by $q^{dt^2+tC_d}$ choices. Since $\dim_k \operatorname{Ext}^1_R(k,k) = d$, composition series and the long exact sequence give

$$\dim_k \operatorname{Ext}^1_R(k, F_t) \le td, \qquad \dim_k \operatorname{Ext}^1_R(k, W_0) \le td + 1.$$

Applying the long exact sequence in its first argument to a composition series of N_a then gives

$$\dim_k \operatorname{Ext}^1_R(N_a, W_0) \le a(td+1).$$

Thus the second extension costs at most $q^{da^2+a(td+1)}$ choices. Summing over $t + a = h$, the total is bounded by

$$\sum_{t=0}^{h} q^{\,d(t^2+a^2)+tC_d+a(td+1)} \le (h+1)q^{\,hC_d+dh^2+h}.$$

These extension representatives cover all normalized module types. Every unmarked Lie isomorphism class admits a normalized presentation, so choices of normalization introduce no additional factor in this upper bound.

For the remaining bounds, dimensions of Hom_R are subadditive across a short exact sequence in either argument. The filtration factors are one K and h copies of k, and

$$\dim_k \operatorname{Hom}_R(K,k) = d-1, \quad \dim_k \operatorname{Hom}_R(k,K) = 1, \quad \dim_k \operatorname{End}_R(K) = w_0.$$

Applying subadditivity in both arguments proves

$$\dim_k \operatorname{End}_R(W) \le w_0 + h(d-1) + h + h^2 = w_0 + hd + h^2.$$

The same filtration gives $\dim_k \operatorname{Soc}_R(W) = \dim_k \operatorname{Hom}_R(k, W) \le h+1$. Ordinary CE cohomology is dimension-subadditive across its long exact sequence. Combining Lemma C.1 with $\dim H^1(U_0, k) = d$ and $\dim H^2(U_0, k) = C_d$ proves (169). □

Proposition C.3 (The fixed-surplus upper bound)**.** *Let $B_{c,r}(q)$ count metabelian 3-Engel k-Lie isomorphism classes of class c and dimension $\ell(c) + r$. For $r \ge 1$, put*

$$Q_r(d) = (2r+1)C_d + (r^2+r+1)d + r^3 + r + \binom{r}{2}(r+1).$$

Then

$$B_{c,r}(q) \le (r+1)^2 q^{\,rw_0+Q_r(d)}. \tag{171}$$

More precisely, the branch with derived dimension $w_0 + h$, where $0 \le h \le r$ and $s = r - h$, contributes at most

$$(h+1)q^{\,sw_0+(2h+1)C_d+d(h^2+2sh+s+1)+sh^2+r+\binom{s}{2}(h+1)}. \tag{172}$$

In particular, for fixed q, r,

$$\log_q B_{c,r}(q) \le r2^{c-1} + O_{q,r}(c^2).$$

Proof. The separate minimum bounds in Lemma C.1 force quotient and derived dimensions

$$(d+s, w_0+h), \qquad s+h=r, \quad s,h\geq 0.$$

Choose and normalize a d-generator core, then complete its quotient basis by $v_1,\dots,v_s$. For the fixed core R-module W, write $P_i = t_i\cdot$ and $T_a = \operatorname{ad} v_a|_W$. Metabelianity makes every T_a commute with all P_i, so $T_a \in \operatorname{End}_R(W)$. The action tuple therefore has at most

$$q^{\,s(w_0+hd+h^2)}$$

choices. The commuting and square-zero conditions select a subset of these tuples.

The core bracket cocycle $\beta_{ij} = [e_i, e_j]$ is closed for the ordinary CE complex on U_0. Changing the core section by vectors of W adds a two-coboundary. By (169), the core cocycle can consequently be represented by at most $q^{d+(h+1)C_d}$ choices. Such section changes leave W, the P_i, and the T_a unchanged, so the module and action census continues to cover the resulting representatives.

Next put $A_{ia} = [e_i, v_a]$. The core/core/extra Jacobi equations are

$$P_iA_{ja} + P_jA_{ia} = T_a\beta_{ij}.$$

For fixed actions and core cocycle, two solutions for a given a differ by an ordinary one-cocycle. Replacing v_a by $v_a + w_a$, $w_a \in W$, adds the one-coboundary $(P_iw_a)_i$. Thus all mixed tuples can be represented by at most

$$q^{\,s((h+1)d+1)}$$

choices. These changes may also replace $[v_a, v_b]$ by $[v_a, v_b] + T_aw_b + T_bw_a$; those remaining brackets are counted after the mixed tuples have been fixed.

Write $Z_{ab} = [v_a, v_b]$. The core/extra/extra Jacobi equations give

$$P_iZ_{ab} = T_aA_{ib} + T_bA_{ia}.$$

Once the preceding data are fixed, two solutions for Z_{ab} differ by a vector in $\bigcap_i \ker P_i = \operatorname{Soc}_R(W)$. Therefore all extra pairs together have at most $q^{\binom{s}{2}(h+1)}$ choices. The remaining Jacobi equations and the generation and class conditions select a subset of the displayed data.

Multiplying these counts by (167) gives exponent

$$\begin{aligned} &hC_d + dh^2 + h + s(w_0+hd+h^2) \\ &\qquad + d + (h+1)C_d + s((h+1)d+1) + \binom{s}{2}(h+1) \\ &= sw_0 + (2h+1)C_d + d(h^2+2sh+s+1) + sh^2 + r + \binom{s}{2}(h+1), \end{aligned}$$

which proves (172). Every Lie algebra under consideration admits one of these presentations; multiple choices of a core only enlarge this upper count. Finally

$$h^2+2sh+s+1 \leq r^2+r+1, \quad sh^2 \leq r^3, \quad \binom{s}{2}(h+1) \leq \binom{r}{2}(r+1),$$

and $\sum_{h=0}^{r}(h+1) \leq (r+1)^2$. Summing the branches proves (171) and its logarithmic consequence. □

C.1 Proof of the transition scale

Write $n = 2^d$, $C_d = \binom{d}{2}$, $w_0 = n-2$, and $D_{\mathrm{par}} = n-d-3$. We use B_h for the branch with derived dimension w_0+h, so $B_0 = B^{\min}_{d+1,r}(q)$ and $B_{d+1,r}(q) = \sum_{h=0}^{r} B_h$.

A uniform frame estimate. For $1 \le r \le D_{\mathrm{par}}$, Theorem 8.3 gives

$$\log_q B_{d+1,r}(q) \ge rn - d^2 - 2dr - r^2 - 3r - 1. \tag{173}$$

Indeed, after extracting $q^{rD_{\mathrm{par}}}$ from the frame numerator, its remaining product is at least

$$\prod_{j\ge 1}(1-q^{-j}) \ge 1 - \sum_{j\ge 1} q^{-j} \ge \frac{2}{3}.$$

The denominator is at most $q^{dr+d^2+r^2}$, proving the estimate.

The minimum-derived branch below the transition scale. For $h \ge 1$, put $s = r - h$. The exponent in (172), apart from sw_0, is at most

$$h\big(3C_d + 3dr + 2r^2 + r\big).$$

Here $2h+1 \le 3h$, $h^2 + 2sh + s + 1 \le 3rh$, $sh^2 \le r^2 h$, and $\binom{s}{2}(h+1) \le r^2 h$. Since $\log_q(h+1) \le h/2$, comparison with (173) gives, with

$$T = 3C_d + d^2 + 5dr + 3r^2 + 2r + 2,$$

the bound

$$\frac{B_h}{B_{d+1,r}(q)} \le q^{-h(w_0 - T)}. \tag{174}$$

When $a = w_0 - T > 0$, summing a geometric series yields

$$1 - \pi_{d,r}(q) \le \frac{q^{-a}}{1-q^{-a}}. \tag{175}$$

If $r = o(\sqrt{n})$, then eventually $r \le D_{\mathrm{par}}$ and $T = o(n)$, so this tends to zero. Moreover the coarse upper bound (171) has an error $O(rd^2 + r^2 d + r^3)$. Together with (173), it gives $\log_q B_{d+1,r}(q) \sim rn$ in the same range.

A cubic lower family and a small characteristic quotient. Fix one minimum class-$d+1$ algebra L_0 from Lemma C.1. Its derived module is K, and

$$\dim_k\big(L_0/\gamma_3(L_0)\big) = d + \dim_k(K/\mathfrak{m}K) = 2d - 1.$$

For $u = \lfloor 2r/3 \rfloor$ and $w = r - u$, each alternating map $b : \Lambda^2 k^u \to k^w$ defines a two-step algebra N_b on $k^u \oplus k^w$, with central second summand. No surjectivity condition is needed. All $q^{w\binom{u}{2}}$ labelled brackets give algebras $L_b = L_0 \oplus N_b$ counted by $B_{d+1,r}(q)$. Since $\gamma_3(N_b) = 0$,

$$L_b/\gamma_3(L_b) \simeq \big(L_0/\gamma_3(L_0)\big) \oplus N_b$$

has dimension $m = r + 2d - 1$. Distinct tensors give distinct brackets on this fixed labelled quotient space. Each quotient isomorphism class has at most $|\operatorname{GL}_m(q)|$ such labelled structures. Isomorphic L_b's have isomorphic characteristic quotients. Consequently,

$$B_{d+1,r}(q) \ge \frac{q^{w\binom{u}{2}}}{|\operatorname{GL}_{r+2d-1}(q)|}, \qquad \log_q B_{d+1,r}(q) \ge \frac{2}{27} r^3 - O\big((r+d)^2\big). \tag{176}$$

This argument uses the classical alternating-tensor count for two-step Lie brackets; see also the coordinate-explicit enumeration in [35]. It neither transfers a group-counting theorem to characteristic two nor assumes cancellation of direct summands.

The minimum-derived branch above the transition scale. Taking $h = 0$ in (172) gives, for every $r \geq 1$,

$$B_0 \leq q^{rw_0 + C_d + d(r+1) + r + \binom{r}{2}}. \tag{177}$$

If $r/\sqrt{n} \to \infty$, then $r \gg d$ and $n = o(r^2)$. Dividing (177) by (176) gives

$$\log_q \pi_{d,r}(q) \leq rn - \frac{2}{27}r^3 + O(r^2 + rd + d^2) = -\big(2/27 - o(1)\big)r^3.$$

This proves (90).

The uniform order of growth. The explicit upper bound in Proposition C.3 is $O(rn + r^3)$ on the logarithmic scale, since $d^2 = O(n)$ and $r^2 d \leq (r^3 + rd^2)/2$. For $r \leq \sqrt{n}$, (173) is at least a fixed positive multiple of rn, uniformly in r, once d is large. For $r \geq \sqrt{n}$, (176) is at least a fixed positive multiple of r^3, again uniformly in r. The larger of rn and r^3 is at least half their sum. This proves (88) and completes the proof of Theorem 9.1.

D Exact principal orbits and multiplicative enumeration

Let $k = \mathbb{F}_q$, $q \geq 4$, have characteristic two, and use $d, C_d, D_{\mathrm{par}}, w_0, R, K, I$ as in Section 8.3. Put $\tau = t_0 \cdots t_{d-1}$, $\tau_j = \prod_{i \neq j} t_i$, and $V_r = \langle v_1, \ldots, v_r \rangle_k$. The binary-map interpretation is used only at $q = 4$.

D.1 All ordinary cocycles

For each $p_a \in \mathcal{P}_d$, choose $a_{0a} = 0$, $a_{ia} \in \mathfrak{m}$, with

$$p_a = \sum_{i=1}^{d-1} t_i a_{ia}. \tag{178}$$

Assigning each monomial to its smallest positive index fixes this choice. Let $b = \eta\beta_\eta + \beta_\kappa + \beta_\lambda$ be the core representative from Lemma B.3. Use the core one-cocycles

$$\zeta_*(e_0) = 0, \quad \zeta_*(e_i) = t_0 t_i \ (i \geq 1), \qquad \zeta_j(e_i) = \mathbf{1}_{i=j}\tau_j \quad (0 \leq j < d).$$

Their classes form a basis of $H^1(U_0, K)$, where $U_0 = k^d$.

Lemma D.1. *For any tuple $\mathbf{p} \in \mathcal{P}_d^r$,*

$$\begin{aligned} H^2(U_0 \oplus V_r, K) &\simeq H^2(U_0, K) \oplus H^1(U_0, K) \otimes V_r^* \oplus H^0(U_0, K) \otimes \Lambda^2 V_r^*, \\ \dim H^2(U_0 \oplus V_r, K) &= d + C_d + r(d+1) + \binom{r}{2}, \\ \dim H^1(U_0 \oplus V_r, K) &= d + 1 + r. \end{aligned}$$

Explicit representatives are, with $\zeta^a \in \operatorname{span}\{\zeta_*, \zeta_0, \ldots, \zeta_{d-1}\}$ and $\theta_{ab} \in k$,

$$\begin{aligned} \beta(e_i, e_j) &= b(e_i, e_j), \\ \beta(e_i, v_a) &= \sum_j a_{ja} b(e_i, e_j) + \zeta^a(e_i), \\ \beta(v_a, v_b) &= \sum_{i<j} (a_{ia}a_{jb} + a_{ib}a_{ja}) b(e_i, e_j) + \sum_i (a_{ia}\zeta^b(e_i) + a_{ib}\zeta^a(e_i)) + \theta_{ab}\tau. \end{aligned} \tag{179}$$

These representatives are unique modulo coboundaries. Intrinsic generation is equivalent to $\eta \neq 0$.

Proof. In the ordinary Koszul cochain complex, replace e_i^* by $e_i^* + \sum_a a_{ia} v_a^*$, using (178). This triangular invertible R-linear change converts the differential $\sum_i t_i e_i^* \wedge + \sum_a p_a v_a^* \wedge$ to the core differential, with zero differential on $\Lambda(V_r^*)$. It is an operation on cochains, not a k-linear change of Lie quotient generators. Expanding the resulting representatives gives (179). The core cohomology bases and $H^0(U_0, K) = k\tau$ prove the dimensions and uniqueness. Every mixed coefficient has degree at least two, so modulo $\mathfrak{m}K$ only the core star remains. It spans $K/\mathfrak{m}K$ exactly when $\eta \neq 0$; apply Nakayama. □

Scalar multiplication of K normalizes $\eta = 1$. The minimum derived-module result covers every object with $h = 0$. Each extra action lies in $\operatorname{End}_R K = R/I$; square zero forces its constant term to vanish, and subtracting its linear part from the corresponding quotient generator gives a unique tuple in $\mathcal{P}_d^r$. Thus the normalized fibers cover the whole minimum-derived stratum.

D.2 The complete effective action

We use ordinary abelian-extension equivalence and section changes as in [5, Theorems 1.1–1.2]. The intrinsic kernel of $U \to \operatorname{rad}(R/I)/\operatorname{rad}(R/I)^2$ is V_r, whether or not the tuple is independent. Every isomorphism has quotient matrix $A_g = \left(\begin{smallmatrix} B & 0 \\ \alpha & \Delta \end{smallmatrix}\right)$, and action intertwining gives (84). The nilpotent implicit-function principle [3, Sections 1.2–1.3] also applies: iteration from zero increases the radical valuation of a difference, hence gives a unique target tuple. Its associated substitution has invertible linear part B.

Preserving $I = (z)$, $z = t_1 \cdots t_{d-1}$, in degree $d-1$ forces $B(H) = H$. Conversely, this condition suffices. The substitution σ_g is an automorphism of R and preserves K, since all p'_a lie in $\mathfrak{m}^2 \subset K$. In $\sigma_g(z)$, terms with at least two additional parameter factors have degree at least $d+1$ and vanish. The remaining terms give

$$\sigma_g(z) = (\det M)z + \xi\tau,$$

so $\sigma_g(I) = I$. No product between distinct parameters has been discarded where its degree could survive.

All module intertwiners are $u\sigma_g|_K$, $u \in (R/I)^\times$. Positive-degree multiplication is null-homotopic, by $\delta\iota_i + \iota_i\delta = t_i$. Consequently only the constant term u_0 acts on ordinary cohomology. The degree-one star transforms by u_0/a; its normalization forces $u_0 = a$. Section changes add precisely two-coboundaries. This proves that (85) is necessary and sufficient.

The formulas give a genuine group action. Under composition, two canonical R-lifts may differ by terms in I. Their maps on K intertwine the same automorphism of R/I, so their ratio is a module unit. Their leading maps agree, making its constant term one; its cohomological action is trivial. Thus the induced fiber maps compose as the actual quotient matrices, without assuming that the two R-lifts agree pointwise on K. This proves Theorem 8.7.

D.3 A summed fixed-point bound

Write

$$F_{d,r} = \prod_{j=0}^{r-1} (1 - q^{j - D_{\mathrm{par}}}),$$

which tends to one for fixed q, r. The frame base has size $q^{rD_{\mathrm{par}}} F_{d,r}$. All its cohomology fibers have the same cardinality, so a fixed-base bound also bounds the fixed proportion in $\mathcal{Y}_{d,r}^{\mathrm{fr}}$.

Suppose $(B, \Delta) \neq (I, I)$. The fixed equation is triangular by coefficient degree: in degree j its linear part is $\Lambda^j B \otimes I_r - I \otimes \Delta^T$, and all nonlinear terms depend on lower degrees. On the full exterior algebra tensored with k^r, put $T = \Lambda B \otimes I_r - I \otimes \Delta^T$. Then

$$T(L_v \otimes I_r) - (L_{Bv} \otimes I_r)T = L_{Bv - v} \otimes \Delta^T.$$

If $B \neq I$, this gives $\operatorname{rank} T \geq r2^{d-2}$. If $B = I$ but $\Delta \neq I$, then $\operatorname{rank} T \geq 2^d$. Removing degrees zero and one and quotienting by I loses at most $r(d+3)$. The common lower bound $2^{d-2} - r(d+3)$ therefore suffices. Since $|G_{d,r}| < q^{d^2+rd+r^2+1}$, the sum of these fixed proportions is at most

$$F_{d,r}^{-1} q^{d^2+rd+r^2+1-2^{d-2}+r(d+3)}. \tag{180}$$

It remains to control pure shears, where $B = I$, $\Delta = I$, and $\alpha \neq 0$. Let $k_0 = \operatorname{rank} \alpha \leq r$. For counting, enlarge $\mathcal{P}_d^r$ to $(\mathfrak{m}^2 R)^r$, adding $2r$ dimensions. Invertible linear changes of ring variables and parameter components put the displacement map into the form

$$(x_1, \ldots, x_{k_0}, y) \longmapsto (x_1 + p_1, \ldots, x_{k_0} + p_{k_0}, y).$$

These changes need not preserve I; it becomes a two-dimensional space I'. Set $S = k[y_1, \ldots, y_m]/(y_i^2)$, $m = d - k_0$, and expand

$$p_a = \sum_{T \subseteq \{1,\ldots,k_0\}} x_T b_{a,T}, \qquad a_a = b_{a,\varnothing} \in \mathfrak{m}_S^2.$$

A necessary condition for a fixed parameter is obtained by setting all active variables x_i to zero:

$$\sum_{T \neq \varnothing} \left(\prod_{i \in T} a_i \right) b_{a,T} \in \operatorname{pr}_S(I') \qquad (1 \leq a \leq r). \tag{181}$$

This is an exact specialization; it retains every product involving different parameters. The target space has dimension at most two.

Condition on all a_a. If some a_i, $1 \leq i \leq k_0$, has a nonzero quadratic coefficient, multiplication by a_i on S has rank at least 2^{m-2}: restrict inputs to variables outside a nonzero quadratic monomial and project outputs to that monomial's full coefficient. The resulting map multiplies by a unit of the complementary ring. Restricting inputs to $b_{a,\{i\}} \in \mathfrak{m}_S$ loses at most one rank. Fixing all other coefficients in one equation of (181) therefore imposes codimension at least $2^{m-2} - 3$. The chosen a_i is held fixed independently of this coefficient block, so this is an affine-linear count with no independence assumption between the different fixed equations.

If all of $a_1, \ldots, a_{k_0}$ have zero quadratic part, the enlarged parameter space has codimension $k_0 \binom{m}{2}$. For $d \geq r + 2$, use $m \geq d - r$ and sum over fewer than q^{rd} nonzero matrices α. The total pure-shear fixed proportion is at most

$$F_{d,r}^{-1} q^{rd+2r} \left(q^{3-2^{d-r-2}} + q^{-\binom{d-r}{2}} \right). \tag{182}$$

For fixed q, r, both (180) and (182) tend to zero. They bound the sum of all nonidentity fixed proportions, not only a probability of nonfree points. Burnside's lemma gives

$$|\mathcal{Y}_{d,r}^{\mathrm{fr}}/G_{d,r}| = \frac{q^{rD_{\mathrm{par}}} F_{d,r}\, q^{f_{d,r}}}{|G_{d,r}|}(1 + o(1)).$$

Nonfree parameter and orbit proportions also tend to zero, by the same stabilizer-sum argument as in Corollary 11.13.

D.4 The whole layer and its typical automorphism order

Rank-deficient tuples number at most $q^{(r-1)D_{\mathrm{par}}+r(r-1)}$; multiplying by the entire fiber size costs only $q^{O_r(d^2)}$ beyond this exponent. The branches with $h \geq 1$ likewise contribute at most $q^{(r-1)2^d+O_{q,r}(d^2)}$, by Proposition C.3. Both are negligible compared with the displayed orbit main term. Substituting the group order proves (86) for all objects in the layer.

An automorphism inducing the identity on U has a principal-unit module block, with q^{w_0-1} choices. Every such unit lifts, because it acts trivially on H^2. Its section blocks form a torsor under $Z^1(U,K)$, whose dimension is

$$(w_0-1)+(d+1+r)=w_0+d+r.$$

Thus each faithful minimum-derived object satisfies

$$|\operatorname{Aut}_k(L_y)| = q^{2w_0+d+r-1}|\operatorname{Stab}_{G_{d,r}}(y)|.$$

Generic freeness and the negligible complementary strata prove (87). At $q=4$, apply the isomorphism correspondence of Theorem 3.2.

E The critical constant and coupled derived defects

The action-space normalization uses the standard Gaussian enumeration of subspaces over finite fields; see, for example, [35, Section 2]. Ordinary extension equivalence is used in the form of a module intertwiner together with cocycles modulo coboundaries [5, Theorems 1.1–1.2]. The large principal-unit kernel follows from the usual contraction identity in the Chevalley–Eilenberg complex: if T_i is the action of e_i, then $\delta\iota_{e_i}+\iota_{e_i}\delta=T_i$. The uniform comparison here combines these tools with the intrinsic near-minimum module and a coupled one-dimensional central-defect family. In particular, the critical constant comes from comparing complete isomorphism-type counts.

E.1 Action subspaces and the lower side of the critical constant

Put $n=2^d$, $w_0=n-2$, $C_d=\binom{d}{2}$, and $D_{\mathrm{par}}=n-d-3$. Let B_h count the class-$d+1$ objects of total dimension $d+w_0+r$ with derived dimension w_0+h, and write $s=r-h$. All counts are unlabelled k-linear isomorphism counts, where $k=\mathbb{F}_q$ has characteristic two and $q\geq 4$.

Lemma E.1 (Normalizing the action subspace)**.** *Fix a normalized core R-module W in the h-branch, with core actions $P_0,\dots,P_{d-1}$. These actions are linearly independent. Set*

$$E=\operatorname{End}_R(W),\qquad S_0=\operatorname{span}_k\{P_0,\dots,P_{d-1}\}.$$

The possible tuples of s additional actions, up to changes of the additional quotient basis and addition of core quotient directions, have a covering set of size at most

$$\sum_{j=0}^{\min(s,L)}\begin{bmatrix}L\\ j\end{bmatrix}_q,\qquad L=w_0+hd+h^2-d.$$

If $2s\leq L$, this is at most $\frac{3}{2}(s+1)q^{s(L-s)}$.

Proof. The filtration in Lemma C.2 has one factor K. A linear relation among the P_i on W would hold on that factor, contradicting the absence of a nonzero linear form in $\operatorname{Ann}_R K=I$. Thus $\dim S_0=d$, and $\dim E/S_0\leq L$.

For each subspace $F\subset E/S_0$, choose once and for all an ordered basis and lifts to E. Given an actual tuple $(T_1,\dots,T_s)$, let F be the span of its images and put $j=\dim F$. A change in $\mathrm{GL}_s(k)$ makes those images the chosen basis followed by zeros. Adding suitable core directions to the additional quotient generators then makes their actual actions equal to the chosen lifts followed by zeros. This is an invertible block change of quotient coordinates fixing the core. It keeps the R-module W and all core actions unchanged.

The brackets transform along with these coordinates. Once the actions have been normalized, the same core, mixed, and extra-pair cocycle covering used in Proposition C.3 applies. Some preselected lifts may give no admissible Lie algebra; including them only enlarges an upper bound. The argument chooses presentations and never divides a parameter count by a group with unknown stabilizers.

There are $\begin{bmatrix} L \\ j \end{bmatrix}_q$ subspaces of dimension j in an L-dimensional space. The usual product formula and $\prod_{i\ge 1}(1-q^{-i}) \ge 2/3$ give $\begin{bmatrix} L \\ j \end{bmatrix}_q \le \frac{3}{2} q^{j(L-j)}$. If $2s \le L$, the exponent increases for $0 \le j \le s$, proving the last assertion. □

In particular, replacing the raw action-tuple census in Proposition C.3 by this covering gives

$$B_h \le \tfrac{3}{2}(h+1)(s+1)q^{\,sw_0+(2h+1)C_d+d(h^2+2sh+1)+sh^2+r+\binom{s}{2}(h+1)-s^2}, \qquad 2s \le w_0+hd+h^2-d. \tag{183}$$

The core section changes leave the actions fixed, and the later changes of additional lifts affect only the remaining brackets, as in that proposition. Thus the new normalization does not invalidate the successive ordinary-cohomology quotients used there.

Lemma E.2 (The minimum-derived count at growing surplus)**.** *Uniformly for* $1 \le r \le D_{\mathrm{par}}/2$*,*

$$\log_q B_0 = rn - \tfrac{1}{2}r^2 + O(rd + d^2). \tag{184}$$

More explicitly,

$$\log_q B_0 \ge rn - \tfrac{1}{2}r^2 - dr - C_d - \tfrac{5}{2}r - 3,$$
$$\log_q B_0 \le rn - \tfrac{1}{2}r^2 + C_d + d - \tfrac{5}{2}r - 1 + \log_q\big(\tfrac{3}{2}(r+1)\big).$$

Proof. For $W = K$, subtracting the linear parts of the additional actions places them in the D_{par}-dimensional space $\mathcal{P}_d$. The standard Koszul elimination of redundant generators gives

$$H^2(U_0 \oplus k^r, K) \simeq H^2(U_0, K) \oplus \big(H^1(U_0, K) \otimes (k^r)^*\big) \oplus \big(\mathrm{Soc}_R(K) \otimes \Lambda^2(k^r)^*\big).$$

Indeed, write each $p_a = \sum_i t_i a_{ia}$ with $a_{ia} \in \mathfrak{m}$, and change cochain generators by $e_i^* \mapsto e_i^* + \sum_a a_{ia} v_a^*$. All mixed coefficients have degree at least two, so intrinsic generation is still exactly the nonzero core-star condition. After normalizing that coefficient to one, every action tuple has q^{f_r} cocycle classes, where

$$f_r = d + C_d + r(d+1) + \binom{r}{2} - 1.$$

Apply the action-subspace normalization directly in $\mathcal{P}_d$:

$$B_0 \le q^{f_r} \sum_{j=0}^{r} \begin{bmatrix} D_{\mathrm{par}} \\ j \end{bmatrix}_q \le \tfrac{3}{2}(r+1) q^{f_r + r(D_{\mathrm{par}} - r)}.$$

This is the stated upper bound.

For the lower bound take every independent ordered r-frame in $\mathcal{P}_d$, together with all its normalized cocycle classes. The vector implicit-recovery equation in Theorem 8.3 makes the target frame unique for each fixed quotient matrix. There are at most $q^{dr} |\mathrm{GL}_d(q)| |\mathrm{GL}_r(q)|$ such matrices. For a fixed quotient matrix, any two module intertwiners differ by a unit of R/I. Principal units act trivially on ordinary cohomology, by the core contraction homotopies.

Thus there are at most $q-1$ different effects on the target cocycle class. This is a fiber bound; it requires neither a free action nor an explicit full orbit classification. Consequently

$$B_0 \geq \frac{q^{f_r}\prod_{j=0}^{r-1}(q^{D_{\mathrm{par}}}-q^j)}{(q-1)q^{dr}|\operatorname{GL}_d(q)||\operatorname{GL}_r(q)|}.$$

The frame product is at least $\frac{2}{3}q^{rD_{\mathrm{par}}}$, while the denominator is at most $q^{dr+d^2+r^2+1}$. Taking logarithms and using $\log_q(2/3)\geq -1$ gives the stated lower bound. □

Proposition E.3 (Minimum-derived dominance below the constant)**.** *Put*

$$Z(d,r)=2d^2+3dr+2r+6.$$

For $1\leq r\leq D_{\mathrm{par}}/2$ *and* $1\leq h\leq r$,

$$\frac{B_h}{B_0}\leq q^{\,h[-n+\frac{1}{2}r^2-\frac{1}{2}h^2+Z(d,r)]}. \tag{185}$$

In particular, if $r/\sqrt{n}\to\alpha<\sqrt{2}$ *as* $d\to\infty$*, then* $B_0/\sum_h B_h\to 1$.

Proof. The condition $r\leq D_{\mathrm{par}}/2$ implies the condition on s in (183) for every h. Subtract the lower exponent in Lemma E.2 from the exponent there, and substitute $s=r-h$. After removing $h(-n+r^2/2-h^2/2)$, the remainder is

$$d^2h+d^2-dh^2+2dhr-dh+dr+\tfrac{1}{2}hr+\tfrac{5}{2}h+r+3.$$

For $h\geq 1$, dropping the negative terms bounds this by $h(2d^2+3dr+3r/2+11/2)$. Also $\log_q[\frac{3}{2}(h+1)(s+1)]\leq (r+1)/2$, since $\log_q(t+1)\leq t/2$ for nonnegative integers t. Absorbing this into $hZ(d,r)$ proves (185).

If $a=n-r^2/2-Z(d,r)>0$, a geometric sum gives

$$1-\frac{B_0}{\sum_h B_h}\leq\sum_{h\geq 1}\frac{B_h}{B_0}\leq\frac{q^{-a}}{1-q^{-a}}.$$

For the stated sequence, $a/n\to 1-\alpha^2/2>0$. No monotonicity assumption on the population proportions is used. □

E.2 A coupled central defect and the upper side of the constant

Retain the notation of Appendix E.1: $n=2^d$, $w_0=n-2$, $C_d=\binom{d}{2}$, and $D_{\mathrm{par}}=n-d-3$, with $d\geq 4$ and $k=\mathbb{F}_q$ of characteristic two, $q\geq 4$. Here B_h counts the class-$d+1$ objects of total dimension $d+w_0+r$ and derived dimension w_0+h. Put $A=R/I$, and write $\tau=t_0\cdots t_{d-1}$. All cohomology below is ordinary Chevalley–Eilenberg cohomology of an abelian quotient, with its specified module action.

Proposition E.4 (One coupled derived defect)**.** *For* $s=r-1$ *with* $2\leq s\leq D_{\mathrm{par}}$*, put*

$$f_s=d+C_d+s(d+1)+\binom{s}{2}-1.$$

Then

$$B_1\geq\frac{q^{f_s}\displaystyle\prod_{j=0}^{s-1}(q^{D_{\mathrm{par}}}-q^j)\big(q^{\binom{s}{2}}-1\big)}{q^{ds+d+2}|\operatorname{GL}_d(q)|\,|\operatorname{GL}_s(q)|}. \tag{186}$$

Every algebra in the family used for this lower bound is indecomposable. In particular,

$$\log_q B_1 \geq sn - ds - C_d - d - 3s - 4. \tag{187}$$

For $3 \leq r \leq D_{\mathrm{par}}/2$, it follows that

$$\frac{B_0}{B_1} \leq q^{n-r^2/2+dr+d^2+r/2+\log_q(\frac{3}{2}(r+1))}. \tag{188}$$

Proof. Apply Proposition F.1 with $h = 1$, restricting its central cocycle to $\Lambda^2 V_s$ and setting all other central coefficients to zero. A nonzero form on that subspace gives exactly the earlier coupled slice. Its complete isomorphism fibre is bounded by $(q-1)^2 q^{ds+d} |\mathrm{GL}_d(q)||\mathrm{GL}_s(q)|$, which is at most the denominator in (186). The indecomposability proof in that proposition applies to this subfamily. Extracting the frame product and the nonzero-form factor, using bounds 2/3 and 3/4, gives (187). Subtracting it from the upper bound in Lemma E.2 gives (188). The general construction retains further central cocycles; its stronger bound is used in Corollary E.7. □

Remark E.5 (Indecomposability of the binary solutions). At $q = 4$, the corresponding binary quadratic solutions are indecomposable under binary-linear conjugacy. Indeed, suppose a solution is conjugate to $f_1 \oplus f_2$ on two nonzero binary subspaces. Its polar Lie algebra splits accordingly, so both factors are metabelian. The intrinsic derived ideal, quotient, and their scalar actions also split. In the proof of Theorem 3.2, choose the bases and normalized-section lifts separately in the two factors. Scalar-weight projection, the actions P_i, and the cocycle refinement Q then preserve these blocks. The auxiliary bracket (18) is consequently the direct sum of the two auxiliary Lie algebras, contradicting Proposition E.4. A nonzero linear factor gives a nonzero abelian auxiliary factor and is excluded by the same argument. This statement concerns direct sums of solutions. It does not assert indecomposability of the original polar Lie algebra, nor require an extension of the correspondence to arbitrary homomorphisms.

E.3 Principal orbits when the surplus grows

Fix the field $k = \mathbb{F}_q$, $q \geq 4$, of characteristic two. Retain $d, n = 2^d, D_{\mathrm{par}} = n-d-3, w_0 = n-2$, and write $P = \mathrm{rad}(R/I)^2$, a vector space of dimension D_{par}. Let B_0 count the minimum-derived stratum in dimension $d + w_0 + r$. The exact orbit model of Theorem 8.7 remains valid for every r.

Lemma E.6 (Pure shears through invariant subspaces)**.** *Let $\mathcal{F}_{d,r}$ be the ordered independent r-frames in P, and let N be the group of automorphisms of R/I whose linear part is the identity. For $\sigma \in N$, set*

$$U_\sigma = \mathrm{span}_k\{\sigma(t_i) - t_i : 0 \leq i < d\}, \qquad F_\sigma = \ker(\sigma - 1 : P \to P).$$

Then the summed fixed proportion of all nonidentity pure shears is

$$\sum_{\alpha \neq 0} \frac{|\operatorname{Fix}_{\mathcal{F}_{d,r}}(\alpha)|}{|\mathcal{F}_{d,r}|} = \sum_{\substack{\sigma \in N \\ \sigma \neq 1}} \frac{|\{Q \leq P : \dim Q = r,\ U_\sigma \leq Q \leq F_\sigma\}|}{\begin{bmatrix} D_{\mathrm{par}} \\ r \end{bmatrix}_q}. \tag{189}$$

Proof. Every choice $u_i \in P$ gives an automorphism $t_i \mapsto t_i + u_i$ of R/I. To see the lifting assertion, use the canonical representatives in $\mathfrak{m}^2 R$. Their substitution has identity linear part, hence is an automorphism of R. It preserves K; the image of $z = t_1 \cdots t_{d-1}$ lies in $kz + k\tau = I$, since terms with two positive-degree increments have degree at least $d+1$. Thus it preserves I, and the induced map is determined by the u_i's. Conversely every member of N has these increments.

For a fixed ordered frame $\mathbf{p}$, the coefficients of $u_i = \sum_a \alpha_{ai} p_a$ are unique. The pure-shear base equation is exactly $\sigma(p_a) = p_a$ for every a. Hence fixed pairs $(\alpha, \mathbf{p})$ correspond to pairs $(\sigma, \mathbf{p})$ with $U_\sigma \le \operatorname{span}(\mathbf{p}) \le F_\sigma$. The number of ordered bases is the same for every r-subspace. Dividing the pair count gives (189). □

For $2 \le j \le d-1$, put

$$a_j = \dim(P/\operatorname{rad}(R/I)^j), \qquad \rho_j = \max\{0, 2^{d-j-1} - d - 3\}.$$

If j is the least order of a nonzero increment of σ, then

$$\operatorname{rank}(\sigma - 1 : P \to P) \ge \rho_j. \tag{190}$$

Indeed lift σ to R, write $T = \sigma - 1$, and use

$$TL_{t_i} - L_{\sigma(t_i)} T = L_{\sigma(t_i) - t_i}.$$

An increment of order j has multiplication rank at least 2^{d-j}: restrict inputs to the variables complementary to one nonzero degree-j monomial and project to its full coefficient; the resulting multiplier has nonzero constant term. The commutator therefore gives $\operatorname{rank} T \ge 2^{d-j-1}$. Restricting to $\mathfrak{m}^2 R$ loses at most $d+1$ rank, and passage to the quotient by I loses at most two more. This proves (190).

If $k_0 = \dim U_\sigma$, the number of possible increment tuples of order at least j and rank k_0 is at most

$$\tfrac{3}{2} q^{k_0(D_{\mathrm{par}} - a_j + d - k_0)}.$$

For each actual tuple the Gaussian ratio in (189) is zero unless $U_\sigma \le F_\sigma$; otherwise it is at most

$$\tfrac{3}{2} q^{-k_0(D_{\mathrm{par}} - r) - (r - k_0)\rho_j}.$$

Thus the contribution from order j, rank k_0, is at most

$$\tfrac{9}{4} q^{k_0(r - a_j + d - k_0) - (r - k_0)\rho_j}. \tag{191}$$

This estimates the sum of stabilizers, rather than only the probability of a nontrivial stabilizer.

Let $j_0 = \lfloor d/2 \rfloor$, $a_0 = a_{j_0}$, and $\rho_0 = \rho_{j_0}$. Symmetry and the central-binomial estimate give $a_0 \sim n/2$, while $\rho_0 \ge \sqrt{n}/2 - d - 3$. For $r \ge 2d$, summing (191) separately over $j < j_0$ and $j \ge j_0$ gives the uniform bound

$$S_{\mathrm{pure}} \le \tfrac{9}{4} d^2 \left(q^{d(r+d) - (r-d)\rho_0} + q^{r + d - a_0} \right). \tag{192}$$

For the first group, drop $-k_0 a_j - k_0^2$ and use $k_0 \le d$, $\rho_j \ge \rho_0$. For the second group use $a_j \ge a_0$, drop the nonpositive rank term, and use $k_0(r - a_0 + d - k_0) \le r + d - a_0$ once $a_0 > r + d$. In particular (192) tends to zero uniformly on

$$2d \le r \le D_{\mathrm{par}}/(4d). \tag{193}$$

For completeness the nonpure linear parts admit a bound on the same range. Put $F_{d,r} = \prod_{i<r}(1 - q^{i - D_{\mathrm{par}}})$, which tends to one uniformly there. For $B \ne I$, retain the stronger exterior rank $r 2^{d-2}$ in Appendix D. The resulting sum is at most

$$F_{d,r}^{-1} q^{d^2 + rd + r^2 + 1 - r 2^{d-2} + r(d+3)}. \tag{194}$$

For $B = I$, put $t = \operatorname{rank}(\Delta - I)$. In every retained graded monomial the diagonal linear block is $\Delta^T - I$, so the total rank is $D_{\mathrm{par}} t$. There are at most $\frac{3}{2} q^{t(2r-t)}$ such matrices Δ. Including all q^{rd} shears and summing over $t \ge 1$ gives

$$S_\Delta \le \tfrac{3}{2} r F_{d,r}^{-1} q^{rd + 2r - 1 - D_{\mathrm{par}}}. \tag{195}$$

Both displays tend to zero uniformly in (193).

Proof of Theorem 9.4. Every normalized cohomology fiber has size $q^{f_{d,r}}$, so each fixed-base proportion also bounds the corresponding fixed proportion in the whole frame-fiber space. Equations (192)–(195) show that the summed nonidentity fixed proportion is $o(1)$, uniformly. Burnside therefore gives the frame count divided by $|G_{d,r}|$ with multiplicative error $1 + o(1)$, and makes nonfree orbit proportions negligible by the same stabilizer-sum argument.

It remains to cover rank-deficient action tuples. Normalizing their action subspace, while retaining all r quotient directions and all $q^{f_{d,r}}$ cocycles, bounds their class count by

$$q^{f_{d,r}} \sum_{j<r} \begin{bmatrix} D_{\mathrm{par}} \\ j \end{bmatrix}_q \le \tfrac{3}{2} r q^{f_{d,r}+(r-1)(D_{\mathrm{par}}-r+1)}.$$

Its ratio to the faithful orbit main term is at most

$$\tfrac{3}{2} r F_{d,r}^{-1} q^{-D_{\mathrm{par}}+rd+d^2+2r} = o(1),$$

uniformly in (193). This proves (93). The exact full-automorphism kernel calculation in Appendix D is valid for every faithful frame, so generic effective freeness gives the stated order. □

The independent subcritical estimate of Proposition E.3 transfers (93) to the entire dimension layer on the subrange $2d \le r \le \sqrt{2n} - 3d - 3$. No such whole-layer conclusion is made above the critical constant.

Corollary E.7 (A sharper upper-side localization)**.** *If $r \ge \sqrt{2n} - d + 1$ eventually, then $\pi_{d,r}(q) \to 0$. In the portion of* (193) *with $r \ge 3$, put*

$$T = \binom{d+r-1}{2} - (d-1).$$

The full-central-cocycle coupled family gives

$$\pi_{d,r}(q) \le (1 + o(1)) \frac{(q^{D_{\mathrm{par}}} - q^{r-1})(q-1)q^2(q^d-1)}{(q^r-1)(q^T-1)}. \tag{196}$$

Proof. Divide (93) by the independent coupled lower bound (197), with $h = 1$ and $s = r - 1$. Its full central-cocycle count is $q^{d-1}(q^T - 1)$; its effective module-block bound is $(q-1)^2 q^d$. Use $|\operatorname{GL}_m|/|\operatorname{GL}_{m-1}| = q^{m-1}(q^m - 1)$. This gives (196). The logarithm of its right-hand side is

$$n - \tfrac{1}{2}(r+d)^2 + \tfrac{1}{2}(r+d) + 2d - 3 + \log_q(q-1) + o(1).$$

The quadratic expression decreases for $r+d \ge 1$. At $r = \sqrt{2n} - d + 1$ it equals $-\sqrt{2n}/2 + 2d - 3$, which tends to minus infinity. This proves the assertion on the indicated range. On all remaining indices with $r > D_{\mathrm{par}}/(4d)$, one has $r/\sqrt{n} \to \infty$; invoke (90). □

F Coupled central defects and the postcritical population

Fix $k = \mathbb{F}_q$ of characteristic two, $q \ge 4$, and retain $R, K, I, A = R/I$ from Appendix E. Put $d = c - 1 \ge 4$, $n = 2^d$, $w_0 = n - 2$, $C_d = \binom{d}{2}$, $D_{\mathrm{par}} = n - d - 3$, and $\tau = t_0 \cdots t_{d-1}$. Write B_h for the branch of $B_{d+1,r}(q)$ with derived dimension $w_0 + h$, and put $s = r - h$. Ordinary cohomology always means the Chevalley–Eilenberg cohomology for the specified abelian quotient action. No restricted cohomology is used.

F.1 A family with arbitrary central defect

Proposition F.1 (The coupled central-defect family)**.** *Suppose* $1 \le s \le D_{\mathrm{par}}$, $r = s+h$, *and*

$$T_{d,s} = \binom{d+s}{2} - (d-1), \qquad 1 \le h \le T_{d,s}.$$

Set

$$f_s = d + C_d + s(d+1) + \binom{s}{2} - 1.$$

There is an indecomposable family in the branch B_h *giving*

$$B_h \ge \frac{q^{f_s+h(d-1)} \prod_{j=0}^{s-1} (q^{D_{\mathrm{par}}} - q^j) \prod_{j=0}^{h-1} \left(q^{T_{d,s}} - q^j\right)}{(q-1)q^{dr} |\operatorname{GL}_d(q)||\operatorname{GL}_s(q)||\operatorname{GL}_h(q)|}. \tag{197}$$

Consequently the following estimate is uniform in this range:

$$\log_q B_h \ge sn + h\binom{d+s}{2} - \tfrac{1}{2}s^2 - dr - C_d - h^2 - \tfrac{5}{2}s - 3. \tag{198}$$

At $q = 4$*, the corresponding binary quadratic solutions are indecomposable under binary-linear conjugacy.*

Proof. Let $Z = k^h$, $W = K \oplus Z$, and $U = U_0 \oplus V_s$, where $U_0 = k^d$ has basis $e_0, \dots, e_{d-1}$. Choose an ordered independent s-frame $\mathbf{p} = (p_1, \dots, p_s)$ in $\mathcal{P}_d$. On K, the core and added quotient generators act by t_i and p_a, respectively; they kill Z. Take every K-valued ordinary two-cohomology class with normalized core-star coefficient one. The complete Koszul decomposition of Lemma D.1 shows that there are exactly q^{f_s} such classes, retaining all core, mixed, and extra-pair coefficients.

Write $\mathcal{N} \subset \Lambda^2 U$ for the span of all basis pairs except the $d-1$ star pairs $e_0 \wedge e_i$, $1 \le i < d$. Thus $\dim \mathcal{N} = T_{d,s}$. Independently choose an alternating map

$$\gamma : \Lambda^2 U \longrightarrow Z$$

whose restriction to $\mathcal{N}$ is surjective; its star values are arbitrary. It is a Z-valued cocycle because the quotient acts trivially on Z. The combined cocycle defines an abelian Lie extension of U by W.

To check its derived ideal, the specified action already gives $[U, W] = \mathfrak{m}K \subset L'$, as a pure submodule, without first assuming that $W = L'$. Every nonstar K-component in the complete cocycle formula (179) lies in $\mathfrak{m}K$: the nonstar core terms have degree at least two, including the λ-terms of degree $d - 2 \ge 2$; mixed terms have degree at least two; extra-pair terms have degree at least three, apart from their degree-d socle term. Subtracting these pure $\mathfrak{m}K$-components from nonstar brackets gives all values of $\gamma|_{\mathcal{N}}$ as pure central vectors. Its surjectivity supplies all of Z. The central star components may then be subtracted. Their K-components q_i satisfy $q_i \equiv t_i \pmod{\mathfrak{m}K}$, and hence generate all of K by Nakayama. More generally, for any central map γ in this slice,

$$\dim_k L' = w_0 + \operatorname{rank}(\gamma|_{\mathcal{N}}).$$

Indeed, modulo $[U, W] = \mathfrak{m}K$, the star brackets have independent $K/\mathfrak{m}K$-components, whereas every nonstar K-component is zero. The bracket image therefore has dimension $d - 1 + \operatorname{rank}(\gamma|_{\mathcal{N}})$ in that quotient. Thus surjectivity on $\mathcal{N}$ is precisely the condition for $L' = W$, independently of the star values. For our surjective choice, from the third term

onwards the lower central series is $\gamma_j(L) = \mathfrak{m}^{j-2}K$, so its class is $d+1$. The commuting quotient actions square to zero, and the extension is metabelian 3-Engel. Its dimension is $d + s + w_0 + h = d + w_0 + r$.

We bound complete isomorphism fibres rather than assume a free parameter action. The action algebra on W is faithfully A, and the quotient action itself is injective: the degree-one terms detect U_0, and independence of the p_a's detects V_s. The kernel of $U \to \operatorname{rad}(A)/\operatorname{rad}(A)^2$ is intrinsically V_s. Thus an isomorphism between two members has a quotient matrix

$$\begin{pmatrix} B & 0 \\ \alpha & \Delta \end{pmatrix}, \qquad B \in \mathrm{GL}_d(k), \quad \Delta \in \mathrm{GL}_s(k).$$

There are at most $q^{ds}|\mathrm{GL}_d(q)||\mathrm{GL}_s(q)|$ such matrices. For each fixed matrix, the vector implicit-recovery equation (81) fixes at most one target frame. It uses only the faithful algebra A and the radical degree of the added actions, so adding the trivial summand Z does not alter its conclusion.

The full module automorphism group has blocks

$$\operatorname{Aut}_R(K \oplus Z) = \left\{ \begin{pmatrix} u & \tau v \\ \lambda & C \end{pmatrix} : u \in A^\times,\ v \in k^{1\times h},\ \lambda \in \operatorname{Hom}_R(K, Z),\ C \in \mathrm{GL}_h(k) \right\}.$$

Indeed, $\operatorname{Hom}_R(Z, K)$ consists of socle-valued maps, and $\dim \operatorname{Hom}_R(K, Z) = h(d-1)$. Since $\tau \in \mathfrak{m}K$, one has $\lambda(\tau) = 0$. Every automorphism must have $u_0 \neq 0$: on $W/\mathfrak{m}W = (K/\mathfrak{m}K) \oplus Z$, its upper-right block is zero and its upper-left block is $u_0 I_{d-1}$. For a unit u, the Schur complement is just C, since $u^{-1}\tau = u_0^{-1}\tau$ and $\lambda(\tau) = 0$. This proves the displayed block description and gives

$$|\operatorname{Aut}_R(W)| = (q-1)|\mathrm{GL}_h(q)|q^{w_0-1+hd}.$$

The subgroup $\operatorname{diag}(1+a, I_h)$, $a \in \operatorname{rad}(A)$, has order q^{w_0-1} and acts trivially on all of $H^2(U, W)$. To see this, lift $a = \sum_i t_i b_i$ in $\mathfrak{m}R$ and use the ordinary contraction identity

$$\delta\Big(\sum_i b_i \iota_{e_i}\Big) + \Big(\sum_i b_i \iota_{e_i}\Big)\delta = a.$$

On W, positive-degree multiplication is the map $\operatorname{diag}(a, 0)$. Hence each displayed principal unit differs from the identity by a null-homotopic map. The effective module effects on cohomology number at most $(q-1)|\mathrm{GL}_h(q)|q^{hd}$.

For a fixed quotient matrix and target frame, all intertwiners differ by an element of $\operatorname{Aut}_R(W)$, and section changes are two-coboundaries. Each effective module effect gives at most one target cohomology class. The selected K-valued classes and Z-valued surjections inject into $H^2(U, K) \oplus H^2(U, Z)$, so each such class represents at most one member of the chosen family. An effect need not preserve our slice: an image outside it contributes no target. This bounds every complete isomorphism fibre by the denominator in (197). The number of surjections $k^{T_{d,s}} \to k^h$ is the second product in its numerator; the first product counts independent action frames. The arbitrary star values contribute $q^{h(d-1)}$. This proves the exact lower bound.

Every family member is indecomposable. The endomorphism ring $\operatorname{End}_R K = A$ is local, so K is indecomposable and is not isomorphic to the trivial module k. If $L = L_1 \oplus L_2$, then $W = L_1' \oplus L_2'$ as R-modules. By finite-length Krull–Schmidt, at most one derived summand contains the indecomposable factor K; any other nonzero derived summand is a sum of trivial modules. All core and polynomial extra actions kill that summand. Its Lie factor also kills the other derived summand by the direct-sum condition. Faithfulness of L/W then puts the whole factor inside the abelian ideal W, contradicting its nonzero derived ideal. If one derived summand is zero, its Lie factor kills W; faithfulness puts it inside W, which lies

in the other factor, so it is zero. At $q = 4$, suppose the corresponding quadratic solution were conjugate to $f_1 \oplus f_2$ on two nonzero binary subspaces. Its derived ideal, quotient, and their intrinsic scalar actions split accordingly. Choose the bases and normalized-section lifts block by block in Theorem 3.2. The scalar-weight projections, operators P_i, and quadratic refinement Q preserve these blocks, so the resulting auxiliary Lie algebra is the direct sum of their two nonzero auxiliary algebras. This contradicts the Lie indecomposability just proved. A nonzero linear factor likewise gives a nonzero abelian auxiliary factor and is excluded. This proves indecomposability of the quadratic solution; it makes no assertion about arbitrary decompositions of its original polar Lie algebra.

Finally, both products after extracting their leading powers of q are at least $\prod_{j\ge1}(1 - q^{-j}) \ge 2/3$. Use $(q-1) < q$ and $|\mathrm{GL}_j(q)| \le q^{j^2}$. Substituting $D_{\mathrm{par}} = n - d - 3$ and f_s gives (198), with the constant -3 covering the extracted product factors. For $h = 1$, $s = r - 1$, the resulting bound is $\log_q B_1 \ge sn - 3s - d - 4$. Relative to the extra-pair-only family, the larger central cocycle space contributes the factor q^{ds+C_d} at leading order. □

F.2 Uniform branch comparison

For real $\rho \ge 0$ and $0 \le \beta \le \rho$, define

$$U(\rho,\beta) = \rho + \left(\frac{\rho^2}{2} - 1\right)\beta - \frac{\beta^3}{2} = (\rho-\beta)\left(1 + \beta^2 + \frac{\beta(\rho-\beta)}{2}\right),$$
$$L(\rho,\beta) = \rho - \beta + \frac{\beta}{2}(\rho-\beta)^2.$$

These are coarse upper and lower branch comparisons. Appendix G sharpens the upper covering and determines the exact leading count of the whole population.

Lemma F.2 (Uniform normalized comparison exponents)**.** *Fix $M > 0$. Uniformly for integers $0 \le h \le r \le M\sqrt{n}$, putting $\rho = r/\sqrt{n}$, $\beta = h/\sqrt{n}$, one has*

$$\log_q B_h \le n^{3/2}U(\rho,\beta) + O_M(dn + d^2\sqrt{n} + n). \tag{199}$$

If $r/\sqrt{n} \to \alpha$, and $0 < \beta_0 < \alpha$ is fixed, then for $h = \lfloor \beta_0\sqrt{n} \rfloor$,

$$\liminf_{d\to\infty} n^{-3/2} \log_q B_h \ge L(\alpha,\beta_0). \tag{200}$$

Proof. For the upper bound the condition on the action-subspace census in (183) holds eventually, uniformly in these branches, because $2s = O_M(\sqrt{n})$ whereas its right-hand dimension is at least $n - 2 - d$. The leading terms in that exponent are

$$sn + sh^2 + \tfrac{1}{2}hs^2 = n^{3/2}U(\rho,\beta).$$

All remaining terms are uniformly $O_M(dn+d^2\sqrt{n}+n)$, including the logarithm of its prefactor. This error is $o(n^{3/2})$.

For the lower bound, $s = r - h \sim (\alpha - \beta_0)\sqrt{n}$. Thus eventually $s \le D_{\mathrm{par}}$ and $h \le \binom{s}{2}$, so Proposition F.1 applies. In (198), the leading terms are $sn + hs^2/2$. Its other terms are $O_M(dn+d^2\sqrt{n}+n)$; rounding h changes the normalized leading expression by $o(1)$. Continuity of L gives the liminf assertion. No rate of convergence for $r/\sqrt{n}$ is required. □

F.3 Proof of the typical surplus order

Proof of Proposition 9.6. Put $\delta = \alpha^2/2 - 1 > 0$, and choose

$$\beta_0 = \frac{\delta}{2\alpha}, \qquad G = \frac{\delta^2}{4\alpha}.$$

Here $0 < \beta_0 < \alpha/4$, and

$$L(\alpha, \beta_0) - \alpha = \delta\beta_0 - \alpha\beta_0^2 + \tfrac{1}{2}\beta_0^3 \geq G > 0.$$

The lower branch $h = \lfloor \beta_0\sqrt{n} \rfloor$ therefore shows that the total population has size at least $q^{(\alpha+G+o(1))n^{3/2}}$.

Set $\epsilon_h = \delta/(16\alpha)$. For $0 \leq \beta \leq \epsilon_h$,

$$U(\alpha, \beta) \leq \alpha + \delta\epsilon_h = \alpha + G/4.$$

By the uniform upper estimate, continuity, and $\log_q(r+1) = O(d)$, the combined count of all branches with $h \leq \epsilon_h\sqrt{n}$ is at most $q^{(\alpha+G/4+o(1))n^{3/2}}$. Their proportion is at most $q^{-(3G/4-o(1))n^{3/2}}$.

For the other endpoint put $\epsilon_s = \alpha/[4(1+6\alpha^2)]$. Eventually $\rho = r/\sqrt{n} \leq 2\alpha$. If $s \leq \epsilon_s\sqrt{n}$, the factorized form of U, with $\beta \leq 2\alpha$, gives

$$U(\rho, \beta) \leq \epsilon_s(1+6\alpha^2) = \alpha/4.$$

The combined proportion of these branches is consequently at most $q^{-(3\alpha/4+G-o(1))n^{3/2}}$. Adding the two exceptional proportions and absorbing their factor two proves the theorem with $\eta = \min\{G/4, \alpha/4\}$. The constants depend on the fixed supercritical α, and no statement at a converging critical ratio is used in the proof. □

G Socle-visible actions and the exact leading population law

The residual action entropy in the branch upper bound can be coupled to a loss of ordinary two-cohomology. This coupling closes the leading-order comparison of Appendix F for all bounded values of $r/2^{d/2}$, including the supercritical regime. Fix $k = \mathbb{F}_q$ of characteristic two, $q \geq 4$, and retain $R, K, d, n = 2^d, w_0 = n-2$ and $C_j = \binom{j}{2}$. Write B_h for the branch with derived dimension $w_0 + h$ and quotient dimension $d + s$, where $h + s = r$. All cohomology in this appendix is ordinary Chevalley–Eilenberg cohomology of the indicated abelian quotient. The one-direction cohomology calculation is the classical codimension-one exact sequence of [11, Proposition 1].

G.1 Restriction to the core socle

Lemma G.1 (The socle-restriction kernel)**.** *Let W be any normalized bounded-defect core module supplied by Lemma C.2. Put*

$$S = \mathrm{Soc}_R(W), \qquad m = \dim_k S, \qquad t = h + 1 - m.$$

Then $1 \leq m \leq h+1$, and the restriction map $\mathrm{End}_R(W) \to \mathrm{End}_k(S)$ has kernel of dimension at most

$$e_{h,m} = w_0 + h(d-1) + tm. \tag{201}$$

Proof. Use the two exact sequences in (170), written here as

$$0 \longrightarrow F \longrightarrow W_0 \longrightarrow K \longrightarrow 0, \qquad 0 \longrightarrow W_0 \longrightarrow W \longrightarrow N \longrightarrow 0, \qquad \dim F + \dim N = h.$$

Let $a = \dim(S \cap F)$, $b = \dim(S \cap W_0)$, and $\epsilon = b - a$. The image of $S \cap W_0$ in K is annihilated by the maximal ideal of R, hence lies in $\mathrm{Soc}_R K = k\tau$. Thus $\epsilon \in \{0, 1\}$. The quotient by S has the more precise filtration

$$0 \longrightarrow F/(S \cap F) \longrightarrow W_0/(S \cap W_0) \longrightarrow K/k^{\epsilon} \longrightarrow 0, \qquad W/(W_0 + S) \simeq N/\operatorname{im}(S \to N).$$

Here k^ϵ is either zero or the socle $k\tau$ of K. The two small factors have combined length

$$(\dim F - a) + (\dim N - m + b) = h - m + \epsilon = t - 1 + \epsilon.$$

Refine them by composition series. Subadditivity of $\mathrm{Hom}_R(-, W)$, and precomposition with the quotient map $K \longrightarrow K/k^\epsilon$, give

$$\dim \mathrm{Hom}_R(W/S, W) \leq \dim \mathrm{Hom}_R(K, W) + (t - 1 + \epsilon)m.$$

Target subadditivity for the original two extensions gives

$$\dim \mathrm{Hom}_R(K, W) \leq \dim \mathrm{End}_R(K) + h \dim \mathrm{Hom}_R(K, k) = w_0 + h(d - 1).$$

The kernel of restriction is precisely $\mathrm{Hom}_R(W/S, W)$. Since $\epsilon \leq 1$, the result follows. This argument uses K as a subquotient; it does not assume that K embeds in W. □

Lemma G.2 (The action–cohomology loss)**.** *Let the extra quotient $V = k^s$ act on W through commuting square-zero R-endomorphisms $T_1, \dots, T_s$. Denote their restrictions to S by Q_a, and put*

$$\rho = \max_{v \in k^s} \mathrm{rank}\Big(\sum_a v_a Q_a\Big), \qquad A_{d,h,s} = d + C_d + s(d + 1) + C_s + hC_{d+s}.$$

Then $0 \leq \rho \leq \lfloor m/2 \rfloor$, and

$$\dim_k H^2(U_0 \oplus V, W) \leq A_{d,h,s} - (t + \rho)C_s. \tag{202}$$

Proof. Take core cohomology first in the ordinary CE double complex. Its total degree-two term bounds the resulting two-cohomology by

$$\dim H^0(V, H^2(U_0, W)) + \dim H^1(V, H^1(U_0, W)) + \dim H^2(V, S).$$

Lemma C.2 bounds the first two terms by $d + C_d + hC_d$ and $s(d + 1 + hd)$, respectively.

If $s \geq 1$, choose an actual k-rational quotient direction attaining rank ρ on S. On this one-dimensional abelian quotient both degree-zero and degree-one cohomology have dimension $m - \rho$. Taking its cohomology first, then that of a complementary $(s - 1)$-space, gives

$$\dim H^2(V, S) \leq \big(C_{s-1} + s - 1\big)(m - \rho) = C_s(m - \rho).$$

For $s = 0$ this term is zero directly. Summing gives (202). Neither spectral sequence is assumed to collapse. All linear combinations of the Q_a's are square zero, so the stated range of ρ follows. Its definition uses the finite base field, not an algebraic closure. □

G.2 Rank-sensitive action coverings

Lemma G.3 (A common-flag rank covering)**.** *The number of ordered commuting square-zero s-tuples of m-square matrices whose span has maximum rank ρ over k is at most*

$$K_{m,s,\rho} := [m]_q!\, 2^{ms} q^{s\rho(m-\rho)}, \qquad [m]_q! = \prod_{j=1}^{m} \frac{q^j - 1}{q - 1}. \tag{203}$$

For $\rho = 0$ one may replace $K_{m,s,0}$ by one. Moreover

$$\log_q K_{m,s,\rho} \leq s\rho(m - \rho) + \tfrac{1}{2}m(m - 1) + ms \log_q 2 + m \log_q (1 - q^{-1})^{-1}. \tag{204}$$

Proof. Let $a_{m,j}$ count strictly upper triangular square-zero matrices of rank j. The classical rank recurrence is

$$a_{m,j} = q^j a_{m-1,j} + (q^{m-j} - q^{j-1}) a_{m-1,j-1}.$$

Indeed a last column v appended to A must lie in $\ker A$. If $\operatorname{rank} A = j$, the rank remains j for the q^j choices in $\operatorname{im} A$; if $\operatorname{rank} A = j-1$, it increases for the $q^{m-j} - q^{j-1}$ choices outside $\operatorname{im} A$. Induction gives $a_{m,j} \le \binom{m}{j} q^{j(m-j)}$, hence

$$\sum_{j=0}^{\rho} a_{m,j} \le 2^m q^{\rho(m-\rho)}.$$

Commuting nilpotent operators have a common complete flag over k: their common kernel contains a rational line, after which one argues inductively on the quotient. They are strictly triangular relative to that flag. There are $[m]_q!$ complete flags. For each flag, ignoring mutual commutation and counting each generator independently gives (203). This is a covering argument and assumes no free flag action. The estimate $[m]_q! \le q^{m(m-1)/2}(1-q^{-1})^{-m}$ gives (204). □

Classical input. The triangular rank recurrence is Kirillov's, as recorded in [13, p. 1, equation *(Sasha)*]. The common-flag covering and the displayed inequality are elementary consequences. The population bound below couples this classical matrix count to the socle-restriction kernel and the rank-sensitive ordinary-cohomology loss.

Proposition G.4 (A full branch covering)**.** *Put*

$$S_e(s) = \sum_{j=0}^{\min(s,e)} \begin{bmatrix} e \\ j \end{bmatrix}_q, \qquad M_h(d) = (h+1) q^{hC_d + dh^2 + h}.$$

For every $h, s \ge 0$, the full branch satisfies

$$B_h \le M_h(d) q^{A_{d,h,s}} \sum_{m=1}^{h+1} S_{e_{h,m}}(s) \sum_{\rho=0}^{\lfloor m/2 \rfloor} K_{m,s,\rho}\, q^{-(h+1-m+\rho)C_s}. \tag{205}$$

Terms $\rho > 0$ may be omitted when $s = 0$.

Proof. The module census in Lemma C.2 supplies at most $M_h(d)$ representatives. Fix one of them, of socle dimension m, and let E_0 be the kernel of $E = \operatorname{End}_R(W) \to \operatorname{End}_k(S)$. Choose a vector-space section of $E \to E/E_0$. Each action then has a kernel coordinate and a visible socle coordinate.

Normalize the kernel-coordinate map $k^s \to E_0$ by $\mathrm{GL}_s(k)$: for its image choose a prescribed basis, followed by zeros. This supplies $S_{\dim E_0}(s)$ choices, at most $S_{e_{h,m}}(s)$ by Lemma G.1. The same quotient-coordinate change transforms the visible tuple; allowing all such tuples still covers every algebra. For a fixed socle tuple the visible coordinate has a unique lift to the chosen complement of E_0, so no additional factor is required. Lemma G.3 supplies its count.

For each action tuple, ordinary two-cohomology covers every abelian Lie extension up to section changes. Lemma G.2 bounds the number of these classes. The class and derived-generation conditions impose further restrictions and may be dropped in an upper covering. Every unlabelled Lie type admits a normalized presentation, so multiplying these coverings, then summing over m and ρ, proves the result. No orbit size is divided out and no module splitting is assumed. □

G.3 The branch envelope

Define the continuous function

$$\Psi(\beta,\gamma)=\begin{cases}\gamma+\frac{1}{2}\beta\gamma^2, & \beta\le\gamma/2,\\ \gamma+\frac{1}{3}\beta^2\gamma+\frac{1}{6}\beta\gamma^2+\frac{1}{12}\gamma^3, & \beta\ge\gamma/2.\end{cases}\tag{206}$$

Proposition G.5 (Uniform leading branch bound)**.** *For each fixed $A_*>0$, uniformly for integers $h,s\ge 0$ with $h+s\le A_*\sqrt{n}$,*

$$\log_q B_h \le n^{3/2}\Psi(h/\sqrt{n},s/\sqrt{n})+O_{q,A_*}(dn).\tag{207}$$

Proof. In (205), put $\beta=h/\sqrt{n}$, $\gamma=s/\sqrt{n}$, $\zeta=m/\sqrt{n}$, and $\xi=\rho/\sqrt{n}$. The module census costs $O_{A_*}(dn)$. Moreover

$$e_{h,m}=n+(h+1-m)m+O_{A_*}(d\sqrt{n}),$$

so eventually $2s\le e_{h,m}$ uniformly. The Gaussian estimate of Appendix E gives $S_e(s)\le\frac{3}{2}(s+1)q^{s(e-s)}$. The flag and triangular prefactors in (204) cost $O_{q,A_*}(n)$. All remaining terms involving d, the difference $C_s-s^2/2$, and the number of summands contribute $O_{q,A_*}(dn)$. The resulting leading expression is

$$F(\beta,\gamma,\zeta,\xi)=\gamma+\tfrac{1}{2}\beta\gamma^2+\gamma(\beta-\zeta)\zeta-\tfrac{1}{2}(\beta-\zeta+\xi)\gamma^2+\gamma\xi(\zeta-\xi).$$

The true range $0\le\zeta\le\beta+n^{-1/2}$ can be replaced by $0\le\zeta\le\beta$ at a cost $O_{A_*}(n)$, since F has uniformly bounded first derivatives on the relevant compact set. The range of ξ is $0\le\xi\le\zeta/2$.

For $\gamma>0$, complete squares in

$$F/\gamma=1+\zeta(\beta+\gamma/2)-\zeta^2+\xi(\zeta-\gamma/2)-\xi^2.$$

The maximizing ξ is $(\zeta-\gamma/2)_+/2$. If $\beta\le\gamma/2$, the maximizing ζ is β; if $\beta\ge\gamma/2$, it is $2\beta/3+\gamma/6$. The resulting values are exactly (206). For $\gamma=0$ all leading expressions are zero. This proves the uniform estimate. □

G.4 The exact leading law and the typical allocation

Put

$$\varphi(\alpha)=\begin{cases}\alpha, & 0\le\alpha\le\sqrt{2},\\ \dfrac{\alpha^3+9\alpha+(\alpha^2+6)^{3/2}}{27}, & \alpha\ge\sqrt{2},\end{cases}\tag{208}$$

and

$$\beta_*(\alpha)=\begin{cases}0, & 0\le\alpha\le\sqrt{2},\\ \dfrac{2\alpha-\sqrt{\alpha^2+6}}{3}, & \alpha>\sqrt{2}.\end{cases}\tag{209}$$

Theorem G.6 (The full leading population law)**.** *For every fixed $A_*>0$, uniformly for integers $1\le r\le A_*\sqrt{n}$,*

$$\log_q B_{d+1,r}(q)=n^{3/2}\varphi(r/\sqrt{n})+O_{q,A_*}(dn).\tag{210}$$

In particular, if $r/\sqrt{n}\to\alpha>0$, then

$$\frac{\log_q B_{d+1,r}(q)}{n^{3/2}}\longrightarrow\varphi(\alpha).$$

For each $\epsilon > 0$, there is $\delta > 0$ such that along this sequence the proportion of types satisfying

$$\left| \frac{\dim_k L' - w_0}{\sqrt{n}} - \beta_*(\alpha) \right| \geq \epsilon$$

is at most $q^{-\delta n^{3/2}}$ for all sufficiently large d. Thus the allocation of surplus to the derived ideal has a unique limiting value. This assertion concerns scaled defects and makes no claim of exact minimum-derived dominance in a shrinking window.

Proof. Fix $\alpha \geq 0$ and write $\gamma = \alpha - \beta$. On the second branch of Ψ, equivalently $0 \leq \gamma \leq 2\alpha/3$, its value is

$$E_\alpha(\gamma) = \gamma + \alpha^2\gamma/3 - \alpha\gamma^2/2 + \gamma^3/4.$$

It is strictly increasing because

$$E'_\alpha(\gamma) = 1 + \tfrac{3}{4}(\gamma - 2\alpha/3)^2 > 0.$$

Hence its maximum occurs at the boundary with the first branch. On the first branch the value is

$$L_\alpha(\gamma) = \gamma + \tfrac{1}{2}\alpha\gamma^2 - \tfrac{1}{2}\gamma^3, \qquad 2\alpha/3 \leq \gamma \leq \alpha.$$

Its unique maximum is at $\gamma = \alpha$ when $\alpha \leq \sqrt{2}$, and otherwise at $\gamma_* = (\alpha + \sqrt{\alpha^2 + 6})/3$. For $\alpha > \sqrt{2}$ this point is strictly greater than $2\alpha/3$. Substitution gives $\varphi(\alpha)$, and the unique complementary value is $\beta_*(\alpha)$.

Apply Proposition G.5 and sum the $r + 1$ branches. This gives the upper bound in (210) with its stated uniform error.

For the lower bound put $\alpha = r/\sqrt{n}$ and choose $h = \lfloor \beta_*(\alpha)\sqrt{n} \rfloor$. Then $0 \leq h \leq r$. If $h = 0$, Lemma E.2 gives $\log_q B_0 \geq rn - O_{A_*}(dn)$. When this rounding occurs with $\alpha > \sqrt{2}$, replacing the real optimizer by zero changes the leading expression by at most $O_{A_*}(n)$, by the uniform Lipschitz bound for L_α on $0 \leq \alpha \leq A_*$.

If $h \geq 1$, put $s = r - h$. Uniformly in this case, $s \leq A_*\sqrt{n}$, and the supercritical optimizer has $s \geq 2r/3 - O(1)$. For all sufficiently large d, $1 \leq s \leq D_{\mathrm{par}}$ and $h \leq C_{d+s} - (d - 1)$. Thus Proposition F.1 applies and gives

$$\log_q B_h \geq sn + \tfrac{1}{2}hs^2 - O_{q,A_*}(dn).$$

Changing the maximizing real h by at most one costs $O_{A_*}(n)$. This is the required lower bound, uniformly through $\alpha = \sqrt{2}$.

Finally, the maximizer of the continuous compact-domain envelope is unique. For fixed $\alpha > 0$ and $\epsilon > 0$, its value on the closed subset $|\beta - \beta_*(\alpha)| \geq \epsilon$ has a strictly positive gap from $\varphi(\alpha)$, unless that subset is empty. Continuity preserves a smaller fixed gap when $r/\sqrt{n}$ is sufficiently close to α. The uniform branch upper, the total lower, and $\log_q(r+1) = o(n^{3/2})$ then prove the exponential concentration estimate. □

What the argument establishes. For $\alpha > \sqrt{2}$, the typical derived surplus is $\beta_*(\alpha)\sqrt{n} + o(\sqrt{n})$; its leading value is positive. The matching lower is furnished by the full-central family, whereas the upper covers every bounded-defect derived module, including nonsplit modules. Equality of the leading exponents does not assert that the full-central family has density one, or that all typical algebras are indecomposable. These finer properties require information below the $n^{3/2}$ scale.

H The first positive derived defect

H.1 The modules and actions at one derived defect

Put $n = 2^d$, $w_0 = n - 2$, $C_d = \binom{d}{2}$, $D_{\mathrm{par}} = n - d - 3$, and $s = r - 1$, with $d \geq 4$. All fields below have characteristic two and fixed order $q \geq 4$. Use R, K, I from (165) and put $A = R/I$, $J = \mathfrak{m}R$, and $\tau = t_0 \cdots t_{d-1}$.

The third module needed below is denoted E. As a vector space it is $K \oplus kz$. On the monomial basis y_S, $S \cap \{1, \ldots, d-1\} \neq \varnothing$, its action agrees with multiplication in K, except that

$$t_1 y_{\{1\}} = z, \qquad \mathfrak{m}z = 0.$$

In particular, the positive generators $y_i = y_{\{i\}}$, $i > 0$, satisfy $t_1 y_1 = z$, $t_i y_i = 0$ for $i > 1$, and $t_i y_j + t_j y_i = 0$ for distinct positive indices. The t_0-direction is free on the K-part.

Lemma H.1 (Three module types at one defect)**.** *Every normalized core module of derived defect one is isomorphic, after a linear change of normalized core coordinates, to exactly one of*

$$K \oplus k, \qquad J, \qquad E.$$

For the core quotient $U_0 = k^d$, the first and third modules have

$$\dim H^0 = 2, \qquad \dim H^1 = 2d + 1, \qquad \dim H^2 = d + 2C_d.$$

For J, these dimensions are respectively $1, d+1, d+C_d$.

Proof. The two extensions in (170), with $h = 1$, give two cases. If the core derived module is K, the full module is an extension of k by K. Since $\mathrm{Ext}^1_R(k, K)$ has dimension one, it is either split or the unique nonsplit type. The sequence $0 \to K \to J \to k \to 0$ is nonsplit: its socle is $k\tau \subset K$, so no trivial summand lifts the quotient. This gives the first two modules.

Otherwise the core derived module is the whole module, and there is an extension $0 \to kz \to W \to K \to 0$. The minimal presentation of K, obtained by extending the presentation of the augmentation ideal in $k[t_1, \ldots, t_{d-1}]/(t_i^2)$, identifies its extension class with coefficients q_i, q_{ij} in the relations

$$t_i y_i = q_i z, \qquad t_i y_j + t_j y_i = q_{ij} z \quad (1 \leq i < j < d).$$

There are $(d-1) + \binom{d-1}{2} = C_d$ coefficients; the t_0-direction introduces no further relation. The ordinary-cohomology connecting map sends the normalized core-star cocycle to $\sum_{i<j} q_{ij} z\, e_0^* \wedge e_i^* \wedge e_j^*$. All the remaining core cohomology representatives have coefficient degree at least two and contribute zero to this map. Since the star coefficient is nonzero, a core cocycle can lift only if every $q_{ij} = 0$. The diagonal vector (q_i) is either zero, giving $K \oplus k$, or can be normalized to $(1, 0, \ldots, 0)$ by $\mathrm{GL}_{d-1}(k)$ and rescaling z. Here the field is perfect, so the Frobenius twist causes no obstruction. The latter module is E. These types are distinct, as their top/socle dimensions are respectively $(d, 2), (d, 1), (d-1, 2)$.

For E, the connecting maps on $H^0(U_0, K)$, $H^1(U_0, K)$, and $H^2(U_0, K)$ all vanish. The top socle representative lifts to a socle vector. The displayed $H^1(K)$ representatives can be taken to be

$$\zeta_*(e_0) = 0, \quad \zeta_*(e_i) = t_0 t_i \ (i > 0), \qquad \zeta_j(e_i) = \mathbf{1}_{i=j}\tau/t_j \ (0 \leq j < d).$$

The first is the connecting image of the socle of R/K, and the last d classes map to the standard Koszul basis for $H^1(U_0, R)$. This also makes the dimension $d + 1$ and the coefficient-degree bound explicit. All these coefficients have degree at least two; the only additional action in E occurs when t_1 multiplies the degree-one vector y_1. In the basis of Lemma B.3, the

remaining $H^2(K)$ representatives are β_κ, with coefficient degree two, and β_λ, with coefficient degree $d-2 \geq 2$. Consequently the only potentially affected representative is the star, whose connecting value is zero because $q_{ij} = 0$. The long exact sequence therefore gives the three claimed dimensions, also immediate for $K \oplus k$. For J, use $0 \to J \to R \to k \to 0$. The core Koszul complex of R has $\dim H^i = \binom{d}{i}$; its representatives have coefficient degree $d-i$. Thus augmentation induces zero on H^i for $i = 0, 1, 2 < d$, and the same long exact sequence gives the dimensions for J. □

For $K \oplus k$, let N_i send the top generator t_i of K to z, killing $\mathfrak{m}K$ and z, and let $\Theta(z) = \tau$, $\Theta(K) = 0$. For E, define $N_i(y_j) = \mathbf{1}_{i=j} z$, $N_i(\mathfrak{m}E) = 0$, and

$$\Theta(y_1) = \tau/t_1, \qquad \Theta(z) = \tau, \qquad \Theta(y_S) = 0 \quad (S \neq \{1\}).$$

These are R-module maps.

Lemma H.2 (Admissible extra actions)**.** *On $K \oplus k$ or E, every additional action compatible with a generating normalized core cocycle has, after subtracting its core linear part, the form*

$$T_a = p_a + v_a\Theta, \qquad p_a \in \mathcal{P}_d, \quad v_a \in k.$$

On J, every additional action is polynomial, in a space of dimension $D_{\mathrm{par}}+1$, after subtracting its core linear part.

Proof. For the split module the usual block description gives, after nilpotency removes the two constant diagonal blocks, $T = p + \sum_i a_i N_i + v\Theta$. For E, one has

$$\mathrm{End}_R(E) = A \oplus \mathrm{Span}_k\{N_i : 1 \leq i < d\} \oplus k\Theta.$$

For completeness its dimension is $w_0 + d$. In $0 \to k \to E \to K \to 0$, the map $\mathrm{End}(K) \to \mathrm{Ext}^1(K, k)$ sends multiplication by u to $u_0 q$. Consequently $\dim \mathrm{Hom}(K, E) = w_0 + d - 2$. Restriction $\mathrm{End}(E) \to \mathrm{Soc}(E)$ has image dimension two, realized by the identity and Θ. This proves the dimension; the displayed independent maps give all endomorphisms. Its residue scalar is p_0, which is zero for a square-zero action. Subtracting core linear multiplication leaves $p \in \mathcal{P}_d$.

The core/core/extra Jacobi equations require $T_*[\beta_0] = 0$ in $H^2(U_0, W)$. Positive-degree polynomial multiplication induces zero on cohomology. The map Θ also induces zero on core H^2. Indeed its effect on the lifted star for E is $(\tau/t_1)e_0^* \wedge e_1^*$, which is the boundary of the one-cochain having value $\tau/(t_0 t_1)$ on e_1. It kills all higher K-coefficient representatives. Its effect on a central two-form is a τ-valued form, also a boundary: for $i \neq j$, the cochain with value τ/t_i on e_j has precisely that boundary. These observations also cover the split module.

By contrast, N_i sends the normalized star to the central class $ze_0^* \wedge e_i^*$. These classes are independent in $H^2(U_0, W)$: the inclusion from $H^2(U_0, k)$ is injective because the preceding $H^1(K)$ connecting map is zero. All other normalized core representatives are killed by N_i. Jacobi therefore forces every $a_i = 0$. For J, finite duality gives $\mathrm{End}_R(J) = R/(\tau)$; nilpotency removes its constant term, and core linear subtraction leaves dimension $D_{\mathrm{par}} + 1$. □

Lemma H.3 (The cohomology cost of a nonpolynomial action)**.** *For $W = K \oplus k$ or E, polynomial actions in s extra directions give*

$$\dim H^2(U_0 \oplus k^s, W) = d + 2C_d + s(2d+1) + 2\binom{s}{2}.$$

If at least one coefficient v_a is nonzero, the dimension is instead

$$d + 2C_d + s(2d+1) + \binom{s}{2}.$$

Thus the exact loss is $\binom{s}{2}$, not a loss in the first one-direction H^2 calculation.

Proof. Eliminate polynomial actions by the ordinary R-linear Koszul change of cochain generators. Since Θ is R-linear, this leaves the residual actions $v_a\Theta$. If nonzero, a quotient basis change makes them $\Theta, 0, \dots, 0$. On core cohomology, Θ_* has rank one on H^0 and rank zero on H^1, H^2. The H^2 assertion was proved above. On H^1, lifted K-representatives have degree at least two and are killed; central representatives map to τe_i^*, the boundary of τ/t_i. On H^0, it sends z to τ. The classical codimension-one exact sequence [11, Proposition 1], or its mapping cone, consequently gives

$$H^0 : 1, \qquad H^1 : (2d+1)+1, \qquad H^2 : (d+2C_d)+(2d+1)$$

as its dimensions. Tensoring with the exterior cochains in the remaining $s-1$ zero directions gives the second formula; the all-zero residual action gives the first. □

H.2 Polynomial base orbits and the one-defect count

For E, every identity-linear substitution $\sigma(t_i) = t_i + u_i$, $u_i \in \mathfrak{m}^2 R$, preserving K, I, admits a semilinear lift to E. Choose polynomial lifts $\widetilde{a} \in E$ for $a \in K$. For $a, b \in K$, the defining relations give

$$a\widetilde{b} + b\widetilde{a} = 0, \qquad a\widetilde{a} = q(a_{[1]})z,$$

where q is the diagonal Frobenius-linear form. The map $y_i \mapsto \widetilde{\sigma(t_i)}$, $z \mapsto z$, extended σ-semilinearly, respects all presentation relations because σ has identity linear part. It induces $\sigma|_K$ on the quotient and the identity on the kernel, so is invertible. The split module has the analogous lift. Both modules have annihilator precisely I: the quotient K forces the annihilator to lie in I, and the explicit action shows that I kills E (and also $K \oplus k$). Thus all of these statements depend only on the classes of the polynomials modulo I, as required by the base action.

It follows that the base group

$$G_0 = k^{ds} \rtimes \mathrm{GL}_s(k)$$

acts by isomorphisms between the corresponding polynomial-action presentation families. A genuine action on a chosen cohomology slice is unnecessary for the following upper covering: choose one base in each base orbit, transport all presentations to it, then normalize their core-star coefficient and central star values. The latter $d-1$ values are uniquely removed by $1 + \sum_i a_i N_i$, which commutes with polynomial actions. Each resulting fiber has at most

$$q^{F_s}, \qquad F_s = 2C_d + s(2d+1) + 2\binom{s}{2}$$

representatives. No orbit-size division on that fiber is used. The uniform base stabilizer-sum bounds proved for Theorem 9.4 apply to the subgroup G_0. Hence independent polynomial frames have

$$(1+o(1))\frac{\prod_{j<s}(q^{D_{\mathrm{par}}} - q^j)}{q^{ds}|\,\mathrm{GL}_s(q)|}$$

base orbits, uniformly in $2d \le s \le D_{\mathrm{par}}/(4d)$. The two polynomial module types therefore contribute at most

$$q^{sn-3s+2C_d+O_q(1)} \tag{211}$$

classes in that range.

The other presentations have smaller covering bounds. Action-subspace normalization in the $(D_{\mathrm{par}}+1)$-dimensional space $\mathcal{P}_d + k\Theta$, combined with Lemma H.3 and the same star normalizations, gives at most

$$\tfrac{3}{2}(s+1)q^{sn+ds-s^2/2-3s/2+2C_d}$$

classes with a nonpolynomial action, for each of the two module types. Here the star normalization is valid even though it need not fix the individual action tuple. If $N = \sum a_i N_i$, then $N^2 = 0$, $N\Theta = 0$, and $\Theta N = \sum a_i M_{\tau/t_i}$. Conjugation by $1 + N$ therefore sends

$$p + v\Theta \longmapsto p + v \sum_i a_i M_{\tau/t_i} + v\Theta.$$

It preserves the polynomial-plus-Θ space and the coefficient v, because $\tau/t_i \in \mathcal{P}_d$. One first normalizes the core star and its central star values, and then chooses a basis for the resulting action subspace. This covers every type by a normalized fiber of dimension $2C_d+s(2d+1)+\binom{s}{2}$; no stabilizer assumption is used here. For J, using all core and extra cohomology without a generation restriction gives the upper bound

$$\tfrac{3}{2}(s+1)q^{sn-s^2/2-3s/2+d+C_d}.$$

Rank-deficient polynomial tuples are bounded by $q^{F_s}\sum_{j<s}\begin{bmatrix} D_{\mathrm{par}} \\ j \end{bmatrix}_q$; relative to the faithful polynomial main covering, their ratio is $q^{-D_{\mathrm{par}}+ds+2s+O_q(\log s+d^2)}$. Indeed every tuple of rank $j < s$ can first be put, by an extra quotient basis change, into a tuple consisting of a basis of its j-dimensional polynomial action space and $s - j$ zero directions. All s quotient directions and the complete fiber are retained; a choice of a Gaussian action subspace covers every such presentation. Thus the independent-frame count and this rank-deficient covering together exhaust the polynomial cases.

Theorem H.4 (One-defect enumeration at the transition scale)**.** *Fix $0 < a < b < \infty$. Uniformly for integers $a\sqrt{n} \le r \le b\sqrt{n}$, as $d \to \infty$, with $s = r - 1$,*

$$\log_q B_1 = sn - 3s + O_{q,a,b}(d^2), \qquad \log_q(B_1/B_0) = -n + \binom{r+d}{2} + O_{q,a,b}(d^2).$$

The nonpolynomial, J-module, and rank-deficient presentations have total type proportion $O_{q,a,b}(q^{-\kappa n})$, for some $\kappa = \kappa(a,b) > 0$. Almost all one-defect types consequently admit a faithful polynomial presentation with module $K \oplus k$ or E.

Proof. The full-central-cocycle family in Proposition F.1, with $h = 1$, gives $\log_q B_1 \ge sn - 3s - d - 4$. For all sufficiently large d, the entire indicated range lies in $2d \le s \le D_{\mathrm{par}}/(4d)$. Thus (211) supplies a uniform upper bound $sn - 3s + 2C_d + O_q(1)$ for the faithful polynomial cases. Relative to the displayed lower bound, the logarithm of the nonpolynomial covering is at most

$$-\tfrac{1}{2}s^2 + ds + \tfrac{3}{2}s + 2C_d + d + 4 + \log_q\big(\tfrac{3}{2}(s+1)\big),$$

and the J covering has logarithmic ratio at most

$$-\tfrac{1}{2}s^2 + \tfrac{3}{2}s + 2d + C_d + 4 + \log_q\big(\tfrac{3}{2}(s+1)\big).$$

Since $s \ge a\sqrt{n}/2$ eventually, each is at most $-\kappa n$ for a fixed positive κ, uniformly in the range. The rank-deficient covering has logarithmic ratio $-D_{\mathrm{par}} + ds + 2s + O_q(d^2 + \log s) \le -n/2$ eventually, also uniformly. Summing these coverings proves the first formula and the assertion about the exceptional presentations.

Finally Theorem 9.4, including the Gaussian and general-linear product estimates, gives uniformly

$$\log_q B_0 = rn - dr - \tfrac{1}{2}r^2 - \tfrac{5}{2}r + C_d - (d-1)^2 - \log_q(q-1) + O_q(1).$$

Subtract this from the first formula, use $s = r - 1$, and absorb the remaining terms of order d^2. The result is the stated formula for B_1/B_0. □

In particular, the comparison changes sign within $O_{q,a,b}(d^2/\sqrt{n})$ of $r = \sqrt{2n} - d + \frac{1}{2}$. For every fixed $\varepsilon > 0$, the ratio B_1/B_0 tends to zero on the side $r \le \sqrt{2n} - d + \frac{1}{2} - \varepsilon$, and tends to infinity on the side $r \ge \sqrt{2n} - d + \frac{1}{2} + \varepsilon$, within the range of the theorem. This compares the first positive-defect branch with the minimum-derived branch. It does not by itself control defects $h \ge 2$ or establish a whole-layer lower-side threshold.

I The discrete critical transition

Fix a finite characteristic-two field $k = \mathbb{F}_q$, $q \ge 4$. Retain R, K, I from (165), and put

$$n = 2^d, \quad w_0 = n - 2, \quad D_{\mathrm{par}} = n - d - 3, \quad C_j = \binom{j}{2}, \quad A = R/I, \quad \tau = t_0 \cdots t_{d-1}, \quad d \ge 4.$$

The branch B_h has derived dimension $w_0 + h$, quotient dimension $d + s$, and $h + s = r$. All cohomology is ordinary Chevalley–Eilenberg cohomology. Write

$$T = \sqrt{2n}, \quad R_d^* = T - d + \tfrac{1}{2}, \quad x_{d,r} = r - R_d^*, \quad E_{d,r} = -n + C_{d+r}, \quad z'(L) = \dim_k(Z(L) \cap L').$$

I.1 Maximum core socle and polynomial rigidity

Proposition I.1 (The modules at maximum core socle)**.** *Every normalized core module W with $\dim W = w_0 + h$ and $\dim \operatorname{Soc}_R W = h + 1$ is, after a linear core-coordinate change, a module $E_Q = K \oplus Z$, $Z = k^h$, as follows. The core kills Z and acts by usual monomial multiplication on K, except that*

$$t_i y_i = q_i \in Z \quad (1 \le i < d), \qquad t_i y_j + t_j y_i = 0 \quad (i \ne j),$$

where y_i lifts t_i. The map $Q : k^{d-1} \to Z$, $Q(e_i) = q_i$, is classified by its rank $0 \le u \le \min(h, d-1)$. Moreover $\operatorname{Ann}_R E_Q = I$.

Proof. In the two extensions (170), $0 \to F \to W_0 \to K \to 0$ and $0 \to W_0 \to W \to N \to 0$, one has

$$\dim \operatorname{Soc} W \le \dim \operatorname{Soc} W_0 + \dim \operatorname{Soc} N \le \dim \operatorname{Soc} F + 1 + \dim \operatorname{Soc} N \le h + 1.$$

Equality forces F, N to be trivial and $\operatorname{Soc} W \to N$ to be surjective. A section into the socle splits the second extension, leaving $0 \to Z \to W \to K \to 0$. The minimal presentation of K has relations $t_i y_i$ and $t_i y_j + t_j y_i$; the t_0-factor is free. Thus this extension has diagonal values q_i and mixed values q_{ij}. The core-star connecting class is $\sum_{i<j} q_{ij} e_0^* \wedge e_i^* \wedge e_j^*$; all other core representatives have coefficient degree at least two and map to zero. The nonzero generating star therefore forces $q_{ij} = 0$. A positive core change M transforms the diagonal map by $M^{[2]}$; Frobenius is bijective over k, so row and column changes leave only its rank. Distinct ranks have distinct top dimensions $d - 1 + h - u$. Finally $E_Q/Z = K$ forces its annihilator into I, and the displayed monomial action shows that I kills E_Q. □

Lemma I.2 (Low core cohomology)**.** *The connecting maps from $H^j(U_0, K)$ to $H^{j+1}(U_0, Z)$ vanish for $j = 0, 1, 2$. Consequently*

$$\dim H^0(U_0, E_Q) = h + 1, \quad \dim H^1(U_0, E_Q) = d + 1 + hd, \quad \dim H^2(U_0, E_Q) = d + C_d + hC_d,$$

and $H^2(U_0, Z) \to H^2(U_0, E_Q)$ is injective.

Proof. The socle vector τ lifts. Representatives for $H^1(U_0, K)$ are $\zeta_*(e_i) = t_0 t_i$ for $i > 0$, $\zeta_*(e_0) = 0$, and $\zeta_j(e_i) = \mathbf{1}_{i=j}\tau/t_j$. Their coefficients have degree at least two, as do the κ- and λ-representatives in Lemma B.3. These lift unchanged, while the star lifts because the mixed relations vanish. The long exact sequence proves the assertions; no higher-degree connecting map is being used. □

Proposition I.3 (Polynomial actions and the normalized fiber)**.** *An R-endomorphism of E_Q that kills its socle and satisfies $T_*[\beta] = 0$ for a generating core cocycle is multiplication by a unique element of* $\operatorname{rad} A$*. Subtracting core linear parts therefore puts every extra action in $\mathcal{P}_d = \operatorname{rad}(A)^2$, of dimension D_{par}. For s such actions, put*

$$A_{d,h,s} = d + C_d + s(d+1) + C_s + hC_{d+s}, \qquad F_{d,h,s} = A_{d,h,s} - 1 - h(d-1).$$

The full ordinary two-cohomology has dimension $A_{d,h,s}$, and normalized generating presentations are covered, up to module isomorphism, by

$$q^{F_{d,h,s}} \tag{212}$$

fiber representatives.

Proof. Since T kills Z, it induces multiplication by $p \in A$ on K; killing τ forces $p_0 = 0$. Subtract that actual polynomial endomorphism. The difference is $N_\Lambda(y_i) = \Lambda_i \in Z$, killing Z and nonsingleton monomials. Positive multiplication is null-homotopic on the core complex, whereas

$$(N_\Lambda)_*[\beta] = \sum_i \Lambda_i e_0^* \wedge e_i^*.$$

These classes are independent by Lemma I.2, so compatibility forces $N_\Lambda = 0$. Faithfulness on the quotient K gives uniqueness.

Write $p_a = \sum_i t_i a_{ia}$, $a_{ia} \in \mathfrak{m}R$. The triangular R-linear change of exterior cochain generators eliminates the polynomial actions, giving

$$H^2(U_0 \oplus k^s, E_Q) \simeq H^2(U_0, E_Q) \oplus \big(H^1(U_0, E_Q) \otimes (k^s)^*\big) \oplus \big(\operatorname{Soc}_R(E_Q) \otimes \Lambda^2(k^s)^*\big).$$

This gives $A_{d,h,s}$. Scalar multiplication normalizes the nonzero star coefficient to one. The commuting automorphisms $1 + N_\Lambda$ then translate its $h(d-1)$ central star coordinates independently. Their action on this value-one fiber is free, since restriction to core cohomology detects every nonzero Λ. Its orbit representatives have the cardinality (212). No further stabilizer is divided out. □

I.2 Transport over polynomial base orbits

Lemma I.4 (The linear core stabilizer and base covering)**.** *For a fixed rank-u representative Q, put*

$$H_Q = \{M \in \mathrm{GL}_{d-1}(k) : M^{[2]} \ker Q = \ker Q\}.$$

Uniformly for $2d \le s \le D_{\mathrm{par}}/(4d)$, its faithful polynomial presentations have at most

$$(1 + o(1))\frac{q^{F_{d,h,s}} \prod_{j<s}(q^{D_{\mathrm{par}}} - q^j)}{q^{ds} |\mathrm{GL}_s(k)||H_Q|} \tag{213}$$

types, with error independent of h, Q. Moreover

$$|H_Q| = q^{u(d-1-u)} |\mathrm{GL}_u(k)||\mathrm{GL}_{d-1-u}(k)|, \quad \log_q |H_Q| = (d-1)^2 - u(d-1-u) + O_q(1). \tag{214}$$

Writing $V_h = sn - 3s + (h-1)C_{d+s}$, the logarithm of (213) *is at most*

$$V_h + (2-h)(d-1) + u(d-1-u) + O_q(1). \tag{215}$$

After summing the module ranks,

$$\log_q B_h^{\mathrm{pol}} \le V_h + 2(d-1) + \log_q(\min(h, d-1)+1) + O_q(1). \tag{216}$$

The same bound includes rank-deficient polynomial tuples when $h \le d+O(1)$ and $r = T+O(d)$.

Proof. An identity-linear substitution $\sigma(t_i) = t_i + u_i$, $u_i \in \mathfrak{m}^2 R$, preserves K, I: for $z = t_1 \cdots t_{d-1}$, its only surviving increment terms give $\sigma(z) = z + a\tau$, $\sigma(\tau) = \tau$. For polynomial lifts in E_Q, the defining relations give $a\widetilde{b} + b\widetilde{a} = 0$ and $a\widetilde{a} = Q(a^{[2]}_{[1]})$. Thus $y_i \mapsto \widetilde{\sigma(t_i)}$, fixing Z, is a semilinear automorphism. For $M \in H_Q$, an invertible $C : Z \to Z$ with $CQ = QM^{[2]}$ exists: define it first on $\operatorname{im} Q$, then extend over a complement. The maps $y_i \mapsto \sum_j M_{ji} y_j$, $Z \mapsto CZ$ lift the linear substitution $\operatorname{diag}(1, M)$.

Hence the genuine polynomial base action restricts to $G_Q = k^{ds} \rtimes (H_Q \times \mathrm{GL}_s(k))$. Transport to one base per orbit and apply (212). This needs neither canonical semilinear lifts nor an action on a chosen fiber slice. The whole-base stabilizer sum in Theorem 9.4 bounds the subgroup sum term by term, uniformly in h, Q. Burnside gives (213). Frobenius identifies H_Q with the parabolic fixing a kernel of dimension $d-1-u$, proving (214). Taking logarithms gives (215); use $(2-h)(d-1) + u(d-1-u) \le 2(d-1)$ and sum the ranks.

For deficient frames use the full fiber times $\sum_{j<s} \begin{bmatrix} D_{\mathrm{par}} \\ j \end{bmatrix}_q$. Relative to the faithful covering, including its linear core denominator, this is at most

$$q^{-D_{\mathrm{par}} + ds + 2s + (d-1)^2 + O_q(\log(s+1))} = q^{-n+o(n)}.$$

This holds uniformly in the stated critical ranges and proves the last assertion without an orbit-size assumption on deficient frames. □

The principal product formula will be used in the form

$$\log_q B_0 = rn - 3r - C_{d+r} + d - 1 - \log_q(q-1) + O_q(1). \tag{217}$$

Corollary I.5 (The refined first-defect comparison)**.** *Uniformly for $r/\sqrt{n}$ in a fixed compact positive interval, with $s = r-1$,*

$$\log_q B_1 = sn - 3s + O_q(d), \tag{218}$$

$$\log_q(B_1/B_0) = E_{d,r} + O_q(d). \tag{219}$$

More precisely, $E_{d,r} - 2d + O_q(1) \le \log_q(B_1/B_0) \le E_{d,r} + d + O_q(1)$. The compact interval may affect the sufficiently large lower bound on d, but not these bounded-error constants.

Proof. For $h = 1$, the two module ranks in (215) give constants $d-1$ and $2d-3$. The exceptional one-defect coverings in Appendix H.2 are exponentially smaller on this scale. The coupled lower bound and the polynomial upper therefore give $sn - 3s - d - 4 \le \log_q B_1 \le sn - 3s + 2d - 3 + O_q(1)$. Subtract (217) to obtain all assertions. □

I.3 Uniform control of the other module and action strata

For a normalized core write $m = \dim \operatorname{Soc}_R W$, $t = h + 1 - m$, and let ρ be the maximum rank of its visible extra action on that socle. Since

$$Z(L) \cap L' = \operatorname{Soc}_R W \cap \bigcap_a \ker Q_a,$$

the condition $z'(L) < h+1$ is equivalent to $t > 0$ or $\rho > 0$, independently of the chosen normalized core. Denote these types by B_h^{bad}. Put

$$U = hC_d + dh^2 + h + A_{d,h,s} + s\big(w_0 + h(d-1) - s\big).$$

For $2s \le w_0$, $h \le s/4$, and $s \ge 40$, the full branch covering (205) implies

$$\log_q B_h \le U + O_q(\log(r+2)), \tag{220}$$

$$\log_q B_h^{\mathrm{bad}} \le U - s^2/16 + O_q(\log(r+2)). \tag{221}$$

Indeed use $S_e(s) \le \frac{3}{2}(s+1)q^{s(e-s)}$ with the upper ambient dimension $e = e_{h,m} \ge w_0$. Relative to $m = h+1, \rho = 0$, the exponent changes by

$$t(sm - C_s) + \rho\big(s(m-\rho) - C_s\big).$$

For $t \ge 1$, the first term is at most $-t(s^2/4 - 3s/2)$. For $\rho = 0$ there is no flag factor. For $\rho \ge 1$, the second term is at most $-s^2/4 + s/2$, and the common-flag prefactor has logarithm at most

$$\tfrac{1}{2}m(m-1) + \tfrac{1}{2}ms + \tfrac{1}{4}m \le 5s^2/32 + 11s/16 + 1/4.$$

Here $\log_q 2 \le 1/2$ and $\log_q(1-q^{-1})^{-1} \le 1/4$. The combined loss is at least $s^2/16$ for $s \ge 40$. All census and summation prefactors cost only $O_q(\log(r+2))$, proving both estimates.

Subtracting (217) gives

$$\log_q(B_h/B_0) \le \Delta_{d,n}(r,h) + O_q(\log(r+2)), \tag{222}$$

where the exact polynomial is

$$\begin{aligned}\Delta_{d,n}(r,h) = {} & h^3/2 - (r+d-1)h^2 \\ & + h\big[-n + r^2/2 + (2d-1/2)r + d^2 - 2d + 5/2\big] \\ & + r(2d+1) + d^2 - d + 1.\end{aligned}$$

Lemma I.6 (Uniform critical cutoffs)**.** *Fix $A_0 \ge 1/2$ and restrict to $r = T - d + a$, $|a| \le A_0$. There are positive constants κ, depending only on the fixed parameters, such that*

$$\sum_{h \le r/5} B_h^{\mathrm{bad}}/B_0 \le q^{-\kappa n}, \tag{223}$$

$$\sum_{h \ge r/5} B_h/B_0 \le q^{-\kappa n^{3/2}}, \tag{224}$$

$$\sum_{h \ge d + \lceil A_0 + 3/2 \rceil} B_h/B_0 \le q^{-\kappa T}. \tag{225}$$

On the one-sided range $-A_0 \le a \le 1/2$, the last cutoff can be replaced by $d+2$.

Proof. For $h \le r/5$, one has $s \ge 4r/5$, and $h^3/2 - rh^2 \le -9rh^2/10$. Completing a square in the remaining $O_{A_0}(dr)h + O_{A_0}(dr)$ terms gives $\Delta \le O_{A_0}(d^2T) = o(n)$. Thus (221) proves the first estimate. For $h \ge r/5$, the scaled defect stays a fixed distance from zero. The unique envelope maximizer at $r/\sqrt{n} = \sqrt{2}$, its uniform $O(dn)$ error, and the principal lower bound prove the second estimate. This use of the continuous envelope is confined to a fixed macroscopic gap.

For the third estimate substitute $r = T - d + a$ in Δ. Its coefficient of T is $-h^2 + (d + a - 1/2)h + 2d + 1$. At $h = d + k$, this equals $(a + 3/2 - k)d - k^2 + (a - 1/2)k + 1$. Take $k = \lceil A_0 + 3/2 \rceil$. This is negative eventually; if its coefficient of d vanishes, its constant is $1 - 2k \le -3$. The remaining terms are $O_{A_0}(d^3) = o(T)$. The derivative of Δ in h is negative uniformly for $d + k \le h \le r/5$: its terms are $-2Th + 3h^2/2 + (d + O_{A_0}(1))T + O_{A_0}(dh + d^2)$, and $h \le T/4$ eventually. Hence this entire interval has $\Delta \le -\Omega_{A_0}(T)$. Summing and adding the macroscopic tail proves the claim. For $a \le 1/2$, take $k = 2$; the coefficient becomes $(a - 1/2)d + 2a - 4 \le -3$. □

I.4 The discrete critical profile

Define $f_x(h) = hx - h(h-1)$ for integers $h \geq 0$, and $M(x) = \max_{h\geq 0} f_x(h)$.

Lemma I.7 (Fixed-defect logarithms)**.** *For fixed $h \geq 1$ and $A_* > 0$, uniformly for $|x_{d,r}| \leq A_*$,*

$$\log_q B_h = V_h + O_{q,h,A_*}(d). \tag{226}$$

Moreover

$$\log_q(B_h/B_0) = \mathcal{D}_h(r) + O_{q,h,A_*}(d), \tag{227}$$

$$= Tf_{x_{d,r}}(h) + O_{q,h,A_*}(d), \tag{228}$$

where $\mathcal{D}_h(r) = hE_{d,r} - h(h-1)(r+d-(h+1)/2) + 3h$. For $h = 0$, the branch comparison is identically zero.

Proof. The coupled lower (198) simplifies exactly to

$$\log_q B_h \geq V_h - dh - h^2 - 3. \tag{229}$$

For fixed h its frame and central-surjectivity conditions hold eventually, since $s = T + O(d)$ and $C_{d+s} - (d-1) \gg h$. The polynomial upper (216) differs from V_h by $O_q(d + \log(h+1))$. The bad count is $q^{-\Omega(n)}B_0$, smaller than this fixed-defect lower family. This proves (226). Use $C_{d+r-h} = C_{d+r} - h(d+r) + h(h+1)/2$ and (217) to obtain $\mathcal{D}_h$. Finally $r + d = T + 1/2 + x$ and $E_{d,r} = Tx + x^2/2 - 1/8$, giving (228). □

Theorem I.8 (The critical logarithmic profile)**.** *For each fixed $A_* > 0$, uniformly for integer r with $|x_{d,r}| \leq A_*$,*

$$\log_q(B_{d+1,r}(q)/B_0) = TM(x_{d,r}) + O_{q,A_*}(d). \tag{230}$$

The profile is $M(x) = 0$ for $x \leq 0$, and $M(x) = jx - j(j-1)$ on $2j-2 \leq x \leq 2j$, $j \geq 1$.

Proof. Lemma I.6 discards $h \geq H_d = d + \lceil A_* + 2 \rceil$ and the bad derived-center stratum. For the remaining polynomial families, (216) gives uniformly

$$\log_q(B_h^{\mathrm{pol}}/B_0) \leq Tf_x(h) + O_{q,A_*}(h^3 + d + \log(h+1)). \tag{231}$$

Choose a fixed integer $J_{A_*} \geq 2A_* + 4$. For $h \geq J_{A_*}$, $f_x(h) \leq -h^2/2$. Since $H_d = O_{A_*}(d)$, the last upper bound is at most $-Th^2/4$ on $J_{A_*} \leq h < H_d$, eventually. Thus

$$\sum_{h \geq J_{A_*}} B_h/B_0 \leq q^{-\kappa_{q,A_*}T}. \tag{232}$$

Every maximizing integer lies below J_{A_*}. For these finitely many branches apply Lemma I.7, then take the maximum and sum. The difference $f_x(h+1) - f_x(h) = x - 2h$ gives the stated piecewise-linear profile. □

Theorem I.9 (The derived-defect staircase)**.** *Let an integer sequence satisfy $x_{d,r} \to x \in \mathbb{R}$. For a uniformly chosen type in the complete dimension layer:*

(i) *if $x < 0$, then $h = 0$ except for proportion $q^{-\kappa T}$;*

(ii) *if $2j - 2 < x < 2j$, $j \geq 1$, then $h = j$ except for proportion $q^{-\kappa T}$;*

(iii) *if $x = 2j$, $j \geq 0$, then $h \in \{j, j+1\}$ except for proportion $q^{-\kappa T}$.*

Here $\kappa > 0$ may depend on the fixed limit and field. At a tie limit,

$$\log_q(B_{j+1}/B_j) = T(x_{d,r} - 2j) + O_{q,j}(d). \tag{233}$$

Consequently $(x_{d,r} - 2j)T/d \to -\infty$ selects $h = j$, and convergence to $+\infty$ selects $h = j + 1$, each with probability tending to one. The weights inside the remaining $O_{q,j}(d/T)$ tie neighborhoods are not determined.

Proof. Choose $A_* > |x| + 1$ and use the finite cutoff J_{A_*} above. At a nontie limit, one integer maximizer has a fixed positive score gap over all other indices below J_{A_*}. At $x = 2j$, exactly $j, j + 1$ maximize, and every other score is lower by at least two. Continuity, the fixed-defect estimates, $d = o(T)$, and (232) turn these gaps into the stated exponential proportions. The maximizing score is nonnegative, so the tail estimate relative to B_0 remains negligible relative to the full count. Subtracting the two adjacent fixed-defect logarithms proves (233); its sign under the variable-distance conditions proves the final assertions. □

I.5 Shrinking localization of minimum-derived dominance

Theorem I.10 (Two-branch support through the lower splice)**.** *Uniformly for integers*

$$T - 3d - 3 \le r \le R_d^* + 1/4, \tag{234}$$

there is $\kappa_q > 0$ such that

$$\sum_{h \ge 2} B_h/B_0 \le q^{-\kappa_q T}. \tag{235}$$

In particular, for $|r - R_d^*| \le 1/4$, $\pi_{d,r} = (1 + B_1/B_0)^{-1}(1 + O_q(q^{-\kappa_q T}))$. The proportion with $z'(L) < h + 1$ is at most $q^{-\kappa_q n}$.

Proof. For the bounded upper end of this strip, the preceding estimates already discard the bad stratum, the macroscopic tail, and $h \ge d + 3$. To extend them to its lower end, use the exact upper polynomials rather than any monotonicity of π:

$$\partial_r \Delta = h(r - h + 2d - 1/2) + 2d + 1 > 0, \qquad \partial_r \mathcal{D}_h = h(r + d + 1/2 - h) > 0 \quad (1 \le h \le r).$$

For $h \le r/5$, evaluate Δ at $\bar{r} = T - d + 3/4$; its raw maximum is $O(d^2 T)$, whereas the bad penalty is $\Omega(n)$. At $h \ge d + 3$, the same evaluation gives the raw cutoff $-\Omega(T)$. For $h \ge r/5$ the uniform macroscopic envelope gap applies throughout the strip. Finally for $2 \le h \le d+2$, evaluate $\mathcal{D}_h$ at $\bar{r}$ in the strong polynomial upper. Its value is $Th(5/4 - h) + O(h^3 + d + \log(h + 1)) \le -Th^2/4$, eventually. All uses of base enumeration have $s = T + O(d)$, uniformly within $2d \le s \le D_{\mathrm{par}}/(4d)$. Summing proves both assertions, and the formula for π follows from the exact branch decomposition. □

Corollary I.11 (A shrinking outer window)**.** *For every integer sequence $r = r(d) \ge 1$,*

$$r \le R_d^* - 4d/T \text{ eventually} \implies \pi_{d,r} \to 1, \qquad r \ge R_d^* + 4d/T \text{ eventually} \implies \pi_{d,r} \to 0.$$

The outer window has width $8d/T = o(1)$. Its boundary density is not specified by these bounds.

Proof. For $\delta = r - R_d^*$, $E_{d,r} = T\delta + \delta^2/2 - 1/8$. The precise bounds in Corollary I.5 give, at the two stated endpoints, respectively $\log_q(B_1/B_0) \le -3d + O_q(1)$ and $\log_q(B_1/B_0) \ge 2d + O_q(1)$. On left-side indices with $r \le T - 3d - 3$, use the old outer localization. On the remaining left indices use Theorem I.10 and the increasing polynomial $E_{d,r}$ to obtain $B_1/B_0 \to 0$. On right-side indices with $r \ge T - d + 1$, use the old upper localization. On the remaining right indices, $r = T - d + O(1)$; the same increasing polynomial and $\pi \le B_0/B_1$ suffice. This is a splice of uniform estimates, with no monotonicity assumption on π. □

Corollary I.12 (The odd-d integer lattice)**.** *Along odd d, put $r = T - d + \ell$ for a fixed integer ℓ. Almost every type has $h = \max\{0, \lceil \ell/2 \rceil\}$, with exceptional proportion $q^{-\kappa_\ell T}$. For arbitrary integer sequences on this subsequence, $r \le T - d$ implies $\pi_{d,r} \to 1$, while $r \ge T - d + 1$ implies $\pi_{d,r} \to 0$.*

Proof. Here $T = 2^{(d+1)/2}$ is integral and $x_{d,r} = \ell - 1/2$, which never equals a tie value. Apply Theorem I.9. For the final assertions every integer lies at least $1/2$ from R_d^*, whereas $d/T \to 0$; apply Corollary I.11. □

J Proofs for central reconstruction

We prove the refinement and morphism assertions of Section 13. All functions have zero constant term. Cochain inputs and polynomial degrees are over $\mathbb{F}_2$; coefficient scalars are over $\mathbb{F}_4$.

J.1 Refinement identities and the complete criterion

Lemma J.1 (Norm and defect identities)**.** *For the operators in* (117)*,*

$$T^3 = \mathrm{id}, \quad N\mathcal{L} = 0, \quad \Delta_g \delta = N, \quad N\Delta_g = \omega N, \quad \Delta_g^2 = \Delta_g + N. \tag{236}$$

For any quadratic q, with $\alpha = \mathrm{pol}\, q$,

$$Nq(x) = \alpha(x, g(x)). \tag{237}$$

The third additive difference at zero of this function is

$$\alpha(a, B(b,c)) + \alpha(b, B(a,c)) + \alpha(c, B(a,b)) = d_{\mathrm{CE}}\alpha(a,b,c). \tag{238}$$

In particular Nq is quadratic whenever α is a CE cocycle.

Proof. The relation $g^2 x = gx + x$ gives $g^3 = \mathrm{id}$ and $T^2 t = Tt + t$ for linear t, proving the first two identities. Multiplying the polynomials in T, using $\omega^2 + \omega = 1$, proves the remaining three. Polarization gives $q(x) + q(gx) + q(x + gx) = \alpha(x, gx)$, proving (237).

The right side has degree at most three. In its third difference, terms involving the linear part of g cancel, while its quadratic part contributes exactly the three terms in (238). Vanishing of this difference removes all degree-three Boolean ANF coefficients. This proves the last assertion. □

Lemma J.2 (The quadratic defect is a CE cocycle)**.** *Let $[\alpha] \in K_g$, choose a quadratic refinement q, and put $p = \Delta_g q$. Then p is quadratic,* $\mathrm{pol}\, p$ *is a CE cocycle, and its class belongs to K_g. This class is independent of the refinement and CE representative.*

Proof. By definition p is quadratic. The norm identities give $Np = \omega Nq$, which is quadratic by Lemma J.1. Taking third differences therefore gives $d_{\mathrm{CE}}\, \mathrm{pol}\, p = 0$. Also $\Delta_g p = \Delta_g^2 q = p + Nq$ is quadratic, so the resulting class lies in K_g.

Changing q by linear ℓ changes $\mathrm{pol}\, p$ by ℓB, a CE coboundary. Changing α by tB can be realized by changing q by δt; its defect is unchanged, because $\Delta_g \delta t = Nt = 0$. These observations prove all choice independence, as well as the well-definedness of Θ. □

Proof of Theorem 13.2. Necessity follows from $\Delta_g q = 0$. Conversely, suppose $\Theta[\alpha] = \Psi[\alpha] = 0$, and put $p = \Delta_g q$. Its polar is a coboundary, so choose linear ℓ such that $\mathrm{pol}\, p = \ell B$.

Replacing q by $q' = q + \ell$ makes $p' = \Delta_g q'$ have zero polar, since $\mathrm{pol}(\Delta_g \ell) = \ell B$. Hence p' is linear. The identities

$$0 = Np' = N\Delta_g q' = \omega N q', \qquad \Delta_g p' = \Delta_g^2 q' = p' + Nq' = p'$$

now give $\Delta_g(q' + p') = 0$. The adjustment $\ell + p'$ is linear, so the new eigen-refinement has exactly the prescribed polar cocycle α. Lemma J.2 supplies the remaining assertions about Ψ. □

The degree-four component of the first obstruction has the explicit fourth difference

$$\alpha(B(a,b), B(c,d)) + \alpha(B(a,c), B(b,d)) + \alpha(B(a,d), B(b,c)). \tag{239}$$

Indeed only the products of two disjoint quadratic parts of g contribute. Thus Θ includes the central lifting condition for (7), together with any degree-three leakage. Its vanishing still leaves the quadratic test Ψ.

J.2 The cyclic domain and the fixed splitting

Proof of Theorem 13.3. For $[\alpha] \in K_g$, the refinement Tq is quadratic, and $\mathrm{pol}(Tq) = \mathrm{pol}(\Delta_g q) + \omega^2 \alpha$ is CE-closed by Lemma J.2. Changing q by linear ℓ changes this cocycle by the coboundary ℓB. A CE representative change via δt changes Tq by $T\delta t = \omega^2 \delta t$, again with coboundary polar. This proves well-definedness.

The norm formula yields $T^2 q = q + Tq + Nq$, so all three iterates are quadratic. Their polars are CE-closed, since $N(T^j q) = Nq$ and (238) applies. Thus τ preserves K_g and $\tau^3 = \mathrm{id}$. Polarizing $\Delta_g q = Tq + \omega^2 q$ gives the formula for Ψ. The complete criterion identifies its kernel with $H_g(A)$. Finally the polynomial $1 + \omega X + \omega^2 X^2$ has values $0, 0, 1$ at $1, \omega, \omega^2$, respectively. The three roots are distinct, proving the projector assertion.

Pullback by a binary-linear solution homomorphism commutes with composition by g, quadratic degree bounds, polarization, and CE coboundaries. Coefficient maps commute with the same operations. This proves naturality without invertibility assumptions. □

The space K_g and its cyclic operator precede the choice of coefficient marking. One can define Θ using the high-degree part of Tq alone. To change a CE representative by tB, use the refinement change Tt, with $T^2 t = Tt + t$. All of these constructions are defined over $\mathbb{F}_2$. Extension to $\mathbb{F}_4$ then selects the ω^2-eigenspace for the prescribed scalar kernel.

Details of the fixed splitting and dimension law. For a CE cocycle α, Lemma J.1 makes $\alpha(x, gx)$ an invariant quadratic. A coboundary evaluates to $tB(x, gx) = 0$ by (5). For invariant quadratic r, $Nr = r$, and (238) proves that its polar is CE-closed. Hence $O_0[\mathrm{pol}\, r] = r$, proving the section assertion on all ordinary cohomology.

An invariant refinement represents a τ-fixed class. Conversely, for a fixed class $[\alpha] \in K_g$, its norm $r = Nq$ is invariant quadratic and

$$[\mathrm{pol}\, r] = (\mathrm{id} + \tau + \tau^2)[\alpha] = [\alpha].$$

Thus the section image is precisely $(K_g)_1$, establishing Proposition 13.4.

Binary descent of K_g and τ means that coefficient Frobenius interchanges its ω- and ω^2-eigenspaces. This proves $\dim K_g = i_g + 2h_g$. The independent reduced monomials $u_i u_j^2$ on the intrinsic abelianization prove $i_g \geq d^2$. Trivial coefficient extension gives $\dim_{\mathbb{F}_4} H^2(L_g, \mathbb{F}_4) = \dim_{\mathbb{F}_2} H^2(L_g, \mathbb{F}_2)$, so rank–nullity proves (126) and (127). For the linear example, every quadratic remains quadratic under composition. Its fixed monomials are $u_i u_j^2$; after linear section gauge, its ω^2-sector is spanned by the $\binom{d}{2}$ products $u_i u_j$, $i < j$. This supplies the sharpness values in the main text.

Finally O_0 restricted to K_g is its fixed-sector retraction. Hence $K_g \cap \ker O_0 = (K_g)_\omega \oplus H_g(A)$. The fixed splitting makes $\ker O_0 \to H^2_{\mathrm{CE}}/K_g$ onto, with this intersection as kernel. Quotienting by $H_g(A)$ proves (128). □

J.3 The full marked Hom sets

Lemma J.3 (Unique correction of a marked Lie homomorphism)**.** *For a map R satisfying (129), its defect d_R is binary-linear, kills W, and satisfies $d_R g = \omega^2 d_R$. The correction in Theorem 13.6 is unique in its K^--coset and respects composition.*

Proof. The polar of (130) is exactly the left side of (129); thus d_R is linear. Using $qg = \omega^2 q$, $q'g' = \omega^2 q'$, $Sg = g'S$, and $g^2 = g + \mathrm{id}$, direct expansion gives

$$d_R g = \omega^2 Cq + t(g + \mathrm{id}) + \omega tg + \omega^2 q' S = \omega^2 d_R.$$

Polarizing this equality yields $d_R B = 0$, so d_R descends to V/W and is Frobenius-semilinear there. Every $k \in K^-_{g,A'}$ has $\delta k = (\omega^2 + \omega)k = k$. Replacing t by $t + k$ changes its defect by exactly k, proving the existence and uniqueness of $t + d_R$.

For two composable marked Lie maps R, R', with base and coefficient blocks (S, C) and (S', C'), expansion gives

$$d_{R'R} = C' d_R + d_{R'} S. \tag{240}$$

Both coefficient composition and base pullback preserve the Frobenius-semilinear spaces. The corrected section block of a composite is consequently the composite of the corrected section blocks. This proves the assertion for all linear maps S, C, including maps with kernel or non-surjective image. □

Proof of Theorem 13.6. Define a Lie-data object to be (g, A, α), where α is an actual CE cocycle whose class lies in $H_g(A)$. A morphism is a triple (S, C, t) satisfying (129); quotient its section block by $t \sim t + k$, $k \in K^-_{g,A'}$. Since $kB = 0$, the Lie condition is preserved. Composition is well defined by the preservation of K^- under the two operations just noted. For general Hom sets this is an additive congruence.

Polarization maps every actual quadratic lift to a Lie-data object and every solution homomorphism to its coset. For fixed source and target refinements, Lemma J.3 supplies exactly one solution arrow in each Lie-data Hom coset, proving full faithfulness. Every Lie-data object has an admissible refinement with its *exact* polar: first choose one with the prescribed CE class, then add δt to correct its polar by any needed coboundary tB. This proves essential surjectivity.

No simultaneous refinement choices are needed for the forward functor. Choosing one on each Lie-data object gives a quasi-inverse; the corrected arrows in the identity Lie-data cosets supply the natural isomorphisms between such quasi-inverses. This proves the stated equivalence, together with its full morphism scope. □

For a fixed actual lift, correction also gives the split exact sequence

$$1 \longrightarrow K^-_{g,A} \longrightarrow \mathrm{Aut}_{\text{marked Lie}}(L_{F_q}; g, A) \longrightarrow \mathrm{Aut}_{\text{marked solution}}(F_q) \longrightarrow 1. \tag{241}$$

The middle group consists of Lie automorphisms covering an allowed solution automorphism of the quotient and a scalar-compatible coefficient automorphism. The splitting is the inclusion of actual solution automorphisms, and the kernel consists of the corrected-to-zero section blocks. This interpretation specifies the marking retained in (241).

There is a second exact sequence,

$$1 \longrightarrow K^+_{g,A} \longrightarrow \mathrm{Aut}_{\text{marked solution}}(F_q) \longrightarrow \mathrm{Stab}_{\mathrm{Aut}(g)\times \mathrm{GL}_{\mathbb{F}_4}(A)}([q]) \longrightarrow 1. \tag{242}$$

Here the stabilizer action on extension classes is $q \mapsto CqS^{-1}$. Equality of the transformed classes is exactly the existence of a linear section block. Its homogeneous solutions are $\ker \delta = K^{+}_{g,A}$: polarizing $tg = \omega t$ gives $tB = 0$, and on the abelianization the same equation is $\mathbb{F}_4$-linearity. The second sequence has no asserted splitting. Its automorphisms preserve the chosen central kernel; when that kernel is characteristic, this preservation is automatic for every solution automorphism.

J.4 The tower and graph interfaces

Proof of Corollary 13.7. Proposition 2.3 makes each lower-central term invariant under the solution. In X_i, the subspace A_i is central, so the induced solution is linear there and defines its $\mathbb{F}_4$-scalar structure. A binary-linear section gives (118), componentwise when $\dim_{\mathbb{F}_4} A_i > 1$. Conversely admissible central lifts construct the next solution.

A solution homomorphism is a polar Lie homomorphism and preserves all lower-central terms, so it induces arrows at all stages. Inductively recover the actual arrow at X_{i-1}. Form the upstairs marked Lie Hom set over that recovered base arrow, and apply Lemma J.3. Its compositional identity proves by induction the full recursive Hom equivalence. Different finite tower lengths may be padded by zero kernels and stabilized stages. The recursive prescription retains the base arrow covered at the next step throughout the construction. □

Proof of Proposition 13.8. The sum of the three graph vectors at $x, y, x+y$ equals $(0, B(x, y))$. These vectors give $0 \oplus W$. Conversely the exact quadratic expansion shows that every sum of graph vectors is $(x, gx+w)$ for some $w \in W$. This proves (132) and its dimension formula. The functional equation gives $\mathcal{J}(x, gx) = (gx, g^2x)$, proving scalar stability. Also $\mathcal{J}(0, w) = (w, w)$, whence $W \oplus W \subseteq M_g$.

The first kernel is immediate. For the second, the condition is $gx + w = 0$; invariance of W and bijectivity of g give $x \in W$, proving $\ker(p\mathcal{J}) = W \oplus 0$. The first projection identifies the stated quotient with V/W and transports $\mathcal{J}$ to its induced scalar g. The graph marking and its projection recover g by the formula in the proposition.

For clarity, a binary-linear intertwiner $Sg = g'S$ induces the $\mathcal{J}$-linear map $(S \oplus S)|_{M_g}$ preserving graph and projection. Conversely a scalar-linear carrier map φ, together with binary-linear S satisfying $p'\varphi = Sp$ and $\varphi\Gamma_g \subseteq \Gamma_{g'}$, gives $Sg = g'S$ on graph points. Both its coordinates are then forced by S, and the graph spans the carrier. Thus retaining these markings also retains every original linear intertwiner. □

Proof of Proposition 13.9. Direct calculation gives

$$\ell_t \mathcal{J}(x, y) = t(x) + \omega^2 t(y) = \omega^2 \ell_t(x, y).$$

Its graph restriction is $tg + \omega t = \delta t$. Hence its kernel on M_g is $\ker \delta = K^{+}_{g,A}$, because the graph spans M_g.

Every antilinear functional on the scalar subspace M_g extends to $V \oplus V$ by extending an $\mathbb{F}_4$-basis and assigning zero on a complement. Write an ambient additive functional as $u(x) + v(y)$. Its antilinearity is equivalent to $u = \omega v$, so it is exactly ℓ_t with $t = v$. This proves surjectivity. The map itself, its kernel, and its graph restriction are canonical; only the extension used to prove surjectivity invokes a complement. The formulas commute with coefficient maps and pullback by binary-linear intertwiners. □

K Uniform reconstruction and conjugacy proofs

This appendix proves the uniform formulas of Section 14. The lifting spaces and cyclic operators are those of Section 13; ordinary CE cohomology always uses the underlying binary polar Lie algebra and the trivial coefficient line $k = \mathbb{F}_4$.

All vector spaces in this appendix are over $k = \mathbb{F}_4$, unless explicitly viewed over $\mathbb{F}_2$. We first establish the realization and free-object ingredients used in Section 14.

K.1 An explicit nonrestrictable class-four seed

Proposition K.1. *On k^9, with coordinates $(a, b, c, z, w, v, h, k, r)$, the map*

$$\begin{aligned}
a' &= \omega a, & b' &= \omega b, & c' &= \omega c,\\
z' &= \omega z + ab, & w' &= \omega w + ac, & v' &= \omega v + bc,\\
h' &= \omega h + av + cz, & k' &= \omega k + aw + a^2 c,\\
r' &= \omega r + zw + ah + bk + a^2 v
\end{aligned}$$

is a binary-quadratic solution of $f^2 = f + \mathrm{id}$. Its polar Lie algebra has lower-central dimensions $(18, 12, 6, 2, 0)$, has nonzero second derived ideal of dimension two, and is not restrictable even as a binary Lie algebra.

Proof. The first three new cross-term cocycles are ab, ac, bc. For $av + cz$, substitution gives two equal ωabc terms, which cancel. The cocycle $aw + a^2 c$ is the familiar class-three identity, using $1 + \omega = \omega^2$. For the final cocycle $q = zw + ah + bk + a^2 v$, substitution leaves the desired $\omega^2 q$: the additional ωacz and ωabw terms each occur twice, while the coefficient of $a^2 bc$ is $1 + \omega + \omega^2 = 0$. Thus every central stage satisfies $q \circ g = \omega^2 q$, proving the functional equation as ordinary polynomial identities over $\mathbb{F}_4$.

Give a, b, c weight one, z, w, v weight two, h, k weight three, and r weight four. Every polar term has output weight at least the sum of its input weights; hence the class is at most four. The brackets include

$$[a, b] = z, \quad [b, c] = v, \quad [a, c] = w + k, \quad [a, w] = k, \quad [c, z] = h, \quad [a, h] = [z, w] = r.$$

Their scalar variants give explicitly

$$L' = \mathrm{span}_k\{z, w, v, h, k, r\}, \qquad \gamma_3 = \mathrm{span}_k\{h, k, r\}, \qquad \gamma_4 = \mathrm{span}_k\{r\}.$$

In particular $z, w \in L'$ and $[z, w] = r \neq 0$, so the algebra is nonmetabelian and has class exactly four. On L', only the z-w bracket survives, giving its two-dimensional second derived ideal.

Finally $(\mathrm{ad}\, a)^2 c = k$. For an arbitrary vector u, the w-coordinate of $[u, c]$ is u_a, and its k-coordinate is u_a^2. Thus $[u, c] = k$ is impossible: its zero w-coordinate forces $u_a = 0$, which also makes its k-coordinate zero. Therefore $(\mathrm{ad}\, a)^2$ is not inner. No restricted square exists, even over $\mathbb{F}_2$. This is an intrinsic obstruction, independent of any change of scalar marking or coordinates. □

The eight-coordinate quotient before adjoining r is a quotient of the free class-three k-Lie normal form on three generators. Its degree-two basis is $z = [a, b], w = [a, c], v = [b, c]$, and its only nonzero degree-three brackets are

$$[a, v] = [c, z] = h, \qquad [a, w] = k.$$

The Jacobi relation at (a, b, c) is respected. The normal-form twist is $P(u)Q(u) = a^2 c\, k$, since the two $abc\, h$ terms cancel. Hence its functional normal form is exactly the eight-coordinate base used above.

K.2 The free class-three base and its linear symmetry

Let U have dimension d, put

$$Z = \Lambda^2 U, \qquad T = (U \otimes Z)/\operatorname{im}\partial_3,$$

where ∂_3 is the usual Jacobi map. Write $u \cdot z$ for the class of $u \otimes z$, and choose the standard cross-term refinement $Q : U \to Z$. The free class-three auxiliary Lie algebra has functional normal form

$$g_d(u, z, t) = (\omega u, \omega z + Q(u), \omega t + u \cdot z + u \cdot Q(u)). \tag{243}$$

The Jacobi relations make $u \cdot Q(u)$ binary-quadratic. Its original polar derived ideal is $W = Z \oplus T$, its third lower-central term is T, and $g_d|_W = \omega\,\mathrm{id}$.

Every $M \in \mathrm{GL}(U)$ gives a genuine solution automorphism

$$S_M(u, z, t) = \big(Mu, (\Lambda^2 M)z + b_M(u), (M \otimes \Lambda^2 M)t\big), \quad b_M(u) = Q(Mu) + (\Lambda^2 M)Q(u).$$

The function b_M is additive and Frobenius-linear, so S_M is binary-linear. Direct substitution verifies $S_M g_d = g_d S_M$. Moreover $b_{MN} = b_M \circ N + (\Lambda^2 M) b_N$, so $M \mapsto S_M$ is a group homomorphism realizing a $\mathrm{GL}(U)$-subgroup. On W/T, its action is $\Lambda^2 M$.

K.3 The exact realizable derived-commutator module

Let $H_{g_d}(k)$ be the admissible quadratic central-extension space, embedded in ordinary binary Lie cohomology. For a class represented by q, write $\alpha = \operatorname{pol} q$. Its restriction to $W \times W$ descends to an alternating k-bilinear form on $Z = W/T$. Denote it by $\rho[q]$. Define the characteristic-two wedge map

$$\mu : \Lambda^2(\Lambda^2 U^*) \longrightarrow \Lambda^4 U^*, \qquad \xi \wedge \eta \longmapsto \xi \wedge \eta.$$

It is well defined: wedge of two two-forms is symmetric and alternating in characteristic two.

Theorem K.2 (The universal class-four nonmetabelian sector)**.** *For every d, the map ρ is well defined and $\mathrm{GL}(U)$-equivariant, and*

$$\operatorname{im}\rho = \ker\mu. \tag{244}$$

Consequently

$$\dim_k \operatorname{im}\rho = \binom{\binom{d}{2}}{2} - \binom{d}{4} = 3\binom{d}{3} + 2\binom{d}{4}.$$

The kernel of ρ is exactly the subspace of metabelian central extensions. Every nonzero derived-commutator sector is therefore realized by a nonmetabelian polar solution of class four.

Proof. Every central Lie extension of a class-three algebra has class at most four. Its commutator of a lifted degree-three vector with a lifted derived vector lies in the fifth lower-central term and is zero. Thus $\alpha(T, W) = 0$, so its derived restriction descends to Z. A CE coboundary vanishes on $W \times W$, because the base derived ideal is abelian. This proves independence of the extension representative.

Since $g_d|_W = \omega\,\mathrm{id}$, the restriction $q|_W$ is a quadratic eigenfunction of scalar weight two. Its polar is therefore k-bilinear; all mixed and both-Frobenius polar types have the wrong scalar weights. This proves the stated target for ρ. The four-variable polar identity of the lifted solution, evaluated on four degree-one inputs, gives

$$\rho[q](u \wedge v, w \wedge t) + \rho[q](u \wedge w, v \wedge t) + \rho[q](u \wedge t, v \wedge w) = 0.$$

This is exactly $\mu\rho[q] = 0$. The explicit subgroup S_M proves equivariance.

For $d \geq 3$, take the quotient of the free class-three base onto the eight-coordinate base of Proposition K.1. In a basis e_1, e_2, e_3, it sends the three wedge generators to z, w, v, and sends the degree-three generators

$$e_1 \cdot (e_2 \wedge e_3),\ e_3 \cdot (e_1 \wedge e_2) \longmapsto h, \qquad e_1 \cdot (e_1 \wedge e_3) \longmapsto k,$$

with all other generators zero. The Jacobi relation maps to $h + 0 + h = 0$. Extend by zero on further input generators. The map intertwines the normal forms directly, because it preserves Q, the action and $P(u)Q(u)$. Pulling back the seed cocycle gives an admissible class whose derived restriction is $(e_1^* \wedge e_2^*) \wedge (e_1^* \wedge e_3^*)$.

Its $\mathrm{GL}(U)$-span contains every shared-index wedge pair. For four distinct indices, replacing the shared covector by a sum produces, after subtracting shared-index terms, the sum of two disjoint pairings. For example,

$$((e_1^* + e_3^*) \wedge e_2^*) \wedge ((e_1^* + e_3^*) \wedge e_4^*)$$

gives $(e_1^* \wedge e_2^*) \wedge (e_3^* \wedge e_4^*) + (e_2^* \wedge e_3^*) \wedge (e_1^* \wedge e_4^*)$ modulo shared terms. The analogous shear supplies an independent second relation among the three disjoint pairings.

The target decomposes by index support. Each three-index support has three shared-pair basis elements, all killed by μ. Each four-index support has three disjoint pairings, on which μ has rank one; its two-dimensional kernel is spanned by the preceding differences. There are no other supports. Hence the shared-pair orbit spans all of $\ker \mu$, proving (244) and the dimension formula.

Finally the base is metabelian and the new kernel is central, so the second derived ideal of a lift is exactly the image of $\alpha(W, W)$. Thus $\rho = 0$ is equivalent to metabelianity. When it is nonzero the class is at least four and at most four, completing the proof. □

K.4 The tensor-algebra realization

The surjectivity in (244) also has a route through a classical restricted-square construction. Let $\mathcal{A} = T_+^{\leq 4}(U)/S$, where degrees at least five are zero and $S \subset U^{\otimes 4}$ is spanned by u^4, $u \in U$. Since S is in top degree it is an ideal, every $x \in \mathcal{A}$ satisfies $x^4 = 0$, and $x \mapsto \omega x + x^2$ solves the functional equation. Indeed only the degree-one part of x contributes to its fourth power. For $\varphi(x) = \omega x + x^2$,

$$\varphi^2(x) = \omega^2 x + x^2 + x^4.$$

The coefficient functions of u^4 span one symmetric tensor word sum for each multiset of four indices. To see this over $\mathbb{F}_4$, reduce the monomial u_i^4 to u_i; the other exponent patterns have each exponent at most three. The resulting distinct reduced monomials are independent functions on $\mathbb{F}_4^d$. Thus in each tensor multidegree S is the line spanned by its full symmetric word sum.

Modulo degree four, use the binary-linear orbit section

$$s(u) = \sum_i \big(u_i a_i + (u_i + u_i^2) a_i^2\big).$$

It sends e_i to a_i and ωe_i to $\omega a_i + a_i^2$. Adjoin the ordinary free-Lie degree-two and degree-three subspaces. On these subspaces the functional map is multiplication by ω; expanding $s(u)^2$ shows that $f(s(u)) + s(\omega u)$ is a sum of ordinary commutators of lengths two and three. Hence the resulting binary subspace is f-stable. Its polar image contains the degree-two commutators, up to degree-three corrections, and their brackets with the generators supply all degree-three words. These words are independent modulo precisely antisymmetry and

Jacobi: on a repeated-index support the image has a nonzero two-word expression, and on a three-index support two independent four-word expressions remain after their single Jacobi relation. Its auxiliary algebra is therefore the free class-three algebra. Equivalently, the normal form is (243), with degree-two generators changed only by degree-three terms. This preserves their quotient Z. Pull back the associative central extension over this binary subspace, then push out its degree-four kernel by any k-linear functional.

In a three-index support, a shared-pair double commutator has eight of the twelve tensor words, whereas S has only their full symmetric sum, so the double commutator survives. In a four-index support, the three disjoint double commutators have disjoint eight-word supports, and their sum is the twenty-four-word symmetric sum in S. These are exactly the single four-variable relation. Thus the double-commutator map into the degree-four kernel has exactly one relation on every four-index support and none on a shared-index support. Any linear functional on its image extends to the full degree-four kernel. The resulting scalar pushouts realize exactly $\ker \mu$.

This gives a second proof of realization, through the functional normal form and the classical restricted-square construction. The seed's nonrestrictability and this realization concern different levels: an f-stable binary subspace of the associative model need not be closed under its ambient scalar action or square.

K.5 The complete admissible dimension

Proposition K.3. *For the free class-three base,*

$$\dim_k H_{g_d}(k) = \binom{d}{2} + 6\binom{d}{3} + 5\binom{d}{4}.$$

Proof. The kernel of ρ consists of metabelian lifts. Under the complete metabelian correspondence, these are precisely central extensions of the ordinary free class-three algebra inside the variety of metabelian Lie algebras with square-zero action on the derived ideal. The correspondence may be made over the fixed base and central kernel: choose its linear scalar splitting compatibly with the kernel, so both the action matrices and alternating coefficients project to the prescribed base normal form.

Let $\mathfrak{F}_d = U \oplus W_d$ be the free object in that variety. Its defining extra identities first occur in bracket length four, so $\mathfrak{F}_d/\gamma_4$ is the ordinary free class-three Lie algebra. Choosing lifts of its free generators in a central extension induces a map from $\mathfrak{F}_d$. The resulting functional on $G = \gamma_4\mathfrak{F}_d/\gamma_5\mathfrak{F}_d$ does not depend on central changes of generator lifts. Conversely every functional $G \to k$ gives such an extension by central pushout of $\mathfrak{F}_d/\gamma_5$.

There are no extra lower-degree extension parameters: the base is free through degree three. More explicitly, for a nonzero scalar central-extension class its kernel lies in the derived ideal; if N is generated by arbitrary lifts of the d free generators, then $E = N + A$ and $E' = N'$, so $A \subset N$ and the map is onto. The zero functional gives the split class. In a compatible normal-form splitting, the central entries of the actions and alternating coefficients vary linearly with this functional. Products of two such entries vanish because the new kernel is central. Hence these constructions are linear under central pushout and Baer sum. Thus $\ker \rho \simeq G^*$.

The coefficient-degree-two term of the established Hilbert polynomial of W_d is $\binom{d}{2} + 3\binom{d}{3} + 3\binom{d}{4}$. Adding the image dimension $3\binom{d}{3} + 2\binom{d}{4}$ proves the formula. □

The dimension formula combines the universal metabelian module, the associative realization, and the functional correspondence. The following subsections determine the cyclic fixed sector and separate the remaining degree leakage.

K.6 The fixed sector and the cyclic spectrum

Theorem 14.1 now follows from the all-class filtration proof in Appendix T. In particular no separate class-three coefficient argument is needed: every invariant quadratic factors through abelianization, where its scalar-invariant basis has d^2 elements.

Corollary K.4 (The quadratic-stable cyclic spectrum)**.** *For any finite quadratic base with abelianization dimension d, write $h = \dim_{\mathbb{F}_4} H_g(\mathbb{F}_4)$. Then*

$$\dim K_1 = d^2, \qquad \dim K_\omega = \dim K_{\omega^2} = h, \qquad \dim K = d^2 + 2h.$$

For the free class-three base,

$$h = \binom{d}{2} + 6\binom{d}{3} + 5\binom{d}{4}.$$

The admissibility projector has rank h. The high-degree obstruction satisfies

$$\operatorname{rank}\Theta = \dim H^2_{\mathrm{CE}}(L_g, \mathbb{F}_4) - (d^2 + 2h);$$

with the complete free-base ordinary dimension given in Theorem 14.6.

Proof. Proposition 13.4 identifies K_1 with invariant quadratics. For surjectivity apply the cyclic norm to a quadratic refinement; for injectivity, if an invariant n has polar tB, then $n+T_g t$ is linear and its norm is n, forcing $n = 0$. Theorem 14.1 gives $\dim K_1 = d^2$. Coefficient Frobenius commutes with the binary CE differential, the degree condition and the canonical cyclic action, and exchanges eigenvalues ω and ω^2. Their dimensions therefore agree, and the second is h by the general admissibility theorem. The free-base value is the separate universal-module and associative-bridge count. □

This determines the quadratic-stable spectrum. The following subsections pass to actual conjugacy types and detect a uniform part of the degree leakage, now completed by Theorem 14.6.

K.7 Universal central kernels and actual conjugacies

Proof of Theorem 14.5. If a nonzero scalar central-extension class had class at most three, its polar algebra would be metabelian. The metabelian correspondence would make it a central extension, still of class at most three, of the ordinary free class-three Lie algebra on the given generators. Lifting those free generators splits that extension. The functional class would then be zero, a contradiction. A central extension of a class-three base has class at most four; its nonzero fourth lower central term therefore equals its one-dimensional marked kernel.

To control all base automorphisms, form the universal central extension with coefficient space $A = H_{g_d}(\mathbb{F}_4)^*$ and the tautological cohomology class. Its fourth lower-central term is all of A. Otherwise a nonzero linear functional annihilating that term would produce a nonzero scalar extension class of class at most three, contradicting the preceding paragraph.

Every base automorphism lifts to this universal extension, paired with the dual cohomology action on its kernel. This uses equality of extension classes and the usual section-block equation; it requires only the existence of each lift. Since $A = \gamma_4 \subseteq E'$, the universal extension has the same abelianization as the base. If the base automorphism is IA, its lift therefore acts identically on that abelianization. In a class-four algebra it then fixes every fourfold commutator: changing an input by a derived element raises its length to at least five. Since these commutators span the whole kernel A, its action on A, and therefore on H_{g_d}, is trivial.

Thus the base action factors through $\mathrm{GL}(d, 4)$, which is actually realized by the explicit normal-form automorphisms. Because the scalar kernel of every nonzero extension is the

characteristic ideal γ_4, any solution conjugacy between two such objects induces a base automorphism and a marked scalar kernel isomorphism. Conversely the ordinary central-extension orbit condition supplies exactly such a conjugacy. Intrinsic ω-intertwining makes the base and kernel maps $\mathbb{F}_4$-linear; no additional field-Frobenius semilinear factor is being omitted.

There are $4^{h_d}-1$ nonzero vectors, scalar multiplication is free, and every full orbit has size at most $3|\operatorname{GL}(d,4)|$. These elementary bounds prove the assertions for N_d, without assuming a free GL-action. The metabelian classes form the invariant subspace $\ker\rho$ of dimension $h_d - r_d$. Comparing its scalar orbit upper bound with the total orbit lower bound gives the stated proportion (with the displayed factor three a harmless weakening). Since r_d grows quartically and $\log_4|\operatorname{GL}(d,4)|$ quadratically, that proportion tends to zero. □

K.8 A pure CE calculation and independent leakage sectors

In this subsection put $b_2(d) = \dim_k H^2_{\mathrm{CE}}(L_{g_d}, k)$.

Lemma K.5 (The two-generator ordinary cohomology)**.** *For the free class-three two-generator base,* $b_2(2) = 10$ *and* $\operatorname{rank}\Theta_2 = 4$.

Proof. Write its normal form as

$$g(a,b,c,d,e) = (\omega a, \omega b, \omega c + ab, \omega d + ac + a^2 b, \omega e + bc + ab^2).$$

Over k, the binary linear coordinate functions have basis $A = a, A' = a^2, B = b, B' = b^2, Z = c, Z' = c^2, D = d, D' = d^2, E = e, E' = e^2$. Write juxtaposition below for exterior product, not pointwise multiplication of these functions. The CE differential is

$$\begin{aligned} dA = dA' = dB = dB' = 0, \qquad & dZ = AB, \qquad & dZ' = A'B', \\ dD = AZ + A'B, \qquad & dD' = A'Z' + AB', & \\ dE = BZ + AB', \qquad & dE' = B'Z' + A'B. & \end{aligned}$$

Put $U^* = \langle A, A', B, B'\rangle$, $Z^* = \langle Z, Z'\rangle$, and $T^* = \langle D, D', E, E'\rangle$. For a closed two-form, its $\Lambda^2 T^*$ part is zero: the $U^*Z^*T^*$ component of its differential detects it injectively, since the four U^*Z^* terms in dT^* are independent. Then its Z^*T^* part is zero, detected by the $\Lambda^2 U^*T^*$ component because dZ, dZ' are independent.

All six $\Lambda^2 U^*$ forms are closed. The differential from U^*Z^* onto $\Lambda^3 U^*$ has rank four, and hence kernel dimension four. Finally the projected differential

$$U^*T^* \oplus \Lambda^2 Z^* \longrightarrow \Lambda^2 U^* Z^*$$

has rank eleven. Its six-dimensional kernel is explicitly spanned by

$$AD,\ A'D',\ BE,\ B'E',\ BD + AE,\ B'D' + A'E'.$$

Indeed their projected differentials vanish; the other image coordinates consist of ten separate basis coordinates and the one combination $ABZ' + A'B'Z$. The remaining $\Lambda^3 U^*$ differentials of these six forms can all be cancelled using U^*Z^*, whose differential is onto. Thus the cocycle space has dimension $6 + 4 + 6 = 16$. The six nonzero one-form differentials are independent, so the coboundary space has dimension six and $b_2(2) = 10$. The established invariant and admissible formulas give $\dim K_{g_2} = 2^2 + 2\binom{2}{2} = 6$, whence $\operatorname{rank}\Theta_2 = 4$. □

Proposition K.6 (Independent two-generator leakage sectors)**.** *For the free class-three rank-d base,* $d \geq 2$,

$$\operatorname{rank}\Theta_d \geq 4\binom{d}{2},$$

and hence

$$b_2(d) \geq d^2 + 6\binom{d}{2} + 12\binom{d}{3} + 10\binom{d}{4}.$$

The two-generator value is proved in the preceding lemma; no higher-rank CE enumeration is required.

Proof. Write $\mathcal{L}_d = H^2_{\mathrm{CE}}(L_{g_d}, k)/K_{g_d}$. This quotient is contravariantly functorial under binary-linear solution intertwiners: CE pullback and composition of quadratic refinements preserve quadratic stability. In particular its dimension is $\operatorname{rank}\Theta_d$.

The free property gives, for every two-element generator subset S, an inclusion $\iota_S : g_2 \to g_d$ and a retraction $\pi_S : g_d \to g_2$, with $\pi_S\iota_S = I$. They are genuine binary-linear solution morphisms directly in (134): the coordinate inclusions and projections preserve Q, the action and its twist. If $S \neq T$, the composite $\pi_S\iota_T$ factors through the free rank-zero or rank-one base. The rank-one base is the linear solution on $\mathbb{F}_4$, so its leakage space is zero. Therefore the cross pullback $\iota_T^*\pi_S^* : \mathcal{L}_2 \to \mathcal{L}_2$ vanishes, whereas the diagonal one is the identity. It follows that

$$\bigoplus_{|S|=2} \mathcal{L}_2 \xrightarrow{\sum \pi_S^*} \mathcal{L}_d \xrightarrow{(\iota_S^*)_S} \bigoplus_{|S|=2} \mathcal{L}_2$$

has composite identity. This proves the direct-summand lower bound $4\binom{d}{2}$. Substitution into $b_2(d) = d^2 + 2h_d + \operatorname{rank}\Theta_d$ gives the displayed polynomial lower bound. □

L Filtered inverse degrees and optimal readout kernels

L.1 The degree of the filtered inverse

In Theorem 15.1, choose a binary basis e_a adapted to Γ_i, assigning weight $w_a = i$ to a lift of a basis vector of Γ_i/Γ_{i+1}. For Boolean coordinates x_a, the exact square-free expansion is

$$N\Big(\sum_a x_a e_a\Big) = \sum_a x_a N(e_a) + \sum_{a<b} x_a x_b [e_a, e_b].$$

Here $N(e_a) \in \Gamma_{w_a+1}$, and $[e_a, e_b] \in \Gamma_{w_a+w_b}$. A coordinate of output weight i consequently has linear input weights at most $i-1$ and quadratic input-weight sums at most i. Each input has strictly smaller weight than i. Solving $y = x + N(x)$ in increasing weights is triangular, with identity diagonal. Inductively, each inverse coordinate of weight i has weighted Boolean degree at most i. Reducing $x_a^2 = x_a$ only decreases weight. Therefore its ordinary degree is at most

$$\max\{|S| : \sum_{a\in S} w_a \leq i\} \leq i \leq c.$$

This is a degree bound on Boolean functions; it does not treat the functional equation as an identity of unreduced formal polynomials.

For completeness, the inverse iteration in the theorem has initial residual $H_0(y) + y = N(y) \in \Gamma_2$. If $r_j = H_0(x_j) + y \in \Gamma_{j+2}$, then $x_{j+1} + x_j = r_j$, and

$$H_0(x_{j+1}) + y = N(x_{j+1}) + N(x_j) \in \Gamma_{j+3}.$$

Thus the $(c-1)$-st residual vanishes. For class one, $N = 0$ and the initial value $x_0 = y$ is already the inverse.

L.2 Checking the inverse on the seven-class chain

For the prefixes of (115), take $J_0 = \omega\,\mathrm{id}$ and write inverse coordinates in the target variables. Successive substitution gives

$$a,\ b,\ c + ab,\ d + ac,\ e + bc,\ h + ae + bd,$$

followed by

$$r + ah + cd + a^3b^2, \qquad s + bh + ce + a^2b^3.$$

For $a = a_0 + \omega a_1$, $a^3 = a_0 + a_1 + a_0a_1$. Hence the term a^3b^2 has nonzero cubic Boolean part, which cannot cancel against $ah + cd$. The last inverse coordinates have degree three. In the metabelian normal form, by contrast, the two displayed maps in Section 15 are quadratic in both directions. The degree observation refers to the specified conjugators, without asserting a converse characterization of metabelianity.

L.3 All optimal carrier kernels

The carrier and kernel argument needs only $f^2 = f + \mathrm{id}$, rather than quadraticity. To see this, let W be the binary span of all additivity defects $f(x+y)+f(x)+f(y)$, and let M_f be the graph span. Then $M_f \cap (0 \times V) = 0 \times W$: every graph-sum relation in its first coordinate is a sum of additivity defects, and every defect gives such a relation. Stability under $\mathcal{J}(x, y) = (y, x + y)$ gives $W \times W \subseteq M_f$. Adding $(w, f(w))$ for $w \in W$ shows $f(W) \subseteq W$. The induced map on $U = V/W$ is linear, with square plus itself equal to the identity. It supplies the required $\mathbb{F}_4$-scalars, and the carrier has the same description as in the quadratic case. Thus Theorem 15.3 remains valid in this wider class when w_B has this additivity-defect meaning.

At the optimum $m = n - d$, the readout kernels are precisely binary sections of subspaces $R \subset U$ with $U = R \oplus \omega R$. Every binary basis of such an R is an $\mathbb{F}_4$-basis of U. The number of these real forms is $|\,\mathrm{GL}(d, 4)|/|\,\mathrm{GL}(d, 2)|$, and each has 2^{dw_B} binary sections into V. Therefore the number of optimal kernels is

$$2^{dw_B}\frac{|\,\mathrm{GL}(d, 4)|}{|\,\mathrm{GL}(d, 2)|}.$$

For a fixed labelled output space $\mathbb{F}_2^{n-d}$, the number of optimal readout matrices is this expression times $|\,\mathrm{GL}(n - d, 2)|$. These count readouts, without identifying them under automorphisms of the original solution.

For an optimal state readout in the nilpotent setting, $\mathcal{O}_H$ is a bijection onto its 2^n-element output space. Hence $L_H : M_f \to Y \oplus Y$ is onto. Its binary kernel dimension is $(n + w_B) - n = w_B$, and its canonical $\mathbb{F}_4$-dimension is $w_B/2$. Injectivity on graph states means exactly that this kernel contains no nonzero difference of two graph points. This explains why its positive dimension does not obstruct state recovery.

M Automorphisms of the fourteen-dimensional chain member

Proof of Proposition 12.4. Put $k = \mathbb{F}_4$, and retain g, A, B from (115). Polarization gives $Z(L_{F_{14}}) = kr$. Thus every binary-linear automorphism commuting with F_{14} preserves kr, and commutation with ω on this line forces its restriction to be multiplication by some $\lambda \in k^*$. It induces $\phi \in \mathrm{Aut}(g)$ on the quotient, so it has exactly the form $S_{\phi,\lambda,t}$ in the proposition. Substitution into $SF_{14} = F_{14}S$ gives the displayed equation for t. The group law is

$$S_{\phi,\lambda,t}S_{\psi,\mu,u} = S_{\phi\psi,\lambda\mu,\,t\circ\psi+\lambda u}.$$

By Table 3, H_g has basis $[A], [B]$, and $\ker\delta = \mathrm{Hom}_k(k^2, k)$, where $\delta t = t \circ g + \omega t$. Hence a lift exists exactly when $[A \circ \phi] = \lambda[A]$; its scalar is then unique and its fiber consists of

$4^2 = 16$ maps. This proves the exact sequence. We compute the stabilizer and then exhibit a section over P.

The metabelian correspondence identifies g with the auxiliary k-Lie algebra

$$\mathfrak{l} = kx + ky + Rc_0, \qquad R = k[t_1, t_2]/(t_1^2, t_2^2),$$

with $[x, y] = c_0$, $[x, uc_0] = t_1 uc_0$, $[y, uc_0] = t_2 uc_0$, and $[Rc_0, Rc_0] = 0$. This is the free two-generator object in the metabelian category with $(\operatorname{ad} v)^2 \mathfrak{l}' = 0$: the universal module has one generator and no triangle relations, and its four monomials are independent. Any two generator images inducing an invertible matrix on the abelianization therefore extend uniquely to an endomorphism. The lower central filtration shows that they generate $\mathfrak{l}$, so this endomorphism is invertible. Consequently

$$|\operatorname{Aut}(g)| = |\mathrm{GL}_2(k)|\,|\mathfrak{l}'|^2 = 180 \cdot 4^8.$$

Its IA subgroup N_{12}, consisting of maps inducing the identity on the abelianization, has order 4^8.

We next check that N_{12} acts trivially on H_g. Consider the full sixteen-dimensional extension with central cocycles (A, B). Every $\phi \in \operatorname{Aut}(g)$ lifts to it: its invertible action on the full quotient H_g supplies the central linear map, and the remaining difference is a splitting coboundary. Its polar algebra has class five and $\gamma_5 = Z = kr + ks$, by Table 2. If $\phi \in N_{12}$, a lift is the identity modulo γ_2. It is therefore the identity on γ_5, since replacing any input in a fivefold commutator by an element of γ_2 gives an element of $\gamma_6 = 0$. The central linear map of the lift is thus the identity, proving the assertion on H_g. This argument uses the full two-dimensional central extension, without selecting a characteristic line in its center.

Let $M \in \mathrm{GL}_2(k)$ be the matrix on the column $(a, b)^T$. The induced action on the extension quotient is

$$\begin{pmatrix} [A \circ \phi] \\ [B \circ \phi] \end{pmatrix} = (\det M)^2 M \begin{pmatrix} [A] \\ [B] \end{pmatrix}. \tag{245}$$

It suffices to check swaps, diagonals, and upper shears, which generate $\mathrm{GL}_2(k)$. The swap interchanges A, B, and the diagonal $\operatorname{diag}(\alpha, \eta)$ acts by $(A, B) \mapsto (\eta^2 A, \alpha^2 B)$. For the upper shear with parameter $u \in k$, use the original-coordinate automorphism

$$(a, b, c, d, e, h) \mapsto (a + ub, b, c + ub^2, d + ue, e, h).$$

Its pullback sends

$$A \mapsto A + uB + u^2 ab + u^3 b^2, \qquad B \mapsto B + u^2 b^2.$$

The last terms vanish in H_g, because $ab = \delta c$ and $b^2 = \delta(b^2)$. Triviality of the IA action now proves (245) for every ϕ. It follows that the line $\langle [A] \rangle$ is fixed exactly when $M \in P$, in which case its scalar is η^2. Thus the stabilizer has order $36 \cdot 4^8$, and the lift fiber gives $|\operatorname{Aut}(F_{14})| = 36 \cdot 4^{10}$.

Finally P has the following simultaneous lift to the original coordinates of F_{14}:

$$\begin{aligned} a' &= \alpha a, & b' &= \gamma a + \eta b, \\ c' &= \alpha\eta c + \alpha\gamma a^2, & d' &= \alpha^2 \eta d, \\ e' &= \alpha\eta^2 e + \alpha\eta\gamma d, & h' &= \alpha^2\eta^2 h, \\ r' &= \eta^2 r + \gamma^2 a^2. \end{aligned}$$

Substitution verifies commutation with F_{14} and multiplication of the triangular matrices. This is a section, proving $\operatorname{Aut}(F_{14}) = N_{14} \rtimes P$. The determinant-one subgroup of P is $k^+ \rtimes k^* \cong A_4$;

the scalar matrices give a complementary central C_3. Therefore $P \cong C_3 \times A_4$. The kernel N_{14} fits into the central extension

$$1 \longrightarrow \mathrm{Hom}_k(k^2, k) \longrightarrow N_{14} \longrightarrow N_{12} \longrightarrow 1.$$

No splitting of this last extension is required or asserted. □

N Proof of global polar nilpotence

Throughout this appendix, $L = L_f$, $B(x, y) = [x, y]$, and only the unconditional identities of Section 2 are assumed. In particular B is a Lie bracket, $[x, f(x)] = 0$, and $f^2(x) = f(x) + x$. No nilpotence hypothesis is used until the final invocation of Theorem 3.4.

N.1 Two biderivations in the enveloping algebra

For an alternating bilinear map $C : L \times L \to L$, write $C_x(y) = C(x, y)$. A biderivation means that every C_x is a derivation; in characteristic two an alternating bilinear map is symmetric, so this also gives the derivation rule in the first variable.

Lemma N.1. *The maps D, G, S of (26) are alternating binary-bilinear biderivations with values in L'. With $A = \operatorname{ad} x$ and $F = \operatorname{ad} f(x)$, one has*

$$S_x = F^2 + AF + A^3 + A^2. \tag{246}$$

They also satisfy

$$S(x, f(x)) = 0, \qquad \sum_{\mathrm{cyc}} S([x, y], z) = 0. \tag{247}$$

Proof. The polarization of $D(x, y) = [f(x), y] + [x, [x, y]]$ in its first variable is zero: the extra terms are $[[x, z], y] + [x, [z, y]] + [z, [x, y]]$, which vanish by Jacobi. Thus D is bilinear. The commuting identity gives $D(x, x) = 0$, and $D_x = \operatorname{ad} f(x) + (\operatorname{ad} x)^2$ is a derivation in characteristic two. Hence D is an alternating biderivation.

In the ordinary enveloping algebra $U(L)$, $(x + y)^2 = x^2 + y^2 + [x, y]$. The polar term in $f(x + y)$ cancels this last summand, so $R_x = x^2 + f(x)$ is binary-linear. Consequently $G(x, y) = [R_x, R_y]$ and $S = G+B+D$ are alternating bilinear. The identities $[x^2, y] = [x, [x, y]]$ and $[x^2, y^2] = [x, [y, [x, y]]]$ give

$$\begin{aligned} G(x, y) = {} & [x, [y, [x, y]]] + [x, [x, f(y)]] \\ & + [y, [y, f(x)]] + [f(x), f(y)]. \end{aligned} \tag{248}$$

In particular these commutators lie in L', not merely in $U(L)$.

To calculate S_x, put $b = [x, y]$ and $c = [f(x), y]$. The polarized commuting identity, first at (x, y) and then at $(f(x), y)$, gives

$$[x, f(y)] = c + [x, b] + [y, b], \qquad [f(x), f(y)] = b + c + [f(x) + y, c].$$

Substitution in (248) gives

$$G(x, y) = b + c + [x + f(x), c] + [x, [x, b]],$$

and hence (246). Since $[x, f(x)] = 0$, the operators A, F commute. Thus S_x commutes with A, or

$$S(x, [x, y]) = [x, S(x, y)]. \tag{249}$$

The commuting elements $x, f(x)$ also make R_x and $R_{f(x)} = f(x)^2 + f(x) + x$ commute. Therefore $S(x, f(x)) = 0$. Its third additive polarization, using that the polar of f is B, is the second identity in (247). Polarizing (249) in x at $x + z$ yields

$$S(x, [z, y]) + S(z, [x, y]) = [x, S(z, y)] + [z, S(x, y)].$$

The cyclic identity identifies the left side with $S([x, z], y)$. This is the derivation rule. Hence S, and then $G = S + B + D$, are biderivations. □

Lemma N.2. *The biderivation G satisfies*

$$G([x, y], z) = [G(x, y), z].$$

If L is perfect and centerless, there is a centroid operator C such that $G(x, y) = C[x, y]$.

Proof. For every alternating biderivation Q in characteristic two, expansion of the derivation identities gives $\sum_{\text{cyc}} Q([x, y], z) = 0$. Another expansion gives

$$\sum_{\text{cyc}} [Q(x, y), z] = Q([x, y], z) + [Q(x, y), z].$$

Jacobi for R_x, R_y, R_z, together with $[R_x, w] = D(x, w)$, gives $\sum_{\text{cyc}} D(G(x, y), z) = 0$. Put $T_x = x + R_x$. Its mixed commutators cancel, so

$$[T_x, T_y] = B(x, y) + G(x, y), \qquad [w, T_z] = (B + D)(w, z).$$

Jacobi for the three T's therefore reads

$$0 = \sum_{\text{cyc}} (B + D)((B + G)(x, y), z).$$

The B-Jacobi sum, the biderivation sum $\sum D(B(x, y), z)$, and the preceding R-Jacobi sum vanish. The remaining sum is $\sum B(G(x, y), z) = 0$, which, by the displayed expansion, is the asserted identity. In particular no adjoint cocycle property of D has been assumed here.

When $L = L'$, define

$$C\left(\sum_i [x_i, y_i]\right) = \sum_i G(x_i, y_i).$$

If $\sum_i [x_i, y_i] = 0$, the asserted identity says that the proposed image commutes with every $z \in L$. Centerlessness makes that image zero, so C is well defined. The same identity gives $C[w, z] = [Cw, z] = [w, Cz]$; these are the centroid identities. □

N.2 Simple algebras with a larger centroid

Assume for a contradiction that L is a nonabelian simple polar algebra. Its centroid K is a finite field. By Lemma N.2, $G = \kappa B$ for some $\kappa \in K$. Every binary derivation d of L is K-linear. Indeed, $[d, c]$ is again in the centroid for $c \in K$, and $c \mapsto [d, c]$ is a derivation of the finite separable field K; such a derivation is zero. Thus every D_x is K-linear. Symmetry then makes D K-bilinear in both arguments.

Jacobi in $U(L)$, at R_x, R_y, z, gives

$$[D_x, D_y] = \operatorname{ad} G(x, y) = \kappa \operatorname{ad}[x, y]. \tag{250}$$

Choose $a \in K$ with $a^2 = \kappa$, and put $E(x, y) = D(x, y) + a[x, y]$. This is an alternating, commutative bilinear product whose left multiplications commute: the mixed commutators cancel

by $[D_x, \operatorname{ad} y] = \operatorname{ad} D(x,y)$, and the two remaining terms cancel by (250). Commutativity and commuting left multiplications imply associativity, since $(x*y)*z = z*(x*y) = x*(z*y) = x*(y*z)$. Alternation therefore gives

$$E_x^2 = 0, \qquad D_x^2 = \kappa(\operatorname{ad} x)^2. \tag{251}$$

Write $A = \operatorname{ad} x$ and $H = D_x^2 + AD_x + A^2$. Substituting $F = D_x + A^2$ in (246), and using $G = S + B + D$, gives

$$G_x = H + A^4 + D_x + A. \tag{252}$$

Apply this identity at λx, and subtract λ times its value at x. Scalar bilinearity of D, G yields

$$(\lambda^2 + \lambda)H + (\lambda^4 + \lambda)A^4 = 0 \qquad (\lambda \in K). \tag{253}$$

If $|K| \geq 8$, a polynomial of degree at most four vanishing at all elements of K has zero coefficients, so $A^4 = 0$. If $K = \mathbb{F}_4$, choose $\lambda \notin \mathbb{F}_2$. Then $H = 0$, and (252) gives $D_x = A^4 + (\kappa+1)A$. Substitution in $H = D_x^2 + AD_x + A^2 = 0$, using (251), gives $A^5 = 0$. Engel's theorem makes L nilpotent in both cases, contradicting nonabelian simplicity.

N.3 The binary-centroid case

It remains that $K = \mathbb{F}_2$. Now $a = \kappa \in \{0, 1\}$. For fixed x, set $N = E_x = D_x + \kappa A$. Equations (251) and (252) become

$$N^2 = 0, \quad [N, A] = 0, \quad N(A + I) = A^4 + A^2 + A. \tag{254}$$

Squaring the last equality yields

$$A^8 + A^4 + A^2 = 0. \tag{255}$$

First there are no nonzero ad-nilpotent elements of L. For if A is nilpotent, (255) gives $A^2(I + A^2 + A^6) = 0$. The second factor is invertible, so $A^2 = 0$. Equation (254) then gives $N = A$. For every y,

$$\operatorname{ad} E(x,y) = [\operatorname{ad} x, E_y] = [E_x, E_y] = 0.$$

Centerlessness gives $E_x = 0$, and hence $A = 0$, so $x = 0$.

The absence of ad-nilpotent elements is not yet a semisimplicity assertion. The next argument removes the square-zero product itself. Associativity gives $E_x E_y = E_{E(x,y)}$, so this product of derivations is again a derivation. Its Leibniz cross terms yield

$$[E_x u, E_y v] + [E_y u, E_x v] = 0.$$

Set $u = x$, and use $E_x x = 0$ and symmetry. It follows that $\operatorname{im} E_x$ is an abelian subalgebra for the original bracket. Fix $N = E_x$ and let $z = Ny$. Because $N^2 = 0$ and $\operatorname{im} N$ is abelian, the derivation rule gives

$$\operatorname{ad}_z(L) \subseteq \ker N, \qquad \operatorname{ad}_z(\ker N) \subseteq \operatorname{im} N, \qquad \operatorname{ad}_z(\operatorname{im} N) = 0.$$

For the first inclusion, $N[z, v] = [Nz, v] + [z, Nv] = 0$. For the second, if $Nv = 0$, then $[Ny, v] = N[y, v]$. The third is exactly the abelianness of $\operatorname{im} N$. Thus $(\operatorname{ad} z)^3 = 0$. The preceding paragraph forces $z = 0$, so $E_x = 0$ for every x.

Equation (254) now reduces to

$$(\operatorname{ad} x)^4 + (\operatorname{ad} x)^2 + \operatorname{ad} x = 0.$$

The polynomial $t^4 + t^2 + t$ has derivative one. Every adjoint operator is therefore semisimple over an algebraic closure: L is anisotropic. Premet's theorem says that a finite-dimensional anisotropic ordinary Lie algebra over a perfect C_1-field of positive characteristic is commutative [34, Theorem 3(ii)]. The finite field $\mathbb{F}_2$ is perfect and C_1, so this contradicts nonabelian simplicity. Equivalently, the displayed identity implies $(\operatorname{ad} x)^8 = \operatorname{ad} x$, and [34, Corollary 1] gives the same conclusion.

N.4 Conclusion

All possible finite centroids of a nonabelian simple polar algebra have been excluded. Corollary 2.5 now implies that every finite quadratic polar algebra is solvable. Theorem 3.4 then proves Theorem 4.1.

The specific interface with the classical results is the pair of biderivations in Lemma N.1 and the adjoint-closed map in Lemma N.2. Premet's theorem is applied only after semisimplicity has been proved, rather than inferred from the weaker absence of ad-nilpotent elements. No finite nilpotent model, Engel identity for the full class of solutions, or scalar extension of the nonlinear map f enters this argument.

O Proofs for the canonical general structure

O.1 Central splitting and the freeness criterion

We first prove Lemma 5.1 independently of any uniform dimension bound or free-object construction in higher rank. For a binary-linear intertwiner T, polarization shows that $T[x, y] = [Tx, Ty]$. The smallest subsolution containing marked points ξ_i is the Lie algebra generated by $\xi_i, g(\xi_i)$: the generating set is stable under g modulo its binary span, and invariance propagates through sums by quadratic polarization and through brackets by (6).

Proof of Lemma 5.1. By Theorem 4.1, every finite target has nilpotent polar algebra. Induct on the binary dimension of such a target (V, f). If its polar algebra is abelian, f is linear and is its intrinsic $\mathbb{F}_4$-scalar. The required map is the $\mathbb{F}_4$-linear map from the marked basis of G/G'.

Otherwise the target center Z is nonzero by nilpotence. Proposition 2.3 makes Z invariant, with linear scalar $f|_Z = \omega$. The smaller solution V/Z has, by induction, a map $h : G \to V/Z$ with the prescribed marked values. Its pullback

$$P_h = \{(x, v) \in G \oplus V : h(x) = v + Z\}$$

is a subsolution of the product and is a central extension of G by (Z, ω). To see its splitting criterion directly, choose a binary-linear section and write its map as

$$(x, z) \longmapsto (g(x), \omega z + q(x)).$$

The functional equation is exactly $qg = \omega^2 q$; changing the section by a linear $t : G \to Z$ changes q by $tg + \omega t$. Thus its class belongs to $H_g(k) \otimes_k Z = 0$. A linear solution section therefore exists. Project it to obtain an intertwining lift $S : G \to V$.

The differences between the prescribed target points and $S\xi_i$ lie in Z. A k-linear map $G/G' \to Z$ can assume exactly these differences on the marked basis. Its inflation t satisfies $tg = \omega t$, so centrality makes $S + t$ an intertwiner with the exact prescribed values. Uniqueness follows because those values determine the images of $\xi_i, g\xi_i$, which Lie-generate G. □

The freeness criterion uses the full elementary central-extension space (28). The sixteen-dimensional coordinate map (115) has $136 = 16 + \binom{16}{2}$ zero-constant binary monomials of degree at most two with coefficients in k. The equation $qF_{16} + \omega^2 q = 0$ therefore has 272 binary unknowns. The final certificate of Table 3 gives

$$\operatorname{rank} = 244, \qquad \dim_{\mathbb{F}_2} \ker = 28, \qquad \dim_{\mathbb{F}_2} \operatorname{im} \delta = 28.$$

It follows that $H_{F_{16}}(k) = 0$. Appendix X describes the exact row, kernel, and section-image checks. The sparse bracket table below also verifies that the points a, b, together with their

images, Lie-generate the sixteen-dimensional space: successive bracket closure has binary dimensions $4, 8, 16$. Its abelianization is the two-dimensional k-space on a, b. The freeness criterion is therefore applicable. By Theorem 4.1, every finite target is nilpotent, so every prescribed pair in a finite solution receives a homomorphism from this free solution. Its image is precisely the subsolution generated by that pair.

O.2 Identities on the free two-generator solution and their propagation

We give the coordinate part of Theorem 5.2 in a form that can be checked without the intermediate-prefix classification. Use the k-coordinate axes $e_a, e_b, e_c, e_d, e_e, e_h, e_r, e_s$ of (115). For arbitrary $\alpha, \beta \in k$, the nonzero polar products between distinct axes are

(i,j)	$[\alpha e_i, \beta e_j]$
(a,b)	$\alpha\beta e_c + \alpha^2\beta e_d + \alpha\beta^2 e_e$
(a,c)	$\alpha\beta e_d + \alpha\beta^2 e_r$
(b,c)	$\alpha\beta e_e + \alpha\beta^2 e_s$
(a,e)	$\alpha\beta e_h + \alpha^2\beta e_r$
(b,d)	$\alpha\beta e_h + \alpha^2\beta e_s$
(a,h)	$\alpha\beta e_r$
(b,h)	$\alpha\beta e_s$
(c,d)	$\alpha\beta e_r$
(c,e)	$\alpha\beta e_s$.

(256)

All omitted products, including those on a single k-axis, are zero; reversing the inputs uses symmetry. The entries record the binary-bilinear bracket in the displayed k-coordinates.

Let the first two coordinates of x be a, b, and those of y be A, B, and put $z = aB + Ab$. Substitution of (256) in the definitions of D and (248) gives

$$D(x,y) = \omega[x,y] + z^2 e_h, \qquad G(x,y) = \omega^2[x,y] + z^2 e_h. \tag{257}$$

Here the scalar multiplication on the right is the displayed coordinate scalar. One economical way to verify both formulas is to use bilinearity of B, D, G: for each row of (256), the residuals $D + \omega B$ and $G + \omega^2 B$ are zero, except in row (a,b), where each equals $\alpha^2\beta^2 e_h$. The single-axis residuals are zero. These formulas give all binary coefficients and prove $G = B + D$ on the entire free solution.

For the nonhomogeneous identity, put $w = [x,y]$ and $v = [x,w]$. If C is the c-coordinate of y, the table gives

$$\begin{aligned} &w_c = z, \quad w_d = aC + Ac + a^2B + A^2b, \quad w_e = bC + Bc + aB^2 + Ab^2, \\ &v_c = 0, \quad v_d = az, \qquad\qquad v_e = bz, \qquad\qquad v_h = aw_e + bw_d. \end{aligned}$$

The third adjoint has only its final two coordinates, and they are

$$\begin{aligned} ([x,v])_r &= a^2bz + a^2w_e + abw_d + acz = az^2, \\ ([x,v])_s &= ab^2z + abw_e + b^2w_d + bcz = bz^2. \end{aligned}$$

The equalities follow by substituting w_d, w_e, z and cancelling paired terms. The only nonzero coordinates of $[w,v]$ are $w_c v_d = az^2$ and $w_c v_e = bz^2$, respectively. Hence $[x,[x,[x,y]]] = [[x,y],[x,[x,y]]]$ for all inputs in this model.

The table gives lower central dimensions $(16, 12, 10, 6, 4, 0)$, with final term $ke_r + ke_s$. The third adjoint just computed lies in that center, so the fourth adjoint is zero. At $x = e_a, y = e_b$ its third adjoint is $e_r \neq 0$. Every pair in an arbitrary finite solution is the image of this

marked pair under the freeness criterion. All three two-variable identities therefore descend, proving (29)–(31), including sharpness.

It remains to prove the three-variable identity, without any three-generator finite input. By Lemma N.1, S_x, F^2, A^2 are derivations, where $A = \operatorname{ad} x$ and $F = \operatorname{ad} f(x)$. Equation (246) therefore makes $AD_x = AF + A^3$ a derivation. Its Leibniz cross term is

$$[[x, y], D(x, z)] + [D(x, y), [x, z]] = 0.$$

Substitute $D_x = F + A^2$. The terms involving F vanish by (7) at $(x, f(x), y, z)$, since $[x, f(x)] = 0$. The remaining terms are

$$[[x, y], [x, [x, z]]] + [[x, [x, y]], [x, z]] = [x, [[x, y], [x, z]]],$$

as required. This completes the proof of Theorem 5.2.

O.3 The Lie pencil and canonical scalars

Proof of Theorem 5.3. For every alternating biderivation, the cyclic sum $\sum D([x, y], z)$ is zero. Jacobi for R_x, R_y, R_z, and the identity $G = B + D$, consequently give

$$0 = \sum_{\text{cyc}} D(G(x, y), z) = \sum_{\text{cyc}} D(D(x, y), z).$$

This is Jacobi for D. Put $T_x = x + R_x$. Now $[T_x, T_y] = B(x, y) + G(x, y) = D(x, y)$, and Jacobi for the three T's gives $\sum(B + D)(D(x, y), z) = 0$. Subtract Jacobi for D to obtain $\sum[D(x, y), z] = 0$. For a biderivation the expansion used in Lemma N.2 identifies this with $D([x, y], z) = [D(x, y), z]$. Thus both brackets and the mixed Jacobi coefficient vanish, proving the pencil assertion after any scalar extension. Finally Jacobi at R_x, R_y, z gives $[D_x, D_y] = \operatorname{ad} G(x, y)$, which is (34). □

We next construct J. For any expressions $w = \sum_i [x_i, y_i]$ and $u = \sum_i D(x_i, y_i)$, (33) gives the ambient identity

$$D_w = \operatorname{ad}_u \quad \text{on all of } L. \tag{258}$$

If $w = 0$, then $u \in W \cap Z(L) = C$. If $w \in C$, center invariance gives $D_w = 0$, so again $u \in C$. This proves both independence of the bracket expression and descent to the quotient $E = W/C$.

Jacobi for D gives $[D_x, D_y] = D_{D(x,y)}$. Combine this with (34) and (258). Representing the successive J-images in W, the result is

$$\operatorname{ad}_{J^2([x,y]+C)} = \operatorname{ad}_{[x,y]+D(x,y)}.$$

Equality here means equality of ambient adjoints; its two representatives differ by C. Bracket values span W, so $J^2 + J = \mathrm{id}$ on E. If $w \in W$, $z \in L$, and u represents $J(w + C)$, then

$$J([w, z] + C) = D(w, z) + C = [u, z] + C.$$

This proves commutation with every ambient adjoint. Taking $z \in W$ and using symmetry proves both centroid identities.

O.4 The canonical restricted square

Both W and $Z(L)$ are invariant. For $c \in C$, $f(w+c) = f(w) + f(c)$, with $f(c) \in C$; hence $\bar{f}$ is well defined. Its polar is the quotient bracket, as is that of $P = \bar{f} + J$.

For $w \in W$, choose $u \in W$ representing $J(w + C)$. By (258), the vector $p(w) = f(w) + u$ represents $P(w + C)$, and

$$\mathrm{ad}_{p(w)} = (\mathrm{ad}_w)^2 \quad \text{on all of } L. \tag{259}$$

We prove scalar semilinearity in this ambient representation, where the kernel on W is exactly C. Set

$$A = \mathrm{ad}_w, \qquad F = \mathrm{ad}_{f(w)}, \qquad Q = D_w = F + A^2 = \mathrm{ad}_u\,.$$

The operators A, F commute. Applying (259) to u gives

$$\mathrm{ad}_{p(u)} = Q^2 = F^2 + A^4.$$

On the other hand $J^2 = I + J$ and (258) give, for $v \in W$, $\mathrm{ad}_{J^2(v+C)} = \mathrm{ad}_v + D_v$, with representatives understood. For $v = p(w) = f(w) + u$,

$$\mathrm{ad}_v = A^2, \qquad D_{f(w)} = F + A + F^2, \qquad D_u = A + Q.$$

Here the middle equality uses $f^2(w) = f(w) + w$, and the last uses $J^2 = I + J$. Linearity of D consequently gives

$$\mathrm{ad}_{J^2 P(w+C)} = A^2 + (F + A + F^2) + (A + Q) = F^2.$$

The difference from $\mathrm{ad}_{p(u)}$ is $A^4 = 0$, by (31). Thus $P(J(w + C)) = J^2 P(w + C)$. The other nontrivial scalar follows from $J^2 = I + J$, the polar rule, and $[v, Jv] = J[v, v] = 0$. Together with (259), these are exactly the characteristic-two restricted Lie axioms over k.

Finally $[v, P(v)] = 0$ follows from restrictedness. In expanding the equation for $\bar{f} = J + P$, therefore,

$$\begin{aligned} \bar{f}^{\,2}(v) &= J^2 v + JP(v) + P(Jv) + P(P(v)) + [Jv, P(v)] \\ &= J^2 v + P(v) + P(P(v)). \end{aligned}$$

But $\bar{f}^{\,2}(v) = \bar{f}(v) + v = J^2 v + P(v)$, so $P^2 = 0$. Every operation is intrinsic. An isomorphism, or a surjection, of solutions preserves the global-center quotients and their bracket and D-values, hence preserves J, P. This proves Theorem 5.4 with its stated naturality.

The unreduced derived ideal also admits a restricted square, although the above proof does not canonically select it. Choose a binary basis w_i of W and representatives u_i for $J(w_i + C)$, prescribe $p(w_i) = f(w_i) + u_i$, and extend by the restricted addition rule. The identity (259) then holds on all of W. Central choices may change this square, so the canonical k-structure asserted in the theorem remains that on E.

O.5 The reduced adjoint representation

Proof of Theorem 5.13. The adjoint action preserves W, C, so it induces the Lie homomorphism ρ on E. Commutation with J, already proved, makes each operator k-linear. For $w \in W$, symmetry and (258) give

$$D(x, w) + C = D(w, x) + C = [J(w + C), x] = J\rho(x)(w + C).$$

Thus $D_x|_E = J\rho(x)$. Since $D_x = \mathrm{ad}_{f(x)} + (\mathrm{ad}_x)^2$, we obtain $\rho(fx) = \rho(x)^2 + \omega\rho(x)$. The fourth-power identity used in the scalar-envelope argument follows from (31). More precisely, Theorem 5.7 gives $(\mathrm{ad}_x)^3(W) = 0$, and hence the associative matrix identity $\rho(x)^3 = 0$ on E. Its proof in Appendix P uses only the canonical restricted structure and its faithful ambient

adjoints, rather than this matrix realization. Matrix cubes here are distinct from the Lie-adjoint cubes appearing in (49).

The kernel K is a Lie ideal, and the polynomial rule makes it f-stable. Its defining condition is $[K, W] \subseteq C$. Since $[K, K] \subseteq W$, it follows that $\gamma_3(K) \subseteq C$ and $\gamma_4(K) = 0$.

To check restricted derivations, let $\delta = \rho(x)$ and $u = w + C \in E$. The polarized commuting identity gives

$$\begin{aligned}\delta(\bar{f}(u)) &= \rho(fx)u + \delta^2 u + [u, \delta u] \\ &= (\delta^2 + \omega\delta)u + \delta^2 u + [u, \delta u] = \omega\delta u + [u, \delta u].\end{aligned}$$

Subtract $\delta(Ju) = \omega\delta u$. The result is $\delta(Pu) = [u, \delta u]$, the restricted-derivation rule. These derivations form a restricted k-Lie subalgebra: the rule is preserved by scalar sums, commutators, and squares, either by the restricted identities or by direct application of the derivation rule. □

O.6 The scalar envelope and its square-zero layer

The matrix image $X = \rho(L)$ carries the solution $F(A) = \omega A + A^2$. Its own special biderivation is

$$D_X(A, B) = [F(A), B] + [A, [A, B]] = \omega[A, B], \tag{260}$$

because $\mathrm{ad}_{A^2} = (\mathrm{ad}_A)^2$ in the associative matrix algebra. In particular the binary derived space X' is already k-linear. The scalar span $N = kX = X + \omega X$ is bracket closed. For $A \in X$, $A^2 = F(A) + \omega A \in N$; expanding a square of a scalar sum then proves that N is square closed. Hence it is exactly the restricted closure of X. Further,

$$N' = kX' = X' = \rho(L') = \mathrm{Inn}(E), \qquad N'' = \mathrm{ad}_E(E').$$

The equality with inner derivations uses the surjection $W \to E$ and does not identify E with its inner derivation algebra.

We first check the ordinary three-Engel identity on the actual binary image. For $A, B \in X$, put

$$K_{21} = A^2B + ABA + BA^2, \quad K_{12} = AB^2 + BAB + B^2A,$$
$$Q = A^2B^2 + AB^2A + B^2A^2, \quad H = BAB^2 + B^2AB, \quad R = B^2AB^2.$$

The zero cube of $A + B$ gives $K_{21} = K_{12} =: K$. For each $\lambda \in \{\omega, \omega^2\}$, the square companion $\lambda B + B^2$ is an actual element of X. Expanding its sum with A gives $0 = K + Q + \lambda H + R$. Subtracting the two equations yields $H = 0$, namely $(\mathrm{ad}_B)^3 A = 0$.

For $A, B \in X$, now expand in the associative matrix algebra:

$$(A + \lambda B)^4 = C_1\lambda + C_2\lambda^2 + C_3\lambda^3.$$

The constant and fourth coefficients vanish since $A^4 = B^4 = 0$. The two outer coefficients are $C_1 = (\mathrm{ad}_A)^3 B$ and $C_3 = (\mathrm{ad}_B)^3 A$. At $\lambda = 1$, the fourth power is again zero, giving $C_2 = C_1 + C_3$. Substitution of $\lambda = \omega$ proves (49).

The ordinary three-Engel identity just proved makes both outer coefficients zero. Thus (49) vanishes for every A, B, and every element of $N = X + \omega X$ has fourth matrix power zero. The restricted fourth-power expansion of $U + \lambda V$, for $U, V \in N$, has degree at most three after removing its zero pure fourth powers. Its vanishing at all four scalars of k makes its linear coefficient $(\mathrm{ad}_U)^3 V$ zero. Hence N is an ordinary three-Engel k-Lie algebra.

For the further reduction $\mathcal{M} = N'' \cap Z(N')$, an inner derivation $\mathrm{ad}_E e$ centralizes $\mathrm{Inn}(E)$ precisely when $[e, E] \subseteq Z(E)$, that is, $e \in Z_2(E)$. Since $N'' = \mathrm{ad}_E(E')$, a representative may be chosen in $E' \cap Z_2(E)$; changing a representative adds an element of $Z(E) \subseteq Z_2(E)$. This proves (47). Every such matrix maps E into $Z(E)$ and kills $Z(E)$, so the product of any two

is zero. Both N'' and $Z(N')$ are ideals of N, the latter by Jacobi and invariance of N'. Their intersection $\mathcal{M}$ is consequently an ideal, and its zero matrix square makes it a restricted ideal. This proves Theorem 5.14, including the optional square-zero reduction $\mathcal{M}$.

O.7 The completion and its kernel

All operations on $\mathcal{T} = N/\mathcal{M}$ are restricted Lie operations. In particular the square descends because $\mathcal{M}$ is a restricted ideal; an associative quotient is neither asserted nor needed. With $P_{\mathcal{T}}(T) = T^{[2]}$, its polarization is the quotient bracket and $P_{\mathcal{T}}^2 = 0$. The restricted rules give

$$F^2(T) = \omega^2 T + T^{[2]} + T^{[4]} = F(T) + T.$$

This proves (50). As a Lie quotient of N, the restricted algebra $\mathcal{T}$ is also ordinary three-Engel.

Proof of Theorem 5.15. The polynomial rule for ρ proves that $q(x) = \rho(x) + \mathcal{M}$ intertwines the quadratic maps. Its kernel I is an f-stable Lie ideal. Since $\mathcal{M}$ is abelian, $[I, I] \subseteq K \cap W$, where $K = \ker \rho$. For $w \in K \cap W$, the inner derivation $\rho(w)$ is zero, so $w + C \in Z(E)$. For $i \in I$, the operator $\rho(i) \in \mathcal{M}$ is inner and hence kills $Z(E)$. Consequently $[i, w] \in C$. Thus $\gamma_3(I) \subseteq C$ and $\gamma_4(I) = 0$. This uses both the abelian image and its inner-action property.

As $\mathcal{M} \subseteq X' \subseteq X$, the image of q is $Y = X/\mathcal{M}$, and

$$\mathcal{T}' = N'/\mathcal{M} = X'/\mathcal{M} \subseteq Y.$$

It follows that Y is a binary Lie ideal of $\mathcal{T}$. Intertwining gives its F-stability, and $N = kX$ gives $kY = \mathcal{T}$. The induced map on the binary quotient $\mathcal{T}/Y$ is well defined by the quadratic addition rule and the ideal property. Its bracket is zero, so it is linear and satisfies $F^2 = F + \mathrm{id}$. This proves the two exact sequences and the abelian-cokernel assertion. □

Naturality under surjections extends to the envelope and completion. A surjective solution homomorphism $S : L \to L_1$ induces a surjective restricted k-Lie map $S_E : E \to E_1$. The rule

$$\Phi\left(\sum_i \lambda_i \rho(x_i)\right) = \sum_i \lambda_i \rho_1(S x_i)$$

is well defined, since $S_E A = \Phi(A) S_E$ and surjectivity detects a zero target operator. The same intertwining proves that Φ preserves brackets and squares. It is surjective, so it sends $N'' \cap Z(N')$ into $N_1'' \cap Z(N_1')$, giving the required maps on the quotients. For a nonsurjective solution map even preservation of the original global center can fail; the argument supplies no such map on reduced-derived quotients.

The restricted ideal generated by the fourth powers of N is zero. Thus $\rho : L \to N$ already intertwines the quadratic rules f and $A \mapsto \omega A + A^2$. The quotient by $\mathcal{M}$ is a further square-zero-ideal reduction that records the canonical kernel and cokernel displayed above.

O.8 The central obstruction on the whole derived ideal

Proof of Proposition 5.5. On the binary space underlying k^{11}, use coordinates

$$(a, b, c, d, z, v, w, s_1, s_2, s_3, t).$$

Define $f = \omega\,\mathrm{id} + q$, where the first four coordinates of q are zero and

$$\begin{aligned}
&q_z = ab + cd, && q_v = ad, && q_w = bd,\\
&q_1 = az + cv + a^2 b, && q_2 = bz + cw + ab^2, && q_3 = aw + bv,\\
&q_t = as_2 + bs_1 + cs_3.
\end{aligned}$$

All have binary degree at most two, since squaring in k is binary-linear. We verify the equation on every input, by showing $qf = \omega^2 q$. The first three nonzero coordinates are immediate. For example,

$$q_1(fx) = \omega^2(az + cv) + \omega a(ab + cd) + \omega c(ad) + a^2 b = \omega^2 q_1(x).$$

The two acd-terms cancel and $\omega + 1 = \omega^2$. The same argument gives the formula for q_2; the extra terms for q_3 are twice ωabd. Finally

$$q_t(fx) = \omega^2 q_t(x) + \omega(aq_2 + bq_1 + cq_3),$$

where the parenthesis is zero: each of abz, acw, bcv, a^2b^2 occurs twice. Thus $f^2(x) = \omega^2 x + q(x) = f(x) + x$.

Use upper-case letters for the coordinates of a second vector. The complete nonzero polar components are

$$\begin{aligned}
&B_z = aB + Ab + cD + Cd, && B_v = aD + Ad, \qquad B_w = bD + Bd,\\
&B_1 = aZ + Az + cV + Cv + a^2B + A^2b,\\
&B_2 = bZ + Bz + cW + Cw + aB^2 + Ab^2,\\
&B_3 = aW + Aw + bV + Bv,\\
&B_t = aS_2 + As_2 + bS_1 + Bs_1 + cS_3 + Cs_3.
\end{aligned}$$

The derived ideal is $W_0 = \langle z, v, w, s_1, s_2, s_3, t\rangle_k$, and its bracket is zero. Indeed the products $(c,d), (a,d), (b,d), (a,z), (b,z), (a,w), (a,s_2)$, with varying scalar, span the seven displayed axes. The same table gives

$$\gamma_3(L) = \langle s_1, s_2, s_3, t\rangle_k, \quad \gamma_4(L) = kt, \quad \gamma_5(L) = 0.$$

The center is exactly kt. Bracketing a central vector with s_1, s_2, s_3 first forces $b = a = c = 0$, and bracketing with a then forces $d = 0$. Within W_0, bracketing with a, b, c forces all six coordinates other than t to vanish. Thus $W_0 \cap Z(L) = kt$, as asserted.

The complete biderivation formula is

$$D(x,y) = \omega B(x,y) + (aB + Ab)^2 e_t. \tag{261}$$

Here ω denotes the displayed coordinate scalar, without any centroid assumption on the whole original polar bracket. To verify it by binary bilinearity, split the space into the top $U = \langle a, b, c, d\rangle_k$ and W_0. For top inputs, the lower polar table gives $D = \omega B$ except possibly in the t-component; that component reduces to

$$Aab^2 + Ba^2b + a(aB^2 + Ab^2) + b(a^2B + A^2b) = a^2B^2 + A^2b^2.$$

For top $x = (a, b, c, d)$, its action on W_0 is

u	z	v	w	s_1	s_2	s_3	t
$[x,u]$	$as_1 + bs_2$	$cs_1 + bs_3$	$cs_2 + as_3$	bt	at	ct	$0.$

The square of this operator is zero by pair cancellation. Thus $D(x,y) = \omega B(x,y)$ for $x \in U, y \in W_0$. On $W_0 \times W_0$ both sides vanish, proving (261).

Let e_i be the coordinate unit vectors, and take the binary exterior two-chain

$$\xi = e_a \wedge e_b + e_c \wedge e_d + e_a \wedge e_z + e_b \wedge e_z.$$

The four bracket values are $z + s_1 + s_2, z, s_1, s_2$, so $B(\xi) = 0$. Equation (261) gives $D(\xi) = e_t \neq 0$. A linear factor $J_W B = D$ would send this zero to a nonzero vector, a contradiction. $\square$

The counterexample retains the auxiliary k-structure of the metabelian normal form. What fails is the stronger prescription $J_W B = D$ on the whole original derived ideal. After reduction modulo kt, its scalar from (36) is precisely the displayed ω. Equivalently, with W_0 regarded as a trivial Lie module, D is a two-cocycle not of the form $T \circ B$; the cycle ξ detects its nonzero class. The general factorization on E and its restricted square are therefore consistent with this obstruction.

P Global stability and the universal restricted derived quotient

The proof of global stability uses the canonical restricted structure of $E = W/(W \cap Z(L))$, but no exact scalar factor on the whole derived ideal. We first obtain a low-class stability lemma directly from the two-orbit identities. Faithful ambient adjoints then control derived inputs in arbitrary class. A character projection on the abelianization and the cyclic norm complete the argument. Finite free objects enter only afterwards, to construct exact restricted covers and the universal quotient.

P.1 The exact cochain criterion and a low-class lemma

Let A be an alternating binary-bilinear cochain, with any finite binary coefficient space, and choose a zero-constant quadratic refinement q_A. Write

$$\mathcal{C}_A(x,y,z) = \Delta_x \Delta_y \Delta_z (q_A f)(0).$$

This is independent of the linear part of q_A. It vanishes identically exactly when $q_A f$ is quadratic. Direct polarization gives

$$\begin{aligned}\mathcal{C}_A(x,y,z) = {} & A(B(x,y),B(z,y)) + A(B(x,z),fy) \\ & + A(fx,B(z,y)) + A(fz,B(x,y)) \\ & + A(B(x,z),B(x,y)+B(z,y)).\end{aligned} \tag{262}$$

The expression is a third additive difference, not an assumed trilinear form.

For $A = B, D$, adjoint closedness converts (262) into commutators. Substitute $fv = R_v + v^2$ in $U(L)$, where $R_v = v^2 + f(v)$ is linear. For $A = B$, the three R-terms have sum $\sum_{\mathrm{cyc}} D(B(x,y),z) = 0$: expanding an alternating biderivation gives six terms that cancel in pairs. For $A = D$, their sum vanishes by Jacobi for D, the identity $D_{D(x,y)} = \mathrm{ad}_{B(x,y)+D(x,y)}$, and adjoint closedness. Thus

$$0 = \mathcal{C}_B(x,y,z) = \sum_{\mathrm{cyc}} \big([x,[x,B(y,z)]] + [B(x,z),B(x,y)]\big), \tag{263}$$

$$\mathcal{C}_D(x,y,z) = \sum_{\mathrm{cyc}} \big([x,[x,D(y,z)]] + [D(x,z),B(x,y)]\big). \tag{264}$$

The first left side is zero because f itself refines B and $f^2 = f + \mathrm{id}$.

We shall also use the exact bracket correction

$$f([u,v]) + D(u,v) = K(u,v), \qquad K(u,v) = [u,[u,[u,v]]] + [v,[v,[u,v]]] + [u,[v,[u,v]]]. \tag{265}$$

Here is an enveloping-algebra derivation. The function $R_x^2 + x^2$ has polar D, since $[R_x, R_y] = B(x,y) + D(x,y)$. As $[x, fx] = 0$, it equals $x^4 + f(x)^2 + x^2$. The identity $f(x)^2 = R_{f(x)} + f^2(x)$ then gives

$$\mathrm{pol}(x \mapsto x^4)(u,v) = B(u,v)^2 + f(B(u,v)) + D(u,v).$$

The ordinary fourth-power expansion gives the same polar as $B(u,v)^2 + K(u,v)$, proving (265). By (30), for $u \in \gamma_a L$ and $v \in \gamma_b L$,

$$K(u,v) \in \gamma_{2a+2b} L. \tag{266}$$

Indeed the three terms have depths at least $3a+2b$, $2a+3b$, and $2a+2b$, respectively.

Lemma P.1 (A low-class stability bound)**.** *For every finite solution, $\mathcal{C}_D(L,L,L) \subseteq \gamma_8 L$. Consequently D is stable whenever $\mathrm{cl}(L) \le 7$, in particular on every subsolution generated by two marked points.*

Proof. Write the six summands of (263) as $t_i = [u_i, v_i]$. For the nested terms take $u_i = x$, $v_i = [x,[y,z]]$, of depths $(1,3)$; for the other terms take $u_i = [x,z]$, $v_i = [x,y]$, of depths $(2,2)$, with cyclic permutations. Each $t_i \in \gamma_4 L$ and $\sum_i t_i = 0$. Quadratic expansion gives

$$\sum_i f(t_i) = \sum_{i<j} [t_i, t_j].$$

Adjoint closedness and the biderivation rule identify $\sum_i D(u_i,v_i)$ with $\mathcal{C}_D$. Applying (265) therefore gives

$$\mathcal{C}_D = \sum_i K(u_i,v_i) + \sum_{i<j} [t_i,t_j] \in \gamma_8 L.$$

The two-point assertion uses the free sixteen-dimensional solution, whose class is five, as established in Appendix O. □

P.2 An order-three lift and its central square defect

Put $C = W \cap Z(L)$, $E = W/C$, and $j = f|_C$. Choose a binary-linear lift $J : W \to W$ of the canonical scalar on E, with $J^2 + J = \mathrm{id}$ and $J|_C = j$. Such a lift exists by splitting the binary vector sequence $0 \to C \to W \to E \to 0$ and transporting the two given $\mathbb{F}_4$-scalar structures. Equation (258) gives

$$[Jw, z] = D(w,z) \qquad (w \in W, z \in L). \tag{267}$$

Define

$$S = f|_W + J, \qquad r = S^2, \qquad a = D + JB. \tag{268}$$

The cochain a takes values in C, is an ordinary trivial-coefficient CE cocycle, and satisfies $a(x, fx) = 0$. The map S has polar $B|_{W^2}$ and satisfies $\mathrm{ad}_{S(w)} = (\mathrm{ad}_w)^2$ on all of L; it is a binary restricted square. It induces the canonical P on E, and $S(C) = 0$. In particular $r(W) \subseteq C$.

Lemma P.2. *The map $r : W \to C$ is quadratic, with polar $A = a|_{W\times W}$, and vanishes on C.*

Proof. The identities $[w, Jw] = D(w,w) = 0$ and ambient compatibility give $[Jw, Sw] = (\mathrm{ad}_w)^2(Jw) = 0$. Expansion of $f^2 = f + \mathrm{id}$ on W, with $f = J + S$, gives $S(Jw) + JS(w) + S^2(w) + S(w) = 0$. Polarization yields

$$\mathrm{pol}(S^2)(u,v) = B(Ju, Jv) + JB(u,v) + B(u,v).$$

This is bilinear, proving quadraticity. Express u, v as sums of original bracket values. Their J-images differ from the corresponding D-sums only by central terms. The identity

$$[D(x,y), D(z,t)] = [B(x,y), (B+D)(z,t)]$$

follows from adjoint closedness and $D_{D(z,t)} = \mathrm{ad}_{B(z,t)+D(z,t)}$. It gives $B(Ju,Jv) = B(u,v) + D(u,v)$ exactly. Substitution proves the asserted polar. Finally $S(C) = 0$ gives $r(C) = 0$. □

Proposition P.3 (The central square-height class)**.** *The function $r = S^2$ descends to $r_E : E \to C$, satisfies $r_E \bar{f} = j r_E$, and defines a class*

$$[r_E] \in H_{\bar{f}}(C, j^2) := \frac{\{q : E \to C : \deg_2 q \le 2,\ q(0) = 0,\ q\bar{f} = jq\}}{\{t\bar{f} + j^2 t : t \in \mathrm{Hom}_{\mathbb{F}_2}(E, C)\}}. \tag{269}$$

This class is independent of the order-three lift J. It is natural under solution isomorphisms and surjective solution maps, with pullback on the quotient and the induced map on center coefficients. The coefficient operator in this central-extension space is j^2, the reverse of the original center operator.

Proof. The cochain a annihilates a central input, and $r(C) = 0$, so the quadratic addition rule makes r constant on cosets of C. The functional equation gives $S(Jw) = J^2 S(w) + r(w)$ and hence $S(J^2 v) = JS(v) + r(v)$. Since $S(C) = 0$ and $r(Sw) = S^3 w = S(rw) = 0$, these identities give $r(Jw) = jr(w)$. Moreover $a(Jw, Sw) = [J^2 w, Sw] = 0$. Thus $r(fw) = r(Jw + Sw) = jr(w)$, proving the eigencondition.

For another lift $J' = J + t$, the map $t : W \to C$ vanishes on C. Expanding the two scalar polynomials gives $tJ = j^2 t$, and $S' = S + t$ gives $r' = r + tS = r + (tf + j^2 t)$. Both t and this gauge equation descend to E, proving independence.

For a surjective solution map $T : L \to L_1$, choose lifts J, J_1 and put $u = TJ + J_1 T : W \to C_1$. It vanishes on C, satisfies $uJ = j_1^2 u$, and obeys $TS = S_1 T + u$. Since $S_1(C_1) = 0$, it follows that $Tr = r_1 T + uS = r_1 T + (uf + j_1^2 u)$. This is precisely the asserted equality of the pushed-forward and pulled-back classes. □

The class retains a central square defect; its definition does not identify its vanishing with global stability of D, nor assert that a representative on E extends to an admissible refinement on all of L.

Whenever a refinement on all of L is needed, fix a binary-linear extension $\widetilde{J} : L \to W$ of J. Then $\widetilde{J}f$ refines JB and is stable, since $\widetilde{J}f^2 = \widetilde{J}(f + \mathrm{id})$ is quadratic. Choose a C-valued quadratic refinement q_a with $q_a|_W = r$; the matching polars make this possible by a linear adjustment. Put $h = q_a f$.

P.3 Faithful ambient coefficients and the cube on W

The map

$$\iota : E \longrightarrow \mathrm{End}_{\mathbb{F}_2}(L), \qquad \iota(w + C) = \mathrm{ad}_w$$

is injective and preserves brackets and restricted squares, by (259). For $z \in L$, put $d = \mathrm{ad}_z$ and let δ be its induced k-linear derivation on E. Ambient compatibility gives $dS(w) = [w, dw]$, so δ is a restricted derivation of (E, P).

Fix $u \in E$ and write $T = \iota(u)$, $v = Pu + \delta u$. Squaring twice in $\mathrm{End}_{\mathbb{F}_2}(L)$ gives

$$(d + T)^2 = d^2 + \iota(v), \qquad (d + T)^4 = \iota(Pv + \delta^2 v).$$

The fourth power is zero by (31), since $d + T$ is an actual ambient adjoint. Use $P^2 = 0$ and $\delta^2 Pu = [u, \delta^2 u]$ to obtain, by injectivity,

$$P\delta u + \delta^3 u + [u, \delta^2 u] + [Pu, \delta u] = 0.$$

For each $\lambda \in k$, a representative of λu in W again gives an actual ambient adjoint. Substitution gives

$$\lambda\delta^3 u + \lambda^2\big(P\delta u + [u, \delta^2 u]\big) + \lambda^3 [Pu, \delta u] = 0.$$

This degree-three polynomial vanishes at all four field elements; therefore

$$\delta^3 u = 0, \qquad P\delta u = [u, \delta^2 u], \qquad [Pu, \delta u] = 0. \tag{270}$$

The coefficient extraction takes place in E, after injectivity of ι; it assumes no k-scalar multiplication on the whole ambient matrix algebra.

Proof of Theorem 5.7. The first identity in (270) says $d^3(W) \subseteq C$. Let $q = d^3|_W$. Successive applications of the restricted-derivation identity give

$$dSw = [w, dw], \qquad d^2Sw = [w, d^2w], \qquad d^3Sw = [dw, d^2w] + [w, d^3w].$$

The last term vanishes by centrality, while (30) gives $[dw, d^2w] = d^3w$. Thus $qS = q$ pointwise. Iterating once gives $qS^2 = q$, but $S^2(W) \subseteq C$ and $q(C) = 0$. Hence $q = 0$. □

P.4 The exact coset formula

For $z \in L$, continue to write $d = \mathrm{ad}_z|_W$, and put

$$k_z(w) = dJw + Jdw = a(w, z) \in C.$$

One has $k_z(C) = 0$, $k_z J = j^2 k_z$, and

$$dSw = [w, dw], \qquad k_z(Sw) = A(w, dw), \qquad d_{f(z)} = dJ + d^2, \qquad k_{f(z)} = k_z d + j^2 k_z. \quad (271)$$

For the second identity, $D_z = dJ = d_{f(z)} + d^2$ is an S-restricted derivation, as are both summands. Apply it to Sw, use the first identity and $[w, Jdw] = D(w, dw)$, and rearrange. The remaining identities follow by commutator expansion, using $dk_z = 0$. The exact lift relation also gives

$$A(Ju, v) = j^2 A(u, v) \qquad (u, v \in W). \quad (272)$$

Indeed both sides equal $B(u, v) + D(u, v) + JD(u, v)$.

Quadratic expansion of $h = q_a f$ now gives

$$h(w + z) + h(w) = r(dw) + A(fw, dw) + k_{f(z)}(fw) + k_{f(z)}(dw) + h(z).$$

Use $fw = Jw + Sw$, (271), and (272). The two $j^2 A(w, dw)$-terms cancel; the non-affine part in w is

$$F_d(w) = r(dw) + A(Sw, dw) + A(w, d^2w). \quad (273)$$

We prove that this function is zero exactly. Put

$$U_0 = S(dw), \qquad V_0 = [Sw, dw], \qquad T_0 = [w, d^2w], \qquad R_0 = S(d^2w).$$

Equations (270) show that $U_0 + V_0 + T_0 \in C$ and $R_0 \in C$; for the latter apply $P\delta v = [v, \delta^2 v]$ at $v = \delta(w + C)$. The polar formula for S^2 is equivalent to

$$S([p, q]) = [Sp, Sq] + [Sp + Sq, [p, q]] + A(p, q).$$

Expand $S(U_0 + V_0 + T_0) = 0$ with this identity. The two $[U_0, V_0]$-terms cancel and $[U_0, T_0] + [V_0, T_0] = 0$ by centrality. Consequently

$$F_d(w) = [Sw, R_0] + [Sw + R_0, T_0] = [Sw, T_0].$$

On the other hand, $r(w) \in C$ implies $0 = d^2 r(w) = [Sw, d^2 Sw] = [Sw, T_0]$. This proves $F_d = 0$.

The remaining affine terms give the precise formula

$$h(w + z) + h(w) + h(z) = j^2 k_{f(z)}(w) + k_{f(z)}(dw) = jk_z(w) + k_z d^2 w.$$

Since $dk_z = 0$, one has $[J, d^3] = k_z d^2$. Theorem 5.7 therefore removes the last term:

$$h(w + z) + h(w) + h(z) = j\, a(w, z) \qquad (w \in W, z \in L). \quad (274)$$

In particular all third differences of h with a derived input vanish.

P.5 A character lemma and the cyclic-norm conclusion

For the coefficient space C, scalar multiplication by ω is the original center operator $j = f|_C$.

Lemma P.4 (Plane-wise quadratics in a nontrivial character)**.** *Let U, A be finite $k = \mathbb{F}_4$-spaces, and let $\phi : U \to A$ have binary degree at most four. Suppose its restriction to every k-subspace of dimension at most two is quadratic. If $\phi(\omega u) = \chi\phi(u)$, where $\chi = \omega$ or ω^2, then ϕ is quadratic.*

Proof. The three inputs of a third difference span at most a three-dimensional k-space, so it suffices to work in k^3. Coefficient functionals reduce to $A = k$, and output Frobenius interchanges the two characters. Take $\chi = \omega^2$. Use the function-polynomial basis with each coordinate exponent in $\{0, 1, 2, 3\}$. Its binary degree is the sum of the binary digit weights: $1, X, X^2, X^3$ have degrees $0, 1, 1, 2$, and tensor products give a basis adapted to the degree filtration. Modulo quadratics, restriction to the coordinate planes leaves only

$$aX^3YZ + bXY^3Z + cXYZ^3 + dXY^2Z^2 + eX^2YZ^2 + gX^2Y^2Z.$$

On $Z = \lambda X$, the forbidden cubic coefficients are $b\lambda$ and $d\lambda^2 + g\lambda$. Taking $\lambda = 1, \omega$ gives $b = d = g = 0$. On $Z = \lambda Y$ one obtains $a = e = 0$, and on $X = \lambda Y$ one obtains $c = 0$. □

Proof of Theorem 5.6. Equation (274) makes $\Delta_w h$ affine-linear for every $w \in W$. In an adapted binary basis no monomial of degree at least three can therefore contain a W-coordinate. Writing $\pi : L \to U = L/W$, we have

$$h = \phi\pi + q_0,$$

where $q_0 : L \to C$ is quadratic and $\phi : U \to C$ has degree at most four, all with zero constant. Every k-plane in U is covered by the subsolution generated by two marked lifts of its basis. Lemma P.1 makes D stable there, and $\widetilde{J}f$ makes JB stable there. Thus h, and hence ϕ, restricts to a quadratic function on that plane; use any binary-linear section of the covering map. No preservation of this subsolution by J is required.

Put $\phi_0 = \phi + \phi\omega + \phi\omega^2$, where scalars act on the input. This is the exact invariant-character projection. The other two character components are quadratic by Lemma P.4. Absorb them into q_0 and rename ϕ_0 as ϕ. Now $\phi(\omega u) = \phi(u)$ exactly.

Let $b_0 = \operatorname{pol} q_0$. Since $\phi\pi$ is unchanged by a W-shift, (274) gives $b_0(w, z) = j\, a(w, z)$. All bracket values lie in W, so

$$\sum_{\text{cyc}} b_0(B(x, y), z) = j \sum_{\text{cyc}} a(B(x, y), z) = 0.$$

Thus b_0 is an ordinary CE cocycle.

For a zero-constant quadratic q with polar b, the cyclic norm is $Nq(x) = q(x) + q(fx) + q(f^2x) = b(x, fx)$. As $a(x, fx) = 0$, we have $Nq_a = Nh = 0$. The quotient map satisfies $\pi f = \omega\pi$, so invariance gives $N(\phi\pi) = 3\phi\pi = \phi\pi$. Therefore

$$\phi\pi = Nq_0 = b_0(x, fx).$$

The right side has degree at most three, and its third additive difference is precisely the CE differential of b_0, which is zero. It is quadratic. A linear section of π shows that ϕ is quadratic, and hence so is $h = q_a f$. Finally $D = a + JB$, with stable refinement $\widetilde{J}f$ for the second summand. This proves stability of D, independently of the chosen refinement. □

P.6 Exact restricted structure on finite free covers

Proof of Theorem 5.8. By Theorem 6.14, a finite free solution (F, g) has $H_g(k) = 0$. The cyclic decomposition of Theorem 13.3 and coefficient Frobenius make both nontrivial eigenspaces of K_g zero. Its remaining fixed sector is detected injectively by O_0, by Proposition 13.4. Every scalar component of D_F is a CE cocycle, is stable by Theorem 5.6, and has zero checksum. Its class is therefore zero. For each $\mathbb{F}_2$-valued coordinate component of D_F, choose a k-valued primitive and compose it with a binary-linear projection $k \to \mathbb{F}_2$ taking 1 to 1. This gives an $\mathbb{F}_2$-valued primitive. Assembling these coordinate primitives gives a binary-linear map $T : F \to F'$ with $D_F = TB_F$. Set $J_F = T|_{F'}$. It is unique because bracket values span F'.

We give the remaining argument for any exact factor $JB = D$. Adjoint closedness implies

$$J[w, z] = D(w, z) = [Jw, z] \qquad (w \in W, z \in L),$$

so J is an ambient centroid on W. For $A = \mathrm{ad}_x$, $F_x = \mathrm{ad}_{f(x)}$, calculation on a bracket value gives

$$(J^2 + J + \mathrm{id})B(x, y) = (F_x^2 + AF_x + A^3 + A^2)y = 0.$$

The last equality is (246) with $G = B + D$. Thus $J^2 + J = \mathrm{id}$ on W.

Put $P = f|_W + J$. Its polar is the original bracket and its ambient adjoint is the square of the original adjoint. Expansion of the functional equation, using $[w, Jw] = [Jw, Pw] = 0$, gives

$$P^2(w) = P(Jw) + J^2P(w).$$

The right side is additive because its polar is $B(Ju, Jv)+J^2B(u, v) = 0$. Moreover $P([u, v]) = K(u, v)$ by (265). For every f-closed Lie subalgebra H, this implies $P(\gamma_i H) \subseteq \gamma_{i+2}H$ for $i \geq 2$; the quadratic cross terms on sums have depth $2i \geq i + 2$. The two-point-generated subsolution containing x, y has class at most five. Hence $P^2([x, y]) \in \gamma_6 H = 0$. Additivity of P^2 and spanning by bracket values give $P^2 = 0$ on W. The preceding equation now gives $P(Jw) = J^2P(w)$. Together with the polar and adjoint identities, this is exactly the restricted-square structure over k. □

The higher-rank finite free objects used here are obtained from the preliminary quartic bound of Proposition 6.1 through Theorem 6.14. That construction uses only the reduced-derived theory and its matrix action; the proof of global stability above uses only the independently certified two-point free object.

P.7 The tensor quotient and exact saturation formula

Proofs of Theorems 5.10 and 5.11. First use only the tensor presentation (42). Since B maps onto W, its relation shows that $j_f(W)$ is stable under ω, and hence is the whole Q_f. The map $k \otimes W \to E$, $\lambda \otimes w \mapsto \lambda(w + C)$, annihilates the relation space. Thus $N_f \subseteq C$. The original bracket descends, and (258) shows that scalar multiplication commutes with every ambient adjoint: for $w = B(c)$,

$$j_f([D(c), x]) = j_f(D(B(c), x)) = \omega j_f([B(c), x]).$$

The descended bracket is therefore k-bilinear. Scalar extension of a map h with $hD = \omega hB$ kills exactly the displayed relation generators, proving the stated universal property.

Now choose any finite free presentation $\pi : F \twoheadrightarrow L$. Use J_F, P_F from Theorem 5.8, and put $H = \ker \pi \cap F'$, $S_0 = H + J_F H$. The scalar polynomial makes S_0 the least J_F-stable subspace containing H. For $h \in H, x \in F$,

$$[J_F h, x] = D_F(h, x) = [g(h), x] + [h, [h, x]] \in H.$$

Thus $[S_0, F] \subseteq H$. The space H is g-stable, and restricted semilinearity gives

$$g(J_F h) = J_F h + J_F^2 g(h) \in S_0.$$

Quadratic expansion, with $[H, J_F H] \subseteq H$, proves $g(S_0) \subseteq S_0$. Hence S_0 is also P_F-stable and is an ideal. The quotient F'/S_0 is a restricted k-Lie algebra with square of height two and induced map $g = J_F + P_F$.

The composite $F' \to W \xrightarrow{j_f} Q_f$ sends $J_F w$ to $\omega j_f(\pi w)$: check this on bracket values, which span F'. It kills H and $J_F H$, so factors through F'/S_0. Conversely F'/S_0 is a scalar-compatible quotient of $W = F'/H$, and the tensor universal property gives the reverse map. The two maps agree on the generating images of W, hence are inverse. This proves (45), including f-stability of N_f and its precise central kernel. It also proves the restricted identities for $P_Q = f_Q + J_Q$. The map to E is now a restricted quotient map, since it agrees with the original projection on every $j_f(w)$.

For an arbitrary linear solution homomorphism T, scalar extension preserves the tensor relations because T preserves B, D. It therefore defines $Q(T)$ without a condition on the image of the global center. Its defining formula preserves brackets and f_Q, and its k-linearity then preserves P_Q. The same formula proves identity and composition laws. □

Proof of Corollary 5.12. If $Tf = g^2 T$, polarization and the definition of D give

$$TB_f = B_g(T \wedge T), \qquad TD_f = (D_g + B_g)(T \wedge T).$$

The Frobenius-semilinear map $\lambda \otimes w \mapsto \lambda^2 \otimes Tw$ sends a relation generator to

$$1 \otimes D_g(Tc) + (1 + \omega^2) \otimes B_g(Tc) = 1 \otimes D_g(Tc) + \omega \otimes B_g(Tc).$$

Thus it descends. It preserves the restricted square because reversal changes f_Q, J_Q to f_Q^2, J_Q^2, whereas $f_Q^2 + J_Q^2 = f_Q + J_Q = P_Q$. Taking $T = \mathrm{id}$ proves the statements about N_f and the reversed scalar. Zero maps give the same underlying zero map for either orientation. □

Q Bockstein-closed group extensions from the orbit equation

The polar biderivations also identify an application to central extensions of finite groups. The interface is the explicit binding operator; the extension and cohomology results used after that identification are those of Pakianathan–Yalçın [31, 32]. Write $k = \mathbb{F}_4$, let $n = \dim_{\mathbb{F}_2} L$, and put $G = B + D$.

Q.1 Two binding operators and their matrix representations

For an alternating bilinear map A, set

$$(\mathcal{T}_f A)(x, y) = A(fx, y) + A(x, B(x, y)).$$

The definition of D and the two-orbit identities give

$$\mathcal{T}_f B = D, \qquad \mathcal{T}_f D = B + D = G, \qquad \mathcal{T}_f G = B. \tag{275}$$

For the middle equality, write $A_x = \mathrm{ad}_x$ and $F_x = \mathrm{ad}_{fx}$. The operator difference between its two sides is $F_x^2 + A_x F_x + A_x^3 + A_x^2$, the vanishing two-orbit defect in the proof of (29).

Proposition Q.1 (The binding interface)**.** *Both quadratic bijections $Q_- = f^2 = f + \mathrm{id}$ and $Q_+ = f$ are Bockstein closed and 2-power exact. Their unique binding operators are, respectively, $x \mapsto D_x$ and $x \mapsto G_x$. Both families are simultaneously strictly triangular in a basis adapted to the lower-central series.*

Proof. Equation (275) gives

$$D(Q_-x, y) = B(x, y) + D(B(x, y), x), \qquad G(Q_+x, y) = B(x, y) + G(B(x, y), x).$$

These are precisely the general bilinear binding criterion of [31, Theorem 1.1]. The source and target have equal dimension, each $Q_\pm$ has zero as its unique zero, and its image spans the target because it is bijective. These are the three 2-power-exact conditions. Binding uniqueness follows from [31, Proposition 8.1]. Finally D_x and G_x send every $\gamma_i L$ into $\gamma_{i+1}L$, so global nilpotence supplies a common strictly triangular basis. □

There are also explicit representations in the quadratic-map sense of [32, Definition 2.5]:

Q	$\rho_{\text{domain}}(x)$	$\rho_{\text{target}}(x)$	$\rho_{\text{target}}(Qx)$
Q_-	D_x	G_x	$D_x^2 + D_x$
Q_+	G_x	D_x	$G_x^2 + G_x$.

(276)

The target quadratic map is $A \mapsto A + A^2$ on $\mathrm{End}_{\mathbb{F}_2}(L)$. For example, $D_x^2 = A_x^2 + A_x D_x$, together with (275), gives the first row. Its polarization is $[D_x, D_y] = G_{B(x,y)}$; the second row follows by exchanging D and G. Both linear kernels are $Z(L)$: if $D_x = 0$, then $A_x^2 = 0$ and $A_{fx} = 0$, so center invariance implies $x \in Z(L)$. If $G_x = 0$, then $D_x = A_x$, the same square identity gives $A_x^2 = 0$, and $A_{fx} = A_x$; thus $f^2 x$ is central, with the same conclusion. The representation pairs are faithful on the central quotient. Their domain and target images need not coincide.

Q.2 Group cohomology and integral lifting

Let $\mathcal{G}_Q$ be the classical central extension

$$0 \longrightarrow V \longrightarrow \mathcal{G}_Q \longrightarrow V \longrightarrow 0$$

whose square map is Q. It has order 2^{2n}, exponent dividing four, and class at most two. For $n > 0$ its exponent is four; its class is exactly two when $B \neq 0$. Since Q is bijective, its elements of order dividing two form the central kernel V. Write $w_B = \dim_{\mathbb{F}_2} L'$ and $z_0 = \dim_{\mathbb{F}_2} Z(L)$. The commutator map is B, giving

$$\mathcal{G}_Q^{\mathrm{ab}} \simeq C_4^{n-w_B} \times C_2^{w_B}, \qquad Z(\mathcal{G}_Q) \simeq C_4^{z_0} \times C_2^{n-z_0}.$$

The two groups from Q_- and Q_+ are isoclinic because their commutator pairings agree. No isomorphism between them is needed.

The regular-sequence and cohomology theorems [32, Proposition 5.2 and Theorem 5.5] apply directly to Proposition Q.1. If $q_1, \ldots, q_n$ are the formal homogeneous quadratic coordinates of the group-extension class, then

$$H^*(\mathcal{G}_Q; \mathbb{F}_2) \simeq \mathbb{F}_2[s_1, \ldots, s_n] \otimes \frac{\mathbb{F}_2[x_1, \ldots, x_n]}{(q_1, \ldots, q_n)}, \qquad |s_i| = 2, \quad |x_i| = 1. \tag{277}$$

This graded-algebra isomorphism uses a choice of the s_i. The quotient factor has Hilbert series $(1+t)^n$, so the Poincaré series is $(1-t)^{-n}$. The formal squares x_i^2 in this formula have degree two; they are distinct from the Boolean function convention $x_i^2 = x_i$.

Corollary Q.2 (Linearity detected by uniform integral lifting)**.** *For either $Q = f$ or $Q = f^2$, the specified extension $E(Q)$ admits a uniform integral lifting in the sense of [31, Theorem 5.9] if and only if f is linear.*

Proof. If $B \neq 0$, both bindings are nonzero: $D = 0$ or $G = 0$ would force $B = 0$ by (275). A nonzero strictly triangular operator is not diagonalizable. Binding uniqueness excludes an alternative simultaneously diagonalizable binding, so [31, Theorem 5.9] excludes a uniform integral lifting. If $B = 0$, then $D = G = 0$ and the extension is C_4^n; it has the standard uniform integral lifting through $\mathbb{Z}^n$. □

This application uses the specified extension and that uniform-lifting notion. The general Bockstein criteria, regular-sequence result and cohomology formula retain their classical attribution; the orbit equation supplies the explicit pair of binding operators.

R Proofs for general growth and finite universal objects

The quantitative argument has two stages. The reduced derived action first gives a quartic bound without higher-rank free objects. This constructs the finite free objects. Their exact whole-derived scalar then removes one central step. We keep that order explicit before giving the reverse Engel realization and the sharp ordinary centre-by-metabelian boundary.

R.1 Paired operators and scalar envelopes

Lemma R.1 (The scalar envelope of a paired binary action)**.** *Let $k = \mathbb{F}_4$, and let X be a binary Lie subalgebra of $\mathrm{End}_k(M)$ such that every $A \in X$ satisfies $A^3 = 0$ and $\omega A + A^2 \in X$. Then X is an ordinary 3-Engel Lie algebra. Its scalar span $N = kX$ is closed under matrix squares, every element of N has fourth power zero, and N is an ordinary 3-Engel k-Lie algebra.*

Proof. For $A, B \in X$, put

$$K_{21} = A^2B + ABA + BA^2, \qquad K_{12} = AB^2 + BAB + B^2A,$$
$$Q = A^2B^2 + AB^2A + B^2A^2, \qquad H = BAB^2 + B^2AB, \qquad R = B^2AB^2.$$

The cube of $A + B$ gives $K_{21} = K_{12} =: K$. The two iterates of the companion map are $\omega B + B^2$ and $\omega^2 B + B^2$, using $B^4 = 0$. For either $\lambda \in \{\omega, \omega^2\}$, expansion of the cube of the actual element $A + \lambda B + B^2$ gives

$$0 = K + Q + \lambda H + R.$$

Subtracting the two equations gives $H = 0$, which is $(\mathrm{ad}_B)^3 A = 0$. These substitutions all occur in X.

The scalar span is bracket closed and $A^2 = (\omega A + A^2) + \omega A \in N$ for every $A \in X$. The square expansion makes N square closed. Every element of N is $A + \omega B$ with $A, B \in X$. Write

$$(A + \lambda B)^4 = C_1\lambda + C_2\lambda^2 + C_3\lambda^3,$$

where the pure fourth powers are zero. Here $C_1 = (\mathrm{ad}_A)^3 B = 0$ and $C_3 = (\mathrm{ad}_B)^3 A = 0$. At $\lambda = 1$, the element is actual and its fourth power vanishes, so $C_2 = 0$. Hence every fourth power in N vanishes.

For arbitrary $U, V \in N$, remove the pure fourth powers from $(U + \lambda V)^4$. The remaining polynomial has degree at most three and vanishes at all four scalars of k. Its linear coefficient is $(\mathrm{ad}_U)^3 V$, proving the last assertion. The conclusion requires fourth powers in the scalar span to vanish; no assertion about all of its matrix cubes is needed. □

Lemma R.2 (Removing squared generators)**.** *For matrices $A_1, \ldots, A_d$ over a field of characteristic two, put $H = \mathrm{Lie}_k\langle A_i \rangle$ and $N = \mathrm{Lie}_k\langle A_i, A_i^2 \rangle$. Then $\gamma_j(N) = \gamma_j(H)$ for every $j \geq 2$. In particular their classes agree when $N \neq 0$ is nilpotent.*

Proof. The identities

$$[A_i^2, T] = [T, A_i, A_i], \qquad [A_i^2, A_j^2] = [A_i, A_j, A_j, A_i]$$

hold in every associative algebra of characteristic two. Expand a left-normed Lie word of length at least two in the displayed generators. Its initial bracket, if it contains a square, becomes an ordinary A_i-word of length at least two. Each later square entry acts by two successive adjoints of A_i. Thus every word of length at least $j \geq 2$ belongs to $\gamma_j(H)$. The reverse inclusion follows from $H \subseteq N$. If $H = 0$, all A_i and hence N are zero; otherwise the equality of lower-central terms also covers the abelian case. □

Lemma R.3 (The short commuting action)**.** *Let $d \geq 2$, and let $a_1, \ldots, a_d$ be commuting $k = \mathbb{F}_4$-linear operators. Suppose that every element of the binary span of $a_i, \omega a_i + a_i^2$ has zero cube. Then $a_i^3 = a_i^2 a_j = 0$ for all i, j, and every product of $d+1$ operators from that span is zero.*

Proof. For two of the actual operators a, b, the cube of $a + b$ gives $a^2 b = ab^2 =: M$. Hence $a^2 b^2 = ab^3 = 0$. The cube of $a + \omega b + b^2$ gives

$$0 = \omega a^2 b + a^2 b^2 + \omega^2 a b^2 = M.$$

The remaining monomials in the commuting generators are square-free monomials and single-generator squares. Their degrees are at most d and two, respectively. A product of $d + 1 \geq 3$ operators from the span expands into monomials of degree at least $d + 1$, all zero. □

R.2 The preliminary bound and the exact free-derived bound

Proof of Proposition 6.1. Put $W = L'$, $C = W \cap Z(L)$, and $E = W/C$. Use the intrinsic $k = \mathbb{F}_4$-structure on E from Theorem 5.4. Its actual action image $X = \{\rho_x : x \in L\}$, with $\rho_x = \mathrm{ad}_x|_E$, satisfies

$$\rho_x^3 = 0, \qquad \rho_{f(x)} = \omega \rho_x + \rho_x^2$$

by Theorems 5.7 and 5.13. Lemma R.1 makes $N = kX$ an ordinary 3-Engel k-Lie algebra. If $\xi_1, \ldots, \xi_d$ are point generators and $A_i = \rho_{\xi_i}$, paired generation (Lemma R.7) gives

$$X = \mathrm{Lie}_{\mathbb{F}_2}\langle A_i, \omega A_i + A_i^2 \rangle, \qquad N = \mathrm{Lie}_k\langle A_i, A_i^2 \rangle.$$

By Lemma R.2, N has the same class as the at most d-generator ordinary algebra $H = \mathrm{Lie}_k\langle A_i \rangle$. The larger-field improvement after Theorem 1 of [40] applies to H:

$$q_* := \mathrm{cl}\, N = \mathrm{cl}\, H \leq q(d).$$

This invokes a Lie-class estimate only.

Suppose $N \neq 0$, so $q_* \geq 1$. Then $\gamma_{q_*+1} L \subseteq \ker \rho \cap W$, and the image of this term in E lies in $Z(E)$. Ambient actions on $Z(E)$ commute: their commutator is the inner action of $[x, y] + C \in E$, which vanishes on $Z(E)$. They are the binary span of $a_i = A_i|_{Z(E)}$ and $\omega a_i + a_i^2$. For $d \geq 2$, Lemma R.3 therefore gives

$$\gamma_{q_*+d+2} L \subseteq C, \qquad \gamma_{q_*+d+3} L = 0.$$

Thus $\mathrm{cl}\, L \leq q(d) + d + 2$. If $N = 0$, then $[L, W] \subseteq C$ and $\gamma_4 L = 0$, which is within the same bound. Rank zero is zero by finite nilpotence; a one-point subsolution is the commuting scalar plane because $[x, f(x)] = 0$. This proves the two small-rank conventions.

This proof uses the canonical reduced-derived structure and the full-derived cube theorem, whose proofs use only the independently certified two-point free object. No higher-rank free object has entered the argument. □

The finite-product construction in Lemma R.8 and Theorem 6.14 now uses this preliminary bound. Its proof appears below and is independent of the sharper estimate that follows.

Proof of Theorem 6.2. For $d \geq 2$, take the finite free solution F on d marked points, constructed using C_0, and put $W = F'$. Theorem 5.8 supplies the exact scalar $J = \omega$ on W, commuting with all ambient actions. For $\rho_x = \operatorname{ad}_x|_W$, the exact factor identity and Theorem 5.7 give

$$\rho_{f(x)} = \omega\rho_x + \rho_x^2, \qquad \rho_x^3 = 0.$$

The preceding scalar-envelope argument now applies on all of W. Write q_* for the class of its scalar action algebra. It is the class of an at most d-generator ordinary k-Lie algebra, and hence $q_* \leq q(d)$.

If the action algebra is nonzero, then

$$\gamma_{q_*+1}F \subseteq \ker\rho \cap W = Z(W).$$

This center is a k-subspace and is invariant under every ambient adjoint. On it the actions commute because W acts trivially. Lemma R.3 gives $\gamma_{q_*+d+2}F = 0$, and therefore $\operatorname{cl} F \leq q(d) + d + 1$. If the action algebra is zero, W is central and $\operatorname{cl} F \leq 2$. Every finite target of point rank at most d is a quotient of F, so the same bound holds throughout the category. The rank-zero and rank-one cases were handled in the preliminary proof. □

R.3 The action on an invariant ideal

Proof of Proposition 6.3. The ideal I' is f-stable by the bracket-image identity and quadratic expansion. Thus $J = f_M$ is binary-linear and $J^2 + J = \mathrm{id}$. Put $A = \operatorname{ad}_x|_M$ and $F = \operatorname{ad}_{f(x)}|_M$. Polarized commuting and bracket-image identities give

$$F = AJ + A^2, \qquad JA = FJ + A + FA. \tag{278}$$

The discarded terms are $[y,[x,y]]$ and $[f(y),[x,y]]$ for $y \in I$; both lie in I'. Set $T = JA + AJ$. Substitution gives $(\mathrm{id}+A)T = A^3$. Since $A^4 = 0$ by Theorem 5.2, this gives $T = A^3$. Commutation with J is an idempotent derivation:

$$[J,[J,A]] = J^2A + AJ^2 = T.$$

But $[J,T] = [J,A^3] = TA^2 + ATA + A^2T = A^5 = 0$. Hence $T = 0$, $A^3 = 0$, and $F = JA + A^2$. Lemma R.1 proves the ordinary 3-Engel assertion for the actual binary image. □

R.4 Reverse Engel realization and the same-rank comparison

Proof of Proposition 6.4. Let $k = \mathbb{F}_4$, put $X = \operatorname{ad} H \subseteq \operatorname{End}_k(H)$, and let $A_1, \dots, A_r$ be the adjoints of chosen generators of H. Every actual $A \in X$ has zero matrix cube. The classical operator identities of [40, Lemmas 1–2 and equations (3)–(4)] give, for $A, B \in X$,

$$\begin{gathered} B^2AB^2 = B^2A^2B^2 = 0, \qquad [A,B]^2 = [A^2,B^2], \\ [A,B]A[A,B] = 0. \end{gathered} \tag{279}$$

The last equality is the commutator sandwich in the proof of Lemma 2 of that source with A, B exchanged. These identities apply to the genuine adjoint representation of the ordinary 3-Engel algebra H.

Set $N = X + \operatorname{span}_k\{B^2 : B \in X\}$. Because k is perfect and $B^2 + C^2 = (B+C)^2 + [B,C]$, every element of N is $A + B^2$ with $A, B \in X$. The square-commutator identities make N bracket closed, with $N' = X'$, and

$$(A + B^2)^2 = A^2 + [A,B^2] \in N.$$

To verify the height-two condition, put $T = [A, B^2]$ and $C = [A, B]$. Equations (279) give

$$T^2 = A(B^2AB^2) + AB^4A + B^2A^2B^2 + (B^2AB^2)A = 0,$$
$$(A + B^2)^4 = [A^2, T] = [C^2, A].$$

Partial polarization of the cube in the k-linear space X gives $C^2A + CAC + AC^2 = 0$. The sandwich identity gives $CAC = 0$, so $[C^2, A] = 0$. Thus every fourth power in N is zero.

The standard restricted square $P(Z) = Z^2$ on N therefore satisfies $P^2 = 0$. On the underlying binary space define

$$f_N(Z) = \omega Z + Z^2.$$

Its polar is the original commutator bracket and

$$f_N^2(Z) = \omega^2 Z + Z^2 + Z^4 = f_N(Z) + Z.$$

This is the restricted-square construction of quadratic maps [31, Lemma 9.5 and Example 9.6], here with the displayed ω-coefficient and height-two condition.

Squares of Lie words of length at least two belong to X', by the commutator-square identity. Hence

$$N = X + \mathrm{span}_k\{A_1^2, \dots, A_r^2\} = \mathrm{Lie}_k\langle A_i, A_i^2\rangle.$$

Lemma R.2 gives $\gamma_j N = \gamma_j X$ for $j \geq 2$, so their classes agree when $X \neq 0$.

Now take only the binary subsolution Y generated by the r points A_i inside (N, f_N). Its point rank is at most r. The scalar span kY is Lie closed and contains $y^2 = f_N(y) + \omega y$ for every $y \in Y$. The square polarization formula makes kY square closed. It therefore contains the restricted envelope of the A_i, which is N, and $kY = N$. Bilinearity gives

$$\gamma_j N = k\gamma_j Y \qquad (j \geq 1).$$

The class of Y is thus that of $X = H/Z(H)$, namely $c - 1$ when H has class $c \geq 2$. For an abelian input take $Y = 0$. The scalar-span argument preserves point rank without identifying the intrinsic scalar of a binary abelianization with the ambient scalar on N. □

Related restricted-power calculations occur in [15, Section 5.5.4, p. 78]: on a support tensor algebra over an infinite field of characteristic two, a canonical restricted fourth-power identity is equivalent to the linearized 3-Engel identity. The construction above uses the adjoint algebra of an arbitrary finite-dimensional 3-Engel $\mathbb{F}_4$-algebra. The binary subsolution Y keeps its point-generator rank within the ordinary generator rank of the input algebra.

Proof of Theorem 6.5. The reverse construction gives $e(d) - 1 \leq \kappa(d)$. For the upper bound use the finite free solution at point rank d and repeat the whole-derived proof of Theorem 6.2. Its ordinary action algebra is generated by at most d elements over k, so its class is at most $e(d)$. The remaining commuting center action costs at most $d + 1$ further steps. Rank one is checked separately, with $e(1) = \kappa(1) = 1$.

For completeness, $e(d) \geq d$. Put $m = d - 1$ and let $R = k[t_1, \dots, t_m]/(t_i^2)$. The standard metabelian algebra $k^m \ltimes R$, with commuting generators y_i acting on the abelian ideal R by multiplication by t_i, is generated by $y_1, \dots, y_m, 1$. It is 3-Engel: on R each adjoint squares to zero, and on the quotient its first image lies in R. Its class is $m + 1 = d$, since $\gamma_j = (t_1, \dots, t_m)^{j-1}R$ for $j \geq 2$. For $d = 1$, use the one-dimensional abelian algebra. The two-sided comparison now gives $\kappa = \Theta(e)$ and the stated equivalence of polynomial upper bounds. Dividing it by $e(d)$ proves the final limit when $e(d)/d \to \infty$. □

Remark R.4 (Unbounded principal-ideal class). The current example in [39, Proposition 2.1] already obstructs a uniform principal-ideal argument within the orbit category. For $m \geq 1$, let $A_m = (t_1, \dots, t_m) \subset k[t_i]/(t_i^2)$, and on $kb \oplus A_m^3$ take the inherited restricted square

$$P(b_0, u, v, w) = (0, b_0 w + uv, uw, vw), \qquad f = \omega \,\mathrm{id} + P.$$

This is the $a = 0$ restricted subalgebra of the current presentation in Appendix S; it has $P^2 = 0$. Writing $A_{\geq j}$ for monomials with support size at least j, the smallest solution ideal containing b is $I = kb \oplus A_{\geq 1}U \oplus A_{\geq 2}V \oplus A_{\geq 3}W$. For $j \geq 2$, its lower-central terms are

$$\gamma_j I = A_{\geq 2j-1}U \oplus A_{\geq 2j}V \oplus A_{\geq 2j+1}W.$$

The current products add disjoint supports, giving both inclusions by partitioning any support above the indicated threshold. Thus $\mathrm{cl}\, I = \lfloor (m+1)/2 \rfloor$, which is unbounded. Here an ideal generated inside the ambient algebra is distinct from the one-point subsolution, which is only a scalar plane.

R.5 The ordinary metabelian ceiling

We now work with ordinary $k = \mathbb{F}_4$-Lie algebras and ordinary k-Lie generators. The algebra $H_d = U \oplus W_d$ is defined by (55), using exactly the module and triangle relations of Section 7.

Lemma R.5 (The classical metabelian ceiling)**.** *Every ordinary metabelian* 3*-Engel* k*-Lie algebra generated by at most* $d \geq 2$ *elements is a quotient of* H_d*. Its class is at most* $\max(4, d+1)$*, and* H_d *attains this bound at exact ordinary generator rank* d*.*

Proof. Let A be metabelian, put $W = A'$, and choose a linear section of A/W. On W, write $P_u = \mathrm{ad}_u|_W$ and let $\beta(u, v)$ be the bracket of the chosen lifts. The P_u commute. The 3-Engel identity gives $P_u^3 = 0$ and $P_u^2 \beta(u, v) = 0$. The partial coefficients of $(P_u + \lambda P_v)^3 = 0$ give $P_u^2 P_v = P_u P_v^2 = 0$. The coefficient of λ in

$$P_{u+\lambda v}^2 \beta(u + \lambda v, w) = 0$$

then gives $P_u^2 \beta(v, w) = 0$. These coefficient separations are valid over k: the polynomials have degree at most three, strictly less than $|k| = 4$. Since

$$W = \sum_v P_v W + \mathrm{span}_k\{\beta(v, w) : v, w \in A/W\},$$

we obtain $P_u^2 W = 0$ for every u. Conversely this square-zero action makes every third adjoint zero on A.

For any chosen d generators, the actions therefore factor through $R_d = k[t_i]/(t_i^2)$. Jacobi gives precisely the triangle relations (67), so the universal commutator-module presentation supplies a surjection $H_d \to A$. For H_d,

$$\gamma_2 H_d = W_d, \qquad \gamma_j H_d = \mathfrak{m}^{j-2} W_d \quad (j \geq 2).$$

Proposition 7.1 puts the highest coefficient degree at two for $d = 2, 3$, and at $d - 1$ for $d \geq 4$. It follows that $\mathrm{cl}\, H_d = \max(4, d+1)$. Its abelianization is U, so its ordinary generator rank is exactly d. This is the direct class consequence of the classical commutator module. □

R.6 The complete top central cover

The exterior-coinvariant description of free centre-by-metabelian central kernels is classical; see [20, (2.1) and Theorem 7.2]. Here we first impose the square-zero action relations of W_d and then compute the full top-weight CE quotient.

Central lifts, class, and ordinary generators. For an ordinary CE cocycle $\alpha : H \wedge H \to Z$, the central extension has bracket

$$[(u, a), (v, b)]_\alpha = ([u, v], \alpha(u, v)).$$

If H is 3-Engel, its extension is 3-Engel exactly when

$$\alpha([y, x, x], x) = 0 \qquad (x, y \in H), \tag{280}$$

because its third iterated bracket is $([y, x, x, x], \alpha([y, x, x], x))$. If H has class $c \geq 2$, projection of each lower-central term gives

$$\gamma_{c+1}(E_\alpha) = \operatorname{span}_k \alpha(\gamma_c H, H) \subseteq Z, \qquad \gamma_{c+2}(E_\alpha) = 0. \tag{281}$$

For $h \in \gamma_c H \subseteq Z(H)$, the CE identity implies $\alpha(h, H') = 0$; coboundaries also vanish on these top pairings. Thus a nonzero top pairing raises the class exactly once.

Let S be generated by arbitrary lifts of ordinary generators of H. Then $E_\alpha = S + Z$, and centrality gives $\gamma_j E_\alpha = \gamma_j S$ for $j \geq 2$. If $\dim Z = 1$ and the top pairing is nonzero, then $Z = \gamma_{c+1} E_\alpha \subseteq S$, so $E_\alpha = S$. Its ordinary generator rank therefore equals that of H.

Proof of Theorem 6.6. Write $W = W_d$, $R = R_d$, and $k = \mathbb{F}_4$. The grading used here is Lie weight: $\deg x_i = 1$, $\deg v_{ij} = 2$, and $\deg t_i = 1$.

The top commutator space. Write $t_S = \prod_{i \in S} t_i$. The top graded part is W_{d+1}, where the subscript denotes the total Lie weight. It has a basis

$$T_i = t_{[d]\setminus\{j\}} v_{ij} \qquad (j \neq i), \tag{282}$$

and this expression does not depend on the choice of j. Indeed, the triangle relation on i, j, k, multiplied by $t_{[d]\setminus\{j,k\}}$, proves independence of j and k. If $s \notin \{i, j\}$, then

$$t_{[d]\setminus\{s\}} v_{ij} = 0. \tag{283}$$

To see this choose $k \notin \{i, j, s\}$, which is possible for $d \geq 4$, and multiply the triangle relation on i, j, k by $t_{[d]\setminus\{k,s\}}$. Its first two terms have a squared variable and vanish; the third is the displayed expression. These reductions show that the T_i span the top part.

Let $\theta : R \to k$ extract the coefficient of $t_1 \cdots t_d$. The map on top-degree presentations

$$\varphi(p v_{ij}) = \theta(p t_j) x_i + \theta(p t_i) x_j \qquad (\deg p = d - 1) \tag{284}$$

annihilates each triangle relation: each resulting term occurs twice. It takes T_i to x_i, proving independence. With $\Delta = x_1 \wedge \cdots \wedge x_d$, the canonical version is

$$W_{d+1} \simeq \det(U) \otimes U, \qquad T_i \longmapsto \Delta \otimes x_i.$$

Here naturality can be checked on generators of $\mathrm{GL}(U)$. Diagonal changes have weight $\det(B)B$ on the displayed T_i. Permutations permute their indices. For $x_1 \mapsto x_1 + x_2$, (282) with $j = 2$ gives $T_1 \mapsto T_1 + T_2$, while T_2 is fixed; for $i \notin \{1, 2\}$ the extra term vanishes by (283). Thus the determinant-twisted statement is independent of the basis.

Completeness of the CE quotient. Give a central coefficient space Z weight $d+2$. For an alternating CE cocycle $\alpha : H_d \wedge H_d \to Z$ homogeneous of total weight $d+2$, write

$$\eta(w,u) = \alpha(w,u), \qquad \beta(w,w') = \alpha(w,w').$$

The only possibly nonzero U–W terms are $\eta : W_{d+1} \otimes U \to Z$. The CE equations on x_i, w, w' give

$$\beta(t_i w, w') = \beta(w, t_i w'), \tag{285}$$

and those on x_i, x_j, w give

$$\beta(v_{ij}, w) = \eta(t_i w, x_j) + \eta(t_j w, x_i).$$

Thus every cocycle is uniquely determined by η, with

$$\beta(r v_{ij}, w) = \eta(t_i r w, x_j) + \eta(t_j r w, x_i), \tag{286}$$

where η is zero off W_{d+1}.

Put $B_{ij} = \eta(T_i, x_j)$. Symmetry of β forces $B_{ij} = B_{ji}$. For $j \neq l$, choose i, k so that i, j, k, l are distinct, and put $r = t_{[d]\setminus\{i,k\}}$. Equations (283)–(286) give

$$\beta(v_{ij}, r v_{kl}) = B_{lj}, \qquad \beta(r v_{kl}, v_{ij}) = B_{jl}.$$

This proves the required symmetry, with no condition on the diagonal.

Conversely, let $B : U \times U \to Z$ be any symmetric bilinear map after the displayed choice of Δ, and define η by its matrix B. Define β by (286). This descends from the free R-module to W: multiplication by each triangle relation gives six η terms, in three equal pairs, because the t_i commute. The square-zero ring relations are already respected. The expression can be written explicitly as

$$\begin{aligned}\beta(r v_{ij}, s v_{kl}) = {} & \theta(r s t_i t_l) B_{kj} + \theta(r s t_i t_k) B_{lj} \\ & + \theta(r s t_j t_l) B_{ki} + \theta(r s t_j t_k) B_{li}.\end{aligned} \tag{287}$$

This formula is symmetric under exchanging (ij, r) and (kl, s) because B is symmetric. It is alternating. More strongly,

$$\beta(pv, v) = 0 \qquad (p \in R,\ v \in W). \tag{288}$$

For one cyclic generator $v = r v_{ij}$, the four terms in (287) reduce to two squared-variable terms, which vanish, and $\theta(p r^2 t_i t_j)(B_{ij} + B_{ji}) = 0$. For a sum of cyclic generators the cross terms cancel in pairs, using symmetry and the balanced property (285). This proves (288), and in particular alternation.

Now set $\alpha|_{U\wedge U} = 0$, use η on $W \otimes U$, and use β on $W \wedge W$. All CE equations hold: the U, U, W equations are the definition of β; the U, W, W equations are balance; the W, W, W equations vanish because W is abelian; and the U, U, U equations vanish by the homogeneous weight. Thus every symmetric B extends, uniquely, to a homogeneous CE cocycle.

There are no coboundaries in this weight: H_d has no bracket of weight $d+2$, so $t([a,b])$ has zero component in this total weight. Consequently the homogeneous cohomology space is naturally

$$H^2_{\mathrm{CE}}(H_d, Z)_{d+2} \simeq \mathrm{Hom}_k(\det(U) \otimes \mathrm{Sym}^2 U, Z). \tag{289}$$

Automatic three-Engel admissibility. For $u, s \in U$ and $v, w \in W$, put $x = u + v$ and $y = s + w$. Write P_u for multiplication by the corresponding linear element of R, and $c(s, u) = [s, u]$. In the metabelian base,

$$[y, x, x] = P_u c(s, u) + P_u P_s v,$$

because $P_u^2 = 0$ on W. The central three-Engel obstruction is therefore

$$\begin{aligned}\alpha([y, x, x], x) = {} & \eta(P_u c(s, u), u) + \eta(P_u P_s v, u) \\ & + \beta(P_u c(s, u), v) + \beta(P_u P_s v, v).\end{aligned}$$

The first term has total weight four and vanishes because $d + 2 \geq 6$. Balance and (286) give

$$\beta(P_u c(s, u), v) = \beta(c(s, u), P_u v) = \eta(P_s P_u v, u) + \eta(P_u^2 v, s) = \eta(P_s P_u v, u).$$

It cancels the second term. The last term is zero by (288) with $p = P_u P_s$. Hence every cocycle in (289) satisfies $\alpha([y, x, x], x) = 0$ for every x, y.

Class and exact generator rank. When $\dim Z = 1$, a nonzero symmetric form has a nonzero top pairing. Equation (281) gives class $d + 2$, and the lifted generators generate the extension by the preceding central-lift argument. The ordinary rank is exactly d. The extension's second derived ideal is contained in Z. It is nonzero by Lemma R.5, since its class exceeds the metabelian ceiling $d + 1$. Hence it equals Z. □

R.7 The rank-two endpoint and the sharp boundary

Lemma R.6 (The ordinary two-generator endpoint)**.** *Over $k = \mathbb{F}_4$, the free ordinary 3-Engel Lie algebra on two generators has dimension eight and class five. In particular $e(2) = 5$.*

Proof. Let F have basis a, b, c, u, v, w, z, t and nonzero unordered brackets

$$\begin{gathered}[a, b] = c, \quad [c, a] = u, \quad [c, b] = v, \quad [u, b] = [v, a] = w, \\ [w, a] = [c, u] = z, \quad [w, b] = [c, v] = t.\end{gathered} \tag{290}$$

All omitted brackets of distinct basis vectors vanish, and the bracket is alternating. In particular z, t are central. For Jacobi on distinct basis triples, the only triples with a nonzero individual nested term are $\{a, b, c\}$, $\{a, b, u\}$, and $\{a, b, v\}$. Their two nonzero terms are respectively $w + w$, $z + z$, and $t + t$. All three sums vanish. Triples with a repeated basis vector satisfy Jacobi automatically by alternation in characteristic two. This proves that the displayed table defines a Lie algebra.

For $x = Aa + Bb + Cc + Du + Ev + Fw + Gz + Tt$ and $R_x(y) = [y, x]$,

$$\begin{aligned}R_x(a) &= Bc + Cu + Ew + Fz, & R_x(b) &= Ac + Cv + Dw + Ft, \\ R_x(c) &= Au + Bv + Dz + Et, & R_x(u) &= Bw + Cz, \\ R_x(v) &= Aw + Ct, & R_x(w) &= Az + Bt, \\ R_x(z) &= R_x(t) = 0.\end{aligned}$$

Consequently

$$\begin{aligned}R_x^2(a) &= ABu + B^2 v + BCw + (BD + C^2 + AE)z, \\ R_x^2(b) &= A^2 u + ABv + ACw + (AE + C^2 + BD)t, \\ R_x^2(c) &= ACz + BCt, \\ R_x^2(u) &= ABz + B^2 t, \qquad R_x^2(v) = A^2 z + ABt, \\ R_x^2(w) &= R_x^2(z) = R_x^2(t) = 0.\end{aligned}$$

Substitution of the first table into the second gives $R_x^3 = 0$: the noncentral terms cancel in pairs, and z, t are central. Thus F is ordinary three-Engel. Its lower central dimensions are

$$(8, 6, 5, 3, 2, 0),$$

because the successive spans are F, $\langle c, u, v, w, z, t\rangle$, $\langle u, v, w, z, t\rangle$, $\langle w, z, t\rangle$, $\langle z, t\rangle$, and zero. It has exact ordinary rank two and $F'' = \langle z, t\rangle \neq 0$.

Every pair of elements has this table. Let L be any ordinary three-Engel Lie algebra over k, and fix $a, b \in L$. Define

$$c = [a, b], \quad u = [c, a], \quad v = [c, b], \quad w = [u, b] = [v, a], \quad z = [w, a], \quad t = [w, b].$$

The equality defining w is Jacobi. Engel gives

$$[u, a] = [v, b] = 0, \qquad [z, a] = [t, b] = 0.$$

For example $[z, a] = [v, a, a, a] = 0$, and the other last equality is $[u, b, b, b] = 0$. Jacobi also gives

$$[c, u] = z, \qquad [c, v] = t.$$

Set

$$p = [z, b], \qquad q = [t, a], \qquad r = [c, w], \qquad s = [u, v].$$

Jacobi on $\{c, u, b\}$, $\{c, v, a\}$, and $\{a, b, w\}$ gives

$$p + r + s = 0, \qquad q + r + s = 0, \qquad r + q + p = 0.$$

Since the characteristic is two, these imply

$$p = q, \qquad r = 0, \qquad s = p.$$

It remains to kill p. Apply the three-Engel identity to $y = a$ and $x = b + \lambda c$:

$$\begin{aligned} 0 &= [a, b + \lambda c, b + \lambda c, b + \lambda c] \\ &= [v + \lambda w + \lambda^2 z, b + \lambda c] \\ &= \lambda^2(r + p) + \lambda^3[z, c]. \end{aligned}$$

For $\lambda \neq 0$, division by λ^2 yields $r + p + \lambda[z, c] = 0$. Two distinct nonzero values of λ exist because $|k| > 2$. Subtraction first gives $[z, c] = 0$, then $r + p = 0$. As $r = 0$, it follows that $p = q = r = s = 0$.

Thus z, t commute with a, b, and are central in the subalgebra generated by a, b: the centralizer of either element is a Lie subalgebra containing the generators. Finally

$$[u, w] = [u, [v, a]] = [[u, v], a] + [v, [u, a]] = 0,$$

and similarly $[v, w] = [v, [u, b]] = [[v, u], b] + [u, [v, b]] = 0$. All unordered brackets of a, b, c, u, v, w, z, t are now exactly those in (290). Their span is a Lie subalgebra containing a, b, and hence equals $\langle a, b\rangle_L$. The linear map from F sending its two generators to this arbitrary pair is therefore a Lie homomorphism. It is unique since a, b generate F. This proves the claimed free universal property and the exact class maximum.

□

Proof of Theorem 6.7. Let L have ordinary generator rank at most $d \geq 2$, satisfy the 3-Engel identity, and have $L'' \subseteq Z(L)$. Then $L/Z(L)$ is metabelian, 3-Engel, and generated by at most d elements. Lemma R.5 gives

$$\operatorname{cl} L \leq \operatorname{cl}(L/Z(L)) + 1 \leq \max(4, d+1) + 1 = \max(5, d+2).$$

If the quotient is zero, the class is at most one. Otherwise the first inequality follows by lifting its lower-central series: the next term lies in $Z(L)$, and one more bracket kills it.

For $d \geq 4$, any nonzero one-dimensional top cover from Theorem 6.6 has exact ordinary rank d, class $d+2$, and central second derived ideal. For $d = 2$, the eight-dimensional algebra in Lemma R.6 has class five and central second derived ideal. Its direct sum with a central one-dimensional abelian algebra attains class five at exact rank three. These models prove sharpness at both exact rank and rank at most d. □

R.8 Point generation, derived length, and dimension

Lemma R.7 (Point generation and point rank)**.** *The subsolution generated by $x_1, \dots, x_s$ is*

$$\operatorname{Lie}_{\mathbb{F}_2}\langle x_1, f(x_1), \dots, x_s, f(x_s)\rangle. \tag{291}$$

This identity does not require finite dimension. In a finite solution, the minimum number of point generators is $d(L) = \dim_{\mathbb{F}_4} L/L'$.

Proof. Let H be the right-hand side of (291). The span of its displayed generators is f-stable by $f^2(x) = f(x) + x$. For Lie words use recursively the bracket-image identity

$$f([u,v]) = [f(u), f(v)] + [u,v] + [f(u) + f(v), [u,v]].$$

If the f-values of shorter words belong to H, so does the f-value of their bracket. The quadratic expansion

$$f\Big(\sum_i u_i\Big) = \sum_i f(u_i) + \sum_{i<j} [u_i, u_j]$$

then proves $f(H) \subseteq H$. Conversely any subsolution containing the points contains their orbit partners and their polar brackets, hence contains H.

The images of s point generators span the abelianization over its intrinsic $\mathbb{F}_4$, giving $d(L) \leq s$. Conversely lift an $\mathbb{F}_4$-basis of L/L'. The corresponding paired elements form a binary basis modulo L'. Their Lie subalgebra H satisfies $L = H + L'$. Induction on the lower-central series gives $L = H + \gamma_j(L)$ for every j: replace the entries in $[L, L]$ by $H + \gamma_j(L)$. Finite nilpotence gives $H = L$. Apply the first assertion. □

Proof of Theorem 6.13. By Lemma R.7, an object counted by $\ell_{\max}(s)$ has $d \leq s$. Theorem 6.2 gives $\operatorname{cl} L \leq C(s)$. The standard commutator inclusion $L^{(j)} \subseteq \gamma_{2^j}(L)$ implies

$$\operatorname{dl}(L) \leq \lfloor \log_2 \operatorname{cl} L \rfloor + 1.$$

For $s \geq 1$, $C(s) \leq (s+1)^4$. Hence the right-hand side is at most $4\log_2(s+1) + 1$. The preliminary bound already makes the possible derived lengths a nonempty bounded set of integers, so their maximum exists before the finite free-object construction.

For the lower bound, take L_{s-1} from Theorem 6.11. It has exact point rank s, and Proposition S.4 gives the lower endpoint in (65). The case $s = 1$ is the nonzero linear scalar solution. Finally direct sum with linear scalar planes raises any smaller rank to s while preserving a positive derived length. Since $\ell_{\max}(s) \geq 1$, the maximum at exact rank equals $\ell_{\max}(s)$. □

Proof of (66). For a class-$c \geq 2$ solution, the first lower-central factor has binary dimension $2d$; all remaining $c-1$ nonzero factors have positive even dimension by Proposition 2.7. Thus

$$\dim_{\mathbb{F}_2} L \geq 2d + 2(c-1).$$

Lemma 2.6 gives $d \geq 2$, and the class bound gives $d \geq a(c)$. The explicit quartic bound satisfies

$$c \leq C(d) \leq \frac{(d+1)^4}{2} + d + 1 \leq \frac{(d+2)^4}{2}.$$

Thus $d \geq (2c)^{1/4} - 2$, proving the second inequality. In particular $m(c) \geq 2c + \Omega(c^{1/4})$; the integer inverse bound through $a(c)$ retains the sharper constants. □

R.9 Finite free objects and central stem towers

We use s for point rank and $k = \mathbb{F}_4$ for the coefficient field. All arrows below are binary-linear intertwiners.

Lemma R.8 (A finite bound at fixed point rank)**.** *Put*

$$B(s) = \sum_{j=1}^{C_0(s)} (2s)^j \qquad (s \geq 1).$$

Every finite solution generated by at most s marked points has binary dimension at most $B(s)$. There are finitely many marked isomorphism types of such solutions.

Proof. The paired generation lemma supplies at most $2s$ binary Lie generators, and Proposition 6.1 gives class at most $C_0(s)$. Jacobi expands every bracket word into left-normed words, and only lengths at most $C_0(s)$ are needed. There are at most $(2s)^j$ such words of length j. This proves the dimension bound. On each standard binary vector space of dimension at most $B(s)$ there are finitely many quadratic maps and finitely many ordered s-tuples. Restricting to solutions and generating tuples preserves finiteness. □

Existence and universal property in Theorem 6.14. Choose one representative of each marked isomorphism type in Lemma R.8, and form their finite direct product. Let $\mathfrak{F}_s^{\mathrm{fin}}$ be its subsolution generated by the diagonal marked points. It is finite, and its dimension is at most $B(s)$ by the lemma.

For any desired point values in a finite target, their generated subsolution is marked-isomorphic to one chosen factor. Projection to that factor, followed by its isomorphism and inclusion into the target, gives the required map. A binary-linear intertwiner preserves polar brackets and the f-images of generators, so Lemma R.7 proves uniqueness. The marked classes span the abelianization over k. Mapping to the linear solution k^s with its standard basis proves that they are independent, hence a basis. For $s = 0$, take the zero solution. □

Lemma R.9 (Generation in a stem extension)**.** *If a surjective morphism of finite solutions $E \to G$ has kernel $A \subseteq E'$, arbitrary lifts of point generators of G generate E.*

Proof. The map on abelianizations is an isomorphism. The subalgebra generated by the lifted points and their orbit partners therefore satisfies $E = H + E'$. The nilpotent generation argument in Lemma R.7 gives $E = H$. □

Proposition R.10 (The full quadratic cover is stem)**.** *For a finite solution g, put $A = H_g(k)^*$. A central quadratic extension representing*

$$\mathrm{id} \in H_g(k) \otimes_k H_g(k)^*$$

has its entire kernel A in the derived ideal. It is nontrivial exactly when $H_g(k) \neq 0$.

Proof. The central-lift construction of Proposition 13.1 tensors with coefficient spaces, so the displayed class has a representative. Every nonzero k-linear functional λ on A pushes this extension out to a nonzero scalar quadratic extension class. Its ordinary binary CE class is nonzero by the polarization injection in that proposition.

Suppose $A \cap E' \neq A$. This intersection is a k-subspace: it is f-stable, and $f|_A = \omega\,\mathrm{id}$. Choose a nonzero k-linear $\lambda : A \to k$ vanishing on $A \cap E'$. It extends to a binary-linear functional on E vanishing on E'. In a section $E = V \oplus A$, write this functional as $t(x) + \lambda(a)$. Its vanishing on brackets says $\lambda\alpha(x, y) = t([x, y])$, exhibiting the scalar polar cocycle as a CE coboundary. This contradicts the injection. Hence $A \subseteq E'$. □

Proposition R.11 (Recognition by vanishing quadratic extensions)**.** *Let F be a finite solution generated by marked points whose classes form a k-basis of F/F'. Then F has the finite-target free property on those points if and only if $H_F(k) = 0$.*

Proof. Assume $H_F(k) = 0$. We construct the map to a finite target E, with arbitrary desired marked values y_i, by induction on its binary dimension. If E is abelian, the map exists through F/F', by k-linearity. Otherwise global nilpotence gives a nonzero full center $Z(E)$, which is f-stable and on which f is its intrinsic scalar ω. The induction hypothesis supplies a map $h : F \to E/Z(E)$ with the required marked cosets.

Pull back the central extension $E \to E/Z(E)$ along h. Its coefficient space is $Z(E)$, and its quadratic extension space is $H_F(k) \otimes_k Z(E) = 0$. It therefore admits a binary-linear solution section. Composition gives a lift $S : F \to E$ of h. The discrepancies $z_i = y_i + S\xi_i$ lie in $Z(E)$. There is a unique k-linear $t : F/F' \to Z(E)$ taking the marked basis to the z_i. Inflate it to F. Since its values are central,

$$f_E(Sx + tx) = S(f_F x) + \omega t(x) = (S + t)(f_F x).$$

Thus $S + t$ takes the exact desired marked values. Uniqueness follows from paired Lie generation.

Conversely, given a scalar central extension of a finite free F, choose lifts of its marked points. The universal property gives a solution homomorphism from F into the extension. Its composite with the quotient fixes every generator, hence is the identity. Every scalar extension therefore splits, so $H_F(k) = 0$. □

Termination assertion in Theorem 6.14. For $s = 0$, the starting solution is zero, its extension space is zero, and the procedure terminates immediately. Assume $s \geq 1$. At each nontrivial full-cover step, Propositions R.10 and Lemma R.9 preserve the marked abelianization k^s and point generation. All stages have dimension at most $B(s)$. Each nonzero central coefficient space adds at least two binary dimensions, so there are at most

$$\left\lfloor \frac{B(s) - 2s}{2} \right\rfloor$$

nontrivial steps. At termination the full extension space vanishes; otherwise another nontrivial full cover would exist. Recognition by Proposition R.11 identifies the terminal marked object with $\mathfrak{F}_s^{\mathrm{fin}}$. □

More generally a fixed-rank finite stem tower with $s \geq 1$, starting in E_0, has at most $\lfloor (B(s) - \dim_{\mathbb{F}_2} E_0)/2 \rfloor$ nontrivial stages. Nonstem extensions can enlarge the abelianization and are not subject to this bound. The universal property extends to locally finite infinite targets because finitely many marked points then generate a finite subsolution. Extending it to arbitrary infinite targets would require an additional local-finiteness or residual argument, which is not supplied here.

Finally $\mathrm{dl}(\mathfrak{F}_s^{\mathrm{fin}}) = \ell_{\max}(s)$: every finite solution generated by at most s points is its quotient, and the free object itself has point rank s. The construction by products, the nilpotent generation argument and the central projectivity criterion are classical mechanisms. Their finite scope here comes from the independent preliminary quartic bound and the specific quadratic extension space; none requires a computation of a named terminal stage.

S The current family and its binary trace restriction

S.1 A classical restricted presentation

Put $k = \mathbb{F}_4$, and write

$$R_m = k[t_1, \dots, t_m]/(t_1^2, \dots, t_m^2), \qquad A_m = (t_1, \dots, t_m) \subseteq R_m.$$

For nonempty $B \subseteq [m]$, let $t_B = \prod_{i \in B} t_i$. These monomials form a basis of A_m. Their product is $t_{B \cup C}$ for disjoint supports and zero otherwise. In particular every element of A_m has square zero.

The classical three-dimensional simple algebra $S = W(1;2)^{(1)}$ has basis U, V, W and table

$$[U, V] = U, \qquad [U, W] = V, \qquad [V, W] = W.$$

The inner derivation $\delta = \mathrm{ad}_U$ sends $U \mapsto 0, V \mapsto U, W \mapsto V$. In characteristic two its square is again a derivation. The commuting pair δ, δ^2 spans a restricted derivation algebra with $\delta^{[2]} = \delta^2$ and $(\delta^2)^{[2]} = 0$. These derivations occur in the classical description of $\mathrm{Der}\, S$ in [38, Proposition 3.3(1)].

Let x, y act on $S \otimes A_m$ by δ, δ^2, respectively, and put $[x, y] = 0$. In $\widehat{L}_m = (kx \oplus ky) \ltimes (S \otimes A_m)$, write $U_B = U \otimes t_B$, and similarly for V_B, W_B. Besides the current products on disjoint supports, the nonzero external products are

$$[x, V_B] = U_B, \qquad [x, W_B] = V_B, \qquad [y, W_B] = U_B. \tag{292}$$

The semidirect-product construction proves Jacobi. Its current table is the disjoint-support table of [39, Proposition 2.1]; adjoining δ supplies a square root of that source's external δ^2-action.

Lemma S.1 (Restricted square and exact height)**.** *The map P in* (63) *is a restricted square on $\widehat{L}_m$, and $P^2 = 0$. Thus $\widehat{f}_m = \omega\,\mathrm{id} + P$ satisfies $\widehat{f}_m^2 = \widehat{f}_m + \mathrm{id}$.*

Proof. Polarization of P gives exactly the displayed Lie table, and $P(\lambda X) = \lambda^2 P(X)$. On its monomial basis, $P(x) = y$, with all other basis squares zero. The table gives $(\mathrm{ad}_x)^2 = \mathrm{ad}_y$. Every other basis adjoint squares to zero: any repeated nonempty current support makes the two-step product zero, including on the external vectors. The discrepancy $\mathrm{ad}_{P(X)} + (\mathrm{ad}_X)^2$ is additive and Frobenius-semilinear, so its vanishing on this basis proves the restricted adjoint identity.

Put

$$u' = av + bw + uv, \qquad v' = aw + uw, \qquad w' = vw.$$

The second iterate is

$$P^2(a, b, u, v, w) = (0, 0, a^2 vw + u'v', u'w', v'w') = 0.$$

Indeed $u'v' = a^2 vw$ after the two $auvw$ terms cancel, and the other products contain u^2, v^2, or w^2. The restricted adjoint identity also gives $[X, P(X)] = 0$. Consequently

$$\begin{aligned}(\omega\,\mathrm{id} + P)^2(X) &= \omega^2 X + (\omega + \omega^2)P(X) + P^2(X) + [\omega X, P(X)] \\ &= \omega^2 X + P(X) = (\omega\,\mathrm{id} + P)(X) + X.\end{aligned}$$

This calculation uses classical restricted-square identities; the broader quadratic-map correspondence is already present in [31, Lemma 9.5 and Example 9.6]. □

Lemma S.2 (Exact class of the ambient algebra)**.** *For $m \geq 1$, the class of $\widehat{L}_m$ is $2m+1$, and $\gamma_{2m+1}(\widehat{L}_m) = kU_{[m]}$.*

Proof. The weights

$$\operatorname{wt} x = 1, \quad \operatorname{wt} y = 2, \quad \operatorname{wt} U_B = 2|B| + 1, \quad \operatorname{wt} V_B = 2|B|, \quad \operatorname{wt} W_B = 2|B| - 1$$

give a positive grading: every nonzero bracket adds weights. The maximum is $2m + 1$, so $\gamma_{2m+2}(\widehat{L}_m) = 0$. The left-normed word

$$W_{\{1\}}, x, x, W_{\{2\}}, x, \ldots, W_{\{m\}}, x$$

has length $2m+1$ and value $U_{[m]}$. The first two x-actions give $U_{\{1\}}$, and each later $W_{\{i\}}, x$ pair enlarges its support by i. Scalar multiples in the first current input give the full k-line. That line is the only homogeneous component of weight $2m + 1$, which proves the last assertion as well. □

The ambient Lie algebra is in fact 3-Engel by a direct calculation. For $X = (a, b, u, v, w)$, set $h = a + u$. Its action on the current ideal and its square are

$$T_X = \begin{pmatrix} v & h & b \\ w & 0 & h \\ 0 & w & v \end{pmatrix}, \qquad T_X^2 = \begin{pmatrix} hw & hv + bw & a^2 \\ vw & 0 & hv + bw \\ 0 & vw & hw \end{pmatrix}.$$

Since $h^2 = a^2$ and $u^2 = v^2 = w^2 = 0$, multiplication gives $T_X^3 = 0$. The image of an external vector $(A, B, 0, 0, 0)$ is the current vector $A(v, w, 0) + B(w, 0, 0)$, which T_X^2 kills. Hence $(\operatorname{ad}_X)^3 = 0$ on all of $\widehat{L}_m$. This separate verification also applies to its binary Lie subalgebras.

S.2 The invariant binary trace graph

Proof of Theorem 6.11. On the external plane the map $\widehat{f}_m$ is

$$(a, b) \longmapsto (\omega a, \omega b + a^2).$$

If $b = a + a^2$, its second coordinate is $\omega a + \omega^2 a^2 = \omega a + (\omega a)^2$. Thus the binary-linear graph

$$L_m = \{(a, a + a^2, u, v, w) : a \in k,\ u, v, w \in A_m\}$$

is $\widehat{f}_m$-stable. Restriction gives exactly (61); the functional equation and binary degree at most two are inherited. The term $a^2 w$ remains binary quadratic because $a \mapsto a^2$ is binary-linear. The graph has binary dimension $2 + 6(2^m - 1) = 6 \cdot 2^m - 4$.

For $m \geq 1$, it contains x and the full current ideal. The class-attaining word in Lemma S.2 therefore lies in it, and the ambient upper bound applies. It follows that

$$\operatorname{cl} L_m = 2m + 1, \qquad \gamma_{2m+1}(L_m) = kU_{[m]}.$$

For $m = 0$, the graph is the single linear scalar block, with class one.

For $m \geq 1$, direct calculation gives

$$L'_m = \operatorname{span}_k\{U_B, V_B : B \neq \varnothing\} \oplus \operatorname{span}_k\{W_B : |B| \geq 2\}. \tag{293}$$

The external products with x supply all scalar multiples of U_B, V_B, since the current inputs have arbitrary coefficients in k. Disjoint current products supply all W_B on supports of size at least two. The reverse inclusion follows from the table: no bracket has external component or singleton W component. On the quotient, the induced f_m is multiplication by ω on $a, W_{\{1\}}, \dots, W_{\{m\}}$. Thus the intrinsic abelianization rank is $m+1$. For $m \geq 2$, both $V_{\{1\}}$ and $U_{\{2\}}$ lie in L'_m, and their bracket is the nonzero $U_{\{1,2\}}$.

Finally the original binary algebra is not restrictable for $m \geq 1$. Indeed

$$(\mathrm{ad}_x)^2 V_{\{i\}} = 0, \qquad (\mathrm{ad}_x)^2 W_{\{i\}} = U_{\{i\}}.$$

If $(\mathrm{ad}_x)^2 = \mathrm{ad}_z$, write the external coordinates of z as $(a, a+a^2)$. The singleton $U_{\{i\}}$ coefficient of $[z, V_{\{i\}}]$ is a, because current-current products strictly increase support degree or vanish. It must be zero, so $a = 0$, and the trace relation gives $b = 0$. But then $[z, W_{\{i\}}]$ has no singleton $U_{\{i\}}$ component, a contradiction. A restricted square would make every squared adjoint inner, so none exists on this binary Lie algebra. This does not conflict with the restricted structure on its larger parent. □

Proposition S.3 (Center and relative quotient minimality)**.** *For $m \geq 1$, both $\widehat{L}_m$ and L_m have center exactly $kU_{[m]}$. No nontrivial solution quotient of L_m retains class $2m+1$.*

Proof. For a central vector, bracketing first with x forces $v = w = 0$. Bracketing with $V_{[m]}$ forces $a = 0$, since positive-support current products with $t_{[m]}$ vanish. In the parent, bracketing with $W_{[m]}$ also forces $b = 0$; in the trace graph this already follows from $a = 0$. The remaining u must annihilate A_m, whose annihilator in A_m is $kt_{[m]}$. To see this, multiplication by t_i kills precisely the monomial terms already containing i; the other supports remain distinct. Simultaneous annihilation by all t_i leaves only the full support.

Every nonzero ideal in a finite nilpotent Lie algebra meets its center: take the last nonzero iterated ambient-adjoint image of that ideal. For an f_m-stable ideal, its intersection with the center is also f_m-stable. The center is a single binary irreducible scalar plane, so that intersection is the whole center. It is also the last lower-central term. Any nonzero quotient kernel therefore kills that term and lowers the class. □

Details for Corollary 6.12. For odd $c = 2j+1$, use L_j. For even $c = 2j$, quotient L_j by $kU_{[j]} = \gamma_{2j+1}(L_j)$. This is an f_j-stable central plane, since f_j acts there as ω. The quotient has class exactly $2j$: a nonzero lower-central series in a nilpotent Lie algebra cannot have two consecutive equal terms. Its dimension decreases by two, giving (64).

For every resulting class $c \geq 5$, the derived bracket $U_{\{1,2\}} = [U_{\{1\}}, V_{\{2\}}]$ survives. It has ambient grading weight five, whereas every element of $\gamma_6(\widehat{L}_j)$ has weight at least six. In the even construction the quotient term is $\gamma_{c+1}(L_j) \subseteq \gamma_6(\widehat{L}_j)$, so it cannot kill this bracket. The odd construction has no quotient. Thus all advertised comparisons for $c \geq 5$ are non-metabelian.

With $j = \lceil (c-1)/2 \rceil$, both dimensions are at most $6 \cdot 2^j - 4$. For $c \geq 5$, $j \leq c-3$, giving

$$6 \cdot 2^j - 4 \leq \tfrac{3}{4}\, 2^c - 4 < 2^c + 2c - 6 = \mu(c).$$

The even bound is two dimensions smaller than this common bound. □

S.3 Exact derived length

Proposition S.4 (Derived-support recurrence)**.** *For every $m \geq 0$,*

$$\mathrm{dl}(L_m) = \lceil \log_2(3m+3) \rceil - 1.$$

Proof. For $m \geq 1$, equation (293) gives the support thresholds $(a_1, b_1, c_1) = (1, 1, 2)$ for the three current letters in L'_m. The product of the spans of monomials of degree at least $p, q \geq 1$ is exactly the span of those of degree at least $p+q$. Indeed a support of size at least $p+q$ can be partitioned into disjoint parts meeting the two thresholds. Thus the table gives the exact recurrence

$$(a_{r+1}, b_{r+1}, c_{r+1}) = (a_r + b_r, a_r + c_r, b_r + c_r) \qquad (r \geq 1).$$

Induction gives

$$(a_r, b_r, c_r) = \begin{cases} ((2^{r+1}-1)/3, (2^{r+1}-1)/3, (2^{r+1}+2)/3), & r \text{ odd}, \\ ((2^{r+1}-2)/3, (2^{r+1}+1)/3, (2^{r+1}+1)/3), & r \text{ even}. \end{cases}$$

The minimum threshold is $\lfloor 2^{r+1}/3 \rfloor$. Consequently $L_m^{(r)} = 0$ first when $2^{r+1} \geq 3(m+1)$, which gives the stated formula. For $m = 0$, the solution is nonzero abelian and both sides equal one. □

S.4 Low-degree linearization and a conditional extension theorem

Proposition S.5 (Quadratic inverse for the entire family)**.** *For the displayed binary-linear scalar $J = \omega \,\mathrm{id}$ on $k \oplus A_m^3$, the map $H = f_m + J^2$ is a quadratic permutation with quadratic inverse, and $Hf_m = JH$. Explicitly,*

$$H(a, u, v, w) = (a, u + av + (a + a^2)w + uv, v + aw + uw, w + vw),$$
$$H^{-1}(a, U, V, W) = (a, U + aV + aW + UV, V + aW + UW, W + VW).$$

Proof. In the full restricted presentation, $\widehat{H} = \mathrm{id} + P$ is an involution, because $P^2 = 0$ and $[X, P(X)] = 0$. It maps the external graph $b = a + a^2$ to the linear diagonal $b = a$. Restricting it in these two directions gives precisely the displayed formulas; the diagonal is stable under ambient multiplication by ω. Alternatively, their compositions cancel directly using only $u^2 = v^2 = w^2 = 0$. The general identity $(f + J^2)f = J(f + J^2)$, for $f^2 = f + \mathrm{id}$ and $J^2 + J = \mathrm{id}$, proves the intertwining assertion. □

Proposition S.6 (Central lifts of an involutively linearized base)**.** *Let g be a binary quadratic solution, and suppose a binary-linear J_0, with $J_0^2 + J_0 = \mathrm{id}$, makes $H_0 = g + J_0^2$ an involution. For any finite k-space A and quadratic $q : V \to A$ with $qg = \omega^2 q$, the central lift*

$$F_q(x, a) = (g(x), \omega a + q(x))$$

has a quadratic linearizing permutation

$$H(x, a) = (H_0(x), a + q(x))$$

whose inverse has binary degree at most three. In particular this applies to every central lift of a height-two restricted base $g = \omega \,\mathrm{id} + P$, $P^2 = 0$.

Proof. Write $\alpha = \mathrm{pol}\, q$. The inverse is

$$H^{-1}(y, b) = (H_0(y), b + qH_0(y)).$$

Quadratic expansion and the eigenfunction equation give

$$qH_0(y) = \omega^2 q(y) + q(J_0^2 y) + \alpha(g(y), J_0^2 y).$$

The first two terms have binary degree at most two and the last has degree at most three. Moreover $H_0 g = J_0 H_0$, and $qg + q = \omega q$, so $HF_q = (J_0 \oplus \omega \,\mathrm{id})H$. For a restricted base, $H_0 = \mathrm{id} + P$ is involutive by the calculation in Proposition S.5. □

The preceding proposition controls one central-extension level, with any number of simultaneous central coordinates. It does not identify a noncentral class-three kernel with a central extension, or establish the same inverse bound for arbitrary successive lifts. The trace family supplies a separate infinite collection with quadratic inverse; its class and derived-length formulas concern the original polar bracket, which the nonlinear change of coordinates does not preserve.

T Fixed quadratic rigidity and the complete free-quotient obstruction

Let $f : V \to V$ be a binary-quadratic solution of $f^2 = f + I$, and let $L = (V, B)$ be its polar Lie algebra. All functions below have zero constant term and values in $k = \mathbb{F}_4$. The coefficient space can more generally be any finite-dimensional k-space, by tensoring. Write $\gamma_i = \gamma_i(L)$.

Lemma T.1 (Fixed binary cocycles on a scalar Lie algebra)**.** *Let $\mathfrak{a}$ be a Lie algebra over k, with scalar operator $J = \omega\,\mathrm{id}$, and regard it as a binary Lie algebra. If an alternating binary CE two-cocycle $\eta : \mathfrak{a} \times \mathfrak{a} \to k$ satisfies*

$$\eta(Jx, Jy) = \eta(x, y),$$

then $\eta(\mathfrak{a}', \mathfrak{a}) = 0$.

Proof. Extend the input binary space and bracket to $\mathfrak{a}_{\mathrm{bin}} \otimes_{\mathbb{F}_2} k$, and extend η bilinearly to this space with values in k. The centroid operator J has the distinct eigenvalues ω, ω^2, so the extended Lie algebra is a direct sum of its two eigenspace ideals $\mathfrak{a}_+$ and $\mathfrak{a}_-$. They commute: applying the centroid to a cross bracket gives both distinct scalar multiples of that bracket. The fixed-form identity makes η vanish on each same-sign pair, since $\omega^2 \neq 1$ and $\omega \neq 1$. For $x, y \in \mathfrak{a}_+$, $z \in \mathfrak{a}_-$, its CE equation reduces to $\eta([x, y], z) = 0$. The analogous equation kills the other derived cross pair. Together with same-sign vanishing this gives the assertion on the extended algebra and hence on the original binary algebra. □

Theorem T.2 (First-integral rigidity in every nilpotency class)**.** *For every finite quadratic solution, L is nilpotent by Theorem 4.1. Every binary-quadratic function $r : V \to k$ satisfying $r \circ f = r$ factors through V/L'. If $d = \dim_k(V/L')$ with its intrinsic scalar structure, then*

$$I_f(k) = \{r \in \mathcal{Q}_{\leq 2}(V, k) : r \circ f = r\} \simeq I_{\bar{f}}(k), \qquad \dim_k I_f(k) = d^2.$$

Proof. Put $\alpha = \mathrm{pol}\, r$. The norm identity

$$r(x) + r(fx) + r(f^2x) = \alpha(x, fx)$$

has left side $r(x)$, since r is invariant and three equals one in characteristic two. Taking its third additive difference proves $d_{\mathrm{CE}}\alpha = 0$.

If $\alpha = 0$, then r is linear; invariance and $f^2x = fx + x$ force $r = 0$, so the claim holds. Otherwise nilpotency makes the following integer finite:

$$m = \max\{i + j : \alpha(\gamma_i, \gamma_j) \neq 0\}.$$

Assume $m \geq 3$. Because $2m > m$, the restriction $\alpha(\gamma_m, \gamma_m)$ vanishes. Thus $r|_{\gamma_m}$ is linear. The ideal γ_m is f-invariant. A linear invariant function on it vanishes: for $u \in \gamma_m$,

$$r(u) = r(f^2u) = r(fu + u) = r(fu) + r(u) = 0.$$

For $x \in \gamma_i$, $y \in \gamma_j$ with $i + j = m$, polarizing $r \circ f = r$ gives

$$\alpha(x, y) = \alpha(fx, fy) + r(B(x, y)) + \alpha(fx + fy, B(x, y)). \tag{294}$$

Here $B(x, y) \in \gamma_m$, so the middle term vanishes. The last term vanishes too: its two pair depths are $i + m$ and $j + m$, both larger than m, and f preserves γ_i and γ_j. Consequently $\alpha(fx, fy) = \alpha(x, y)$ on these pairs.

The maximal-depth part of α defines an alternating binary two-form α_m on $\mathrm{gr}_\gamma L = \bigoplus_i \gamma_i/\gamma_{i+1}$: on factors of degrees i, j with $i + j = m$, use α, and on all other pairs use zero. It is well defined since replacing either input by the next lower-central term raises the pair depth above m. The CE equation for α on triples with total depth m is exactly the CE equation for α_m; on other homogeneous triples its terms are zero. Thus α_m is CE-closed.

The graded scalar theorem makes $\mathrm{gr}_\gamma L$ a k-Lie algebra under the operators induced by f. Equation (294) shows that α_m is fixed by these simultaneous scalars. By Lemma T.1, it vanishes if either input belongs to the derived algebra of $\mathrm{gr}_\gamma L$. The lower-central graded algebra is generated by degree one; in particular every component of degree at least two belongs to its derived algebra. When $m \geq 3$, every pair of positive degrees summing to m has such an input. This forces $\alpha_m = 0$, contradicting the definition of m.

It follows that $m \leq 2$. Therefore $\alpha(L', V) = 0$. The restriction $r|_{L'}$ is linear and invariant, so it is zero by the same argument above. Hence $r(x + w) = r(x)$ for every $w \in L'$, proving factorization.

On the abelianization, $\bar{f}$ is multiplication by ω. Its invariant quadratic functions have basis u_i^3, and $u_i^2 u_j, u_i u_j^2$ for $i < j$, of total dimension $d + 2\binom{d}{2} = d^2$.

□

Corollary T.3 (The fixed cohomological sector in every nilpotency class)**.** *For every finite quadratic solution, with coefficient line k,*

$$(K_f)_1 \simeq I_f(k) \simeq I_{\bar{f}}(k), \qquad \dim(K_f)_1 = d^2.$$

Consequently, writing $h_f = \dim_k H_f(k)$,

$$\dim K_f = d^2 + 2h_f, \qquad b_2 = \operatorname{rank}\Theta + d^2 + 2h_f, \qquad h_f = \frac{b_2 - \operatorname{rank}\Theta - d^2}{2}.$$

These identities determine the fixed sector and its contribution to the obstruction in arbitrary nilpotency class. They do not assign the general values of b_2 or $\operatorname{rank}\Theta$.

Proof. The canonical polarization section identifies invariant quadratics with $(K_f)_1$. Binary descent exchanges the two nontrivial cyclic eigenspaces, each of dimension h_f. Apply the theorem and rank–nullity for Θ. □

T.1 Ordinary binary cohomology of the free class-three quotient

Put $k = \mathbb{F}_4$, and let $\mathfrak{l}_d$ be the ordinary free class-three k-Lie algebra on $U = k^d$. Write its homogeneous decomposition as

$$\mathfrak{l}_d = U \oplus Z \oplus T, \qquad Z = \Lambda_k^2 U, \qquad T = (U \otimes Z)/\operatorname{im}\partial_3.$$

The functional normal form of the paper is

$$g_d(u, z, t) = (\omega u, \omega z + Q(u), \omega t + u \cdot z + u \cdot Q(u)).$$

Its original binary polar algebra is not being identified with $\mathrm{Res}_{k/\mathbb{F}_2} \mathfrak{l}_d$. Instead it is a central, nonhomogeneous deformation of that algebra. This distinction is retained throughout the argument.

Lemma T.4 (A degree-three central-deformation criterion)**.** *Over a field F of characteristic two, let $L_0 = U \oplus Z \oplus T$ be a class-three graded Lie algebra with T central, $[U,U] = Z$, $[U,Z] = T$, and $[Z,Z] = 0$. Let $R : \Lambda^2_F U \to T$ be any alternating linear map, and define L_R by replacing the bracket of two degree-one inputs by*

$$[u,v]_R = \beta(u,v) + R(u,v), \qquad \beta(u,v) = [u,v]_0.$$

This is again a Lie algebra. If

$$D : U^* \otimes Z^* \longrightarrow \Lambda^3_F U^*, \qquad D(b)(u,v,w) = b(u,\beta(v,w)) + b(v,\beta(u,w)) + b(w,\beta(u,v)) \quad (295)$$

is onto, then $\dim_F H^2_{\mathrm{CE}}(L_R,F)$ is independent of R. The same conclusion holds for the filtration dimensions obtained by restricting cocycles to the (U,T) and (Z,Z) components.

Proof. The new term is central, so Jacobi is unchanged. For any ordinary CE two-cocycle α, CE on (U,Z,T) forces $\alpha(T,T) = 0$, since $[U,Z]$ spans T. CE on (U,U,T) then forces $\alpha(Z,T) = 0$, since $\beta(U,U)$ spans Z. Thus the remaining cochain components are

$$a \in \Lambda^2 U^*, \quad b \in U^* \otimes Z^*, \quad c \in U^* \otimes T^*, \quad e \in \Lambda^2 Z^*.$$

The (U,U,Z) equation is a fixed homogeneous linear condition $M(c,e) = 0$, independent of R: its prospective R term pairs T with Z and vanishes. The (U,U,U) equation is $Db + C_R c = 0$. All other CE equations now hold automatically. Since D is onto, every $(c,e) \in \ker M$ admits a b, with fiber dimension $\dim\ker D$. The a component is arbitrary. Consequently

$$\dim Z^2(L_R,F) = \binom{\dim U}{2} + \dim\ker D + \dim\ker M,$$

independently of R. Also $[L_R,L_R] = Z \oplus T$, because $[U,Z]$ spans T and $[U,U]$ projects onto Z. Hence $\dim B^2 = \dim Z + \dim T$ is fixed. This proves the assertion, including the stated component filtration. □

Lemma T.5 (The binary free quotient satisfies the criterion)**.** *For $L_0 = \mathrm{Res}_{k/\mathbb{F}_2}\mathfrak{l}_d$, the map in* (295) *is onto for every $d \geq 1$. The original polar algebra L_{g_d} is an L_R as in Lemma T.4.*

Proof. The polar bracket is

$$[(u,z,t),(v,w,s)] = (0,\beta(u,v),u\cdot w + v\cdot z + R(u,v)), \qquad R = \mathrm{pol}(u\cdot Q(u)).$$

The Jacobi relations cancel its distinct-index cubic terms, leaving a binary-bilinear R into the central T. This is exactly the central-deformation form above.

Surjectivity can be checked after the faithfully flat scalar extension from $\mathbb{F}_2$ to k. Since $k \otimes_{\mathbb{F}_2} k \simeq k \times k$, the dual of the binary degree-one space splits as $P \oplus \overline{P}$, and the degree-two dual splits as $\Lambda^2 P \oplus \Lambda^2 \overline{P}$. In these coordinates D is the ordinary exterior wedge. Its pure-type pieces map onto $\Lambda^3 P$ and $\Lambda^3 \overline{P}$. Its mixed pieces

$$\overline{P} \otimes \Lambda^2 P, \qquad P \otimes \Lambda^2 \overline{P}$$

map isomorphically onto the two mixed summands of $\Lambda^3(P \oplus \overline{P})$. These are all four summands. The argument includes $d = 1,2$: the absent pure exterior powers are zero, not additional exceptions. □

Theorem T.6 (The complete ordinary binary CE dimension)**.** *For every $d \geq 1$,*

$$\begin{aligned} b_2(g_d) &:= \dim_{\mathbb{F}_2} H^2_{\mathrm{CE}}(L_{g_d},\mathbb{F}_2) = \dim_k H^2_{\mathrm{CE}}(L_{g_d},k) \\ &= \frac{d^4+d^2}{2} = d^2 + 6\binom{d}{2} + 18\binom{d}{3} + 12\binom{d}{4}. \end{aligned}$$

Consequently the degree-leakage obstruction has exact rank

$$\operatorname{rank}_k \Theta_d = 4\binom{d}{2} + 6\binom{d}{3} + 2\binom{d}{4}. \tag{296}$$

Proof. The preceding lemmas reduce the dimension to the underlying binary graded algebra L_0, without asserting an isomorphism of Lie algebras. Extending scalars gives a direct sum of two conjugate copies of the free class-three k-Lie algebra. CE Künneth gives

$$b_2(L_0) = 2b_2(\mathfrak{l}_d) + d^2,$$

where the cross summand comes from the tensor product of the two d-dimensional abelianizations.

For completeness, let $\mathfrak{f}_d$ be the free Lie algebra, with homogeneous bracket-length components. The presentation $\mathfrak{l}_d = \mathfrak{f}_d/\gamma_4\mathfrak{f}_d$ and the ordinary Lie Hopf formula give

$$H_2(\mathfrak{l}_d, k) \simeq \frac{\gamma_4\mathfrak{f}_d}{[\mathfrak{f}_d, \gamma_4\mathfrak{f}_d]} = \frac{\gamma_4\mathfrak{f}_d}{\gamma_5\mathfrak{f}_d}.$$

Its dimension is the fourth Witt number $(d^4 - d^2)/4$. The free Lie Hall-basis count, Hopf formula and CE Künneth all hold in characteristic two for alternating Lie algebras. In particular the integer Witt formula does not require inverting 4 in k. This yields the asserted ordinary dimension. Finally the already proved quadratic-stable dimension is

$$\dim_k K_{g_d} = d^2 + 2h_d, \qquad h_d = \binom{d}{2} + 6\binom{d}{3} + 5\binom{d}{4}.$$

Rank–nullity now gives (296). For $d = 1$ this says $b_2 = 1$, $\operatorname{rank}\Theta = 0$; for $d = 2$ it recovers $b_2 = 10$ and rank four. □

Classical homology and the binary interface. The Hopf formula and CE Künneth formula used here are classical; see [19, Section 2.2], over a field of arbitrary characteristic. The step specific to the original polar algebra is the invariance of its ordinary second Betti number under the binary Frobenius central deformation. Combining this with the admissible spectrum determines the entire leakage rank. The argument uses a dimension comparison and does not identify the original polar algebra with the restriction of scalars of its auxiliary algebra.

U Orbit geometry, orientation, and normal-element reduction

U.1 Global reversal and linear arrows

All degrees in this appendix are Boolean ANF degrees. A solution is a zero-constant binary-quadratic map $f : V \to V$ satisfying $f^2 = f + I$.

Lemma U.1 (Quadratic parallel maps)**.** *Let $\dim_{\mathbb{F}_2} V = n \geq 2$. If $h : V \to V$ has zero constant term, degree at most two, and $h(x) \in \mathbb{F}_2 x$ for every x, then there are unique $a \in \mathbb{F}_2$ and $\ell \in V^*$ such that*

$$h(x) = (a + \ell(x))x.$$

Proof. In binary coordinates, h_i vanishes identically when $x_i = 0$, so its square-free ANF is

$$h_i(x) = x_i \left(c_i + \sum_{j \neq i} b_{ij} x_j \right).$$

On $x_i = x_j = 1$, the two nonzero coordinates of a parallel vector must agree. Comparing coefficients gives $b_{ik} = b_{jk}$ for distinct i, j, k, and $c_i + b_{ij} = c_j + b_{ji}$. Thus every off-diagonal column k has a common value ℓ_k. For $n = 2$ each such column has just one entry, so the same definition applies. The remaining equations say $c_i + \ell_i = a$, independent of i. Substitution gives the formula. If $(a+\ell(x))x = 0$ for all x, the affine function $a+\ell$ vanishes on all nonzero points; for $n \geq 2$, evaluating at $e_i, e_j, e_i + e_j$ forces $a = 0$ and every $\ell_i = 0$. □

Every nonzero orbit of f is the set of the three nonzero vectors in a binary plane. Let $\mathcal{S}_f$ be this unoriented spread, and write $P_f(x) = x \wedge f(x) \in \Lambda^2_{\mathbb{F}_2} V$. Over $\mathbb{F}_2$, a nonzero decomposable two-vector determines its plane with no nontrivial scalar factor. Hence $P_f = P_g$ is equivalent to $\mathcal{S}_f = \mathcal{S}_g$.

Theorem U.2 (Global orientation rigidity)**.** *If quadratic solutions f, g on V have the same unoriented orbit spread, then $g = f$ or $g = f^2$.*

Proof. For every nonzero x, $g(x)$ is one of $f(x), f(x) + x$, so $h = g + f$ is parallel to x. The lemma gives $g(x) = f(x) + (a + \ell(x))x$. On each three-point orbit of f, the fixed-point-free order-three permutation g has one of the two cyclic orientations. Thus $a + \ell$ is constant on each orbit, and $\ell f = \ell$. Applying this twice and using binary linearity,

$$\ell(x) = \ell(f^2 x) = \ell(fx + x) = \ell(fx) + \ell(x) = 0.$$

Therefore $\ell = 0$, and $a = 0, 1$ gives respectively f, f^2. The zero-dimensional case has a unique map and is separate; every nonzero solution space has even dimension at least two, as required by the lemma. □

Corollary U.3 (Linear spread isomorphisms)**.** *Every binary-linear spread isomorphism $S : \mathcal{S}_f \to \mathcal{S}_g$ intertwines either f with g or f with g^2. Consequently*

$$\mathrm{Aut}_{\mathrm{GL}(V,2)}(\mathcal{S}_f) = \mathrm{GL}(V, 2) \cap N_{\mathrm{Sym}(V)}(\langle f \rangle),$$

and the conjugation map gives

$$1 \longrightarrow C_{\mathrm{GL}(V,2)}(f) \longrightarrow \mathrm{Aut}_{\mathrm{GL}(V,2)}(\mathcal{S}_f) \longrightarrow H \longrightarrow 1, \qquad H \leq C_2.$$

The index is one or two. Index two requires an actual binary-linear conjugacy between f and f^2; its existence is not automatic, and a splitting of this group extension is not asserted.

Proof. Apply the theorem to $S^{-1} g S$ and f. The group assertions are then ordinary conjugation on the cyclic group of order three. □

Degree and morphism boundaries. For unrestricted permutations, the orientations of the spread planes can be chosen independently: there are $2^{(2^n-1)/3}$ order-three solutions on a fixed spread. Whenever a spread admits a quadratic orientation, the degree bound leaves exactly its two global directions. This distinction is already sharp in dimension four. Take a linear solution on $\mathbb{F}_4^2$, with binary coordinates (x_1, x_2, x_3, x_4), and set

$$h(x) = \big(x_1(1 + x_3)(1 + x_4), x_2(1 + x_3)(1 + x_4), 0, 0\big).$$

Then $g = f + h$ reverses just the first coordinate plane, remains an order-three solution with the same spread, and has degree exactly three. Thus no orientation statement of this strength holds for arbitrary cubic solutions or arbitrary set permutations.

The preceding isomorphism statement extends to arbitrary binary-linear maps through the Plücker condition, as follows.

U.2 A separate all-linear-map statement

Theorem U.4 (All Plücker-compatible linear arrows)**.** *Let $T : V \to V'$ be binary-linear, where f, g are quadratic solutions on the source and target. Then*

$$(\Lambda^2 T)P_f = P_g T$$

holds if and only if $Tf = gT$ or $Tf = g^2 T$. For $T \neq 0$ the choice is unique; the zero map satisfies both alternatives.

Proof. The two intertwining alternatives imply the exterior identity. Conversely assume the latter. If $Tx = 0$ but $Tfx = v \neq 0$, applying the identity at fx gives

$$0 = v \wedge T(f^2 x) = v \wedge v = P_g(v),$$

impossible. Hence $\ker T$ is f-invariant. For $Tx \neq 0$, the identity gives the nonzero decomposable two-vector $P_g(Tx)$, so Tx, Tfx span exactly the target orbit plane. In particular a nonzero T has rank at least two. Every source plane therefore maps either to zero or isomorphically to a target plane.

Put $h = Tf + gT$. This is quadratic, zero on $\ker T$, and parallel to Tx elsewhere. Choose coordinates in which T is projection onto its rank-r image, with $r \geq 2$. The parallel-map coordinate proof applies to those r coordinates, with all kernel variables retained as extra independent variables. Comparing two active coordinates shows that coefficients on every kernel variable are the same in each active row. It yields

$$h(x) = (a + \ell(x))Tx$$

for one affine binary scalar function $a + \ell$ on the whole source. The orientation is constant on every source orbit outside $\ker T$, since its image is a target three-cycle. Consequently the zero-constant quadratic function $r(x) = \ell(fx) + \ell(x)$ is supported on $\ker T$.

A zero-constant Boolean quadratic cannot have nonzero support contained in a subspace of codimension at least two. To see this, write the subspace as $x_1 = \cdots = x_r = 0$. A function supported there has ANF divisible by $\prod_{i=1}^{r}(1 + x_i)$. If $r > 2$, its degree exceeds two unless zero. If $r = 2$, the degree bound forces a scalar multiple of that product, whose constant term is nonzero unless the scalar is zero. Thus $r = 0$.

It follows that $\ell f = \ell$, and $f^2 = f + I$ again forces $\ell = 0$. Therefore $Tf = gT$ or $Tf = g^2 T$. If both held, then $gTx = g^2 Tx$ for every x, which forces $T = 0$. This proves uniqueness for nonzero arrows. Composable nonzero arrows have the product orientation whenever their composite is nonzero; a zero composite has no unique sign. □

The Boolean support observation is also recorded in [10, Lemma 2.11]. The range-compatibility theorems in that work concern additive maps on operator spaces; the relative quadratic coefficient argument above establishes the additional statement needed here.

U.3 Normal elements and the intrinsic center

Proposition U.5 (An unoriented polar algebra)**.** *For a quadratically orientable spread, B_f is independent of its quadratic orientation. Moreover, every binary-linear map T satisfying*

$$(\Lambda^2 T)P_f = P_g T, \qquad P_f(x) = x \wedge f(x),$$

is a homomorphism of the original polar Lie algebras, whether its orientation sign is positive or negative.

Proof. Global orientation rigidity gives precisely the two orientations f and $f^2 = f+I$; adding a binary-linear map does not change the polar. The all-linear-map version gives $Tf = gT$ or $Tf = g^2T$. Polarizing the selected equality gives $TB_f(x,y) = B_g(Tx,Ty)$, since $B_{g^2} = B_g$. The zero map satisfies both signs and the same conclusion. □

The underlying category here has spreads admitting quadratic orientations as its objects and Plücker-compatible binary-linear maps as its arrows. The proposition gives a well-defined functor to binary Lie algebras. It does not assert that every Lie homomorphism preserves the spread: this already fails for arbitrary binary-linear maps between abelian polar algebras.

Definition U.6 (Normal element)**.** A member ℓ of a binary line spread $\mathcal{S}$ is normal if, for every other member m, the four-dimensional vector subspace $\ell+m$ is partitioned by members of $\mathcal{S}$. Equivalently, the corresponding projective 3-space is partitioned by spread lines. This is the standard normal-element definition of Rottey–Sheekey [36].

Lemma U.7 (The four-dimensional restriction)**.** *Every quadratic solution on a binary space of dimension at most four has zero polar bracket.*

Proof. Its polar algebra is nilpotent by the global theorem. In a nonabelian nilpotent polar algebra, abelianization has binary dimension at least four: if it had dimension two, lifts $x, f(x)$ would generate modulo the derived ideal, hence generate the nilpotent algebra, and they commute by $B(x, f(x)) = 0$. This would make the algebra abelian. Each nonzero lower central factor has even binary dimension. A nonabelian algebra therefore has dimension at least $4 + 2 = 6$. □

Theorem U.8 (Normal lines are exactly the central orbit planes)**.** *Let f quadratically orient $\mathcal{S}$, with polar algebra L. For every $\ell \in \mathcal{S}$,*

$$\ell \text{ is normal} \quad \Longleftrightarrow \quad \ell \subseteq Z(L).$$

Consequently the orientation-free set

$$N(\mathcal{S}) := \{0\} \cup \bigcup_{\ell \text{ normal}} (\ell \setminus \{0\})$$

is exactly $Z(L)$ and is a binary vector subspace. If $z = \dim_{\mathbb{F}_2} Z(L)$, then z is even and

$$\#\{\text{normal elements of } \mathcal{S}\} = \frac{2^z - 1}{3}.$$

Proof. The center is f-invariant and f is binary-linear there, by the polarized commuting identity. Suppose $\ell \subseteq Z(L)$ and $m \in \mathcal{S}$. For $z \in \ell$, $x \in m$,

$$f(z+x) = f(z) + f(x) + B(z,x) = f(z) + f(x) \in \ell + m.$$

Thus $\ell + m$ is f-invariant, and its nonzero vectors partition into whole orbit planes. This proves normality.

Conversely, let ℓ be normal. For $x \notin \ell$ let $m = \langle x, f(x)\rangle$. Distinct spread planes meet only at zero, so $U = \ell + m$ has binary dimension four. Normality says that U is a union of whole f-orbits; hence $f|_U$ is a quadratic solution on U. The preceding lemma gives $B|_{U\times U} = 0$, and in particular $B(x,\ell) = 0$. If $x \in \ell$, the same conclusion holds because a single orbit plane has zero polar: its restriction is one of the two linear order-three maps. Hence $B(V,\ell) = 0$ and $\ell \subseteq Z(L)$.

Every nonzero central vector lies in a central orbit plane, since $fZ(L) = Z(L)$. This proves $N(\mathcal{S}) = Z(L)$. The restriction to the center is a fixed-point-free binary-linear map of order three, so its dimension is even and its nonzero vectors partition into three-point orbits, giving the count. □

Proposition U.9 (Intrinsic quotient spread)**.** *Let $Z = N(\mathcal{S}) = Z(L)$ and $\pi : V \to V/Z$. Then*

$$\mathcal{S}/Z = \{\pi(m) : m \in \mathcal{S},\ m \not\subseteq Z\}$$

(with repeated images removed) is a binary line spread of V/Z, defined from $\mathcal{S}$ alone. It is quadratically oriented by the induced solution $\bar{f}$, and its polar algebra is $L/Z(L)$.

Proof. A spread plane meeting Z nontrivially is contained in Z, since Z is f-invariant. Every other plane projects injectively to a two-dimensional subspace. Since $B(V, Z) = 0$ and $f(Z) = Z$, the rule $\bar{f}(x + Z) = f(x) + Z$ is well-defined and satisfies $\bar{f}^2 = \bar{f} + I$. Its nonzero orbit plane through $x + Z$ is precisely $\pi(\langle x, f(x)\rangle)$. Thus any two nonzero projected planes are equal or disjoint, and they partition all nonzero quotient vectors. Polarization identifies the quotient polar algebra. □

Theorem U.10 (Geometric reconstruction of the upper central series)**.** *Start with $Z_0 = 0$. Whenever $Z_i \neq V$, take the quotient spread on V/Z_i, form the union of zero with all its normal elements, and let Z_{i+1} be its full inverse image in V. For a quadratically orientable spread this procedure is intrinsically defined and gives exactly the upper central series of L:*

$$Z_{i+1}/Z_i = Z(L/Z_i).$$

If $V \neq 0$ and L has nilpotency class c, it reaches V in exactly c steps. Every nonzero quotient spread encountered has at least one normal element.

Proof. Apply the center and quotient propositions successively. At each step the induced map is again a quadratic solution. Global nilpotence makes each nonzero quotient center nonzero and makes the upper central series exhaust V, with its usual length c. □

Geometric tests for quadratic orientability. A binary line spread without a normal element cannot have a quadratic orientation. More generally, if the recursive procedure above fails to produce a vector subspace, fails to give a quotient spread, or reaches a nonzero quotient without a normal element, that spread is excluded. For quadratically orientable spreads the number of normal elements in every quotient is $1, 5, 21, 85, \ldots$ according to its positive center dimension. These are necessary conditions; recursive normal reduction is not asserted to be sufficient for a quadratic orientation. The Lie center and upper central series are preserved by spread isomorphisms. For general non-surjective arrows a central element need only map to an element central in the image, as for ordinary Lie homomorphisms.

U.4 Comparison with classical normal-element geometry

The definition and quotient interpretation of normal elements are classical finite geometry. The quadratic-degree restriction identifies this incidence condition with the center of the orientation-independent polar algebra, through the four-dimensional restriction lemma.

There is a concrete classical calibration in $\mathrm{PG}(5, 2)$. McDonough–Shaw–Topalova [29] classify nine isomorphism classes of book spreads. Their §1.1 defines a book by five solids through a common spine ℓ partitioning the complement of ℓ, with four spread lines in each page in addition to the spine. A spread in this space is a book spread if and only if it has a normal element: normality provides the five pages from the quotient line spread of V/ℓ; conversely the book pages prove normality of its spine. This equivalence is a short translation of the two definitions. Mateva–Topalova [28] classify all 131044 spread isomorphism classes. Thus exactly $131044 - 9 = 131035$ of those classes have no normal element and are excluded from quadratic orientability by Theorem U.8.

One of the nine classical book spreads has exactly two possible spines: the representative Q_{11115} in their Table 1 has unmarked spread stabilizer order 1728 and marked-book stabilizer order 864. The book-stabilizer orbit of its spine therefore has two members, and the full classification distinguishes this book orbit from the other eight underlying spread types. Hence it has exactly two normal elements. Such a spread cannot be quadratically oriented, since two is not $(2^{2k}-1)/3$ for any integer k. This also shows concretely why the mere existence of a normal element is weaker than the required center geometry.

Corollary U.11 (The six-dimensional geometric range)**.** *Of the* 131044 *binary-linear equivalence classes of line spreads of* $\mathrm{PG}(5,2)$*, exactly two admit a quadratic orientation. They are the Desarguesian spread and the orbit spread of*

$$f(a,b,c) = (\omega a, \omega b, \omega c + ab), \qquad (a,b,c) \in \mathbb{F}_4^3.$$

They have respectively 21 *and* 1 *normal elements. Thus the normal element test excludes exactly* 131035 *classes immediately, and the existing six-dimensional solution classification excludes the other seven nonorientable book-spread classes.*

Proof. Every six-dimensional polar algebra is nilpotent. If its bracket is zero, f is binary-linear with minimal polynomial t^2+t+1, giving one binary-linear conjugacy class and its Desarguesian orbit spread. If the bracket is nonzero, its abelianization has even dimension at least four and less than six, hence exactly four. The complete abelianization-four chain classification gives a unique six-dimensional solution up to binary-linear conjugacy, represented by the displayed map. Its center is precisely the c-coordinate plane. Therefore the two spreads have 21 and 1 normal elements by the center theorem and are inequivalent. This proves the two-class assertion. The cited spread and book-spread enumerations give the exclusion counts. □

U.5 Cubic orbit separation and the CE degree bridge

Let $f : V \to V$ be a binary-quadratic solution of $f^2 = f + I$. The exterior powers below are over $\mathbb{F}_2$.

Theorem U.12 (The canonical plane readout)**.** *The map*

$$\mathcal{P}_f : V \longrightarrow \Lambda^2 V, \qquad \mathcal{P}_f(x) = x \wedge f(x)$$

is an invariant polynomial map of binary degree at most three. It separates exactly the f-orbits:

$$\mathcal{P}_f(x) = \mathcal{P}_f(y) \quad \Longleftrightarrow \quad y \in \{x, f(x), f^2(x)\}.$$

In coordinates its invariant scalar components are $x_i f_j(x) + x_j f_i(x)$, $i < j$.

Proof. The functional equation gives

$$\mathcal{P}_f(fx) = fx \wedge (fx + x) = x \wedge fx.$$

Its coordinates are products of a linear and a quadratic function, so they have binary degree at most three. For $x \neq 0$, x and fx are independent over $\mathbb{F}_2$: neither is zero and $fx = x$ would force $x = 0$. Thus their wedge is a nonzero decomposable tensor determining their binary two-plane uniquely. The three nonzero points of this plane are exactly $x, fx, x + fx = f^2x$. Equality of two such wedges is therefore equivalent to equality of the corresponding orbits. The zero state has wedge zero and no nonzero state does. □

This is the Plücker representation of the binary orbit planes. The plane-encoding fact is classical exterior algebra; the degree bound follows directly from the quadratic functional equation. This representation gives a canonical low-degree readout of the orbit partition.

Corollary U.13 (Exact separating degree for finite solutions)**.** *For a nonzero finite quadratic solution, the smallest binary degree of invariant functions that together separate all f-orbits is two when f is linear, and three when f is nonlinear.*

Proof. There are no nonzero invariant linear functions: applying a linear invariant to $f^2x = fx + x$ forces it to vanish. For linear f, the plane map has degree at most two and separates orbits, proving the first case. For nonlinear nilpotent f, its derived ideal W is nonzero. Every invariant quadratic factors through V/W, so it has the same value at zero and at every nonzero $w \in W$; those states are in different orbits. Quadratic invariants cannot separate all orbits. The theorem supplies a cubic separating family, proving the second case. Global nilpotency applies by Theorem 4.1. □

The cubic part is the ordinary CE differential

Take scalar values in $k = \mathbb{F}_4$, and let $C^2 = \mathrm{Alt}^2_{\mathbb{F}_2}(V, k)$, $Z^2 = \ker(d_{\mathrm{CE}} : C^2 \to C^3)$, and $B^3 = \mathrm{im}(d_{\mathrm{CE}} : C^2 \to C^3)$. Put $N = I + T + T^2$, $Tq = q \circ f$, and define the norm-generated invariant space

$$\mathcal{N}_2(f) = N\mathcal{Q}_{\leq 2}(V, k) \subseteq I_{\leq 3}(f).$$

Proposition U.14 (An exact degree bridge)**.** *There is a natural exact sequence*

$$0 \longrightarrow I_{\leq 2}(f) \longrightarrow \mathcal{N}_2(f) \xrightarrow{\Delta^3} B^3 \longrightarrow 0.$$

The function paired with $\alpha \in C^2$ is $r_\alpha(x) = \alpha(x, fx)$, and $\Delta^3 r_\alpha = d_{\mathrm{CE}}\alpha$. In particular the genuine cubic directions in this canonical separating family are exactly C^2/Z^2. If $n = \dim_{\mathbb{F}_2} V$, $w_B = \dim_{\mathbb{F}_2} L'_f$, and $b_2 = \dim_k H^2_{\mathrm{CE}}(L_f, k)$, then

$$\dim_k \mathcal{N}_2(f) = i_f + \binom{n}{2} - b_2 - w_B.$$

Here $i_f = d^2$ by Theorem 14.1.

Proof. For a quadratic refinement q of α, the norm identity gives $Nq(x) = \alpha(x, fx)$. Its third additive difference is $d_{\mathrm{CE}}\alpha$. Every alternating α has such a refinement; thus the image of Δ^3 on $\mathcal{N}_2(f)$ is exactly B^3. Its kernel consists of invariant functions of degree at most two. Conversely every invariant quadratic r belongs to the norm image because $Nr = r$. This proves exactness.

The kernel of $d_{\mathrm{CE}} : C^1 \to C^2$ is the space of linear functions killing the derived ideal, so $\dim B^2 = w_B$. Hence $\dim Z^2 = b_2 + w_B$, and $\dim B^3 = \binom{n}{2} - b_2 - w_B$. Taking dimensions in the exact sequence gives the formula. □

This exact sequence describes the norm-generated cubic family, not all cubic invariants. For example, for a linear scalar map in dimension $d \geq 3$, the abelian cubic invariants $u_i u_j u_\ell$ lie outside this norm image, whose functions are then all quadratic. The plane readout separates orbits, not the three states within an orbit or the orientation of its cycle. Indeed $\mathcal{P}_{f^2} = \mathcal{P}_f$. Thus it provides an exact orbit readout under a known rule while retaining the usual distinction from state reconstruction and rule conjugacy.

V Explicit polar obstructions

The following calculations illustrate how the polar identities exclude standard simple Lie algebras directly. The matrix example uses the four-variable identity, while the Witt example uses the commuting condition $[x, f(x)] = 0$. Together they give concrete applications of the structural identities from Section 2.

Proposition V.1. *No polar algebra has a Lie subquotient isomorphic to the underlying binary Lie algebra of $\mathfrak{sl}_3(K)$, for a finite field K of characteristic two. In particular it cannot be $\mathfrak{sl}_m(K)$ or $\mathfrak{psl}_m(K)$, $m \geq 3$.*

Proof. In $\mathfrak{sl}_3(K)$, take $a = E_{12}, b = E_{23}, c = E_{31}, d = E_{11} + E_{22}$. The three terms in (7) are $E_{11} + E_{33}, E_{22} + E_{33}, 0$, whose sum is nonzero. A polynomial identity passes to subalgebras and quotients. The upper 3×3 block gives the required subalgebra in $\mathfrak{sl}_m(K)$. For $m \geq 4$ it also embeds in the quotient by scalar matrices, since the block intersects that center trivially. For $m = 3$ the center is zero. □

Lemma V.2 (Two centralizers)**.** *Let L be a Lie algebra over a characteristic-two field K. If $x, y \in L$ satisfy*

$$C_L(x) = Kx, \qquad C_L(x + y) = K(x + y), \qquad [y, [x, y]] \notin K[x, y],$$

then there is no zero-constant binary-quadratic map with polar bracket $[\ ,\ ]$ satisfying $[v, f(v)] = 0$ for all v.

Proof. The commuting condition forces $f(x) = \alpha x$, $f(x + y) = \beta(x + y)$, for some $\alpha, \beta \in K$. Polarization then gives $f(y) = (\alpha+\beta)x+\beta y+[x, y]$. Commuting this value with y contradicts the third hypothesis. No K-linearity of f is used. □

Proposition V.3. *For every finite characteristic-two field K and $r \geq 2$, the derived divided-power Witt algebra $W(1; r)'$ over K, viewed over $\mathbb{F}_2$, is not a polar algebra of a quadratic solution.*

Proof. Use the classical divided-power algebra $A = O(1; r)$, with basis $x^{(a)}$, $0 \leq a < 2^r$, and multiplication

$$x^{(a)}x^{(b)} = \binom{a+b}{a} x^{(a+b)}$$

when $a + b < 2^r$, and zero otherwise. Let $D(x^{(a)}) = x^{(a-1)}$, with $D(1) = 0$. Then $\ker D = K$, and $W(1; r)' = \operatorname{span}_K\{x^{(a)}D : 0 \leq a \leq 2^r - 2\}$; see [25, Section 4.5].

If u is a unit of A, a commuting vector vD satisfies $u^2D(v/u) = 0$, so $C_{AD}(uD) = KuD$. Choose $x = D, y = x^{(2)}D$. Both 1 and $1 + x^{(2)}$ are units, so the first two hypotheses of Lemma V.2 hold also inside $W(1; r)'$. Finally

$$[x, y] = x^{(1)}D, \qquad [y, [x, y]] = x^{(2)}D,$$

because $(x^{(1)})^2 = 0$. These are K-independent. □

W Central-depth bounds and their exact starting systems

This appendix proves Theorem 6.8 and Proposition 6.9. Work over $k = \mathbb{F}_4$ and use left-normed brackets. Set $K = H''$, $M_0 = K$ and $M_{i+1} = [M_i, H]$. Write $F_d = \mathcal{F}_d$ for the relatively free ordinary three-Engel algebra on d degree-one generators, as in Section 6.4. For $U = k^d$, use

$$R = \Lambda(U), \qquad P_d = \langle c_{ij} = c_{ji},\ c_{ii} = 0 \mid t_ic_{jk} + t_jc_{ik} + t_kc_{ij} = 0\rangle_R, \qquad B_d = \operatorname{Sym}^2_R(P_d).$$

The grading is $\deg t_i = 1$, $\deg c_{ij} = 2$. Write $P = P_d$ and $B = B_d$ when the rank is fixed. The tensor product defining B_d is over R; it is distinct from a tensor product over k carrying a diagonal derivation action. The arguments use the classical commutator and Koszul constructions [21, 20, 27].

W.1 The metabelian presentation and its top

In a metabelian three-Engel algebra, the actions on the derived ideal commute. The coefficient of two copies of x and one copy of z in $[y, x+\lambda z, x+\lambda z, x+\lambda z] = 0$ gives

$$[y, z, x, x] + [y, x, z, x] + [y, x, x, z] = 0.$$

The last two terms agree modulo H'', so $t_x^2(H'/H'') = 0$. Jacobi supplies the displayed triangle relation. Conversely $U \oplus P$ with $[u, v] = c(u, v)$ and $[w, u] = t_u w$ is metabelian and three-Engel: after the first bracket, the next two actions give $t_x^2 = 0$. Thus this is the free metabelian object in question.

For $d \geq 4$, $P_n = 0$ for $n > d + 1$. The only possible coefficient in degree $d + 2$ is $t_1 \cdots t_d$, and it kills every c_{ij}: multiply the triangle on i, j, k by $t_i t_j$ for a distinct k, then multiply by the remaining variables. Lower-support cases are covered by the same argument and the degree bound for the active ring. The degree-$d + 1$ elements reduce to

$$T_i = t_{[d]\setminus\{j\}} c_{ij} \quad (j \neq i).$$

Triangles identify the choices of j, and

$$\iota : P \to R \otimes U, \qquad c_{ij} \mapsto t_i \otimes x_j + t_j \otimes x_i$$

takes T_i to $(t_1 \cdots t_d) \otimes x_i$. Therefore $P_{d+1} \cong \det(U) \otimes U$ and the metabelian class ceiling is $d+1$ in these ranks. This argument also shows the finite presentation being used. The argument proves injectivity on the displayed graded top and does not require global injectivity.

W.2 The top symmetric square of the Koszul presentation

The following module statement has a direct, rank-uniform proof:

$$(B_d)_{d+2} \cong \det U \otimes \operatorname{Sym}^2 U \quad (d \geq 5), \qquad (B_d)_n = 0 \quad (n > d + 2,\ d \geq 4). \tag{297}$$

For the second assertion, degree $d + 3$ suffices since B_d is generated in degree four over R. Proceed by induction on the active support. Supports of size at most three have no degree above six: in rank three, a coefficient using all three variables kills every c_{ij} by in the presentation of P_d; ranks one and two are immediate. For active support $r \geq 4$, every exponent in a surviving symbol is at most three. In degree $r + 3$ only the patterns $(3, 2, 1, \ldots, 1)$ and $(2, 2, 2, 1, \ldots, 1)$ remain.

For the first pattern, with repeated indices i, j, the symbols are

$$D = c_{ij} \odot c_{ij}\, t_{[r]\setminus\{j\}}, \qquad X_k = c_{ij} \odot c_{ik}\, t_{[r]\setminus\{k\}} \quad (k \notin \{i, j\}).$$

Triangle (i, j, k) paired with c_{ij} gives $X_k = D$. Triangle (i, j, k) paired with c_{il}, with four distinct indices, gives $X_l = 0$: its other two terms have squared coefficients. Thus all these symbols vanish.

For the second pattern, with repeated indices i, j, k, the symbols are

$$A_i = c_{ij} \odot c_{ik}\, t_{[r]\setminus\{i\}} \quad \text{and cyclic variants,}$$

$$B_{i,a} = c_{ia} \odot c_{jk}\, t_{[r]\setminus\{a\}} \quad (a \notin \{i, j, k\}),$$

with cyclic variants. Triangle (i,j,k) paired with c_{ia} gives $B_{i,a}=0$; triangle (i,j,a) paired with c_{ik} gives $B_{j,a}=A_i$. A spare fourth index exists for $r\ge 4$, so all symbols vanish. This completes the induction and the upper-degree bound.

For degree $d+2$ and $d\ge 5$, that bound also shows that every component with fewer than d active variables is zero. The full-support patterns are precisely $(3,1,\dots,1)$ and $(2,2,1,\dots,1)$.

For the first pattern fix the tripled index i. Its symbols are

$$X_{jk}=c_{ij}\odot c_{ik}\,t_{[d]\setminus\{j,k\}}\quad (j,k\neq i,\ j\neq k).$$

Triangle (i,j,k) paired with c_{ih}, for distinct $j,k,h\neq i$, gives $X_{kh}=X_{jh}$. All symbols are therefore equal.

For the second pattern fix the doubled indices i,j and put $O=[d]\setminus\{i,j\}$. The symbols are

$$\begin{aligned} D&=c_{ij}\odot c_{ij}\,t_O,\\ A_a&=c_{ij}\odot c_{ia}\,t_{[d]\setminus\{i,a\}}, & B_a&=c_{ij}\odot c_{ja}\,t_{[d]\setminus\{j,a\}},\\ C_{ab}&=c_{ia}\odot c_{jb}\,t_{[d]\setminus\{a,b\}}, & E_{ab}&=c_{ij}\odot c_{ab}\,t_{[d]\setminus\{a,b\}}, \end{aligned}$$

where $a,b\in O$, $a\neq b$, and $E_{ab}=E_{ba}$. Triangles give

$$D=A_a+B_a,\qquad C_{ab}=A_a=B_b,\qquad E_{ab}=A_a+A_b.$$

With three distinct indices $a,b,c\in O$, triangle (i,a,b) paired with c_{jc} gives $C_{bc}=C_{ac}$, hence $A_b=A_a$. Since $|O|\ge 3$, all A_a,B_a,C_{ab} have one common value and $D=E_{ab}=0$. Each full-support component has dimension at most one.

To prove nonvanishing and naturality, use the Koszul map

$$\iota:P_d\to R\otimes U,\qquad c_{ij}\mapsto t_i\otimes x_j+t_j\otimes x_i.$$

Multiplication defines

$$\mu:\operatorname{Sym}^2_R(P_d)\to R\otimes\operatorname{Sym}^2 U.$$

In degree $d+2$ its target is $R_d\otimes\operatorname{Sym}^2 U=\det U\otimes\operatorname{Sym}^2 U$. The X_{jk} map to $(t_1\cdots t_d)\otimes x_i^2$ and the A_a map to $(t_1\cdots t_d)\otimes x_ix_j$. This proves (297), including the natural module identification. In characteristic two the symmetric square here is not silently identified with a space of divided powers.

W.3 Peeling an operator into the first outer input

Let $1\le r\le 3$, and assume $M_{r+1}=0$. Jacobi gives $[M_i,H']\subseteq M_{i+2}$, so

$$\Theta_r(w,v;a_1,\dots,a_r)=[\widetilde{w},\widetilde{v},a_1,\dots,a_r]$$

is well defined on $(H'/K)^2\times U^r$. It is balanced in the two derived legs. The partial Engel identity with inputs a,a,v, followed by $a_2,\dots,a_r$, gives

$$\Theta_r(t_aw,v;a,a_2,\dots,a_r)=0.$$

Indeed its other terms belong to M_{r+1}: the term involving $[w,a,a]\in K$ first brackets with $v\in H'$, landing in M_2, and then has $r-1$ outer actions. Polarization exchanges an operator with the first outer input.

Consequently, for $N\ge r+5$, there is a natural epimorphism

$$B_{N-r+1}\otimes U^{\otimes(r-1)}\twoheadrightarrow (M_r)_N. \tag{298}$$

For a square-free coefficient t_S, peel one factor $a \in S$ and send a symbol to

$$t_S(c_{ij} \odot c_{kl}) \otimes a_2 \otimes \cdots \otimes a_r \longmapsto \Theta_r(t_{S\setminus\{a\}} c_{ij}, c_{kl}; a, a_2, \ldots, a_r).$$

The exchange makes the choice independent, kills squared coefficients, and respects each triangle relation by peeling from its common coefficient, of degree $N - r - 4 \geq 1$. This proves a quotient map; it does not put an unproved residual R-action on an exterior quotient.

The second-layer ceiling. In the free graded algebra with $M_2 = 0$, the quotient by M_1 is centre-by-metabelian. Its quotient by the centre is metabelian and has class at most $d + 1$, hence it has class at most $d + 2$. Every homogeneous part in degree $N > d + 2$ is therefore in M_1. But (298) for $r = 1$ is an epimorphism $B_N \twoheadrightarrow (M_1)_N$, and (297) kills B_N. Thus the free depth-two object already has class at most $d+2$ for every $d \geq 4$. This elementary proof needs neither the full top central cohomology classification nor an external unrestricted Engel class bound.

W.4 An explicit collecting identity for the third layer

Work in characteristic two over a field with more than three elements. Brackets are left normed. Set $v = [p, a]$, $b = [p, q]$, and define

$$E_{21}(y; x, z) = [y, x, x, z] + [y, x, z, x] + [y, z, x, x],$$

$$E_{111}(y; x, z, t) = \sum_{\sigma \in S_3} [y, x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}], \qquad (x_1, x_2, x_3) = (x, z, t).$$

Both vanish in an ordinary three-Engel algebra: they are the complete cubic coefficients, whose extraction is valid because $3 < |k|$. The permutations in the second formula are indexed permutations, even when some substituted values happen to coincide.

Write

$$D(x_1, x_2, x_3, x_4; x_5, x_6, x_7) = [[[x_1, x_2], [x_3, x_4]], x_5, x_6, x_7].$$

Let $D_1, \ldots, D_{13}$ be the following displayed values or brackets of values. Each is zero if $[H'', H, H, H] = 0$.

1	$D(p, a, [p, q], q; h, r, r)$
2	$D(p, a, [p, q], q; r, r, h)$
3	$[D(p, q, p, a; r, r, q), h]$
4	$[D(p, a, h, r; p, q, q), r]$
5	$[D(p, a, h, r; q, q, p), r]$
6	$D([p, q], q, h, r; r, p, a)$
7	$D([p, q], q, h, r; r, a, p)$
8	$D(h, r, [p, a], r; p, q, q)$
9	$D(h, r, [p, a], r; q, q, p)$
10	$D(q, r, [p, q], r; p, a, h)$
11	$D(q, r, [p, q], r; a, p, h)$
12	$D(p, a, [p, q], r; q, r, h)$
13	$D(p, q, q, r; [p, a], r, h)$

The following equality holds already in the free ordinary Lie algebra in characteristic two:

$$[[[p, q], q, r], [p, a], h, r] = [E_{21}(v; r, b), q, h] + [E_{111}(r; q, v, [b, r]), h] + \sum_{i=1}^{13} D_i. \tag{299}$$

It is an explicit collecting identity, using only the Lie relations. It can be checked by expanding commutators; `verification/central_depth/verify_sparse_seed.py` does this independently, without a quotient-space or rank calculation. Under the three-Engel and depth-three hypotheses every summand on the right vanishes, proving the seed.

W.5 The third-layer source and its vanishing

In the free depth-three object put $T_d = (M_2(F_d)/M_3(F_d))_{d+3}$. The preceding module calculation and peeling construction give a natural epimorphism

$$\Psi_d : (\det U \otimes \mathrm{Sym}^2 U) \otimes U \twoheadrightarrow T_d, \qquad U = k^d, \quad d \geq 5. \tag{300}$$

Its ingredients are the metabelian presentation over $R = \Lambda(U)$, the central pairing

$$\Theta(w, v; a, b) = [\widetilde{w}, \widetilde{v}, a, b],$$

balance $\Theta(t_a w, v; h, z) = \Theta(w, t_a v; h, z)$, and the exchange

$$\Theta(t_a w, v; h, z) = \Theta(t_h w, v; a, z).$$

It identifies the degree-$d+2$ part of $\mathrm{Sym}^2_R(P_d)$ with $\det U \otimes \mathrm{Sym}^2 U$ via the Koszul multiplication

$$c_{ij} \longmapsto t_i \otimes x_j + t_j \otimes x_i.$$

Only the source in (300) is a symmetric tensor module: T_d is a quotient, whose extra relations must still be proved.

Fix distinct coordinate indices p, q, a, h, and put $S = [d] \setminus \{p, a\}$, with $h \in S \setminus \{q\}$. For any coordinate generator z, the word

$$P^{(d)}_{pq,z} = \Theta(t_{S\setminus\{h\}} c_{pq}, c_{pa}; h, z) \tag{301}$$

is the image of $(t_1 \cdots t_d) \otimes x_p x_q \otimes z$ under (300). This follows directly by multiplying the two Koszul images.

The rank-five case. Equation (299) gives the universal identity

$$[[[p, q], q, r], [p, a], h, r] = 0. \tag{302}$$

It holds in every ordinary three-Engel algebra with $M_3 = 0$. The five formal variables are p, q, a, h, r. The multidegree is $(2, 2, 1, 1, 2)$. The explicit identity above and its free-associative expansion certify all its Lie inputs.

For $d = 5$, (302) is exactly (301) with $z = r$ distinct from p, q. It kills the image of one tensor $x_p x_q \otimes x_r$ with three distinct indices. Permutation matrices kill all such coordinate tensors. A transvection $x_r \mapsto x_r + x_p$, fixing x_p, x_q, then kills $x_p x_q \otimes x_p$; similarly it kills the tensor with last input x_q. Hence every off-diagonal tensor $x_p x_q \otimes z$ is killed.

Finally, for any q, z, choose $p \notin \{q, z\}$. The transvection $x_p \mapsto x_p + x_q$ fixes z and has determinant one. Applied to the already killed tensor $x_p x_q \otimes z$, it kills $x_q^2 \otimes z$. Therefore the entire source of (300) is killed, and $T_5 = 0$.

Propagation to arbitrary rank. Assume $T_d = 0$ for some $d \geq 5$. Add a new generator y. For old p, q, z, choose a, h as in (301). Since (301) is zero in the relatively free depth-three algebra on the old generators, it remains zero after every substitution

$$x_p \longmapsto x_p + \lambda[x_p, y], \qquad \lambda \in k.$$

There are two occurrences of x_p in (301), or three if $z = x_p$. Thus this is a vector-valued polynomial in λ of degree at most three. A polynomial of degree at most three vanishing on all four elements of $\mathbb{F}_4$ has all coefficients zero. In particular its coefficient of λ vanishes. The field-size check is necessary here; an unqualified extraction from a degree-four polynomial on $\mathbb{F}_4$ would be invalid.

The variable p is absent from the operator coefficient $t_{S\setminus\{h\}}$ and from the first outer input h. Modulo $K = H''$, the two head variations are

$$\delta c_{pq} = t_y c_{pq} + t_p c_{qy}, \qquad \delta c_{pa} = t_y c_{pa} + t_p c_{ay}. \tag{303}$$

The two t_y terms cancel by balance. If $z = x_p$, its variation lies in H' and gives zero in the last outer slot of Θ. Changes of head lifts by K also give zero. The surviving coefficient is

$$\Theta(t_p t_{S\setminus\{h\}} c_{qy}, c_{pa}; h, z) + \Theta(t_p t_{S\setminus\{h\}} c_{pq}, c_{ay}; h, z). \tag{304}$$

Under the rank-$(d+1)$ symmetric-module identification, (304) is the image of

$$t_{[d]\setminus\{a\}}\big(c_{qy} \odot c_{pa} + c_{pq} \odot c_{ay}\big) \otimes z.$$

In the notation of the module calculation above these are respectively $C_{a,y}$ and $E_{a,y}$. Their images under Koszul multiplication are $x_p x_q$ and zero. Consequently (304) is exactly

$$\Psi_{d+1}((t_1 \cdots t_d t_y) \otimes x_p x_q \otimes z). \tag{305}$$

It vanishes by coefficient extraction.

For any off-diagonal tensor in rank $d+1$, one can choose the new generator y outside $\{p, q, z\}$ and apply this argument to the remaining d generators. Hence every off-diagonal tensor is killed. The determinant-one transvection used in Step 2 then kills every diagonal tensor. Surjectivity of (300) proves $T_{d+1} = 0$.

This is an induction on arbitrary rank, not an extrapolation from ranks five and six. Equations (303)–(305), including the zero E term, are proved in the rank-uniform module presentation.

The class bound. The previous second-central-layer theorem gives $\operatorname{cl} D_{d,2} = d+2$ for $d \geq 4$. Since $D_{d,3} \to D_{d,2}$ is central, $\operatorname{cl} D_{d,3} \leq d+3$. Any degree-$d+3$ component is contained in $M_2/M_3 = T_d$. The vanishing just proved therefore improves the upper bound to $d+2$. The previously constructed centre-by-metabelian top covers have exact generator rank d and class $d+2$, and already satisfy $H'' \subseteq Z(H)$. They give sharpness. Every at-most-d-generated algebra in the stated variety is a quotient of $D_{d,3}$, completing the third-layer case of Theorem 6.8.

W.6 The next-to-top balanced symmetric square

For $U = k^d$, write $R = \Lambda U$ and

$$P_d = \langle c_{ij} = c_{ji},\ c_{ii} = 0 \mid t_i c_{jk} + t_j c_{ik} + t_k c_{ij} = 0 \rangle_R, \qquad B_d = \operatorname{Sym}^2_R P_d.$$

Use $\deg t_i = 1$, $\deg c_{ij} = 2$. The Koszul multiplication $\mu : B_d \to R \otimes \operatorname{Sym}^2 U$ is induced by $c_{ij} \mapsto t_i x_j + t_j x_i$. For $d \geq 6$ it gives a natural identification

$$(B_d)_{d+1} \simeq \mathcal{K}_d := \ker\left(\Lambda^{d-1} U \otimes \operatorname{Sym}^2 U \xrightarrow{\partial} \det U \otimes U\right), \tag{306}$$

where

$$\partial(r \otimes xy) = (r \wedge x) \otimes y + (r \wedge y) \otimes x.$$

In particular

$$\dim \mathcal{K}_d = \frac{d(d-1)(d+2)}{2}.$$

This is a specific truncated-module computation using classical Koszul and symmetric-power constructions.

Proof of (306)**.** The bound (297) $(B_r)_n = 0$ for $n > r+2$, together with the rank-at-most-three calculation, shows that degree $d+1$ has support of size $d-1$ or d. On a support of size $d-1 \geq 5$ the previous top theorem identifies the component with $\det(k^{d-1}) \otimes \mathrm{Sym}^2(k^{d-1})$ and μ is injective.

For full support, the multidegree has one doubled index i. Put $O = [d]\setminus\{i\}$. Its generators are

$$X_{ab} = c_{ia}c_{ib}\, t_{[d]\setminus\{i,a,b\}}, \qquad Y_{a;bc} = c_{ia}c_{bc}\, t_{[d]\setminus\{a,b,c\}},$$

with distinct indices in O. Triangles give $Y_{c;ab} = X_{ac} + X_{bc}$ and, for four distinct indices in O,

$$X_{ac} + X_{ad} + X_{bc} + X_{bd} = 0. \tag{307}$$

The edge space on O modulo the rectangle relations (307) has dimension at most $|O|$. One way to see this is to dualize: edge weights satisfying every rectangle relation have the form $f_{ab} = u_a + u_b + c$, a space of dimension $|O|$. The images of X_{ab} are

$$Q_i + P_{i,a} + P_{i,b}, \qquad Q_i = t_{\widehat{i}} \otimes x_i^2, \quad P_{i,a} = t_{\widehat{a}} \otimes x_i x_a.$$

They have rank $|O|$: a triangle yields Q_i, and differences $P_{i,a} + P_{i,b}$ have rank $|O| - 1$. Thus μ is injective also on full support. Here $t_{\widehat{j}}$ is the product of all variables except t_j.

The kernel of ∂ has the following complete coordinate description:

- $t_{\widehat{j}} \otimes x_a x_b$ with $a, b \neq j$;
- for each doubled i, Q_i and $P_{i,a} + P_{i,b}$ with $a, b \neq i$.

Each target monomial is either one of the first items, Q_i, or $P_{i,a}$; on the latter ∂ is $(t_1 \cdots t_d) \otimes x_i$. This proves both the completeness of the displayed kernel and (306).

W.7 The depth-four source and nonlinear suspension

If $M_4 = 0$, set

$$\Theta_3(w, v; a, b, c) = [\widetilde{w}, \widetilde{v}, a, b, c].$$

As in the third-layer proof, $[M_j, H'] \subseteq M_{j+2}$ shows that the derived inputs factor through H'/H'' and the outer inputs through H/H'. Balance and the exchange of an operator with the first outer input follow from Jacobi and the partial Engel identity, now with two outer brackets appended. Peeling therefore gives, for $N \geq 8$,

$$(B_d)_{N-2} \otimes U \otimes U \twoheadrightarrow (M_3)_N.$$

At $N = d+3$, (306) gives the natural epimorphism

$$\Phi_d : \mathcal{K}_d \otimes U^{\otimes 2} \twoheadrightarrow T_d := (M_3/M_4)_{d+3}. \tag{308}$$

Assume $T_d = 0$ and add a new generator y. Substitute $x_i \mapsto x_i + \lambda[x_i, y]$ in any old source word. On H'/H'', the coefficient operators do not change, because $[x_i, y] \in H'$. The outer inputs do not change in H/H'. Only the two commutator heads change. Consequently the resulting expression in the central target has degree at most two in λ, regardless of the number of occurrences of i in the original source word. Coefficient extraction over $\mathbb{F}_4$ is therefore valid. This detail prevents a false extraction from a raw degree-four polynomial.

The coefficient of λ induces on Koszul images the map

$$\sigma_{i,y}(z) = (t_y x_i + t_i x_y)\frac{\partial z}{\partial x_i}, \tag{309}$$

with the coefficient $r \in R$ held fixed. For example,

$$\delta_i c_{ij} = t_y c_{ij} + t_i c_{jy}, \qquad \mu(\delta_i c_{ij}) = t_j(t_y x_i + t_i x_y).$$

Triangle relations are respected: their two residual terms cancel by the triangle on (j, k, y). Thus (309) defines a map on the balanced symmetric module, not just on a list of representatives. For u, v in the old generator space,

$$\Phi_{d+1}(\sigma_{i,y}(z) \otimes u \otimes v) = 0. \tag{310}$$

W.8 The suspension images span the next source

Let $n = d + 1 \geq 7$. Let Y be the $\mathrm{GL}_n(k)$-stable span of the tensors in (310), allowing every omitted coordinate y. It lies in $\ker \Phi_n$. We show $Y = \mathcal{K}_n \otimes U^{\otimes 2}$. Fix coordinate outer inputs u, v throughout each construction.

First consider $t_{\widehat{j}} \otimes x_a x_b$, where $a, b \neq j$ and $a \neq b$. Choose $y \notin \{j, a, b, u, v\}$. In the old space use $z = t_{\widehat{j,y}} \otimes x_a x_b$ and apply $\sigma_{a,y}$. Since its coefficient already contains t_a, (309) gives exactly $t_{\widehat{j}} \otimes x_a x_b$. Thus all these tensors belong to Y. Their square counterparts follow by a shear $x_a \mapsto x_a + x_b$, choosing $a \notin \{j, b, u, v\}$ so that the coefficient and the outer inputs are fixed. Hence all $t_{\widehat{j}} \otimes x_b^2$, $b \neq j$, are in Y as well.

For $P_{i,a} + P_{i,b}$ choose $y \notin \{i, a, b, u, v\}$. In the old space the element $Q_i + P_{i,a} + P_{i,b}$ belongs to $\mathcal{K}_d$. Equation (309) with respect to x_i kills Q_i and sends the other two terms to the desired $P_{i,a} + P_{i,b}$ in rank n. Thus all these differences are in Y.

Finally fix the desired Q_j, and choose $i \notin \{j, u, v\}$. The already available missing-support tensor $t_{\widehat{j}} \otimes x_i^2 \otimes u \otimes v$ transforms under $x_i \mapsto x_i + x_j$ into the sum of itself, $t_{\widehat{i}} \otimes x_j^2 \otimes u \otimes v$, and $(Q_i + Q_j) \otimes u \otimes v$. The first two are known to lie in Y, hence so does the last. Scaling x_i by $\omega \in \mathbb{F}_4 \setminus \mathbb{F}_2$ gives $(\omega^2 Q_i + \omega Q_j) \otimes u \otimes v$. The outer inputs are fixed by the choice of i. Since $\omega^2 + \omega = 1$, these two vectors separate Q_i and Q_j. This proves the spanning assertion without assuming semisimplicity. All choices are possible for $n \geq 7$.

Together with (308)–(310), this proves

$$T_d = 0 \implies T_{d+1} = 0 \qquad (d \geq 6). \tag{311}$$

W.9 The finite base and the rank-five boundary

The previous third-layer result gives $\mathrm{cl}\, D_{d,3} = d + 2$ for $d \geq 5$. Since $D_{d,4} \to D_{d,3}$ is central, $\mathrm{cl}\, D_{d,4} \leq d + 3$. In rank six the only full-support degree-nine patterns are

$$(4, 1, 1, 1, 1, 1), \quad (3, 2, 1, 1, 1, 1), \quad (2, 2, 2, 1, 1, 1).$$

Each corresponding capped graded universal extension calculation has zero top kernel. The final cochain matrices have respectively 458, 1299, and 2218 columns and full rank. The exact input tables and independent replay are supplied in `verification/central_depth/`. These are complete homogeneous cubic coefficients over $\mathbb{F}_4$, not samples at binary points.

Components supported on at most five generators vanish because $D_{5,4}$ has class at most eight. Therefore $T_6 = 0$. Induction (311) proves $T_d = 0$ for every $d \geq 6$, and centrality then improves the upper class bound from $d + 3$ to $d + 2$. The attaining models of Theorem 6.6 give equality.

For rank five, the supplied 178-dimensional capped model has

$$(\dim M_0, \ldots, \dim M_4) = (109, 60, 39, 11, 0)$$

and class eight. Its word $[[[p, q], p], [p, a], h, p, r]$ is nonzero. Its quotients by M_1, M_2, M_3 all have class seven and exact generator rank five. They give a concrete $7 \to 7 \to 7 \to 8$ chain of three central lifts. The kernels need not be one-dimensional. The same lower example and the central upper bound prove $\mathrm{cl}\, D_{5,4} = 8$ for the complete relatively free object.

Completeness of the finite starting step. At each weight, the replay enumerates all alternating cochains allowed by the multidegree cap, imposes Jacobi, every coefficient of the cubic Engel identity, and $M_4 = 0$, and computes their common kernel. It also checks that the actual added bracket coordinates are independent and span this entire kernel. Induction on weight therefore identifies the capped universal component, rather than merely verifying a model. The three degree-nine kernels vanish. The independent implementation uses the opposite pivot order and does not import the constructor. Complete cubic coefficients are required over $\mathbb{F}_4$; testing only binary-coordinate inputs would not certify the same variety.

The rank-four boundary. The supplied 40-, 41-, and 42-dimensional bracket tables form central quotients of exact rank four and classes $6, 6, 7$. Direct exact checks impose Jacobi and the complete cubic Engel coefficients. Their second derived ideals have central depths at most one, two, and three, respectively. The second-layer bound and the one-step central upper bound prove the rank-four entries of Proposition 6.9. These lower witnesses and the rank-five model establish every stated boundary value.

X Exact certificates and computational controls

The verification package accompanies the structural theory with independent exact calculations. Its rank certificates complete the finite extension-space steps in Theorem 12.2; the remaining controls reconstruct specified models, actions, complexes, and decoding maps. The global nilpotence and quadratic-stability theorems, all-class bounds, reconstruction theorems, and asymptotic laws have their general proofs in the preceding sections and appendices.

X.1 Encoding and completeness of the rank certificates

Use the binary basis $1, \omega$ of $\mathbb{F}_4$, with $\omega^2 = \omega + 1$. On a binary n-space a zero-constant quadratic scalar function has monomials x_i and $x_i x_j$, $i < j$. An $\mathbb{F}_4$-valued candidate q therefore has $2(n + \binom{n}{2}) = n(n+1)$ binary unknowns. The quotient map g is stored by the binary output vector of each ANF monomial, in the coordinate order of (115).

Each input x gives two binary rows from $q(g(x)) + \omega^2 q(x)$. This function has degree at most four. Its vanishing on all inputs of weights one through four is equivalent to its vanishing everywhere: the zero value is prescribed, and Boolean Möbius inversion recovers every coefficient of degree at most four. Thus these rows specify the complete extension equations.

For each $n \in \{4, 6, 8, 10, 12, 14, 16\}$, the certificate records the quotient coefficients and monomial order, independent rows with their originating input and output bit, a kernel basis, the splitting-change vectors $\delta(t)$, and the representatives of H_g in Table 3. The independent checker reconstructs every row from the quotient coefficients and checks:

(i) the stated rows are linearly independent over $\mathbb{F}_2$;

(ii) every kernel vector annihilates every required row, and row rank plus kernel dimension equals the number of unknowns;

(iii) the splitting-change vectors and stated quotient representatives together span exactly that kernel.

This proves both rank bounds and identifies the extension quotient. In particular the checker verifies

$$\dim_{\mathbb{F}_4} H_g = \frac{\dim_{\mathbb{F}_2} E_g - \dim_{\mathbb{F}_2} \operatorname{im} \delta}{2}$$

and checks that the named quotient basis has this size. The seven binary equation ranks are $14, 30, 58, 92, 132, 184, 244$; the corresponding $\mathbb{F}_4$-dimensions of H_g are $1, 2, 1, 1, 2, 1, 0$. The final zero certifies the terminal central-cover stage of the specified sixteen-dimensional solution. Its free two-generator property follows from Lemma 5.1. The general finite-stem theorem has the separate proof of Theorem 6.14.

X.2 Coverage of the exact controls

The original fifty controls are indexed below, with paths relative to `verification/`. Reports and `verification/REPLAY_GUIDE.txt` give model ranges, case counts, and commands.

Conclusion or interface	Program group	Exact coverage
Central descent and coordinate tables	`check_chain_rank_certificates.py`; root-level interfaces	Complete equation ranks, splitting quotients, transcription, scalar weights, and projective actions.
Metabelian equality cases and compatible extrema	`mu_extremizers/`; root-level family and orbit controls	Universal modules, cocycle actions, the complete 4^9 class-five partition, low-class orbits, masses, and centralizer interfaces.
Fixed surplus and varying fields	`near_minimum_controls.py`; `fixed_surplus/`	Specified models, nonlinear recovery, module and ordinary CE dimensions, and ordinary Lie comparisons over $\mathbb{F}_4, \mathbb{F}_8, \mathbb{F}_{16}$.
Growing populations and critical behavior	`surplus_transition/`; `critical_constant/`; `critical_window/`	Coordinate actions, stabilizer sums, nonsplit defect models, and uniform-bound arithmetic interfaces.
Socle defects and the discrete staircase	`action_cohomology/`; `critical_refinement/`	Module-endomorphism and CE equations, semilinear lifts, generation conditions, and critical score identities.
Intrinsic reconstruction and readout	`structural_reconstruction/`	Transported operations, graph and cyclic-cohomology interfaces, explicit lift equations, and all states of the fourteen-dimensional decoder.
General structure and constructions	`general_structure/`	Full Boolean coefficient identities for the 36-, 44-, and 90-dimensional models; the 22-dimensional obstruction; trace-current models and reduced-derived restricted structures.
Global stability, rank, and central covers	`general_structure/`	Complete ambient-cube and nonfactorization controls; reverse restricted realization, scalar saturation, commuting actions, and ordinary Engel top-cover equations.
Fixed sectors, cohomology, and geometry	`general_structure/`	Quadratic invariance equations, free binary CE ranks, orientation arrows, normal elements, centers, and quotient reduction.

Reports identify exhaustive and selected checks; the spread counts use the cited enumerations.

On its embedded 48-dimensional model, the stability control checks full Boolean composition, all 17 296 basis Jacobi triples, and the lower-central series. The 18 472 nonzero supports of weights at most three, tested on a 42-dimensional derived basis, recover all 775 824 Boolean coefficients of the cubic-linear expression $\mathrm{ad}_z^3(w)$; all vanish. The four-pair cycle has zero B-sum and nonzero central D-sum, distinguishing stability from factorization. Appendix P proves general stability.

The rank–class control reconstructs a two-generator ordinary $\mathbb{F}_4$-3-Engel input of binary dimension ten, its adjoint and restricted envelope of dimensions six and ten, and a six-dimensional binary subsolution Y. It checks all 1024 envelope fourth powers and all 64 inputs of Y, scalar saturation, class preservation with the adjoint's one-step loss, and two-point generation. Finite commuting quotients and integer bounds are checked separately; the general comparison has its proof in Appendix R.

The ordinary top-cover control checks all 65 536 $\mathbb{F}_4$-inputs and eight basis arguments of the class-five rank-two model, together with its complete CE and Engel central-extension equations. At ordinary ranks $d = 4, 5$, it reconstructs the raw Jacobi modules and all top-weight CE and $\mathbb{F}_4$-cubic coefficients, obtaining symmetric top spaces of dimensions $10, 15$. The one-dimensional lifts have exact ranks $4, 5$ and classes $6, 7$. These are finite controls for the general top-cover and centre-by-metabelian results in Appendix R.

X.3 Replaying the package

With CPython 3.14.6 and SymPy 1.14.0, run from the project directory:

```
python verification/run_all.py --output-dir ../PaperG_replay
```

Use a new or empty destination. The runner records logs and `replay_summary.json`, checks source JSON hashes, and signals failures. Use `--no-write` for a temporary copy. Dependencies are pinned in `verification/requirements.txt`; commands and ranges are in `verification/REPLAY_GUIDE.txt`.
Attributions: `general_structure/SOURCE_ATTRIBUTION.txt` under `verification/`. The package accompanies ordinary proofs; no Lean or other proof-assistant formalization is supplied.

X.4 Central-depth identities and the rank-six base

The separate `verification/central_depth/` directory supplies the inputs and independent checks for Appendix W. Its six checking programs were replayed with CPython 3.12.14 using only the standard library. They are run separately from the original fifty-program entry point above; the directory README lists all eight invocations, including the three universal multidegree systems. The five-variable seed has a 22-node proof: two complete cubic Engel coefficients, thirteen depth relations, six bracket operations, and one sum. Its free-associative replay also rejects three deliberately damaged certificates. The positive rank-five model is checked by its complete cubic coefficients, Jacobi equations, lower-central series, and the three displayed central quotients.

For each of the three rank-six multidegree caps, the universal replay checks every extension weight from two through nine. It verifies both the constraint ranks and that all stored new bracket coordinates form the full kernel. This certifies the finite base used in the general fourth-layer argument. The suspension program checks the displayed spanning constructions in ranks seven through ten; the arbitrary-rank step is the symbolic proof. The Jacobian control checks a limitation of that method and is not a claim of a fifth-layer class theorem. The README gives the replay commands and identifies each report.

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